

Supplementary Table S2.1a. Benchmark dataset of STEC genomes used for validation of stx detection pipelines.

The 200 benchmark genomes with their identifiers, accessions and assembly characteristics, the expected STxOP_nt reference operons carried by each genome, and the expected Stx subtypes against which every workflow was scored.

Expected subtypes counts each subtype once per genome irrespective of copy number. Operon-level detail supporting these profiles is given in Supplementary Table S2.1b.

200 rows, 16 columns.

Fastq ID	Fasta ID	Assembly level	Source	CFIA ID	BioSample	GenBank assembly accession	Illumina reads accession	Contigs (n)	Genome size (bp)	Illumina coverage (x)	Expected STxOp_nt operons	Expected subtypes	Expected subtypes (n)	Disrupted operon note	Curation note
2018-SEQ-1757	2021-MIN-0249	Complete	CFIA	CFIAFB20190172	SAMN26419929	GCA_032267725.1	SRR18215048	4	5090190	128.0092	StxOp933_Stx2i_43	Stx2i	1		
2025-SEQ-1251	2023-MIN-0010	Complete	CFIA	CFIAFB20130254	SAMN09668424	GCA_012140745.1	SRR7629286	4		88.45	StxOp3_Stx1a_9; StxOp1089_Stx2a_1	Stx1a; Stx2a	2		
2025-SEQ-1502	2022-MIN-0034	Complete	CFIA	CFIAFB20190266	SAMN40646271	GCA_042949985.1	SRR28490834	9	5252504	154.4	StxOp432_Stx2a_30; StxOp943_Stx2k_40	Stx2a; Stx2k	2		
2025-SEQ-1503	2019-MIN-0073	hybrid	CFIA	CFIAFB20180611	SAMN40646226	GCA_042968915.2	SRR28490853	21	5591229	184.04	StxOp13_Stx1a_19; StxOp399_Stx1c_12; StxOp673_Stx2b_113	Stx1a; Stx1c; Stx2b	3		2 copies of vtx2b
2025-SEQ-1504	2023-MIN-0001	hybrid	CFIA	CFIAFB20180485	SAMN13334854	GCA_027232265.2	SRR22858738	17	5624422	167.63	StxOp399_Stx1c_12; StxOp1348_Stx2b_110	Stx1c; Stx2b	2		
2025-SEQ-1505	2021-MIN-0009	hybrid	CFIA	CFIAFB20190239	SAMN40646259	GCA_042949905.1	SRR28490847	12	5277841	124.24	StxOp933_Stx2i_43	Stx2i	1		
2025-SEQ-1506	2022-MIN-0057	Complete	CFIA	CFIAFB20190167	SAMN40646276	GCA_042968935.1	SRR28490829	6	5347647	176.22	StxOp2108_Stx1c_27	Stx1c	1		
2025-SEQ-1507	2023-MIN-0017	hybrid	CFIA	CFIAFB20180434	SAMN11373407	GCA_026889795.2	SRR22715052	16	5589391	158.76	StxOp13_Stx1a_19; StxOp675_Stx2b_116; StxOp684_Stx2b_221	Stx1a; Stx2b	2		2 alleles of stx2b
2025-SEQ-1509	2022-MIN-0070	hybrid	CFIA	CFIAFB20190168	SAMN40646267	GCA_042951305.1	SRR28490839	11	5491462	151.66	StxOp1035_Stx1c_12	Stx1c	1		
2025-SEQ-1619	GCA_005037795.2	Complete	NCBI	OLC0711	SAMN08768108	GCA_005037795.2	SRR7896254	2	5470829	51.08	StxOp1577_Stx1a_9	Stx1a	1		
2025-SEQ-1621	GCA_005037805.2	Complete	NCBI	OLC0715	SAMN08768106	GCA_005037805.2	SRR7896256	1	5132754	188.59	StxOp3_Stx1a_9; StxOp1089_Stx2a_1	Stx1a; Stx2a	2		
2025-SEQ-1629	2021-MIN-0136	hybrid	CFIA	OLC0726						221.77	StxOp439_Stx2a_107; StxOp842_Stx2d_115	Stx2a; Stx2d	2		
2025-SEQ-1632	GCA_005037725.2	Complete	NCBI	OLC0731	SAMN08768102	GCA_005037725.2	SRR7896252	4	5803315	157.03	StxOp9_Stx1a_15; StxOp429_Stx2a_16	Stx1a; Stx2a	2		
2025-SEQ-1634	GCA_018986775.1	Complete	NCBI	OLC0733	SAMN19011648	GCA_018986775.1	SRR14750048	4	5163097	120.37	StxOp954_Stx2o_41	Stx2o	1		
2025-SEQ-1638	2021-MIN-0144	Complete	CFIA	CFIAFB20080070	SAMN13333400	GCA_027230405.1	SRR22858851			90.89	StxOp702_Stx2c_17	Stx2c	1		
2025-SEQ-1642	2021-MIN-0255	hybrid	CFIA	OLC0982						205.39	StxOp2210_Stx1d_4	Stx1d	1		
2025-SEQ-1666	2019-MIN-0085	hybrid	CFIA	OLC0989	SAMN04420140	TBD	SRR6061329	12		274.3064	StxOp674_Stx2b_114; StxOp4184_Stx2c_724; StxOp1494_Stx2d_115	Stx2b; Stx2c; Stx2d	3		
2025-SEQ-1667	2021-MIN-0169	Complete	CFIA	OLC0992						97.3091	StxOp2209_Stx1d_4	Stx1d	1		
2025-SEQ-1668	2019-MIN-0086	hybrid	CFIA	OLC0993	SAMN04420144	TBD	SRR6061351	31		180.4317	StxOp399_Stx1c_12; StxOp419_Stx2a_1; StxOp1355_Stx2b_194; StxOp670_Stx2b_109	Stx1c; Stx2a; Stx2b	3		2 copies of vtx2b, different variants
2025-SEQ-1675	2017-MER-0367	short read	CFIA	OLC1000						82.7039	StxOp1039_Stx1d_4	Stx1d	1		
2025-SEQ-1688	GCA_018986815.1	Complete	NCBI	CFIAFB20120266	SAMN19011646	GCA_018986815.1	SRR14750046	3	5041163	111.43	StxOp932_Stx2h_49	Stx2h	1		
2025-SEQ-1691	2022-MIN-0050	hybrid	CFIA	OLC1061						118.43	StxOp399_Stx1c_12	Stx1c	1		
2025-SEQ-1703	2025-MIN-0001	Complete	CFIA	OLC1114						100.54	StxOp1043_Stx2a_1; StxOp702_Stx2c_17	Stx2a; Stx2c	2		
2025-SEQ-1763	2022-MIN-0028	hybrid	CFIA	OLC1265						191.36	StxOp3298_Stx2b_23; StxOp4389_Stx2d_112	Stx2b; Stx2d	2		
2025-SEQ-1771	GCA_018986795.1	Complete	NCBI	CFIAFB20140388	SAMN19011647	GCA_018986795.1	SRR14750047	5	5191720	129.12	StxOp936_Stx2j_104	Stx2j	1		
2025-SEQ-1785	2023-MIN-0015	Complete	CFIA	CFIAFB20170267	SAMN13334743	GCA_027234145.1	SRR22858698	6	5331905	102.17	StxOp927_Stx2g_147	Stx2g	1		

Fastq ID	Fasta ID	Assembly level	Source	CFIA ID	BioSample	GenBank assembly accession	Illumina reads accession	Contigs (n)	Genome size (bp)	Illumina coverage (x)	Expected STxOp_nt operons	Expected subtypes	Expected subtypes (n)	Disrupted operon note	Curation note
2025-SEQ-1786	2021-MIN-0153	hybrid	CFIA	CFIAFB20170337	SAMN11373508	GCA_026888995.2	SRR22715048	15	5064535	67.47	StxOp3298_Stx2b_23; StxOp1486_Stx2d_112	Stx2b; Stx2d	2		Only 2d found with entire DB, but 2b, 2d found with 100% identity when only these two alleles are used in KMA DB
2025-SEQ-1788	2023-MIN-0013	hybrid	CFIA	CFIAFB20180484	SAMN11373510	GCA_026888985.1	SRR22715046	21	5317558	94.5	StxOp826_Stx2d_9	Stx2d	1		
2025-SEQ-1789	2022-MIN-0053	Complete	CFIA	CFIAFB20180503	SAMN13334860	GCA_027232025.1	SRR22858741	4	5068790	156.77	StxOp1036_Stx1c_12	Stx1c	1		
2025-SEQ-1790	2021-MIN-0250	Complete	CFIA	CFIAFB20180511	SAMN11373501	GCA_027656605.1	SRR18215053	3	5083333	203.03	StxOp933_Stx2i_43	Stx2i	1		
2025-SEQ-1791	2022-MIN-0054	Complete	CFIA	CFIAFB20180540	SAMN13334869	GCA_027231625.1	SRR22858710	3	4854273	181.79	StxOp1571_Stx2e_146	Stx2e	1		
2025-SEQ-1792	2022-MIN-0056	Complete	CFIA	CFIAFB20190157	SAMN26419928	GCA_032267685.1	SRR18215049	7	5404784	153.82	StxOp399_Stx1c_12; StxOp3320_Stx2b_101	Stx1c; Stx2b	2		
2025-SEQ-1793	2022-MIN-0071	Complete	CFIA	CFIAFB20190194	SAMN40734924	GCA_042986525.1	SRR28538989	5	5147933	161.76	StxOp933_Stx2i_43	Stx2i	1		
2025-SEQ-1795	2017-PAC-0030	Complete	CFIA	CFIAFB20150153	SAMN10746418	GCA_012017125.1	SRR8448490	4	5437080	186.26	StxOp1089_Stx2a_1	Stx2a	1		
2025-SEQ-1796	GCA_018966965.1	Complete	NCBI	CFIAFB20160171	SAMN13334712	GCA_018966965.1	SRR22858824	5	5544299	134.65	StxOp3_Stx1a_9; StxOp1089_Stx2a_1	Stx1a	1	Transposon insertion within stx2a subunit A gene	
2025-SEQ-1798	16-9255	Complete	NCBI	CFIAFB20170192	SAMN07339706	GCA_002844165.1	SRR5816151	2	5479471	165.71	StxOp419_Stx2a_1	Stx2a	1		Use closed genome/reads from published strain
2025-SEQ-1800	2022-MIN-0055	Complete	CFIA	CFIAFB20180560	SAMN13334877	GCA_027232345.2	SRR22858713	5	5020818	151.48	StxOp3625_Stx2b_215; StxOp1448_Stx2c_152	Stx2b; Stx2c	2		
2025-SEQ-1964	2021-MIN-0170	hybrid	CFIA	OLC0995						177.6067	StxOp3298_Stx2b_23; StxOp4178_Stx2c_569	Stx2b; Stx2c	2		
DRR178135	GCA_009002845.1	Complete	NCBI		SAMD00166593	GCA_009002845.1	DRR178135	2	5579570	46.3767	StxOp1577_Stx1a_9; StxOp1089_Stx2a_1	Stx1a; Stx2a	2		2 copies of Stx2a
DRR401569	GCA_026001525.1	Complete	NCBI		SAMD00523005	GCA_026001525.1	DRR401569	3	5428456	321.1602	StxOp22_Stx1a_35; StxOp1089_Stx2a_1; StxOp717_Stx2c_130	Stx1a; Stx2a; Stx2c	3		
DRR401577	GCA_026001745.1	Complete	NCBI		SAMD00523013	GCA_026001745.1	DRR401577	2	5382570	95.9963	StxOp1577_Stx1a_9; StxOp1089_Stx2a_1	Stx1a; Stx2a	2		
ERR3330139	GCA_016667705.1	Contig	NCBI		SAMEA5611643	GCA_016667705.1	ERR3330139	132	5264794	72.4161	StxOp922_Stx2f_726	Stx2f	1		
GCF_002303275	GCA_002303275.1	Complete	NCBI		SAMN05831100	GCA_002303275.1	TBD	2	5087499	88.7469	StxOp417_Stx1e_13	Stx1e	1		
SRR10541011	GCA_011822915.1	Contig	NCBI		SAMN13405539	GCA_011822915.1	SRR10541011	76	4824624	220.2443	StxOp932_Stx2h_49	Stx2h	1		
SRR10911972	GCA_012661225.1	Contig	NCBI		SAMN13880036	GCA_012661225.1	SRR10911972	120	5072842	100.772	StxOp948_Stx2m_48	Stx2m	1		
SRR11485261	GCA_020809385.1	Complete	NCBI		SAMN13045723	GCA_020809385.1	SRR11485261	3	5216538	242.9945	StxOp1301_Stx2a_654; StxOp846_Stx2d_144	Stx2a; Stx2d	2		
SRR11550988	GCA_014815785.1	Contig	NCBI		SAMN14603490	GCA_014815785.1	SRR11550988	88	5040843	38.4179	StxOp399_Stx1c_12	Stx1c	1		
SRR12207194	GCA_014058445.2	Complete	NCBI		SAMN15423745	GCA_014058445.2	SRR12207194	7	5936184	125.3667	StxOp9_Stx1a_15; StxOp429_Stx2a_16	Stx1a; Stx2a	2		
SRR12506954	GCA_014281005.1	Contig	NCBI		SAMN15892896	GCA_014281005.1	SRR12506954	98	5172491	67.0697	StxOp653_Stx2b_23; StxOp1515_Stx2d_115	Stx2b; Stx2d	2		
SRR1272511	GCA_012833475.1	Contig	NCBI		SAMN02566905	GCA_012833475.1	SRR1272511	39	4958904	100.7919	StxOp1580_Stx1a_9	Stx1a	1		
SRR1272522	GCA_012833535.1	Contig	NCBI		SAMN02566895	GCA_012833535.1	SRR1272522	67	4923420	105.462	StxOp432_Stx2a_30	Stx2a	1		
SRR1272883	GCA_012833235.1	Contig	NCBI		SAMN02566902	GCA_012833235.1	SRR1272883	55	4984877	135.9264	StxOp1365_Stx2c_7	Stx2c	1		
SRR12844969	GCA_014656625.1	Complete	NCBI		SAMN16077358	GCA_014656625.1	SRR12844969	2	5040622	210.6944	StxOp829_Stx2d_27	Stx2d	1		
SRR13068663	GCA_015529125.1	Contig	NCBI		SAMN16811693	GCA_015529125.1	SRR13068663	72	5252908	74.318	StxOp1711_Stx1a_9; StxOp1157_Stx2a_6	Stx1a; Stx2a	2		
SRR13132130	GCA_015923545.1	Contig	NCBI		SAMN16911778	GCA_015923545.1	SRR13132130	38	5062407	109.516	StxOp953_Stx2n_126	Stx2n	1		
SRR13209800	GCA_016404485.2	Complete	NCBI		SAMN16993346	GCA_016404485.2	SRR13209800	8	5457891	57.5427	StxOp13_Stx1a_19; StxOp675_Stx2b_116	Stx1a; Stx2b	2		

Fastq ID	Fasta ID	Assembly level	Source	CFIA ID	BioSample	GenBank assembly accession	Illumina reads accession	Contigs (n)	Genome size (bp)	Illumina coverage (x)	Expected STxOp_nt operons	Expected subtypes	Expected subtypes (n)	Disrupted operon note	Curation note
SRR13209817	GCA_016403875.2	Complete	NCBI		SAMN16993373	GCA_016403875.2	SRR13209817	2	5132179	47.7535	StxOp1510_Stx2d_115; StxOp1512_Stx2d_115	Stx2d	1		2 copies Stx2d
SRR13209822	GCA_016403975.2	Complete	NCBI		SAMN16993369	GCA_016403975.2	SRR13209822	3	5177967	76.7128	StxOp1541_Stx2d_134	Stx2d	1		
SRR13240070	GCA_015978445.1	Contig	NCBI		SAMN17058657	GCA_015978445.1	SRR13240070	85	4939135	100.4317	StxOp4_Stx1a_9; StxOp1173_Stx2a_16; StxOp848_Stx2d_164	Stx1a; Stx2a; Stx2d	3		
SRR13428524	GCA_016834695.1	Complete	NCBI		SAMN14648871	GCA_016834695.1	SRR13428524	4	5715173	77.7536	StxOp3_Stx1a_9; StxOp702_Stx2c_17	Stx1a; Stx2c	2		
SRR13450255	GCA_021766875.1	Contig	NCBI		SAMN17367908	GCA_021766875.1	SRR13450255	67	4845555	39.1618	StxOp1173_Stx2a_16; StxOp1228_Stx2a_18	Stx2a	1		2 variants of Stx2a--both guided assembly
SRR13557757	GCA_023278265.1	Complete	NCBI		SAMN18287925	GCA_023278265.1	SRR13557757	1	4656824	134.0519	StxOp1089_Stx2a_1	Stx2a	1		
SRR13605977	GCA_017172955.1	Contig	NCBI		SAMN17730384	GCA_017172955.1	SRR13605977	113	5457244	49.6617	StxOp1594_Stx1a_9; StxOp1574_Stx2g_39	Stx1a; Stx2g	2		
SRR13674450	GCA_017081175.1	Complete	NCBI		SAMN16357494	GCA_017081175.1	SRR13674450	5	5006697	622.4083	StxOp1361_Stx2c_7	Stx2c	1		
SRR13796384	GCA_025991315.1	Contig	NCBI		SAMN18075802	GCA_025991315.1	SRR13796384	110	5519091	63.1096	StxOp1041_Stx1d_4	Stx1d	1		
SRR13957227	GCA_017740375.1	Contig	NCBI		SAMN18290598	GCA_017740375.1	SRR13957227	60	4912197	121.606	StxOp913_Stx2e_102	Stx2e	1		
SRR14071719	GCA_017588055.1	Contig	NCBI		SAMN18492553	GCA_017588055.1	SRR14071719	52	5150237	49.4748	StxOp955_Stx2o_545	Stx2o	1		
SRR14750050	GCA_018986835.1	Complete	NCBI		SAMN19011645	GCA_018986835.1	SRR14750050	2	4917512	85.2846	StxOp432_Stx2a_30	Stx2a	1		
SRR14898862	GCA_019020085.1	Contig	NCBI		SAMN19841336	GCA_019020085.1	SRR14898862	137	5061658	82.9559	StxOp415_Stx1d_4	Stx1d	1		
SRR14934092	GCA_019431315.1	Complete	NCBI		SAMN20003582	GCA_019431315.1	SRR14934092	2	5107075	89.8338	StxOp898_Stx2e_25	Stx2e	1		
SRR14934097	GCA_019431335.1	Complete	NCBI		SAMN20003598	GCA_019431335.1	SRR14934097	3	5230054	105.8904	StxOp898_Stx2e_25	Stx2e	1		
SRR15076491	GCA_019184365.1	Contig	NCBI		SAMN20125940	GCA_019184365.1	SRR15076491	79	5161028	110.2187	StxOp403_Stx1c_26; StxOp1339_Stx2b_109	Stx1c; Stx2b	2		
SRR15497538	GCA_021534925.1	Complete	NCBI		SAMN20824181	GCA_021534925.1	SRR15497538	2	4953042	242.4247	StxOp914_Stx2e_133; StxOp916_Stx2e_146	Stx2e	1		2 different alleles of 2e
SRR15949660	GCA_020041845.1	Contig	NCBI		SAMN21495183	GCA_020041845.1	SRR15949660	126	5494012	190.0605	StxOp399_Stx1c_12	Stx1c	1		
SRR16071833	GCA_021535005.1	Complete	NCBI		SAMN21841571	GCA_021535005.1	SRR16071833	3	5238876	325.9091	StxOp915_Stx2e_136	Stx2e	1		
SRR1610028	GCA_003018775.1	Complete	NCBI		SAMN03098211	GCA_003018775.1	SRR1610028	2	5539654	36.4008	StxOp419_Stx2a_1	Stx2a	1		
SRR1724591	GCA_003018835.2	Complete	NCBI		SAMN03265976	GCA_003018835.2	SRR1724591	2	5380119	156.9626	StxOp1577_Stx1a_9	Stx1a	1		
SRR1773021	GCA_012826915.1	Contig	NCBI		SAMN03293362	GCA_012826915.1	SRR1773021	62	5136156	131.0997	StxOp1575_Stx2l_21	Stx2l	1		
SRR17838893	GCA_021763325.1	Contig	NCBI		SAMN25524145	GCA_021763325.1	SRR17838893	68	5272626	94.2551	StxOp1157_Stx2a_6; StxOp1283_Stx2a_170	Stx2a	1		2 variants Stx2a
SRR17923069	GCA_022051165.1	Contig	NCBI		SAMN25709003	GCA_022051165.1	SRR17923069	93	5103422	154.4078	StxOp4737_Stx2l_465	Stx2l	1		
SRR18191506	GCA_026135385.1	Complete	NCBI		SAMN26036515	GCA_026135385.1	SRR18191506	4	5348285	50.2982	StxOp4_Stx1a_9; StxOp1165_Stx2a_16; StxOp1512_Stx2d_115	Stx1a; Stx2a; Stx2d	3		
SRR18191510	GCA_026135105.1	Complete	NCBI		SAMN26036540	GCA_026135105.1	SRR18191510	9	5822451	45.1949	StxOp9_Stx1a_15	Stx1a	1		
SRR18191519	GCA_026135215.1	Contig	NCBI		SAMN26036536	GCA_026135215.1	SRR18191519	22	5769060	72.0536	StxOp1577_Stx1a_9	Stx1a	1		
SRR18191528	GCA_026135465.1	Complete	NCBI		SAMN26036514	GCA_026135465.1	SRR18191528	6	5257865	50.7798	StxOp1708_Stx1a_9; StxOp1283_Stx2a_170	Stx1a; Stx2a	2		
SRR18191538	GCA_026134365.1	Complete	NCBI		SAMN26036527	GCA_026134365.1	SRR18191538	2	5834018	52.6801	StxOp1577_Stx1a_9	Stx1a	1		
SRR18191541	GCA_026135265.1	Complete	NCBI		SAMN26036526	GCA_026135265.1	SRR18191541	4	5268782	34.9116	StxOp1036_Stx1c_12	Stx1c	1		
SRR18191544	GCA_025988925.1	Complete	NCBI		SAMN26036519	GCA_025988925.1	SRR18191544	5	4943051	53.9007	StxOp4_Stx1a_9; StxOp1168_Stx2a_16	Stx1a; Stx2a	2		
SRR18191549	GCA_025988125.1	Complete	NCBI		SAMN26036579	GCA_025988125.1	SRR18191549	4	5262159	61.4148	StxOp840_Stx2d_99	Stx2d	1		
SRR18191551	GCA_026134405.1	Complete	NCBI		SAMN26036578	GCA_026134405.1	SRR18191551	3	5163781	27.5265	StxOp1036_Stx1c_12	Stx1c	1		
SRR18191553	GCA_025988205.1	Complete	NCBI		SAMN26036577	GCA_025988205.1	SRR18191553	2	5345303	56.652	StxOp71_Stx1a_89; StxOp3298_Stx2b_23; StxOp883_Stx2d_471	Stx1a; Stx2d	2	Frameshift mutation in stx2b subunit A gene (4725483-4726711)	2 copies of Stx2d
SRR18191566	GCA_025988985.1	Complete	NCBI		SAMN26036518	GCA_025988985.1	SRR18191566	5	4987186	44.4649	StxOp457_Stx2a_181	Stx2a	1		

Fastq ID	Fasta ID	Assembly level	Source	CFIA ID	BioSample	GenBank assembly accession	Illumina reads accession	Contigs (n)	Genome size (bp)	Illumina coverage (x)	Expected STxOp_nt operons	Expected subtypes	Expected subtypes (n)	Disrupted operon note	Curation note
SRR18191571	GCA_025988425.1	Complete	NCBI		SAMN26036556	GCA_025988425.1	SRR18191571	2	5430616	34.3785	StxOp1594_Stx1a_9	Stx1a	1		
SRR18191580	GCA_025989005.1	Complete	NCBI		SAMN26036516	GCA_025989005.1	SRR18191580	10	5300421	58.9267	StxOp1040_Stx1d_4	Stx1d	1		
SRR18191598	GCA_026134965.1	Complete	NCBI		SAMN26036558	GCA_026134965.1	SRR18191598	2	5644625	34.7065	StxOp702_Stx2c_17	Stx2c	1		
SRR18191601	GCA_025988955.1	Complete	NCBI		SAMN26036524	GCA_025988955.1	SRR18191601	4	5179346	48.129	StxOp826_Stx2d_9	Stx2d	1		
SRR18191603	GCA_025988885.1	Complete	NCBI		SAMN26036523	GCA_025988885.1	SRR18191603	3	5132730	58.2804	StxOp933_Stx2i_43	Stx2i	1		
SRR18191605	GCA_026134415.1	Complete	NCBI		SAMN26036522	GCA_026134415.1	SRR18191605	4	5087513	53.0513	StxOp1168_Stx2a_16	Stx2a	1		
SRR18191606	GCA_025989075.1	Complete	NCBI		SAMN26036513	GCA_025989075.1	SRR18191606	3	5316065	39.7857	StxOp1035_Stx1c_12	Stx1c	1		
SRR18191610	GCA_025988905.1	Complete	NCBI		SAMN26036520	GCA_025988905.1	SRR18191610	6	5144288	33.5266	StxOp1321_Stx2a_6	Stx2a	1		2 copies of Stx2a
SRR18191616	GCA_025988145.1	Complete	NCBI		SAMN26036585	GCA_025988145.1	SRR18191616	8	5575909	56.0893	StxOp399_Stx1c_12; StxOp1339_Stx2b_109	Stx1c; Stx2b	2		
SRR18191618	GCA_026134125.1	Complete	NCBI		SAMN26036584	GCA_026134125.1	SRR18191618	3	5314568	25.2296	StxOp1530_Stx2d_115	Stx2d	1		
SRR18191628	GCA_025988225.1	Complete	NCBI		SAMN26036569	GCA_025988225.1	SRR18191628	3	5094287	54.0093	StxOp4_Stx1a_9; StxOp1168_Stx2a_16	Stx1a; Stx2a	2		
SRR18191635	GCA_026134865.1	Complete	NCBI		SAMN26036566	GCA_026134865.1	SRR18191635	1	5313712	55.4946	StxOp3298_Stx2b_23; StxOp1494_Stx2d_115	Stx2d	1	Frameshift mutation in stx2b subunit A gene (1741851-1743088)	Contains Stx2b with frameshift mutation (1741851-1743088)
SRR18191644	GCA_026134385.1	Complete	NCBI		SAMN26036517	GCA_026134385.1	SRR18191644	3	5602836	45.5898	StxOp13_Stx1a_19; StxOp1339_Stx2b_109; StxOp675_Stx2b_116	Stx1a; Stx2b	2		2 different alleles of Stx2b
SRR1871147	GCA_012824895.1	Contig	NCBI		SAMN03381320	GCA_012824895.1	SRR1871147	40	4987210	80.6372	StxOp1157_Stx2a_6	Stx2a	1		2 copies Stx2a
SRR19368884	GCA_023537105.1	Contig	NCBI		SAMN28610721	GCA_023537105.1	SRR19368884	40	5114780	85.165	StxOp1540_Stx2d_132	Stx2d	1		
SRR1944449	GCA_012823715.1	Contig	NCBI		SAMN03436236	GCA_012823715.1	SRR1944449	72	5152473	109.1344	StxOp716_Stx2c_106	Stx2c	1		
SRR1950298	GCA_001188105.2	Contig	NCBI		SAMN03255436	GCA_001188105.2	SRR1950298	22	5370171	78.5618	StxOp675_Stx2b_116	Stx2b	1		
SRR1981027	GCA_014447175.1	Contig	NCBI		SAMN03492057	GCA_014447175.1	SRR1981027	87	5024446	86.2295	StxOp1344_Stx2b_109	Stx2b	1		
SRR1981699	GCA_014441045.1	Contig	NCBI		SAMN03492677	GCA_014441045.1	SRR1981699	68	5060556	122.7296	StxOp429_Stx2a_16	Stx2a	1		
SRR1999106	GCA_003018295.1	Complete	NCBI		SAMN03569084	GCA_003018295.1	SRR1999106	3	5431970	69.6157	StxOp1580_Stx1a_9	Stx1a	1		
SRR1999111	GCA_012819655.1	Contig	NCBI		SAMN03569089	GCA_012819655.1	SRR1999111	90	5068439	72.4069	StxOp1133_Stx2a_6	Stx2a	1		
SRR1999883	GCA_012818575.1	Contig	NCBI		SAMN03569523	GCA_012818575.1	SRR1999883	80	5090304	136.7707	StxOp1173_Stx2a_16	Stx2a	1		
SRR1999997	GCA_015518365.1	Contig	NCBI		SAMN03569583	GCA_015518365.1	SRR1999997	125	5335433	133.6197	StxOp923_Stx2g_26	Stx2g	1		
SRR2015112	GCA_003018055.1	Complete	NCBI		SAMN02991246	GCA_003018055.1	SRR2015112	3	5823288	127.3155	StxOp1577_Stx1a_9	Stx1a	1		
SRR2015125	GCA_003018655.1	Complete	NCBI		SAMN02991250	GCA_003018655.1	SRR2015125	1	5167951	40.3419	StxOp4_Stx1a_9	Stx1a	1		
SRR2015133	GCA_012696945.1	Contig	NCBI		SAMN02991263	GCA_012696945.1	SRR2015133	89	4933073	84.0825	StxOp1448_Stx2c_152	Stx2c	1		
SRR2038672	GCA_012816825.1	Contig	NCBI		SAMN03656162	GCA_012816825.1	SRR2038672	65	4997340	78.9158	StxOp1181_Stx2a_16	Stx2a	1		
SRR2056392	GCA_018096145.1	Contig	NCBI		SAMN03734229	GCA_018096145.1	SRR2056392	64	5238312	270.6811	StxOp1250_Stx2a_127	Stx2a	1		
SRR2095948	GCA_018096125.1	Contig	NCBI		SAMN03755592	GCA_018096125.1	SRR2095948	79	5156644	62.9147	StxOp1250_Stx2a_127	Stx2a	1		
SRR2102939	GCA_012810615.1	Contig	NCBI		SAMN03792972	GCA_012810615.1	SRR2102939	76	5039539	149.6647	StxOp935_Stx2j_103	Stx2j	1		
SRR21085779	GCA_024971295.1	Contig	NCBI		SAMN30345600	GCA_024971295.1	SRR21085779	73	4897304	65.6505	StxOp432_Stx2a_30; StxOp4728_Stx2k_40	Stx2a; Stx2k	2		
SRR2120919	GCA_018096065.1	Contig	NCBI		SAMN03840336	GCA_018096065.1	SRR2120919	96	5228622	74.1122	StxOp1173_Stx2a_16	Stx2a	1		
SRR21504745	GCA_025140695.1	Contig	NCBI		SAMN30758315	GCA_025140695.1	SRR21504745	123	4907852	118.1908	StxOp1575_Stx2l_21	Stx2l	1		
SRR21507356	GCA_025137655.1	Contig	NCBI		SAMN30758782	GCA_025137655.1	SRR21507356	104	5217483	51.2097	StxOp1580_Stx1a_9; StxOp1477_Stx2d_38	Stx1a; Stx2d	2		
SRR21620895	GCA_014854575.1	Complete	NCBI		SAMN16279676	GCA_014854575.1	SRR21620895	3	5358674	48.8485	StxOp1040_Stx1d_4	Stx1d	1		
SRR21634505	GCA_014805805.1	Complete	NCBI		SAMN16231001	GCA_014805805.1	SRR21634505	3	5024336	36.6102	StxOp679_Stx2b_135	Stx2b	1		
SRR2177988	GCA_012803775.1	Contig	NCBI		SAMN04002945	GCA_012803775.1	SRR2177988	73	4999023	75.0319	StxOp1173_Stx2a_16	Stx2a	1		
SRR2178103	GCA_012803055.1	Contig	NCBI		SAMN03952639	GCA_012803055.1	SRR2178103	55	4937820	134.8941	StxOp1173_Stx2a_16	Stx2a	1		
SRR23421918	GCA_028605285.1	Contig	NCBI		SAMN33261764	GCA_028605285.1	SRR23421918	80	5021027	62.1629	StxOp13_Stx1a_19; StxOp1250_Stx2a_127	Stx1a; Stx2a	2		

Fastq ID	Fasta ID	Assembly level	Source	CFIA ID	BioSample	GenBank assembly accession	Illumina reads accession	Contigs (n)	Genome size (bp)	Illumina coverage (x)	Expected STxOp_nt operons	Expected subtypes	Expected subtypes (n)	Disrupted operon note	Curation note
SRR23828388	GCA_029206145.1	Contig	NCBI		SAMN33745055	GCA_029206145.1	SRR23828388	73	4894459	83.0414	StxOp1575_Stx2l_21	Stx2l	1		Second variant found with guided assembly
SRR24225801	GCA_030898125.1	Contig	NCBI		SAMN34264827	GCA_030898125.1	SRR24225801	29	5012893	92.9226	StxOp932_Stx2h_49	Stx2h	1		
SRR2480649	GCA_012794415.1	Contig	NCBI		SAMN04088929	GCA_012794415.1	SRR2480649	86	5320956	77.9634	StxOp1035_Stx1c_12	Stx1c	1		
SRR2481365	GCA_012786265.1	Contig	NCBI		SAMN03763795	GCA_012786265.1	SRR2481365	72	5092484	126.9079	StxOp935_Stx2j_103	Stx2j	1		
SRR2544681	GCA_000464955.2	Complete	NCBI		SAMN02147037	GCA_000464955.2	SRR2544681	2	5045919	83.5876	StxOp1160_Stx2a_11	Stx2a	1		
SRR2549981	GCA_012790495.1	Contig	NCBI		SAMN04116613	GCA_012790495.1	SRR2549981	52	5075569	169.1532	StxOp1708_Stx1a_9; StxOp1548_Stx2d_164	Stx1a; Stx2d	2		
SRR25642116	GCA_031666695.1	Contig	NCBI		SAMN36992383	GCA_031666695.1	SRR25642116	99	5069535	205.8247	StxOp3298_Stx2b_23; StxOp1448_Stx2c_152	Stx2b; Stx2c	2		
SRR26362796	GCA_032610235.1	Contig	NCBI		SAMN37790630	GCA_032610235.1	SRR26362796	65	5027343	121.3098	StxOp4_Stx1a_9; StxOp1217_Stx2a_18	Stx1a; Stx2a	2		
SRR26411587	GCA_032808445.1	Contig	NCBI		SAMN37858350	GCA_032808445.1	SRR26411587	70	5110641	70.0548	StxOp1089_Stx2a_1; StxOp1168_Stx2a_16	Stx2a	1		2 variants Stx2a
SRR26805704	GCA_003018235.1	Complete	NCBI		SAMN08579578	GCA_003018235.1	SRR26805704	3	5194649	109.8318	StxOp935_Stx2j_103	Stx2j	1		
SRR27018329	GCA_040131905.1	Contig	NCBI		SAMN38570051	GCA_040131905.1	SRR27018329	56	5329491	73.0467	StxOp1041_Stx1d_4	Stx1d	1		
SRR27135499	GCA_034235735.1	Contig	NCBI		SAMN38716539	GCA_034235735.1	SRR27135499	77	5175944	75.1438	StxOp1593_Stx1a_9; StxOp1157_Stx2a_6	Stx1a; Stx2a	2		
SRR27349220	GCA_034648675.1	Contig	NCBI		SAMN39124370	GCA_034648675.1	SRR27349220	83	5028573	132.6067	StxOp13_Stx1a_19; StxOp1157_Stx2a_6	Stx1a; Stx2a	2		
SRR27399504	GCA_034992965.1	Contig	NCBI		SAMN39224882	GCA_034992965.1	SRR27399504	45	4966469	58.5655	StxOp953_Stx2n_126	Stx2n	1		
SRR27958430	GCA_036595135.1	Contig	NCBI		SAMN39934763	GCA_036595135.1	SRR27958430	64	5148904	79.1964	StxOp4_Stx1a_9; StxOp1096_Stx2a_6; StxOp1168_Stx2a_16	Stx1a; Stx2a	2		2 variants Stx2a
SRR27967202	GCA_019585465.1	Complete	NCBI		SAMN11997563	GCA_019585465.1	SRR27967202	4	5147272	96.5559	StxOp1168_Stx2a_16	Stx2a	1		
SRR28356459	GCA_037194595.1	Contig	NCBI		SAMN40472744	GCA_037194595.1	SRR28356459	97	5153443	104.1078	StxOp1586_Stx1a_9; StxOp1574_Stx2g_39	Stx1a; Stx2g	2		
SRR28420203	GCA_037489925.1	Contig	NCBI		SAMN40580754	GCA_037489925.1	SRR28420203	73	5085992	126.86	StxOp39_Stx1a_54; StxOp1096_Stx2a_6	Stx1a; Stx2a	2		
SRR28618069	GCA_038054095.1	Contig	NCBI		SAMN40921104	GCA_038054095.1	SRR28618069	73	5047051	224.626	StxOp4_Stx1a_9; StxOp1124_Stx2a_6	Stx1a; Stx2a	2		
SRR28723267	GCA_038374585.1	Contig	NCBI		SAMN40996271	GCA_038374585.1	SRR28723267	86	5093874	66	StxOp1173_Stx2a_16	Stx2a	1		
SRR28723282	GCA_038374445.1	Contig	NCBI		SAMN40996277	GCA_038374445.1	SRR28723282	70	5160705	55.6286	StxOp13_Stx1a_19; StxOp1217_Stx2a_18	Stx1a; Stx2a	2		
SRR28837514	GCA_038761315.1	Contig	NCBI		SAMN41117619	GCA_038761315.1	SRR28837514	57	5194724	54.838	StxOp943_Stx2k_40	Stx2k	1		
SRR28994774	GCA_039605075.1	Contig	NCBI		SAMN41327196	GCA_039605075.1	SRR28994774	84	5022899	53.0554	StxOp4_Stx1a_9; StxOp2848_Stx2a_183	Stx1a; Stx2a	2		
SRR29067298	GCA_039664935.1	Contig	NCBI		SAMN41436329	GCA_039664935.1	SRR29067298	88	5040249	60.6951	StxOp943_Stx2k_40	Stx2k	1		
SRR3068663	GCA_012764415.2	Contig	NCBI		SAMN04377066	GCA_012764415.2	SRR3068663	101	4972443	72.1566	StxOp939_Stx2j_225	Stx2j	1		
SRR3136876	GCA_012726435.1	Contig	NCBI		SAMN04429913	GCA_012726435.1	SRR3136876	71	5033631	51.4014	StxOp966_Stx1a_18; StxOp1168_Stx2a_16	Stx1a; Stx2a	2		
SRR3180475	GCA_003018515.1	Complete	NCBI		SAMN04504054	GCA_003018515.1	SRR3180475	2	5955903	154.8619	StxOp1577_Stx1a_9; StxOp1089_Stx2a_1	Stx1a; Stx2a	2		
SRR3180697	GCA_012739175.1	Contig	NCBI		SAMN04505640	GCA_012739175.1	SRR3180697	74	4941872	184.6875	StxOp4647_Stx2f_725	Stx2f	1		
SRR3180701	GCA_012750475.1	Contig	NCBI		SAMN04505644	GCA_012750475.1	SRR3180701	70	5062417	177.6867	StxOp918_Stx2f_89	Stx2f	1		
SRR3189185	GCA_003017975.1	Complete	NCBI		SAMN04516926	GCA_003017975.1	SRR3189185	2	5388459	122.4301	StxOp3_Stx1a_9; StxOp1089_Stx2a_1	Stx1a; Stx2a	2		
SRR3191403	GCA_001695515.1	Complete	NCBI		SAMN04510563	GCA_001695515.1	SRR3191403	2	5866200	44.1139	StxOp1577_Stx1a_9; StxOp702_Stx2c_17	Stx1a; Stx2c	2		
SRR3195851	GCA_001753445.1	Complete	NCBI		SAMN04447635	GCA_001753445.1	SRR3195851	2	5509007	45.0379	StxOp702_Stx2c_17	Stx2c	1		
SRR3213947	GCA_012763755.1	Contig	NCBI		SAMN04530293	GCA_012763755.1	SRR3213947	59	4769705	145.4883	StxOp1040_Stx1d_4	Stx1d	1		

Fastq ID	Fasta ID	Assembly level	Source	CFIA ID	BioSample	GenBank assembly accession	Illumina reads accession	Contigs (n)	Genome size (bp)	Illumina coverage (x)	Expected STxOp_nt operons	Expected subtypes	Expected subtypes (n)	Disrupted operon note	Curation note
SRR3231803	GCA_003018535.1	Complete	NCBI		SAMN04558451	GCA_003018535.1	SRR3231803	4	5634505	166.8983	StxOp3_Stx1a_9; StxOp1089_Stx2a_1	Stx1a; Stx2a	2		
SRR3278959	GCA_012711755.1	Contig	NCBI		SAMN04570731	GCA_012711755.1	SRR3278959	59	5058230	127.3933	StxOp432_Stx2a_30; StxOp4728_Stx2k_40	Stx2a; Stx2k	2		
SRR3290048	GCA_012711215.2	Complete	NCBI		SAMN04578435	GCA_012711215.2	SRR3290048	2	5047088	145.9676	StxOp953_Stx2n_126	Stx2n	1		
SRR3394639	GCA_012703965.1	Contig	NCBI		SAMN04605221	GCA_012703965.1	SRR3394639	64	5085571	136.1857	StxOp1173_Stx2a_16; StxOp1259_Stx2a_128	Stx2a	1		2 variants Stx2a
SRR3394830	GCA_012901845.1	Contig	NCBI		SAMN04621361	GCA_012901845.1	SRR3394830	71	4868833	46.0239	StxOp1036_Stx1c_12	Stx1c	1		
SRR3438052	GCA_012686325.1	Contig	NCBI		SAMN04273143	GCA_012686325.1	SRR3438052	86	5153374	66.7392	StxOp4677_Stx2g_510	Stx2g	1		
SRR3723069	GCA_012894035.1	Contig	NCBI		SAMN05283785	GCA_012894035.1	SRR3723069	55	5054156	139.5597	StxOp4731_Stx2k_519	Stx2k	1		
SRR3794154	GCA_012668825.1	Contig	NCBI		SAMN05368321	GCA_012668825.1	SRR3794154	57	5185136	138.0056	StxOp806_Stx2c_642; StxOp866_Stx2d_350	Stx2c; Stx2d	2		
SRR4176409	GCA_012627465.1	Contig	NCBI		SAMN05733151	GCA_012627465.1	SRR4176409	82	5447126	165.1827	StxOp399_Stx1c_12; StxOp653_Stx2b_23	Stx1c; Stx2b	2		
SRR4430014	GCA_020372765.1	Contig	NCBI		SAMN05718050	GCA_020372765.1	SRR4430014	73	5358571	51.4595	StxOp1041_Stx1d_4	Stx1d	1		
SRR5264142	GCA_012498465.1	Contig	NCBI		SAMN06285391	GCA_012498465.1	SRR5264142	82	5114310	113.6428	StxOp3298_Stx2b_23; StxOp1448_Stx2c_152	Stx2b; Stx2c	2		
SRR5434351	GCA_012482175.1	Contig	NCBI		SAMN06684358	GCA_012482175.1	SRR5434351	75	5037525	65.213	StxOp1040_Stx1d_4	Stx1d	1		
SRR5616352	GCA_012466565.1	Contig	NCBI		SAMN07172024	GCA_012466565.1	SRR5616352	50	4948848	109.4772	StxOp1038_Stx1d_4	Stx1d	1		
SRR5642675	GCA_012463025.1	Contig	NCBI		SAMN07177511	GCA_012463025.1	SRR5642675	76	5083638	65.2303	StxOp941_Stx2j_705	Stx2j	1		
SRR5832137	GCA_020364625.1	Contig	NCBI		SAMN07349904	GCA_020364625.1	SRR5832137	60	4955121	203.1319	StxOp1645_Stx1a_9; StxOp1168_Stx2a_16	Stx1a; Stx2a	2		
SRR6128619	GCA_013342905.2	Complete	NCBI		SAMN07709929	GCA_013342905.2	SRR6128619	2	5138311	97.2812	StxOp952_Stx2n_126	Stx2n	1		
SRR6321371	GCA_014700215.1	Contig	NCBI		SAMN08095911	GCA_014700215.1	SRR6321371	161	4826739	54.6077	StxOp918_Stx2f_89	Stx2f	1		
SRR6767327	GCA_012253565.1	Contig	NCBI		SAMN08595463	GCA_012253565.1	SRR6767327	81	4838819	80.9363	StxOp938_Stx2j_223	Stx2j	1		
SRR6903006	GCA_012224845.2	Complete	NCBI		SAMN08799860	GCA_012224845.2	SRR6903006	1	5248255	55.25	StxOp955_Stx2o_545	Stx2o	1		
SRR7050023	GCA_010231855.1	Complete	NCBI		SAMN08964444	GCA_010231855.1	SRR7050023	2	5651618	56.9821	StxOp1043_Stx2a_1; StxOp702_Stx2c_17	Stx2a; Stx2c	2		only 2c found when entire database is used, 2a and 2c found when only specific alleles used
SRR7163944	GCA_012203485.1	Contig	NCBI		SAMN09207670	GCA_012203485.1	SRR7163944	192	5130790	45.1247	StxOp949_Stx2m_547	Stx2m	1		
SRR7235296	GCA_014629915.1	Contig	NCBI		SAMN09283022	GCA_014629915.1	SRR7235296	61	5415443	101.8132	StxOp4746_Stx2o_41	Stx2o	1		
SRR7612274	GCA_005042145.1	Contig	NCBI	LS #17-3-2	SAMN08797212	GCA_005042145.1	SRR7612274	85	4957319	113.1918	StxOp943_Stx2k_40	Stx2k	1		
SRR7738059	GCA_012124695.1	Contig	NCBI		SAMN09878861	GCA_012124695.1	SRR7738059	50	5012999	56.4961	StxOp953_Stx2n_126	Stx2n	1		
SRR7767477	GCA_012118465.1	Contig	NCBI		SAMN09934175	GCA_012118465.1	SRR7767477	52	4965732	58.1862	StxOp953_Stx2n_126	Stx2n	1		
SRR7767490	GCA_012119015.1	Contig	NCBI		SAMN09934183	GCA_012119015.1	SRR7767490	37	5044257	101.8029	StxOp4596_Stx2e_36	Stx2e	1		
SRR7826316	GCA_003203755.1	Complete	NCBI		SAMN09278797	GCA_003203755.1	SRR7826316	3	5626456	91.0969	StxOp3_Stx1a_9; StxOp1089_Stx2a_1	Stx1a; Stx2a	2		
SRR7826361	GCA_003288295.1	Complete	NCBI		SAMN08822959	GCA_003288295.1	SRR7826361	3	5432691	119.8481	StxOp21_Stx1a_33	Stx1a	1		
SRR7828257	GCA_003112205.1	Complete	NCBI		SAMN08165042	GCA_003112205.1	SRR7828257	4	5261709	100.941	StxOp4_Stx1a_9; StxOp1141_Stx2a_6	Stx1a; Stx2a	2		
SRR7828259	GCA_003176855.1	Complete	NCBI		SAMN08724662	GCA_003176855.1	SRR7828259	3	5272885	69.5035	StxOp1257_Stx2a_127; StxOp589_Stx2a_525; StxOp842_Stx2d_115	Stx2a; Stx2d	2		2 variants of Stx2a
SRR7896248	GCA_005014055.1	Complete	NCBI	NML 12-3002	SAMN08768111	GCA_005014055.1	SRR7896248	4	5131000	101.2465	StxOp439_Stx2a_107; StxOp842_Stx2d_115	Stx2a; Stx2d	2		Only 2d found with entire DB, but 2a, 2d found with 100% identity when only these two alleles are used in KMA DB

Fastq ID	Fasta ID	Assembly level	Source	CFIA ID	BioSample	GenBank assembly accession	Illumina reads accession	Contigs (n)	Genome size (bp)	Illumina coverage (x)	Expected STxOp_nt operons	Expected subtypes	Expected subtypes (n)	Disrupted operon note	Curation note
SRR7896259	GCA_005037775.2	Complete	NCBI	NML 09-2555	SAMN08768109	GCA_005037775.2	SRR7896259	2	5093580	106.271	StxOp1208_Stx2a_16; StxOp1510_Stx2d_115	Stx2a; Stx2d	2		2 copies of Stx2d; both 2a and 2d found when only two alleles used with KMA, only 2d found if whole DB used
SRR7958537	GCA_003903075.2	Contig	NCBI		SAMN10170522	GCA_003903075.2	SRR7958537	142	5036523	71.9066	StxOp942_Stx2j_706	Stx2j	1		
SRR8931387	GCA_011955925.1	Contig	NCBI		SAMN11466918	GCA_011955925.1	SRR8931387	77	5383487	51.6972	StxOp167_Stx1a_191; StxOp1036_Stx1c_12; StxOp1348_Stx2b_110	Stx1a; Stx1c; Stx2b	3		
SRR9007492	GCA_011950125.1	Contig	NCBI		SAMN11569941	GCA_011950125.1	SRR9007492	128	5215947	83.3175	StxOp948_Stx2m_48	Stx2m	1		
SRR9833314	GCA_011901845.1	Contig	NCBI		SAMN12361752	GCA_011901845.1	SRR9833314	130	5036248	94.9856	StxOp937_Stx2j_108	Stx2j	1		
SRR9972670	GCA_018259255.1	Contig	NCBI		SAMN12533168	GCA_018259255.1	SRR9972670	51	4935447	108.1204	StxOp1168_Stx2a_16	Stx2a	1		

Supplementary Table S2.1b. Operon-level detail for the benchmark dataset.

Every stx operon copy detected in the 200 benchmark genomes, with its position, orientation and alignment statistics against the STxOP_nt v1.0.0 reference database.

Percent match is GeneSeekr's reported value, computed as positives divided by the greater of the alignment length and the reference operon length. Strand follows the BLASTN convention: the query is always ascending, so a minus strand indicates that the subject coordinates descend. Sequences are stored in contig (query) orientation, not reference orientation.

Query and subject sequences are given as FASTA records at the end of this table.

293 rows, 22 columns.

Fastq ID	Fasta ID	Assembly level	STxOP_nt operon identifier	Stx subtype	Variant	Operon copy	Query start	Query end	Strand	Alignment length (bp)	Reference operon length (bp)	Percent match	Positives	Mismatches	Gaps	E-value	Bit score	Coverage of stx allele (x)	Relative coverage of stx allele	Functional	Disruption
2018-SEQ-1757	2021-MIN-0249	Complete	StxOp933_Stx2i_43	Stx2i	Stx2i_43		2153681	2154922	-	1242	1242	100	1242	0	0	0	2241	143.48	1.12	Yes	
2025-SEQ-1251	2023-MIN-0010	Complete	StxOp1089_Stx2a_1	Stx2a	Stx2a_1		484450	485690	-	1241	1241	100	1241	0	0	0	2239			Yes	
2025-SEQ-1251	2023-MIN-0010	Complete	StxOp3_Stx1a_9	Stx1a	Stx1a_9		795438	796664	-	1227	1227	100	1227	0	0	0	2214			Yes	
2025-SEQ-1502	2022-MIN-0034	Complete	StxOp432_Stx2a_30	Stx2a	Stx2a_30		473036	474276	+	1241	1241	100	1241	0	0	0	2239	189.1	1.22	Yes	
2025-SEQ-1502	2022-MIN-0034	Complete	StxOp943_Stx2k_40	Stx2k	Stx2k_40		1313209	1314449	-	1241	1241	100	1241	0	0	0	2239	247.28	1.6	Yes	
2025-SEQ-1503	2019-MIN-0073	hybrid	StxOp13_Stx1a_19	Stx1a	Stx1a_19		688482	689708	-	1227	1227	100	1227	0	0	0	2214	72.75	0.4	Yes	
2025-SEQ-1503	2019-MIN-0073	hybrid	StxOp399_Stx1c_12	Stx1c	Stx1c_12		25470	26697	-	1228	1228	100	1228	0	0	0	2215	88.08	0.48	Yes	
2025-SEQ-1503	2019-MIN-0073	hybrid	StxOp673_Stx2b_113	Stx2b	Stx2b_113	1 of 2	749473	750708	-	1236	1236	100	1236	0	0	0	2230	172.04	0.93	Yes	
2025-SEQ-1503	2019-MIN-0073	hybrid	StxOp673_Stx2b_113	Stx2b	Stx2b_113	2 of 2	780873	782108	-	1236	1236	100	1236	0	0	0	2230	172.04	0.93	Yes	
2025-SEQ-1504	2023-MIN-0001	hybrid	StxOp1348_Stx2b_110	Stx2b	Stx2b_110		1517814	1519049	-	1236	1236	100	1236	0	0	0	2230	129.68	0.77	Yes	
2025-SEQ-1504	2023-MIN-0001	hybrid	StxOp399_Stx1c_12	Stx1c	Stx1c_12		726868	728095	+	1228	1228	100	1228	0	0	0	2215	37.74	0.23	Yes	
2025-SEQ-1505	2021-MIN-0009	hybrid	StxOp933_Stx2i_43	Stx2i	Stx2i_43		536833	538074	+	1242	1242	100	1242	0	0	0	2241	177.6	1.43	Yes	
2025-SEQ-1506	2022-MIN-0057	Complete	StxOp2108_Stx1c_27	Stx1c	Stx1c_27		4935657	4936883	+	1227	1227	100	1227	0	0	0	2214	41.08	0.23	Yes	
2025-SEQ-1507	2023-MIN-0017	hybrid	StxOp13_Stx1a_19	Stx1a	Stx1a_19		1163553	1164779	-	1227	1227	100	1227	0	0	0	2214	108.13	0.68	Yes	
2025-SEQ-1507	2023-MIN-0017	hybrid	StxOp675_Stx2b_116	Stx2b	Stx2b_116		4069051	4070286	+	1236	1236	100	1236	0	0	0	2230	163.78	1.03	Yes	
2025-SEQ-1507	2023-MIN-0017	hybrid	StxOp684_Stx2b_221	Stx2b	Stx2b_221		774194	775429	-	1236	1236	100	1236	0	0	0	2230	163.78	1.03	Yes	
2025-SEQ-1509	2022-MIN-0070	hybrid	StxOp1035_Stx1c_12	Stx1c	Stx1c_12		989857	991084	+	1228	1228	100	1228	0	0	0	2215	170.12	1.12	Yes	
2025-SEQ-1619	GCA_005037795.2	Complete	StxOp1577_Stx1a_9	Stx1a	Stx1a_9		3301283	3302509	-	1227	1227	100	1227	0	0	0	2214	61.01	1.19	Yes	
2025-SEQ-1621	GCA_005037805.2	Complete	StxOp1089_Stx2a_1	Stx2a	Stx2a_1		2267399	2268639	+	1241	1241	100	1241	0	0	0	2239	89.63	0.48	Yes	
2025-SEQ-1621	GCA_005037805.2	Complete	StxOp3_Stx1a_9	Stx1a	Stx1a_9		1942198	1943424	-	1227	1227	100	1227	0	0	0	2214	70.27	0.37	Yes	
2025-SEQ-1629	2021-MIN-0136	hybrid	StxOp439_Stx2a_107	Stx2a	Stx2a_107		1808891	1810131	+	1241	1241	100	1241	0	0	0	2239			Yes	
2025-SEQ-1629	2021-MIN-0136	hybrid	StxOp842_Stx2d_115	Stx2d	Stx2d_115		2389788	2391028	-	1241	1241	100	1241	0	0	0	2239			Yes	
2025-SEQ-1632	GCA_005037725.2	Complete	StxOp429_Stx2a_16	Stx2a	Stx2a_16		2263361	2264601	+	1241	1241	100	1241	0	0	0	2239	46.09	0.29	Yes	
2025-SEQ-1632	GCA_005037725.2	Complete	StxOp9_Stx1a_15	Stx1a	Stx1a_15		3398764	3399990	-	1227	1227	100	1227	0	0	0	2214	41.01	0.26	Yes	
2025-SEQ-1634	GCA_018986775.1	Complete	StxOp954_Stx2o_41	Stx2o	Stx2o_41		4929620	4930855	+	1236	1236	100	1236	0	0	0	2230	57.01	0.47	Yes	
2025-SEQ-1638	2021-MIN-0144	Complete	StxOp702_Stx2c_17	Stx2c	Stx2c_17		4718889	4720129	+	1241	1241	100	1241	0	0	0	2239			Yes	
2025-SEQ-1642	2021-MIN-0255	hybrid	StxOp2210_Stx1d_4	Stx1d	Stx1d_4		133369	134595	+	1227	1227	100	1227	0	0	0	2214			Yes	
2025-SEQ-1666	2019-MIN-0085	hybrid	StxOp1494_Stx2d_115	Stx2d	Stx2d_115		131065	132305	-	1241	1241	100	1241	0	0	0	2239	365.67	1.33	Yes	
2025-SEQ-1666	2019-MIN-0085	hybrid	StxOp4184_Stx2c_724	Stx2c	Stx2c_724		176155	177395	+	1241	1241	100	1241	0	0	0	2239	104.22	0.38	Yes	
2025-SEQ-1666	2019-MIN-0085	hybrid	StxOp674_Stx2b_114	Stx2b	Stx2b_114		641767	643002	-	1236	1236	100	1236	0	0	0	2230	121.16	0.44	Yes	
2025-SEQ-1667	2021-MIN-0169	Complete	StxOp2209_Stx1d_4	Stx1d	Stx1d_4		653796	655022	-	1227	1227	100	1227	0	0	0	2214			Yes	
2025-SEQ-1668	2019-MIN-0086	hybrid	StxOp1355_Stx2b_194	Stx2b	Stx2b_194		3312494	3313729	-	1236	1236	100	1236	0	0	0	2230	269.34	1.49	Yes	
2025-SEQ-1668	2019-MIN-0086	hybrid	StxOp399_Stx1c_12	Stx1c	Stx1c_12		931202	932429	+	1228	1228	100	1228	0	0	0	2215	199.28	1.1	Yes	
2025-SEQ-1668	2019-MIN-0086	hybrid	StxOp419_Stx2a_1	Stx2a	Stx2a_1		390007	391247	-	1241	1241	100	1241	0	0	0	2239	66.61	0.37	Yes	
2025-SEQ-1668	2019-MIN-0086	hybrid	StxOp670_Stx2b_109	Stx2b	Stx2b_109		651598	652833	+	1236	1236	100	1236	0	0	0	2230	269.34	1.49	Yes	
2025-SEQ-1675	2017-MER-0367	short read	StxOp1039_Stx1d_4	Stx1d	Stx1d_4		1151	2377	-	1227	1227	100	1227	0	0	0	2214			Yes	
2025-SEQ-1688	GCA_018986815.1	Complete	StxOp932_Stx2h_49	Stx2h	Stx2h_49		3440418	3441653	+	1236	1236	100	1236	0	0	0	2230	74.72	0.67	Yes	
2025-SEQ-1691	2022-MIN-0050	hybrid	StxOp399_Stx1c_12	Stx1c	Stx1c_12		79535	80762	-	1228	1228	100	1228	0	0	0	2215			Yes	
2025-SEQ-1703	2025-MIN-0001	Complete	StxOp1043_Stx2a_1	Stx2a	Stx2a_1		31046	32286	-	1241	1241	100	1241	0	0	0	2239			Yes	

Fastq ID	Fasta ID	Assembly level	STxOP_nt operon identifier	Stx subtype	Variant	Operon copy	Query start	Query end	Strand	Alignment length (bp)	Reference operon length (bp)	Percent match	Positives	Mismatches	Gaps	E-value	Bit score	Coverage of stx allele (x)	Relative coverage of stx allele	Functional	Disruption
2025-SEQ-1703	2025-MIN-0001	Complete	StxOp702_Stx2c_17	Stx2c	Stx2c_17		4232445	4233685	+	1241	1241	100	1241	0	0	0	2239			Yes	
2025-SEQ-1763	2022-MIN-0028	hybrid	StxOp3298_Stx2b_23	Stx2b	Stx2b_23		12487	13722	-	1236	1236	100	1236	0	0	0	2230			Yes	
2025-SEQ-1763	2022-MIN-0028	hybrid	StxOp4389_Stx2d_112	Stx2d	Stx2d_112		698177	699417	+	1241	1241	100	1241	0	0	0	2239			Yes	
2025-SEQ-1771	GCA_018986795.1	Complete	StxOp936_Stx2j_104	Stx2j	Stx2j_104		2244862	2246095	-	1234	1234	100	1234	0	0	0	2226	21.66	0.17	Yes	
2025-SEQ-1785	2023-MIN-0015	Complete	StxOp927_Stx2g_147	Stx2g	Stx2g_147		14576	15817	+	1242	1242	100	1242	0	0	0	2241	57.35	0.56	Yes	
2025-SEQ-1786	2021-MIN-0153	hybrid	StxOp1486_Stx2d_112	Stx2d	Stx2d_112		785	2025	-	1241	1241	100	1241	0	0	0	2239	272.46	4.04	Yes	
2025-SEQ-1786	2021-MIN-0153	hybrid	StxOp3298_Stx2b_23	Stx2b	Stx2b_23		3305773	3307008	-	1236	1236	100	1236	0	0	0	2230	146	2.16	Yes	
2025-SEQ-1788	2023-MIN-0013	hybrid	StxOp826_Stx2d_9	Stx2d	Stx2d_9		894558	895798	+	1241	1241	100	1241	0	0	0	2239	188.66	2	Yes	
2025-SEQ-1789	2022-MIN-0053	Complete	StxOp1036_Stx1c_12	Stx1c	Stx1c_12		4949393	4950621	+	1229	1229	100	1229	0	0	0	2217	62.03	0.4	Yes	
2025-SEQ-1790	2021-MIN-0250	Complete	StxOp933_Stx2i_43	Stx2i	Stx2i_43		4671049	4672290	-	1242	1242	100	1242	0	0	0	2241	130.68	0.64	Yes	
2025-SEQ-1791	2022-MIN-0054	Complete	StxOp1571_Stx2e_146	Stx2e	Stx2e_146		2727009	2728244	-	1236	1236	100	1236	0	0	0	2230	87.55	0.48	Yes	
2025-SEQ-1792	2022-MIN-0056	Complete	StxOp3320_Stx2b_101	Stx2b	Stx2b_101		4237611	4238846	+	1236	1236	100	1236	0	0	0	2230	75.92	0.49	Yes	
2025-SEQ-1792	2022-MIN-0056	Complete	StxOp399_Stx1c_12	Stx1c	Stx1c_12		5329341	5330568	+	1228	1228	100	1228	0	0	0	2215	47.52	0.31	Yes	
2025-SEQ-1793	2022-MIN-0071	Complete	StxOp933_Stx2i_43	Stx2i	Stx2i_43		2253601	2254842	+	1242	1242	100	1242	0	0	0	2241	254.07	1.57	Yes	
2025-SEQ-1795	2017-PAC-0030	Complete	StxOp1089_Stx2a_1	Stx2a	Stx2a_1		4424692	4425932	+	1241	1241	100	1241	0	0	0	2239	200.07	1.07	Yes	
2025-SEQ-1796	GCA_018966965.1	Complete	StxOp1089_Stx2a_1	Stx2a	Stx2a_1															No	Transposon insertion within stx2a subunit A gene
2025-SEQ-1796	GCA_018966965.1	Complete	StxOp3_Stx1a_9	Stx1a	Stx1a_9		5203331	5204557	-	1227	1227	100	1227	0	0	0	2214	51.73	0.38	Yes	
2025-SEQ-1798	16-9255	Complete	StxOp419_Stx2a_1	Stx2a	Stx2a_1		5387822	5389062	+	1241	1241	100	1241	0	0	0	2239	438.48	2.65	Yes	
2025-SEQ-1800	2022-MIN-0055	Complete	StxOp1448_Stx2c_152	Stx2c	Stx2c_152		3185845	3187085	+	1241	1241	100	1241	0	0	0	2239	67.02	0.44	Yes	
2025-SEQ-1800	2022-MIN-0055	Complete	StxOp3625_Stx2b_215	Stx2b	Stx2b_215		1400126	1401361	-	1236	1236	100	1236	0	0	0	2230			Yes	
2025-SEQ-1964	2021-MIN-0170	hybrid	StxOp3298_Stx2b_23	Stx2b	Stx2b_23		2562298	2563533	-	1236	1236	100	1236	0	0	0	2230			Yes	
2025-SEQ-1964	2021-MIN-0170	hybrid	StxOp4178_Stx2c_569	Stx2c	Stx2c_569		5066160	5067400	+	1241	1241	100	1241	0	0	0	2239			Yes	
DRR178135	GCA_009002845.1	Complete	StxOp1089_Stx2a_1	Stx2a	Stx2a_1	1 of 2	1690490	1691730	+	1241	1241	100	1241	0	0	0	2239	80.03	1.73	Yes	
DRR178135	GCA_009002845.1	Complete	StxOp1089_Stx2a_1	Stx2a	Stx2a_1	2 of 2	2447719	2448959	-	1241	1241	100	1241	0	0	0	2239	80.03	1.73	Yes	
DRR178135	GCA_009002845.1	Complete	StxOp1577_Stx1a_9	Stx1a	Stx1a_9		1213063	1214289	+	1227	1227	100	1227	0	0	0	2214	38.86	0.84	Yes	
DRR401569	GCA_026001525.1	Complete	StxOp1089_Stx2a_1	Stx2a	Stx2a_1		1556765	1558005	+	1241	1241	100	1241	0	0	0	2239	124.3	0.39	Yes	
DRR401569	GCA_026001525.1	Complete	StxOp22_Stx1a_35	Stx1a	Stx1a_35		2946526	2947752	-	1227	1227	100	1227	0	0	0	2214	85.74	0.27	Yes	
DRR401569	GCA_026001525.1	Complete	StxOp717_Stx2c_130	Stx2c	Stx2c_130		5221926	5223166	+	1241	1241	100	1241	0	0	0	2239	41.7	0.13	Yes	
DRR401577	GCA_026001745.1	Complete	StxOp1089_Stx2a_1	Stx2a	Stx2a_1		1591391	1592631	+	1241	1241	100	1241	0	0	0	2239	69.56	0.72	Yes	
DRR401577	GCA_026001745.1	Complete	StxOp1577_Stx1a_9	Stx1a	Stx1a_9		5244666	5245892	+	1227	1227	100	1227	0	0	0	2214	90.9	0.95	Yes	
ERR3330139	GCA_016667705.1	Contig	StxOp922_Stx2f_726	Stx2f	Stx2f_726		8576	9811	-	1236	1236	100	1236	0	0	0	2230	56.07	0.77	Yes	
GCF_002303275	GCA_002303275.1	Complete	StxOp417_Stx1e_13	Stx1e	Stx1e_13		1682301	1683527	+	1227	1227	100	1227	0	0	0	2214	213.37	2.4	Yes	
SRR10541011	GCA_011822915.1	Contig	StxOp932_Stx2h_49	Stx2h	Stx2h_49		10295	11530	+	1236	1236	100	1236	0	0	0	2230	213.37	0.97	Yes	
SRR10911972	GCA_012661225.1	Contig	StxOp948_Stx2m_48	Stx2m	Stx2m_48		22194	23429	+	1236	1236	100	1236	0	0	0	2230	92.89	0.92	Yes	
SRR11485261	GCA_020809385.1	Complete	StxOp1301_Stx2a_654	Stx2a	Stx2a_654		366329	367569	+	1241	1241	100	1241	0	0	0	2239	178.11	0.73	Yes	
SRR11485261	GCA_020809385.1	Complete	StxOp846_Stx2d_144	Stx2d	Stx2d_144		2071448	2072688	+	1241	1241	100	1241	0	0	0	2239	99.4	0.41	Yes	
SRR11550988	GCA_014815785.1	Contig	StxOp399_Stx1c_12	Stx1c	Stx1c_12		66663	67890	+	1228	1228	100	1228	0	0	0	2215	28.17	0.73	Yes	
SRR12207194	GCA_014058445.2	Complete	StxOp429_Stx2a_16	Stx2a	Stx2a_16		2222163	2223403	+	1241	1241	100	1241	0	0	0	2239	106.4	0.85	Yes	
SRR12207194	GCA_014058445.2	Complete	StxOp9_Stx1a_15	Stx1a	Stx1a_15		3579261	3580487	-	1227	1227	100	1227	0	0	0	2214	106.46	0.85	Yes	
SRR12506954	GCA_014281005.1	Contig	StxOp1515_Stx2d_115	Stx2d	Stx2d_115		4924	6164	-	1241	1241	100	1241	0	0	0	2239	84.96	1.27	Yes	
SRR12506954	GCA_014281005.1	Contig	StxOp653_Stx2b_23	Stx2b	Stx2b_23		32761	33996	+	1236	1236	100	1236	0	0	0	2230	25.71	0.38	Yes	
SRR1272511	GCA_012833475.1	Contig	StxOp1580_Stx1a_9	Stx1a	Stx1a_9		213121	214347	-	1227	1227	100	1227	0	0	0	2214	94.14	0.93	Yes	
SRR1272522	GCA_012833535.1	Contig	StxOp432_Stx2a_30	Stx2a	Stx2a_30		2914	4154	-	1241	1241	100	1241	0	0	0	2239	108.05	1.02	Yes	
SRR1272883	GCA_012833235.1	Contig	StxOp1365_Stx2c_7	Stx2c	Stx2c_7		1405	2645	-	1241	1241	100	1241	0	0	0	2239	132.76	0.98	Yes	
SRR12844969	GCA_014656625.1	Complete	StxOp829_Stx2d_27	Stx2d	Stx2d_27		2692860	2694100	-	1241	1241	100	1241	0	0	0	2239	151.86	0.72	Yes	
SRR13068663	GCA_015529125.1	Contig	StxOp1157_Stx2a_6	Stx2a	Stx2a_6		3934	5174	-	1241	1241	100	1241	0	0	0	2239	58.31	0.78	Yes	
SRR13068663	GCA_015529125.1	Contig	StxOp1711_Stx1a_9	Stx1a	Stx1a_9		31022	32248	+	1227	1227	100	1227	0	0	0	2214	55.1	0.74	Yes	
SRR13132130	GCA_015923545.1	Contig	StxOp953_Stx2n_126	Stx2n	Stx2n_126		1915	3150	-	1236	1236	100	1236	0	0	0	2230	78.81	0.72	Yes	

Fastq ID	Fasta ID	Assembly level	STxOP_nt operon identifier	Stx subtype	Variant	Operon copy	Query start	Query end	Strand	Alignment length (bp)	Reference operon length (bp)	Percent match	Positives	Mismatches	Gaps	E-value	Bit score	Coverage of stx allele (x)	Relative coverage of stx allele	Functional	Disruption
SRR13209800	GCA_016404485.2	Complete	StxOp13_Stx1a_19	Stx1a	Stx1a_19		1468469	1469695	-	1227	1227	100	1227	0	0	0	2214	43.82	0.76	Yes	
SRR13209800	GCA_016404485.2	Complete	StxOp675_Stx2b_116	Stx2b	Stx2b_116		2665386	2666621	-	1236	1236	100	1236	0	0	0	2230	54.22	0.94	Yes	
SRR13209817	GCA_016403875.2	Complete	StxOp1510_Stx2d_115	Stx2d	Stx2d_115		1348401	1349641	-	1241	1241	100	1241	0	0	0	2239	89.23	1.87	Yes	
SRR13209817	GCA_016403875.2	Complete	StxOp1512_Stx2d_115	Stx2d	Stx2d_115		1091396	1092636	-	1241	1241	100	1241	0	0	0	2239	89.23	1.87	Yes	
SRR13209822	GCA_016403975.2	Complete	StxOp1541_Stx2d_134	Stx2d	Stx2d_134		1843799	1845039	+	1241	1241	100	1241	0	0	0	2239	83.94	1.09	Yes	
SRR13240070	GCA_015978445.1	Contig	StxOp1173_Stx2a_16	Stx2a	Stx2a_16		2803	4043	-	1241	1241	100	1241	0	0	0	2239	108.99	1.09	Yes	
SRR13240070	GCA_015978445.1	Contig	StxOp4_Stx1a_9	Stx1a	Stx1a_9		58227	59453	+	1227	1227	100	1227	0	0	0	2214	90.08	0.9	Yes	
SRR13240070	GCA_015978445.1	Contig	StxOp848_Stx2d_164	Stx2d	Stx2d_164		239	1479	+	1241	1241	100	1241	0	0	0	2239	73.7	0.73	Yes	
SRR13428524	GCA_016834695.1	Complete	StxOp3_Stx1a_9	Stx1a	Stx1a_9		4026347	4027573	-	1227	1227	100	1227	0	0	0	2214	77.16	0.99	Yes	
SRR13428524	GCA_016834695.1	Complete	StxOp702_Stx2c_17	Stx2c	Stx2c_17		3531527	3532767	-	1241	1241	100	1241	0	0	0	2239	86.91	1.12	Yes	
SRR13450255	GCA_021766875.1	Contig	StxOp1173_Stx2a_16	Stx2a	Stx2a_16		216	1456	+	1241	1241	100	1241	0	0	0	2239	37.02	0.95	Yes	
SRR13450255	GCA_021766875.1	Contig	StxOp1228_Stx2a_18	Stx2a	Stx2a_18		216	1456	+	1241	1241	100	1241	0	0	0	2239	37.02	0.95	Yes	
SRR13557757	GCA_023278265.1	Complete	StxOp1089_Stx2a_1	Stx2a	Stx2a_1		1458677	1459917	+	1241	1241	100	1241	0	0	0	2239	119.7	0.89	Yes	
SRR13605977	GCA_017172955.1	Contig	StxOp1574_Stx2g_39	Stx2g	Stx2g_39		867	2108	-	1242	1242	100	1242	0	0	0	2241	47.14	0.95	Yes	
SRR13605977	GCA_017172955.1	Contig	StxOp1594_Stx1a_9	Stx1a	Stx1a_9		967	2193	+	1227	1227	100	1227	0	0	0	2214	42.31	0.85	Yes	
SRR13674450	GCA_017081175.1	Complete	StxOp1361_Stx2c_7	Stx2c	Stx2c_7		4464009	4465249	+	1241	1241	100	1241	0	0	0	2239	549.28	0.88	Yes	
SRR13796384	GCA_025991315.1	Contig	StxOp1041_Stx1d_4	Stx1d	Stx1d_4		31936	33162	-	1227	1227	100	1227	0	0	0	2214	57.95	0.92	Yes	
SRR13957227	GCA_017740375.1	Contig	StxOp913_Stx2e_102	Stx2e	Stx2e_102		293385	294620	-	1236	1236	100	1236	0	0	0	2230	128.66	1.06	Yes	
SRR14071719	GCA_017588055.1	Contig	StxOp955_Stx2o_545	Stx2o	Stx2o_545		95586	96821	-	1236	1236	100	1236	0	0	0	2230	42.51	0.86	Yes	
SRR14750050	GCA_018986835.1	Complete	StxOp432_Stx2a_30	Stx2a	Stx2a_30		2652533	2653773	-	1241	1241	100	1241	0	0	0	2239	86.2	1.01	Yes	
SRR14898862	GCA_019020085.1	Contig	StxOp415_Stx1d_4	Stx1d	Stx1d_4		2558	3784	+	1227	1227	100	1227	0	0	0	2214	71.73	0.86	Yes	
SRR14934092	GCA_019431315.1	Complete	StxOp898_Stx2e_25	Stx2e	Stx2e_25		2270039	2271274	+	1236	1236	100	1236	0	0	0	2230	79.21	0.88	Yes	
SRR14934097	GCA_019431335.1	Complete	StxOp898_Stx2e_25	Stx2e	Stx2e_25		428629	429864	+	1236	1236	100	1236	0	0	0	2230	109.49	1.03	Yes	
SRR15076491	GCA_019184365.1	Contig	StxOp1339_Stx2b_109	Stx2b	Stx2b_109		8408	9643	-	1236	1236	100	1236	0	0	0	2230	102.85	0.93	Yes	
SRR15076491	GCA_019184365.1	Contig	StxOp403_Stx1c_26	Stx1c	Stx1c_26		174206	175433	+	1228	1228	100	1228	0	0	0	2215	98.4	0.89	Yes	
SRR15497538	GCA_021534925.1	Complete	StxOp914_Stx2e_133	Stx2e	Stx2e_133		2005797	2007032	+	1236	1236	100	1236	0	0	0	2230	335.07	1.38	Yes	
SRR15497538	GCA_021534925.1	Complete	StxOp916_Stx2e_146	Stx2e	Stx2e_146		2754019	2755254	-	1236	1236	100	1236	0	0	0	2230	335.07	1.38	Yes	
SRR15949660	GCA_020041845.1	Contig	StxOp399_Stx1c_12	Stx1c	Stx1c_12		174342	175569	+	1228	1228	100	1228	0	0	0	2215	180.17	0.95	Yes	
SRR16071833	GCA_021535005.1	Complete	StxOp915_Stx2e_136	Stx2e	Stx2e_136		2821994	2823229	-	1236	1236	100	1236	0	0	0	2230	302.29	0.93	Yes	
SRR1610028	GCA_003018775.1	Complete	StxOp419_Stx2a_1	Stx2a	Stx2a_1		4190370	4191610	-	1241	1241	100	1241	0	0	0	2239	57.29	1.57	Yes	
SRR1724591	GCA_003018835.2	Complete	StxOp1577_Stx1a_9	Stx1a	Stx1a_9		4310560	4311786	-	1227	1227	100	1227	0	0	0	2214	141.22	0.9	Yes	
SRR1773021	GCA_012826915.1	Contig	StxOp1575_Stx2l_21	Stx2l	Stx2l_21		133544	134784	+	1241	1241	100	1241	0	0	0	2239	141.91	1.08	Yes	
SRR17838893	GCA_021763325.1	Contig	StxOp1157_Stx2a_6	Stx2a	Stx2a_6		582858	584098	+	1241	1241	100	1241	0	0	0	2239	87.8	0.93	Yes	
SRR17838893	GCA_021763325.1	Contig	StxOp1283_Stx2a_170	Stx2a	Stx2a_170		3721	4961	-	1241	1241	100	1241	0	0	0	2239	87.8	0.93	Yes	
SRR17923069	GCA_022051165.1	Contig	StxOp4737_Stx2l_465	Stx2l	Stx2l_465		1322	2562	+	1241	1241	100	1241	0	0	0	2239	150.8	0.98	Yes	
SRR18191506	GCA_026135385.1	Complete	StxOp1165_Stx2a_16	Stx2a	Stx2a_16		4441667	4442907	+	1241	1241	100	1241	0	0	0	2239	51.1	1.02	Yes	
SRR18191506	GCA_026135385.1	Complete	StxOp1512_Stx2d_115	Stx2d	Stx2d_115		1007282	1008522	+	1241	1241	100	1241	0	0	0	2239	45.41	0.9	Yes	
SRR18191506	GCA_026135385.1	Complete	StxOp4_Stx1a_9	Stx1a	Stx1a_9		1908085	1909311	-	1227	1227	100	1227	0	0	0	2214	46.59	0.93	Yes	
SRR18191510	GCA_026135105.1	Complete	StxOp9_Stx1a_15	Stx1a	Stx1a_15		5686555	5687781	-	1227	1227	100	1227	0	0	0	2214	35.04	0.78	Yes	
SRR18191519	GCA_026135215.1	Complete	StxOp1577_Stx1a_9	Stx1a	Stx1a_9		1235328	1236554	-	1227	1227	100	1227	0	0	0	2214	69.28	0.96	Yes	
SRR18191528	GCA_026135465.1	Complete	StxOp1283_Stx2a_170	Stx2a	Stx2a_170		4498431	4499671	+	1241	1241	100	1241	0	0	0	2239	56.2	1.11	Yes	
SRR18191528	GCA_026135465.1	Complete	StxOp1708_Stx1a_9	Stx1a	Stx1a_9		4296265	4297491	+	1227	1227	100	1227	0	0	0	2214	116.91	2.3	Yes	
SRR18191538	GCA_026134365.1	Complete	StxOp1577_Stx1a_9	Stx1a	Stx1a_9		3782603	3783829	+	1227	1227	100	1227	0	0	0	2214	58.63	1.11	Yes	
SRR18191541	GCA_026135265.1	Complete	StxOp1036_Stx1c_12	Stx1c	Stx1c_12		361958	363186	+	1229	1229	100	1229	0	0	0	2217	44.54	1.28	Yes	
SRR18191544	GCA_025988925.1	Complete	StxOp1168_Stx2a_16	Stx2a	Stx2a_16		4780075	4781315	+	1241	1241	100	1241	0	0	0	2239	47.36	0.88	Yes	
SRR18191544	GCA_025988925.1	Complete	StxOp4_Stx1a_9	Stx1a	Stx1a_9		606999	608225	+	1227	1227	100	1227	0	0	0	2214	38.15	0.71	Yes	
SRR18191549	GCA_025988125.1	Complete	StxOp840_Stx2d_99	Stx2d	Stx2d_99		5239584	5240824	+	1241	1241	100	1241	0	0	0	2239	70.62	1.15	Yes	
SRR18191551	GCA_026134405.1	Complete	StxOp1036_Stx1c_12	Stx1c	Stx1c_12		612089	613317	+	1229	1229	100	1229	0	0	0	2217	24.64	0.9	Yes	

Fastq ID	Fasta ID	Assembly level	StxOP_nt operon identifier	Stx subtype	Variant	Operon copy	Query start	Query end	Strand	Alignment length (bp)	Reference operon length (bp)	Percent match	Positives	Mismatches	Gaps	E-value	Bit score	Coverage of stx allele (x)	Relative coverage of stx allele	Functional	Disruption
SRR18191553	GCA_025988205.1	Complete	StxOp3298_Stx2b_23	Stx2b	Stx2b_23		4725483	4726711				100								No	Frameshift mutation in stx2b subunit A gene (4725483-4726711)
SRR18191553	GCA_025988205.1	Complete	StxOp71_Stx1a_89	Stx1a	Stx1a_89		1258299	1259525	-	1227	1227	100	1227	0	0	0	2214	64.98	1.15	Yes	
SRR18191553	GCA_025988205.1	Complete	StxOp883_Stx2d_471	Stx2d	Stx2d_471	1 of 2	10810	12050	-	1241	1241	100	1241	0	0	0	2239	140.19	2.47	Yes	
SRR18191553	GCA_025988205.1	Complete	StxOp883_Stx2d_471	Stx2d	Stx2d_471	2 of 2	5286272	5287512	+	1241	1241	100	1241	0	0	0	2239	140.19	2.47	Yes	
SRR18191566	GCA_025988985.1	Complete	StxOp457_Stx2a_181	Stx2a	Stx2a_181		1742873	1744113	+	1241	1241	100	1241	0	0	0	2239	48.05	1.08	Yes	
SRR18191571	GCA_025988425.1	Complete	StxOp1594_Stx1a_9	Stx1a	Stx1a_9		4808056	4809282	+	1227	1227	100	1227	0	0	0	2214	45.56	1.33	Yes	
SRR18191580	GCA_025989005.1	Complete	StxOp1040_Stx1d_4	Stx1d	Stx1d_4		3522557	3523783	+	1227	1227	100	1227	0	0	0	2214	46.06	0.78	Yes	
SRR18191598	GCA_026134965.1	Complete	StxOp702_Stx2c_17	Stx2c	Stx2c_17		812462	813702	+	1241	1241	100	1241	0	0	0	2239	38.95	1.12	Yes	
SRR18191601	GCA_025988955.1	Complete	StxOp826_Stx2d_9	Stx2d	Stx2d_9		4619344	4620584	+	1241	1241	100	1241	0	0	0	2239	45.02	0.94	Yes	
SRR18191603	GCA_025988885.1	Complete	StxOp933_Stx2i_43	Stx2i	Stx2i_43		1702110	1703351	-	1242	1242	100	1242	0	0	0	2241	66.23	1.14	Yes	
SRR18191605	GCA_026134415.1	Complete	StxOp1168_Stx2a_16	Stx2a	Stx2a_16		3950053	3951293	+	1241	1241	100	1241	0	0	0	2239	66.59	1.26	Yes	
SRR18191606	GCA_025989075.1	Complete	StxOp1035_Stx1c_12	Stx1c	Stx1c_12		5176155	5177382	+	1228	1228	100	1228	0	0	0	2215	37.12	0.93	Yes	
SRR18191610	GCA_025988905.1	Complete	StxOp1321_Stx2a_6	Stx2a	Stx2a_6	1 of 2	926060	927300	-	1241	1241	100	1241	0	0	0	2239	36.56	1.09	Yes	
SRR18191610	GCA_025988905.1	Complete	StxOp1321_Stx2a_6	Stx2a	Stx2a_6	2 of 2	4723759	4724999	+	1241	1241	100	1241	0	0	0	2239	36.56	1.09	Yes	
SRR18191616	GCA_025988145.1	Complete	StxOp1339_Stx2b_109	Stx2b	Stx2b_109		1067638	1068873	-	1236	1236	100	1236	0	0	0	2230	55.73	0.99	Yes	
SRR18191616	GCA_025988145.1	Complete	StxOp399_Stx1c_12	Stx1c	Stx1c_12		630338	631565	+	1228	1228	100	1228	0	0	0	2215	53.27	0.95	Yes	
SRR18191618	GCA_026134125.1	Complete	StxOp1530_Stx2d_115	Stx2d	Stx2d_115		4509933	4511173	+	1241	1241	100	1241	0	0	0	2239	26.61	1.05	Yes	
SRR18191628	GCA_025988225.1	Complete	StxOp1168_Stx2a_16	Stx2a	Stx2a_16		2221305	2222545	-	1241	1241	100	1241	0	0	0	2239	59.52	1.1	Yes	
SRR18191628	GCA_025988225.1	Complete	StxOp4_Stx1a_9	Stx1a	Stx1a_9		4905178	4906404	+	1227	1227	100	1227	0	0	0	2214	53.68	0.99	Yes	
SRR18191635	GCA_026134865.1	Complete	StxOp1494_Stx2d_115	Stx2d	Stx2d_115		5181239	5182479	-	1241	1241	100	1241	0	0	0	2239			Yes	
SRR18191635	GCA_026134865.1	Complete	StxOp3298_Stx2b_23	Stx2b	Stx2b_23		5181239	5182479	-	1241	1241	100	1241	0	0	0	2239			No	Frameshift mutation in stx2b subunit A gene (1741851-1743088)
SRR18191644	GCA_026134385.1	Complete	StxOp1339_Stx2b_109	Stx2b	Stx2b_109		588786	590021	-	1236	1236	100	1236	0	0	0	2230	56	1.23	Yes	
SRR18191644	GCA_026134385.1	Complete	StxOp13_Stx1a_19	Stx1a	Stx1a_19		4832778	4834004	+	1227	1227	100	1227	0	0	0	2214	34.6	0.76	Yes	
SRR18191644	GCA_026134385.1	Complete	StxOp675_Stx2b_116	Stx2b	Stx2b_116		3667869	3669104	+	1236	1236	100	1236	0	0	0	2230	56	1.23	Yes	
SRR1871147	GCA_012824895.1	Contig	StxOp1157_Stx2a_6	Stx2a	Stx2a_6	1 of 2	32754	33994	+	1241	1241	100	1241	0	0	0	2239	60.78	0.75	Yes	
SRR1871147	GCA_012824895.1	Contig	StxOp1157_Stx2a_6	Stx2a	Stx2a_6	2 of 2	627492	628732	+	1241	1241	100	1241	0	0	0	2239	60.78	0.75	Yes	
SRR19368884	GCA_023537105.1	Contig	StxOp1540_Stx2d_132	Stx2d	Stx2d_132		400569	401809	+	1241	1241	100	1241	0	0	0	2239	64.22	0.75	Yes	
SRR1944449	GCA_012823715.1	Contig	StxOp716_Stx2c_106	Stx2c	Stx2c_106		1773	3013	-	1241	1241	100	1241	0	0	0	2239	91.09	0.83	Yes	
SRR1950298	GCA_001188105.2		StxOp675_Stx2b_116	Stx2b	Stx2b_116		17521	18756	-	1236	1236	100	1236	0	0	0	2230	83.92	1.07	Yes	
SRR1981027	GCA_014447175.1	Contig	StxOp1344_Stx2b_109	Stx2b	Stx2b_109		444	1679	+	1236	1236	100	1236	0	0	0	2230	102.68	1.19	Yes	
SRR1981699	GCA_014441045.1	Contig	StxOp429_Stx2a_16	Stx2a	Stx2a_16		12622	13862	-	1241	1241	100	1241	0	0	0	2239	167.88	1.37	Yes	
SRR1999106	GCA_003018295.1	Complete	StxOp1580_Stx1a_9	Stx1a	Stx1a_9		2577961	2579187	+	1227	1227	100	1227	0	0	0	2214	59.68	0.86	Yes	
SRR1999111	GCA_012819655.1	Contig	StxOp1133_Stx2a_6	Stx2a	Stx2a_6		194	1434	+	1241	1241	100	1241	0	0	0	2239	120.93	1.67	Yes	
SRR1999883	GCA_012818575.1	Contig	StxOp1173_Stx2a_16	Stx2a	Stx2a_16		3794	5034	-	1241	1241	100	1241	0	0	0	2239	147.16	1.08	Yes	
SRR1999997	GCA_015518365.1	Contig	StxOp923_Stx2g_26	Stx2g	Stx2g_26		103391	104632	+	1242	1242	100	1242	0	0	0	2241	93.6	0.7	Yes	
SRR2015112	GCA_003018055.1	Complete	StxOp1577_Stx1a_9	Stx1a	Stx1a_9		2083910	2085136	-	1227	1227	100	1227	0	0	0	2214	143.52	1.13	Yes	
SRR2015125	GCA_003018655.1	Complete	StxOp4_Stx1a_9	Stx1a	Stx1a_9		1574898	1576124	-	1227	1227	100	1227	0	0	0	2214	47.26	1.17	Yes	
SRR2015133	GCA_012696945.1	Contig	StxOp1448_Stx2c_152	Stx2c	Stx2c_152		29707	30947	-	1241	1241	100	1241	0	0	0	2239	93.18	1.11	Yes	
SRR2038672	GCA_012816825.1	Contig	StxOp1181_Stx2a_16	Stx2a	Stx2a_16		67564	68804	+	1241	1241	100	1241	0	0	0	2239	93.19	1.18	Yes	
SRR2056392	GCA_018096145.1	Contig	StxOp1250_Stx2a_127	Stx2a	Stx2a_127		16390	17630	+	1241	1241	100	1241	0	0	0	2239	404.56	1.49	Yes	
SRR2095948	GCA_018096125.1	Contig	StxOp1250_Stx2a_127	Stx2a	Stx2a_127		10893	12133	+	1241	1241	100	1241	0	0	0	2239	98.58	1.57	Yes	
SRR2102939	GCA_012810615.1	Contig	StxOp935_Stx2j_103	Stx2j	Stx2j_103		44971	46204	+	1234	1234	100	1234	0	0	0	2226	116.83	0.78	Yes	
SRR21085779	GCA_024971295.1	Contig	StxOp432_Stx2a_30	Stx2a	Stx2a_30		151677	152917	-	1241	1241	100	1241	0	0	0	2239	57.56	0.88	Yes	
SRR21085779	GCA_024971295.1	Contig	StxOp4728_Stx2k_40	Stx2k	Stx2k_40		12452	13692	-	1241	1241	100	1241	0	0	0	2239	57.45	0.88	Yes	
SRR2120919	GCA_018096065.1	Contig	StxOp1173_Stx2a_16	Stx2a	Stx2a_16		10607	11847	+	1241	1241	100	1241	0	0	0	2239	83.8	1.13	Yes	

Fastq ID	Fasta ID	Assembly level	STxOP_nt operon identifier	Stx subtype	Variant	Operon copy	Query start	Query end	Strand	Alignment length (bp)	Reference operon length (bp)	Percent match	Positives	Mismatches	Gaps	E-value	Bit score	Coverage of stx allele (x)	Relative coverage of stx allele	Functional	Disruption
SRR21504745	GCA_025140695.1	Contig	StxOp1575_Stx2l_21	Stx2l	Stx2l_21		1420	2660	-	1241	1241	100	1241	0	0	0	2239	126.43	1.07	Yes	
SRR21507356	GCA_025137655.1	Contig	StxOp1477_Stx2d_38	Stx2d	Stx2d_38		47068	48308	+	1241	1241	100	1241	0	0	0	2239	39.26	0.77	Yes	
SRR21507356	GCA_025137655.1	Contig	StxOp1580_Stx1a_9	Stx1a	Stx1a_9		19436	20662	-	1227	1227	100	1227	0	0	0	2214	44.96	0.88	Yes	
SRR21620895	GCA_014854575.1	Complete	StxOp1040_Stx1d_4	Stx1d	Stx1d_4		1962100	1963326	+	1227	1227	100	1227	0	0	0	2214	28.94	0.59	Yes	
SRR21634505	GCA_014805805.1	Complete	StxOp679_Stx2b_135	Stx2b	Stx2b_135		1343677	1344912	+	1236	1236	100	1236	0	0	0	2230	22.26	0.61	Yes	
SRR2177988	GCA_012803775.1	Contig	StxOp1173_Stx2a_16	Stx2a	Stx2a_16		67328	68568	+	1241	1241	100	1241	0	0	0	2239	60.72	0.81	Yes	
SRR2178103	GCA_012803055.1	Contig	StxOp1173_Stx2a_16	Stx2a	Stx2a_16		67564	68804	+	1241	1241	100	1241	0	0	0	2239	129.61	0.96	Yes	
SRR23421918	GCA_028605285.1	Contig	StxOp1250_Stx2a_127	Stx2a	Stx2a_127		533	1773	+	1241	1241	100	1241	0	0	0	2239	57.25	0.92	Yes	
SRR23421918	GCA_028605285.1	Contig	StxOp13_Stx1a_19	Stx1a	Stx1a_19		249	1475	+	1227	1227	100	1227	0	0	0	2214	77.15	1.24	Yes	
SRR23828388	GCA_029206145.1	Contig	StxOp1575_Stx2l_21	Stx2l	Stx2l_21		25195	26435	-	1241	1241	100	1241	0	0	0	2239	79.93	0.96	Yes	
SRR24225801	GCA_030898125.1	Contig	StxOp932_Stx2h_49	Stx2h	Stx2h_49		422944	424179	+	1236	1236	100	1236	0	0	0	2230	107.47	1.16	Yes	
SRR2480649	GCA_012794415.1	Contig	StxOp1035_Stx1c_12	Stx1c	Stx1c_12		173304	174531	+	1228	1228	100	1228	0	0	0	2215	71.33	0.91	Yes	
SRR2481365	GCA_012786265.1	Contig	StxOp935_Stx2j_103	Stx2j	Stx2j_103		23234	24467	+	1234	1234	100	1234	0	0	0	2226	116.58	0.92	Yes	
SRR2544681	GCA_000464955.2	Complete	StxOp1160_Stx2a_11	Stx2a	Stx2a_11		3090447	3091687	-	1241	1241	100	1241	0	0	0	2239	70.39	0.84	Yes	
SRR2549981	GCA_012790495.1	Contig	StxOp1548_Stx2d_164	Stx2d	Stx2d_164		2896	4136	-	1241	1241	100	1241	0	0	0	2239	193.53	1.14	Yes	
SRR2549981	GCA_012790495.1	Contig	StxOp1708_Stx1a_9	Stx1a	Stx1a_9		356847	358073	+	1227	1227	100	1227	0	0	0	2214	215.78	1.28	Yes	
SRR25642116	GCA_031666695.1	Contig	StxOp1448_Stx2c_152	Stx2c	Stx2c_152		2731	3971	-	1241	1241	100	1241	0	0	0	2239	226.81	1.1	Yes	
SRR25642116	GCA_031666695.1	Contig	StxOp3298_Stx2b_23	Stx2b	Stx2b_23		9437	10672	+	1236	1236	100	1236	0	0	0	2230	160.49	0.78	Yes	
SRR26362796	GCA_032610235.1	Contig	StxOp1217_Stx2a_18	Stx2a	Stx2a_18		72686	73926	+	1241	1241	100	1241	0	0	0	2239	92.7	0.76	Yes	
SRR26362796	GCA_032610235.1	Contig	StxOp4_Stx1a_9	Stx1a	Stx1a_9		174060	175286	+	1227	1227	100	1227	0	0	0	2214	86.89	0.72	Yes	
SRR26411587	GCA_032808445.1	Contig	StxOp1089_Stx2a_1	Stx2a	Stx2a_1		266703	267943	+	1241	1241	100	1241	0	0	0	2239	32.35	0.46	Yes	
SRR26411587	GCA_032808445.1	Contig	StxOp1168_Stx2a_16	Stx2a	Stx2a_16		851	2091	+	1241	1241	100	1241	0	0	0	2239	32.35	0.46	Yes	
SRR26805704	GCA_003018235.1	Complete	StxOp935_Stx2j_103	Stx2j	Stx2j_103		55452	56685	+	1234	1234	100	1234	0	0	0	2226	183.08	1.67	Yes	
SRR27018329	GCA_040131905.1	Contig	StxOp1041_Stx1d_4	Stx1d	Stx1d_4		458940	460166	+	1227	1227	100	1227	0	0	0	2214	67.45	0.92	Yes	
SRR27135499	GCA_034235735.1	Contig	StxOp1157_Stx2a_6	Stx2a	Stx2a_6		3781	5021	-	1241	1241	100	1241	0	0	0	2239	73.85	0.98	Yes	
SRR27135499	GCA_034235735.1	Contig	StxOp1593_Stx1a_9	Stx1a	Stx1a_9		65518	66744	+	1227	1227	100	1227	0	0	0	2214	57.43	0.76	Yes	
SRR27349220	GCA_034648675.1	Contig	StxOp1157_Stx2a_6	Stx2a	Stx2a_6		2466	3706	-	1241	1241	100	1241	0	0	0	2239	112.58	0.85	Yes	
SRR27349220	GCA_034648675.1	Contig	StxOp13_Stx1a_19	Stx1a	Stx1a_19		24050	25276	+	1227	1227	100	1227	0	0	0	2214	122.34	0.92	Yes	
SRR27399504	GCA_034992965.1	Contig	StxOp953_Stx2n_126	Stx2n	Stx2n_126		361276	362511	-	1236	1236	100	1236	0	0	0	2230	54.39	0.93	Yes	
SRR27958430	GCA_036595135.1	Contig	StxOp1096_Stx2a_6	Stx2a	Stx2a_6		1660	2900	-	1241	1241	100	1241	0	0	0	2239	173.92	2.2	Yes	
SRR27958430	GCA_036595135.1	Contig	StxOp1168_Stx2a_16	Stx2a	Stx2a_16		1661	2901	-	1241	1241	100	1241	0	0	0	2239	173.92	2.2	Yes	
SRR27958430	GCA_036595135.1	Contig	StxOp4_Stx1a_9	Stx1a	Stx1a_9		1095570	1096796	+	1227	1227	100	1227	0	0	0	2214	138.03	1.74	Yes	
SRR27967202	GCA_019585465.1	Complete	StxOp1168_Stx2a_16	Stx2a	Stx2a_16		275	1515	+	1241	1241	100	1241	0	0	0	2239	79.23	0.82	Yes	
SRR28356459	GCA_037194595.1	Contig	StxOp1574_Stx2g_39	Stx2g	Stx2g_39		90	1331	+	1242	1242	100	1242	0	0	0	2241	97.85	0.94	Yes	
SRR28356459	GCA_037194595.1	Contig	StxOp1586_Stx1a_9	Stx1a	Stx1a_9		2634	3860	+	1227	1227	100	1227	0	0	0	2214	77.08	0.74	Yes	
SRR28420203	GCA_037489925.1	Contig	StxOp1096_Stx2a_6	Stx2a	Stx2a_6		2629	3869	+	1241	1241	100	1241	0	0	0	2239	98.67	0.78	Yes	
SRR28420203	GCA_037489925.1	Contig	StxOp39_Stx1a_54	Stx1a	Stx1a_54		173180	174406	+	1227	1227	100	1227	0	0	0	2214	100.04	0.79	Yes	
SRR28618069	GCA_038054095.1	Contig	StxOp1124_Stx2a_6	Stx2a	Stx2a_6		190729	191969	+	1241	1241	100	1241	0	0	0	2239	212.6	0.95	Yes	
SRR28618069	GCA_038054095.1	Contig	StxOp4_Stx1a_9	Stx1a	Stx1a_9		86745	87971	+	1227	1227	100	1227	0	0	0	2214	209.67	0.93	Yes	
SRR28723267	GCA_038374585.1	Contig	StxOp1173_Stx2a_16	Stx2a	Stx2a_16		173	1413	+	1241	1241	100	1241	0	0	0	2239	81.39	1.23	Yes	
SRR28723282	GCA_038374445.1	Contig	StxOp1217_Stx2a_18	Stx2a	Stx2a_18		4150	5390	-	1241	1241	100	1241	0	0	0	2239	47.84	0.86	Yes	
SRR28723282	GCA_038374445.1	Contig	StxOp13_Stx1a_19	Stx1a	Stx1a_19		235981	237207	+	1227	1227	100	1227	0	0	0	2214	44.14	0.79	Yes	
SRR28837514	GCA_038761315.1	Contig	StxOp943_Stx2k_40	Stx2k	Stx2k_40		16773	18013	-	1241	1241	100	1241	0	0	0	2239	62.9	1.15	Yes	
SRR28994774	GCA_039605075.1	Contig	StxOp2848_Stx2a_183	Stx2a	Stx2a_183		839	2079	+	1241	1241	100	1241	0	0	0	2239	74.71	1.41	Yes	
SRR28994774	GCA_039605075.1	Contig	StxOp4_Stx1a_9	Stx1a	Stx1a_9		60645	61871	+	1227	1227	100	1227	0	0	0	2214	52.13	0.98	Yes	
SRR29067298	GCA_039664935.1	Contig	StxOp943_Stx2k_40	Stx2k	Stx2k_40		15393	16633	-	1241	1241	100	1241	0	0	0	2239	65.71	1.08	Yes	
SRR3068663	GCA_012764415.2	Contig	StxOp939_Stx2j_225	Stx2j	Stx2j_225		18851	20084	+	1234	1234	100	1234	0	0	0	2226	54.47	0.75	Yes	
SRR3136876	GCA_012726435.1	Contig	StxOp1168_Stx2a_16	Stx2a	Stx2a_16		1415	2655	+	1241	1241	100	1241	0	0	0	2239	68.35	1.33	Yes	
SRR3136876	GCA_012726435.1	Contig	StxOp966_Stx1a_18	Stx1a	Stx1a_18		31585	32811	+	1227	1227	100	1227	0	0	0	2214	36.66	0.71	Yes	
SRR3180475	GCA_003018515.1	Complete	StxOp1089_Stx2a_1	Stx2a	Stx2a_1		3599374	3600614	-	1241	1241	100	1241	0	0	0	2239	238.08	1.54	Yes	

Fastq ID	Fasta ID	Assembly level	STxOP_nt operon identifier	Stx subtype	Variant	Operon copy	Query start	Query end	Strand	Alignment length (bp)	Reference operon length (bp)	Percent match	Positives	Mismatches	Gaps	E-value	Bit score	Coverage of stx allele (x)	Relative coverage of stx allele	Functional	Disruption
SRR3180475	GCA_003018515.1	Complete	StxOp1577_Stx1a_9	Stx1a	Stx1a_9		1811918	1813144	+	1227	1227	100	1227	0	0	0	2214	120.41	0.78	Yes	
SRR3180697	GCA_012739175.1	Contig	StxOp4647_Stx2f_725	Stx2f	Stx2f_725		3450	4685	-	1236	1236	100	1236	0	0	0	2230	160.57	0.87	Yes	
SRR3180701	GCA_012750475.1	Contig	StxOp918_Stx2f_89	Stx2f	Stx2f_89		51888	53123	-	1236	1236	100	1236	0	0	0	2230	178.22	1	Yes	
SRR3189185	GCA_003017975.1	Complete	StxOp1089_Stx2a_1	Stx2a	Stx2a_1		2482658	2483898	-	1241	1241	100	1241	0	0	0	2239	101.01	0.83	Yes	
SRR3189185	GCA_003017975.1	Complete	StxOp3_Stx1a_9	Stx1a	Stx1a_9		2805840	2807066	-	1227	1227	100	1227	0	0	0	2214	93.45	0.76	Yes	
SRR3191403	GCA_001695515.1	Complete	StxOp1577_Stx1a_9	Stx1a	Stx1a_9		3187914	3189140	-	1227	1227	100	1227	0	0	0	2214	72.73	1.65	Yes	
SRR3191403	GCA_001695515.1	Complete	StxOp702_Stx2c_17	Stx2c	Stx2c_17		2900798	2902038	-	1241	1241	100	1241	0	0	0	2239	52.3	1.19	Yes	
SRR3195851	GCA_001753445.1	Complete	StxOp702_Stx2c_17	Stx2c	Stx2c_17		1969897	1971137	+	1241	1241	100	1241	0	0	0	2239	25.37	0.56	Yes	
SRR3213947	GCA_012763755.1	Contig	StxOp1040_Stx1d_4	Stx1d	Stx1d_4		96959	98185	-	1227	1227	100	1227	0	0	0	2214	171.12	1.18	Yes	
SRR3231803	GCA_003018535.1	Complete	StxOp1089_Stx2a_1	Stx2a	Stx2a_1		4779341	4780581	+	1241	1241	100	1241	0	0	0	2239	120.85	0.72	Yes	
SRR3231803	GCA_003018535.1	Complete	StxOp3_Stx1a_9	Stx1a	Stx1a_9		4458596	4459822	+	1227	1227	100	1227	0	0	0	2214	103.87	0.62	Yes	
SRR3278959	GCA_012711755.1	Contig	StxOp432_Stx2a_30	Stx2a	Stx2a_30		37667	38907	-	1241	1241	100	1241	0	0	0	2239	95.97	0.75	Yes	
SRR3278959	GCA_012711755.1	Contig	StxOp4728_Stx2k_40	Stx2k	Stx2k_40		25260	26500	+	1241	1241	100	1241	0	0	0	2239	110.12	0.86	Yes	
SRR3290048	GCA_012711215.2	Complete	StxOp953_Stx2n_126	Stx2n	Stx2n_126		2062856	2064091	+	1236	1236	100	1236	0	0	0	2230	172.57	1.18	Yes	
SRR3394639	GCA_012703965.1	Contig	StxOp1173_Stx2a_16	Stx2a	Stx2a_16		67683	68923	+	1241	1241	100	1241	0	0	0	2239	168.4	1.24	Yes	
SRR3394639	GCA_012703965.1	Contig	StxOp1259_Stx2a_128	Stx2a	Stx2a_128		492459	493699	+	1241	1241	100	1241	0	0	0	2239	168.4	1.24	Yes	
SRR3394830	GCA_012901845.1	Contig	StxOp1036_Stx1c_12	Stx1c	Stx1c_12		65692	66920	+	1229	1229	100	1229	0	0	0	2217	38.91	0.85	Yes	
SRR3438052	GCA_012686325.1	Contig	StxOp4677_Stx2g_510	Stx2g	Stx2g_510		103	1344	-	1242	1242	100	1242	0	0	0	2241	75.57	1.13	Yes	
SRR3723069	GCA_012894035.1	Contig	StxOp4731_Stx2k_519	Stx2k	Stx2k_519		6405	7645	+	1241	1241	100	1241	0	0	0	2239	229.59	1.65	Yes	
SRR3794154	GCA_012668825.1	Contig	StxOp806_Stx2c_642	Stx2c	Stx2c_642		821	2061	-	1241	1241	100	1241	0	0	0	2239	113.52	0.82	Yes	
SRR3794154	GCA_012668825.1	Contig	StxOp866_Stx2d_350	Stx2d	Stx2d_350		13379	14619	-	1241	1241	100	1241	0	0	0	2239	111.26	0.81	Yes	
SRR4176409	GCA_012627465.1	Contig	StxOp399_Stx1c_12	Stx1c	Stx1c_12		391774	393001	+	1228	1228	100	1228	0	0	0	2215	144.08	0.87	Yes	
SRR4176409	GCA_012627465.1	Contig	StxOp653_Stx2b_23	Stx2b	Stx2b_23		46407	47642	+	1236	1236	100	1236	0	0	0	2230	150.72	0.91	Yes	
SRR4430014	GCA_020372765.1	Contig	StxOp1041_Stx1d_4	Stx1d	Stx1d_4		49367	50593	-	1227	1227	100	1227	0	0	0	2214	52.33	1.02	Yes	
SRR5264142	GCA_012498465.1	Contig	StxOp1448_Stx2c_152	Stx2c	Stx2c_152		24127	25367	-	1241	1241	100	1241	0	0	0	2239	111.14	0.98	Yes	
SRR5264142	GCA_012498465.1	Contig	StxOp3298_Stx2b_23	Stx2b	Stx2b_23		9437	10672	+	1236	1236	100	1236	0	0	0	2230	90.75	0.8	Yes	
SRR5434351	GCA_012482175.1	Contig	StxOp1040_Stx1d_4	Stx1d	Stx1d_4		6339	7565	+	1227	1227	100	1227	0	0	0	2214	55.23	0.85	Yes	
SRR5616352	GCA_012466565.1	Contig	StxOp1038_Stx1d_4	Stx1d	Stx1d_4		4289	5515	-	1227	1227	100	1227	0	0	0	2214	82.29	0.75	Yes	
SRR5642675	GCA_012463025.1	Contig	StxOp941_Stx2j_705	Stx2j	Stx2j_705		708	1941	-	1234	1234	100	1234	0	0	0	2226	64.18	0.98	Yes	
SRR5832137	GCA_020364625.1	Contig	StxOp1168_Stx2a_16	Stx2a	Stx2a_16		3288	4528	-	1241	1241	100	1241	0	0	0	2239	160.91	0.79	Yes	
SRR5832137	GCA_020364625.1	Contig	StxOp1645_Stx1a_9	Stx1a	Stx1a_9		442401	443627	+	1227	1227	100	1227	0	0	0	2214	143.04	0.7	Yes	
SRR6128619	GCA_013342905.2	Complete	StxOp952_Stx2n_126	Stx2n	Stx2n_126		2949794	2951029	-	1236	1236	100	1236	0	0	0	2230	49.6	0.51	Yes	
SRR6321371	GCA_014700215.1	Contig	StxOp918_Stx2f_89	Stx2f	Stx2f_89		21194	22429	-	1236	1236	100	1236	0	0	0	2230	59.43	1.09	Yes	
SRR6767327	GCA_012253565.1	Contig	StxOp938_Stx2j_223	Stx2j	Stx2j_223		4714	5947	+	1234	1234	100	1234	0	0	0	2226	51.61	0.64	Yes	
SRR6903006	GCA_012224845.2	Complete	StxOp955_Stx2o_545	Stx2o	Stx2o_545		2085533	2086768	+	1236	1236	100	1236	0	0	0	2230	31.84	0.58	Yes	
SRR7050023	GCA_010231855.1	Complete	StxOp1043_Stx2a_1	Stx2a	Stx2a_1		2292576	2293816	-	1241	1241	100	1241	0	0	0	2239			Yes	
SRR7050023	GCA_010231855.1	Complete	StxOp702_Stx2c_17	Stx2c	Stx2c_17		1766528	1767768	-	1241	1241	100	1241	0	0	0	2239	65.86	1.16	Yes	
SRR7163944	GCA_012203485.1	Contig	StxOp949_Stx2m_547	Stx2m	Stx2m_547		1037	2272	+	1236	1236	100	1236	0	0	0	2230	103.45	2.29	Yes	
SRR7235296	GCA_014629915.1	Contig	StxOp4746_Stx2o_41	Stx2o	Stx2o_41		3396	4631	+	1236	1236	100	1236	0	0	0	2230	75.39	0.74	Yes	
SRR7612274	GCA_005042145.1	Contig	StxOp943_Stx2k_40	Stx2k	Stx2k_40		1891	3131	-	1241	1241	100	1241	0	0	0	2239	88.41	0.78	Yes	
SRR7738059	GCA_012124695.1	Contig	StxOp953_Stx2n_126	Stx2n	Stx2n_126		1756	2991	-	1236	1236	100	1236	0	0	0	2230	49.07	0.87	Yes	
SRR7767477	GCA_012118465.1	Contig	StxOp953_Stx2n_126	Stx2n	Stx2n_126		1629	2864	-	1236	1236	100	1236	0	0	0	2230	46.04	0.79	Yes	
SRR7767490	GCA_012119015.1	Contig	StxOp4596_Stx2e_36	Stx2e	Stx2e_36		253038	254273	+	1236	1236	100	1236	0	0	0	2230	110.48	1.09	Yes	
SRR7826316	GCA_003203755.1	Complete	StxOp1089_Stx2a_1	Stx2a	Stx2a_1		3003285	3004525	-	1241	1241	100	1241	0	0	0	2239	166.34	1.83	Yes	
SRR7826316	GCA_003203755.1	Complete	StxOp3_Stx1a_9	Stx1a	Stx1a_9		3323797	3325023	-	1227	1227	100	1227	0	0	0	2214	113.75	1.25	Yes	
SRR7826361	GCA_003288295.1	Complete	StxOp21_Stx1a_33	Stx1a	Stx1a_33		3191680	3192906	-	1227	1227	100	1227	0	0	0	2214	98.75	0.82	Yes	
SRR7828257	GCA_003112205.1	Complete	StxOp1141_Stx2a_6	Stx2a	Stx2a_6		2472128	2473368	-	1241	1241	100	1241	0	0	0	2239	79.72	0.79	Yes	
SRR7828257	GCA_003112205.1	Complete	StxOp4_Stx1a_9	Stx1a	Stx1a_9		1133150	1134376	+	1227	1227	100	1227	0	0	0	2214	93.34	0.92	Yes	
SRR7828259	GCA_003176855.1	Complete	StxOp1257_Stx2a_127	Stx2a	Stx2a_127		1127462	1128702	+	1241	1241	100	1241	0	0	0	2239	85.1	1.22	Yes	
SRR7828259	GCA_003176855.1	Complete	StxOp589_Stx2a_525	Stx2a	Stx2a_525		1329173	1330413	+	1241	1241	100	1241	0	0	0	2239	85.1	1.22	Yes	

Fastq ID	Fasta ID	Assembly level	STxOP_nt operon identifier	Stx subtype	Variant	Operon copy	Query start	Query end	Strand	Alignment length (bp)	Reference operon length (bp)	Percent match	Positives	Mismatches	Gaps	E-value	Bit score	Coverage of stx allele (x)	Relative coverage of stx allele	Functional	Disruption
SRR7828259	GCA_003176855.1	Complete	StxOp842_Stx2d_115	Stx2d	Stx2d_115		1915126	1916366	-	1241	1241	100	1241	0	0	0	2239	31.45	0.45	Yes	
SRR7896248	GCA_005014055.1	Complete	StxOp439_Stx2a_107	Stx2a	Stx2a_107		2922502	2923742	-	1241	1241	100	1241	0	0	0	2239			Yes	
SRR7896248	GCA_005014055.1	Complete	StxOp842_Stx2d_115	Stx2d	Stx2d_115		2341668	2342908	+	1241	1241	100	1241	0	0	0	2239	88.25	0.87	Yes	
SRR7896259	GCA_005037775.2	Complete	StxOp1208_Stx2a_16	Stx2a	Stx2a_16		2857082	2858322	-	1241	1241	100	1241	0	0	0	2239			Yes	
SRR7896259	GCA_005037775.2	Complete	StxOp1510_Stx2d_115	Stx2d	Stx2d_115	1 of 2	3053767	3055007	-	1241	1241	100	1241	0	0	0	2239	182.45	1.72	Yes	
SRR7896259	GCA_005037775.2	Complete	StxOp1510_Stx2d_115	Stx2d	Stx2d_115	2 of 2	4522494	4523734	-	1241	1241	100	1241	0	0	0	2239	182.45	1.72	Yes	
SRR7958537	GCA_003903075.2	Contig	StxOp942_Stx2j_706	Stx2j	Stx2j_706		116366	117599	+	1234	1234	100	1234	0	0	0	2226	49.63	0.69	Yes	
SRR8931387	GCA_011955925.1	Contig	StxOp1036_Stx1c_12	Stx1c	Stx1c_12		369375	370603	+	1229	1229	100	1229	0	0	0	2217	33.36	0.65	Yes	
SRR8931387	GCA_011955925.1	Contig	StxOp1348_Stx2b_110	Stx2b	Stx2b_110		816	2051	-	1236	1236	100	1236	0	0	0	2230	44.92	0.87	Yes	
SRR8931387	GCA_011955925.1	Contig	StxOp167_Stx1a_191	Stx1a	Stx1a_191		7564	8790	-	1227	1227	100	1227	0	0	0	2214	29.28	0.57	Yes	
SRR9007492	GCA_011950125.1	Contig	StxOp948_Stx2m_48	Stx2m	Stx2m_48		891	2126	+	1236	1236	100	1236	0	0	0	2230	53.94	0.65	Yes	
SRR9833314	GCA_011901845.1	Contig	StxOp937_Stx2j_108	Stx2j	Stx2j_108		38215	39448	-	1234	1234	100	1234	0	0	0	2226	58.28	0.61	Yes	
SRR9972670	GCA_018259255.1	Contig	StxOp1168_Stx2a_16	Stx2a	Stx2a_16		62346	63586	+	1241	1241	100	1241	0	0	0	2239	103.54	0.96	Yes	

The query (contig) and subject (STxOP_nt reference) sequence of every stx operon copy listed above, as FASTA records. Each header gives the genome, the reference operon identifier, the operon copy, and whether the record is the query or the subject sequence.

T CAGTCATTATTAAACTGCACCTTCAGCAAAATCCAGAGCCTGAGTCACAGGATTGGATTGTATTGTTACGGTCATTCTGTAACTGTGCACCTTTGTAACAACGGCTGCAAAATCCACCG
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[illegible]

>2025-SEQ-1251 | StxOp3_Stx1a_9 | subject (1227 bp)

TCAACGAAAAATAACCTTCGCTGAATCCCCCTCCATTATGACAGGCATTAGTTTTAATGGTTACAGTCATCCCCGTAATTTGCGCACTGAGAAGAAGAGACTGAAGATTCCATCTGTTGGTAAATAATCTTTTATCACCCACTTTAAGCTGTAAGAGTATCGTCATCATTATATTTTGTATACTCCACCTTTCCAGTTACACAATCAGGCGTCGCCAGCGCACTTGCTGAAAAAATAAGAAAGCGATGCGAGCTATTAAATAATGTTTTTTTTCATTATTTACCCCCCTCAACTGCTAAGTATTTCTGCGCATCAGAATTGCCCCAGAGTGGATGAATCCACAATATTTTATTGTGCGTAAATCCCAGGACTCTTCCATCTGCCGGACACATAGAAGGAAACTCATCAGATGCCATTCTGGCAACTCGCGATGCATGATGATGACAATTGAGTATTAATGCCACGCTTCCCAGAATTGCATTAAATGCTTCCAAAAGAAATCTCTCTACACGAAACAGAGTCTTGTCATGATAATCAGGCAGGACACTACTCAACCTTTCCCAGTTCAATGTAAGATCAACATCTTCAGCAGTCAATTACATAAGAAACGCCACTGAGATCATCGAGTGTGTACGAAATCCCCCTCTGTATTTGCGGAAACGTAAGAGTGTACAGTAACAAACCGTAAACATCGCTCTTGCCACAGACTGCGTGAAGTTCCACTATCGCAGATTAAATCCAGATAAGAAGTAGTCAACGAATGGCGATTATCTGCGATCCCCGACGACTGATCCCTGCAACACGCTGTAACGTTGGTATAGTCTACTGTCAACGAGCAAATGTAACCGCTGTTGTACCCTGGAAGGTAAACATGTGAAAAATCAGCAAGGCGATAAAAAACATTATTTGTCTGTTAAACAAATCCTGTACATATAAATTATTTCTGTTCAACAAATAAGCCGTAGATTATTAACCCGCCCTTCTCTGGATCTATCCCTCTGACATCAACTGCAAAACAAATTATCCCTGTGCCACTATCAATCATCAGTAAAGACGTACCTCTGATGAAATAGTCTGTAAAGGATACCTATTGCGAGGCGAATGACATTACGCGAATCTACATACGCTCTTGCGAGTCGAGAAGTCTAAGGTAATTCCTTCGCAACCACATTAACTGAAAAGATAACAAAGAAAAAAGTAGCACTCTAAAAATAATTATTTTTCAT

>2025-SEQ-1502 | StxOp432_Stx2a_30 | query (1241 bp)

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>2025-SEQ-1502 | StxOp432_Stx2a_30 | subject (1241 bp)

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>2025-SEQ-1502 | StxOp943_Stx2k_40 | query (1241 bp)

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>2025-SEQ-1502 | StxOp943_Stx2k_40 | subject (1241 bp)

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>2025-SEQ-1503 | StxOp13_Stx1a_19 | query (1227 bp)

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>2025-SEQ-1503 | StxOp13_Stx1a_19 | subject (1227 bp)

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>2025-SEQ-1503 | StxOp399_Stx1c_12 | query (1228 bp)

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TGATGCGAGCTATTAATAATATTTTTTTTCATTTTACCCCCCTCAACTGCTAATAGCCCTGCGTATCAAAATTTGCCCCAGAGTGGATGAGTCCCACAATATTTTATTGTGCGTAATCCCAC
GCATCTTCCATCTACCGGGACACATAGAAGGAAACTCATTAGGTACAATTTCTGGCAACTCGCGATGCATGATGATGACAATTCAGTATTAATGCCACGCTACCCAGAATTGCATTAAACAC
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TTCCGCTATTGCGACATTAAATCCAGATAAGAAGTAGTCAACGAATGGCGATTTATCTGTCATCCCGTACGACTGATCCCGCAACACGCTGTAACGTGGTATAGCTACTGTCAACGAGACA
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>2025-SEQ-1503 | StxOp399_Stx1c_12 | subject (1228 bp)

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TTCCGCTATGCGACATTAAATCCAGATAAGAAGTAGTCAACGAATGGCGATTTATCTGCATCCCCGTACGACTGATCCCGCAACACGCTGTAACGTGGTATAGCTACTGTCAACGAGACA
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>2025-SEQ-1503 | StxOp673_Stx2b_113 | copy 1 of 2 | query (1236 bp)

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CGTGATATATTATTAAGGATATTGCGCCCACTCTGACACCACCCCTCTCCCCGAAACTCCGGAAAGCACATTGCTGATTCTCCCCAGTTCAAGTGTGAGGTCACATCTTCCGGTGTATC
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CATGGCATTTCCACTAACTCCAATTAAACGCGAGATATGATGAAACAGTGAGTGACACTGATTGCAATTCGCGAACGTTCCAGCGCTGCGACACGTTGCAAGAGTGGTATAACTGCTGTC
CGTTGTCATGGAAACAGTTGTACACCGGGCACTGATATGTTGCAAAATCAGAAAAATCTGTAAAGATTTTGTGCCGTATTAACGAATCCGGCCACATATAAATTATTTTGTCTCAAC
AATCAGACGAAGATGGTCAAAACGCGCTGATAGACATCAAGCCCTCGTATATCCACAGCAAAATAACTGCCCGTGGGGTGTGGTTAATAACAGACACCGATGTGGTCCCCTGAGATAT
ATGCTCAAGAGGAGTCGATATTTCTGTCCGTATACTATTTAACGAAGAGACATAACTTTGTTGAGTCGAAAGGTCTATCTGTAATTTCTGGGAATAGGATACCGAAGAAAAAGCCAGCAA
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>2025-SEQ-1503 | StxOp673_Stx2b_113 | copy 1 of 2 | subject (1236 bp)

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>2025-SEQ-1503 | StxOp673_Stx2b_113 | copy 2 of 2 | query (1236 bp)

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CGTTGTGTCATGGAACAGTTGTACACCGGGCAGTGTATATGTGCAAAATCAGAAAATCTGTAGAAAGTATTGTTGCCGTATTAAACGAATCCGGCCACATATAAATTATTTTGTCAAC
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>2025-SEQ-1503 | StxOp673_Stx2b_113 | copy 2 of 2 | subject (1236 bp)

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>2025-SEQ-1504 | StxOp1348_Stx2b_110 | query (1236 bp)

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>2025-SEQ-1504 | StxOp1348_Stx2b_110 | subject (1236 bp)

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CGCTGATATATTATTAAGGATATTTCGCCCCACTCTGACACCAACCTCTCCCCGAAACTCCGGAAGCACATTGCTGATTCTCCCCCAGTTCAGTGTGAGGTCACACTTCTTCGGTGTCTAT
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>2025-SEQ-1504 | StxOp399_Stx1c_12 | query (1228 bp)

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TATCTGGATTTAATGTCGCATAGCGGAACCTCACTGACGCAGTCTGTGGCAAGAGCGATGTTACGGTTTGTTACTGTGACAGCTGAAGCTTTACGTTTTTCGGCAAATTCAGAGGGGATTT
CGTACAACACTTGTGATCTCAGTGGGCGTTCTTATGTAATGACTGCTGAAGATGTTGATCTTACGTTGAACCTGGGGAAGGTTGAGTAGTGCTCCTGCCTGACTATCATGGACAAGACTCT
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>2025-SEQ-1504 | StxOp399_Stx1c_12 | subject (1228 bp)

ATGAAAAATAAATTTTTAGAGTGCCTAACCTTTTTCTTTGTTATCTTTCTGTTAATGTGGTTGCGAAGGAATTTACCTTAGATTTCTCGACAGCAAAAGACGTATGTAGATTTCGCTGAAT
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CGTACAACACTTGTGATCTCAGTGGGCGTTCTTATGTAATGACTGCTGAAGATGTTGATCTTACGTTGAACCTGGGGAAGGTTGAGTAGTGCTCCTGCCTGACTATCATGGACAAGACTCT
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TCTATGTGCCCGGTAGATGGAAGAGTGCCTGGGATTACGCACAATAAAATTTGTGGGACTCATCCACTCTGGGGGCAATTTTGATACGCAGGGCTATTAGCAGTTGAGGGGGTAAAAAT
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AGTTAAAGTGGGAGATAAAGAATTTATTACTAACAGATGGAATCTTCAGTCTCTTCTTCTCAGTGACAAAATTACGGGGATGACGGTAACCATTAACATGCTGTCATAATGGAGG
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>2025-SEQ-1505 | StxOp933_Stx2i_43 | query (1242 bp)

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AGTATACGGACAGCGATATCGACCCCTCTTGAACATATATCTCAGGGAGCTACATCGGTATCCGTTATTAATCATACACCACCAGGAAGTTATATTTCCGTAGGTATACGAGGGCTTGAT
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>2025-SEQ-1505 | StxOp933_Stx2i_43 | subject (1242 bp)

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AGAGTGGGAGAAATATCCTTTAATAATATATCAGCGATACTTGGTACTGTGGCGTTTACTGAAATGCCATCATCAGGGCGCGCTTTTGTTCCGCGCGTGAATGAAGAGAGTCAACCA
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>2025-SEQ-1506 | StxOp2108_Stx1c_27 | query (1227 bp)

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CGTACAACACTTGTGATCTCAGTGGGCGTTCTTATGTAATGACTGCTGAAGATGTTGATCTTACGTTGAACCTGGGGAAGGTTGAGTAGTGCTCCTGCCTGACTATCATGGACAAGACTCT
GTTCTGTTGGGAAGAAATTTCTTTTGGAAAGTGTTAATGCAATTTCTGGGTAGCGTGGCAATTAATACTGAATTTGTCATCATCATGCATCGCGAGTTGCCAGAATTTGTACCTAATGAGTTTCCT
TCTATGTGCCCGGTAGATGGAAGAGTGCCTGGGATTACGCACAATAAAATTTGTGGGACTCATCCACTCTGGGGGCAATTTTGATACGCAGGGCTATTAGCAGTTGAGGGGGTAAAAATG
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GTTAAAGTGGGAGATAAAGAATTTATTACTAACAGATGGAATCTTCAGTCTCTTCTTCTCAGTGACAAAATTACGGGGATGACGGTAACCATTAACATGCTGTCATAATGGAGG
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>2025-SEQ-1506 | StxOp2108_Stx1c_27 | subject (1227 bp)

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CGTACAACACTTGTGATCTCAGTGGGCGTTCTTATGTAAATGACTGCTGAAGATGTTGATCTTACGTTGAACGTGGGGAAGGTTGAGTAGTGCTCCTGCCTGACTATCATGGACAAGACTCT
GTTCTGTTGGGAAGATTTCTTTTGGAAAGTGTTAATGCAATTCCTGGGTAGCGTGGCAATTAATACTGAATTTGTCATCATCATGCATCGCGAGTTGCCAGAATTTGACCTAATGAGTTTCCT
TCTATGTGCCCCGTAGATGGAAGAGTGGCTGGGATTACGCACAATAAAATTTGTGGGACTCATCCACTCTGGGGCAATTTTGATACGCGAGGCTATTAGCAGTTGAGGGGGTAAAAATG
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>2025-SEQ-1507 | StxOp13_Stx1a_19 | query (1227 bp)

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CGATGACACTTAAATAATGTTTTTTTCATTTTACCCCTCAACTGCTAATAGTTCTGCGCATCAGAATTGCCCCAGAGTGGATGAATCCACAATATTTTTATTGCGTAAATCCCACG
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TCCACTATGCGCACTTAAATCCAGATAAGAAGTAGTCAACGAATGGCGATTTATCTGCATCCCCGTACGACTGATCCCTGCAACACGCTGTAACGTGGTATAGTACTGTCAACGACAA
TGTAAACCGCTGTTGTACCTGGAAAGGTAAACATGTGAAAAATCAGCAAAAGCGATGAAAAACATTTATTTGCTGTGTTAAACAAATCCTGTACATATAAAATATTTTCGTTCAACAATAAGCCG
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TAGCACTCTAAAAATAATTATTTTCAT

>2025-SEQ-1507 | StxOp13_Stx1a_19 | subject (1227 bp)

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CGATGACACTTAAATAATGTTTTTTTCATTTTACCCCTCAACTGCTAATAGTTCTGCGCATCAGAATTGCCCCAGAGTGGATGAATCCACAATATTTTTATTGTCGTAATCCCACG
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TCCAAAAGAAATTTCTTCTACACGAACAGAGTCTGTGCCATGATAGTCAGGCAGGACACTACTCAACCTTCCCAAGTTCAATGTAAGATCAACATCTTCAGCAGTCATTACATAAGAAGC
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>2025-SEQ-1507 | StxOp675_Stx2b_116 | query (1236 bp)

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TCAGTGCCCGGTGTGACAACGTGTTCCATGACAACGGACAGCAGTTATACCACTCTGCAACGTGTGCGACGCGTGGAACGTTCCGGAATGCAAAATCAGTCGTCACTCACTGGTTTCATCA
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AGAGTGGGGCGAATATCCTTTAATAATATATCAGCGATACTGGGCACAGTGGCGGTTTACTGAAATGCCATCATCAGGGGGCACGTTCCGTTTCGCTCCGTGAATGAAGAGATACAACCA
GAATGTGAGATAACTGGCGACAGGCGAGTTATAAGGCTAAACAAATACTTTATGGGAAAGTAAATCCGCGAGCTGCTTTTCTGAAATCGCAGGGCTCACTCTTTAAATACATCCGGAGAAATAA
CGGGAGTTAAATATGAAGAAGATATTTGTAGCGGCTTTATTTGCTTTTGTGTTTCTGTTAATGCAATGGCAGCTGATTGTGCAAAAGGTAAAAATTGAGTTCTCTAAGTATAATGAGAATGAT
ACATTCACAGTAAAAGTGGCCGGGAAAGAGTACTGGACTAACCGCTGGAATCTGCAACCGCTACTGCAAAAGCGCACAGTTAACAGGAATGACGGTAACAATCAAATCAAATACCTGTGCG
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>2025-SEQ-1507 | StxOp675_Stx2b_116 | subject (1236 bp)

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CGTCTGGCACTGTCTGAAACTGCTCCTGTTTATACGATGACACCGGAAGAAAGTGGACTCACACTGAACTGGGGGAGAATCAGCAATGTGCTTCGGAGTTTCGGGGAGAGGGTGGTGT
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GAATGTGAGATAACTGGCGACAGGCGAGTTATAAGGCTAAACAAATACTTTATGGGAAAGTAAATCCGCGAGCTGCTTTTCTGAAATCGCAGGGCTCACTCTTTAAATACATCCGGAGAAATAA
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CGTACAAACCTTGATGATCTCAGTGGGCGTTCTTATGTAATGACTCGTGAAGATGTTGATCTTACGTTGAACGTGGGGAAGGTGAGTAGTGCTCTGCTGCATCATCTGGAACAGACTCT
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GAAAAAATAATTATTAAAGCTGCATCACTTTCTTTTTTCAGCAAGTGTGCTGGCTGCGCCAGTCTGTGTAACGTGGGAAGGTGGAGTATACAAAAATAATGATGACGATACCTTTAC
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TCAACGAAAAATAACTTCGCTGAATCCCCCTCCATTATGACAGGCATTAGTTTTAATGGTACAGTCATCCCCGTAATTTGCGCACTGAGAAGAAGAGACGAAGATTCCATCTGTTGGTAAATAATCTTTTATCACCCCACTTAACTGTGAAGAGTTATGCATCATCTGTTATTTGATATACCAATTTCCAGTTACACAATCAGGCGTCCGACAGCGCACTTGCTGAAAAAATAAGAAAGCGATCGAGCATTAATAAATGTTTTTTTTCATTTTACCCTCCCACTGCAATAGTTCTGCGCATCAAGTTGCCCGACAGTGGAATGAATCCCAATAATTTTATGTGCAATATATCCCGACGACTCTTCCATCTCGCCGGACACATAGAAGGAAACTCATCAGATGCCATCTG6GCAACTCG6CATGCTGATGATGACAATTCAGTATTTAATGCCACCTTCCCAGAAATTCGATTTAATGCTTCCAAAAGAAATTTCTCTACACGACGAAGAGCTTTGCTCATGATAGTTCAGGACAGGACGCTACTCAACCTTCCCCAGTCCAATGTGAAGATCAACATCTTCAGCAGTCATTCATAGAAGAACCCCATGAGATCATCCAGTGTGTGACAAAATCCCCCTGTGATTTTGGCGAAAACGTAAAGCTTCAGCTGTCACAGTAAACAAACCGTAACATCGCTCTTGCCACAGACTCGCTCATGAGAGTGTCACTATG6GACATATAATTCAGATAAGAAGATGTCAACGAAATGGCGATTTATCTGCATCCCCGATCAGCTGATCCCTGCAACACGCTGTAAAGCTGGATTATAGCTATGTCACCAACAAATGTAACCGCTGTGTGACTGGAAAGGTAAACATGTGAAAAATCAGCAAAAGCAGTAAAAACCAATTTTGTCTGTAAACAATCTGTGCATATAAGATTTGCTGTGATTAATTAACCGCCCTTCTCTGTGATCATCTCCCTGCATCAACTGCAAAACAAATTTATCCCTGTGCGCAATCAATCATCAGTAAAGACGTACCTCTGTGAATTAATGCTGTGAAATGGAGTACCTATTG6GACGCAATTCAGCTACGCTTTCGAGTGTGCAAGGAAAGTCAAGGTAATTTCCCTCGCCACCCACATTTAACTGAAAAGATAACAAAGAAAAAAGTTAGCATCTCAAAAAATAATTTTTCAT

>2025-SEQ-1619 | StxOp1577_Stx1a_9 | subject (1227 bp)

TCAACGAAAAATAACTTCGCTGAATCCCCCTCCATTATGACAGGCATTAGTTTTAATGGTTACAGTCATCCCCGTAATTTGCGCACTGAGAAGAAGAGACTGAAGATTCCATCTGTTGGTAAATAATTCTTTATCACCCACTTTAACTGTAAAGGTATCGTCATCATTATATTTTGTATACTCCACCTTTCCAGTTACACAATCAGGCGTCGCCAGCGCACTTGCTGAAAAAAATGAAAGCGATGCGAGCTATTAAATAATGTTTTTTTCATTTTACCCTTCAACTGCTAATAGTTTCTGCGCATCAGAATTGCCCCAGAGTGGATGAATCCACAATATTTTATTGTGCGTAATCCCACGGACTCTTCCATCTGCCGGACACATAGAAGGAAACTCATCAGATGCCATTCTGGCAACTCGCGATGCATGATGATGACAATTTCAGTATTAATGCCACGCTTCCCAGAATTGCATTAAATGCTTCCAAAAGAAATTTCTTCTACACGAACAGAGTCTTGTCATGATAGTCAGGCAGGACGCTACTCAACCTTCCCAGTTCAATGTAAGATCAACATCTTCAGCAGTCATTACATAAGAACGCCACTGAGATCATCCAGTGTGTACGAAATCCCCCTCTGTATTTGCCGAAACGTAAAGCTGACAGTGCACAGTAACAAACCGTAAACATCGCTCTTGCCACAGACTGCGTCAGGTGCCACTATCGCGACATTAAATCCAGATAAGAAGTAGTCAACGAATGGCGATTTATCTGCATCCCCGTACGACTGATCCCTGCAACACGCTGTAACGTGGTATAGCTACTGTCAACGAGCAA TGTAAACCGCTGTTGTACTTGGAAAGGTAAACATGTGAAAAATCAGCAAAGCGATAAAAAACATTATTTGTCTGTGTTAAACAAATCCTGTACATATAAATTATTTTCGTTCAACAATAAGCCGTAGATTATTAACCCGCCCTTCTCTGGATCTATCCCTCTGACATCAACTGCAAAACAAATTATCCCTGTGCCACTATCAATCATCAGTAAAGACGTACCTCTGATGAAATAGTCTGTAA TGGAGTACCTATTGCGAGGCGAATGACATTACAGCGAATCTACATACGTCTTTGCGAGTCGAGAAGTCTAAGGTAATTCCTTCGCCACCACATTAACTGAAAAGATAACAAGAAAAAAGT TAGCACTCTAAAAATAATTATTTTCAT

>2025-SEQ-1621 | StxOp1089_Stx2a_1 | query (1241 bp)

ATGAAGTGATATATTATTTAAATGGGTACTGTGCTGTTACTGGGTTTTCTTCGGTATCCTATTCCCAGGAGTTTACGATAGACTTTTCGACCCAAACAAAGTTATGTCTCTTCGTTAAAT AGTATACGGACAGAGATATCGACCCCTCTTGAACATATATCTCAGGGGACCACATCGGTGCTGTTATTAAACACACCCCAACGGGCAGTATTATTTGCTGTGGATATACGAGGGCTTGAT GTCTATACGGCGCGTTTTGACCATCTTCGCTGATTATTGAGCAAAATAAATTTATATGTGGCCGGGTTTCGTTAATACGGCAACAAATACTTTCTACCGTTTTTCAGATTTTACACATATA TCAGTGCCCGGTTGTGACACCGGTTTCCATGACAACGGACAGCAGTTATACCACTCTGCAACGTTGTCGACGCGCTGGAACGTTCCGGAAATGCAAAATCAGTCGTCACTCACTGGTTTCATCA TATCTGGCGTTAATGGAGTTTCAGTGGTAATACAAATGACCAGAGATGCATCCAGAGCAGTTCTGCGGTTTTGTCACGTGTCACAGCAGAAGCCTTACGCTTCAGGCAGATACAGAGAGAATTT CGTCAGGCACTGTCTGAAACTGCTCCTGTGTATACGATGACGCCGGGAGACGTGGACCTCACTCTGAACTGGGGCGAATCAGCAATGTGCTTCCGGAGTATCGGGGAGAGGATGGTGTCA AGAGTGGGGAGAATATCCTTTAATAATATATCAGCGATACTGGGGACTGTGGCCGTTATACTGAATTGCCATCATCAGGGGGCGCGTTCTGTTGCGCGCGTGAATGAAGAGAGTCAACCA GAAATGTCAGATAACTGGCGACAGGCGCTGTTATAAAAAATAACAATACATTATGGGAAAGTAAATACAGCTGCAGCGTTTTCTGAAACAGAAAGTCACAGTTTTTATATACAACGGGTAAATAA AGGAGTTAAGCATGAAGAAGATGTTTATGGCGGTTTTATTTGCATTAGCTTCTGTTAATGCAATGGCGGCGAATTGTGCTAAAGGTAATAATTGAGTTTTCCAAGTATAATGAGGATGACA CATTTACAGTGAAGGTTGACGGGAAAGAACTAGGACCAGTCGCTGGAATCTGCAACCGTTACTGCAAGGTGCTCAGTTGACAGGAATGACTGTCACAATCAAATCCAGTACCTGTGAAAT CAGGCTCCGGATTTGCTGAAGTGCAGTTTAATAATGACTGA

>2025-SEQ-1621 | StxOp1089_Stx2a_1 | subject (1241 bp)

ATGAAGTGATATATTATTTAAATGGGTACTGTGCTGTTACTGGGTTTTCTTCGGTATCCTATTCCCAGGAGTTTACGATAGACTTTTCGACCCAAACAAAGTTATGTCTCTTCGTTAAAT AGTATACGGACAGAGATATCGACCCCTCTTGAACATATATCTCAGGGGACCACATCGGTGCTGTTATTAAACACACCCCAACGGGCAGTATTATTTGCTGTGGATATACGAGGGCTTGAT GTCTATACGGCGCGTTTTGACCATCTTCGCTGATTATTGAGCAAAATAAATTTATATGTGGCCGGGTTTCGTTAATACGGCAACAAATACTTTCTACCGTTTTTCAGATTTTACACATATA TCAGTGCCCGGTTGTGACACCGGTTTCCATGACAACGGACAGCAGTTATACCACTCTGCAACGTTGTCGACGCGCTGGAACGTTCCGGAAATGCAAAATCAGTCGTCACTCACTGGTTTCATCA TATCTGGCGTTAATGGAGTTTCAGTGGTAATACAAATGACCAGAGATGCATCCAGGACGTTCTGCGTTTTGTCACGTGTCACAGCAGAAGCCTTACGCTTCAGGCAGATACAGAGAGAATTT CGTCAGGCACTGTCTGAAACTGCTCCTGTGTATACGATGACGCCGGGAGACGTGGACCTCACTCTGAACTGGGGCGAATCAGCAATGTGCTTCCGGAGTATCGGGGAGAGGATGGTGTCA AGAGTGGGGAGAATATCCTTTAATAATATATCAGCGATACTGGGGACTGTGGCCGTTATACTGAATTGCCATCATCAGGGGGCGCGTTCTGTTGCGCGCGTGAATGAAGAGAGTCAACCA GAAATGTCAGATAACTGGCGACAGGCGCTGTTATAAAAAATAACAATACATTATGGGAAAGTAAATACAGCTGCAGCGTTTTCTGAAACAGAAAGTCACAGTTTTTATATACAACGGGTAAATAA AGGAGTTAAGCATGAAGAAGATGTTTATGGCGGTTTTATTTGCATTAGCTTCTGTTAATGCAATGGCGGCGGATTGTGCTAAAGGTAATAATTGAGTTTTCCAAGTATAATGAGGATGACA CATTTACAGTGAAGGTTGACGGGAAAGAACTAGGACCAGTCGCTGGAATCTGCAACCGTTACTGCAAGGTGCTCAGTTGACAGGAATGACTGTCACAATCAAATCCAGTACCTGTGAAAT CAGGCTCCGGATTTGCTGAAGTGCAGTTTAATAATGACTGA

>2025-SEQ-1621 | StxOp3_Stx1a_9 | query (1227 bp)

TCAACGAAAAATAACTTCGCTGAATCCCCCTCCATTATGACAGGCATTAGTTTTAATGGTTACAGTCATCCCCGTAATTTGCGCACTGAGAAGAAGAGACTGAAGATTCCATCTGTTGGT AAATAATTCTTTATCACCCACTTTAACTGTAAAGGTATCGTCATCATTATATTTTGTATACTCCACCTTTCCAGTTACACAATCAGGCGTCGCCAGCGCACTTGCTGAAAAAAATGAAAG CGATGCGAGCTATTAAATAATGTTTTTTTCATTTTACCCTTCAACTGCTAATAGTTTCTGCGCATCAGAATTGCCCCAGAGTGGATGAATCCACAATATTTTATTGTGCGTAATCCCACG GACTCTTCCATCTGCCGGACACATAGAAGGAAACTCATCAGATGCCATTCTGGCAACTCGCGATGCATGATGATGACAATTTCAGTATTAATGCCACGCTTCCCAGAATTGCATTAAATGCT TCCAAAAGAAATTTCTTCTACACGAACAGAGTCTTGTCATGATAATCAGGCAGGACACTACTCAACCTTCCCAGTTCAATGTAAGATCAACATCTTCAGCAGTCATTACATAAGAACG CCCACTGAGATCATCCAGTGTGTACGAAATCCCCCTCTGTATTTGCCGAAAACTAAAGCTTCAGCTGTACAGTAACAAACCGTAACATCGCTCTTGCCACAGACTGCGTCAGTGAGGT TCCACTATGCGACATTAAATCCAGATAAGAAGTAGTCAACGAATGGCGATTATCTGCATCCCCGTACGACTGATCCCTGCAACACGCTGTAACGTGGTATAGCTACTGTCAACAGACAA TGTAAACCGCTGTTGTACTTGGAAAGGTAAACATGTGAAAAATCAGCAAAGCGATAAAAAACATTATTTGTCTGTTAAACAAATCCTGTACATATAAATTATTTTCGTTCAACAATAAGCCG TAGATTATTAACCCGCCCTTCTCTGGATCTATCCCTCTGACATCAACTGCAAAACAAATTATCCCTGTGCCACTATCAATCATCAGTAAAGACGTACCTCTGATGAAATAGTCTGTAA TGGAGTACCTATTGCGAGGCGAATGACATTACGCGAATCTACATACGTCTTTGCGAGTCGAGAAGTCTAAGGTAATTCCTTCGCAACCACATTAACTGAAAAGATAACAAGAAAAAAGT TAGCACTCTAAAAATAATTATTTTCAT

>2025-SEQ-1621 | StxOp3_Stx1a_9 | subject (1227 bp)

TCAACGAAAAATAACTTCGCTGAATCCCCCTCCATTATGACAGGCATTAGTTTTAATGGTTACAGTCATCCCCGTAATTTGCGCACTGAGAAGAAGAGACTGAAGATTCCATCTGTTGGT AAATAATTCTTTATCACCCACTTTAACTGTAAAGGTATCGTCATCATTATATTTTGTATACTCCACCTTTCCAGTTACACAATCAGGCGTCGCCAGCGCACTTGCTGAAAAAAATGAAAG CGATGCGAGCTATTAAATAATGTTTTTTTCATTTTACCCTTCAACTGCTAATAGTTTCTGCGCATCAGAATTGCCCCAGAGTGGATGAATCCACAATATTTTATTGTGCGTAATCCCACG GACTCTTCCATCTGCCGGACACATAGAAGGAAACTCATCAGATGCCATTCTGGCAACTCGCGATGCATGATGATGACAATTTCAGTATTAATGCCACGCTTCCCAGAATTGCATTAAATGCT TCCAAAAGAAATTTCTTCTACACGAACAGAGTCTTGTCATGATAATCAGGCAGGACACTACTCAACCTTCCCAGTTCAATGTAAGATCAACATCTTCAGCAGTCATTACATAAGAACG CCCACTGAGATCATCCAGTGTGTACGAAATCCCCCTCTGTATTTGCCGAAAACTAAAGCTTCAGCTGTACAGTAACAAACCGTAACATCGCTCTTGCCACAGACTGCGTCAGTGAGGT TCCACTATGCGACATTAAATCCAGATAAGAAGTAGTCAACGAATGGCGATTATCTGCATCCCCGTACGACTGATCCCTGCAACACGCTGTAACGTGGTATAGCTACTGTCAACAGACAA TGTAAACCGCTGTTGTACTTGGAAAGGTAAACATGTGAAAAATCAGCAAAGCGATAAAAAACATTATTTGTCTGTTAAACAAATCCTGTACATATAAATTATTTTCGTTCAACAATAAGCCG TAGATTATTAACCCGCCCTTCTCTGGATCTATCCCTCTGACATCAACTGCAAAACAAATTATCCCTGTGCCACTATCAATCATCAGTAAAGACGTACCTCTGATGAAATAGTCTGTAA TGGAGTACCTATTGCGAGGCGAATGACATTACGCGAATCTACATACGTCTTTGCGAGTCGAGAAGTCTAAGGTAATTCCTTCGCAACCACATTAACTGAAAAGATAACAAGAAAAAAGT TAGCACTCTAAAAATAATTATTTTCAT

>2025-SEQ-1629 | StxOp439_Stx2a_107 | query (1241 bp)

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AGAGTGGGGAGAATATCCTTTAATAATATATCAGCGATACTGAGTACTGTGGCGTTTATACTGAAATGCCATCATCAGGGGCGCGTCTCTGTTGCGCCGTTGAATGAAGAGAGTCAACCA
GAATGTGACAGTAACGTGGCGACAGGCCGTTATAAAAATAACAATACATTATGGGAAAGTAATACAGCAGCAGCGGTTTCTGAAACAGAAAGTCACAGTTTTTATATACAACGGGTAAATAA
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>2025-SEQ-1629 | StxOp439_Stx2a_107 | subject (1241 bp)

ATGAAGTGATATATTATTTAAATGGGTACTGTGCCTGTTACTGGGTTTTCTTCGGTATCCTATTTCCGGGAGTTTACGATAGACTTTTCGACCCAACAAAGTTATGTCTCTCGTTAAAT
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AGAGTGGGGAGAATATCCTTTAATAATATATCAGCGATACTGAGTACTGTGGCGTTTATACTGAAATGCCATCATCAGGGGCGCGTCTCTGTTGCGCCGTTGAATGAAGAGAGTCAACCA
GAATGTGACAGTAACGTGGCGACAGGCCGTTATAAAAATAACAATACATTATGGGAAAGTAATACAGCAGCAGCGGTTTCTGAAACAGAAAGTCACAGTTTTTATATACAACGGGTAAATAA
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CATTTACAGTGAAGGTTGACGGGAAAGAACTGGAACAGTCGCTGGAATCTGCAACCGTTACTGCAAAAGTGTCTAGCTGACAGGAATGACTGTCAACAATCAAATCCAGTACCTGTGAAT
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>2025-SEQ-1629 | StxOp842_Stx2d_115 | query (1241 bp)

TCAGTCATTATTTAACTGCACTTCAGCAAAATCCGGAGCCTGATTACAGGTAAGTGGATTGATTGTGACAGTCATTCTCTGTCACTGAGCAGTTCGAGTAACGGTTGCAGATTCAGCG
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CTGGTCATTGTATTAACTGAACTCCAATTAAACGCCAGATATGATGAAACAGTGAGTGACAGTGAATTTGCAATTCGGGAACGTTCCAGCGCTGCGACACGTTGACAGTGTTGATATAACTG
CTGTCGTTGTGATGGAACCGTTGTGACACCGGGCAGTATATATGTGTAATACTGAAAAACGCGTAGAAAGTATTTGTTGCCGTATTAACGAACCCGGGCCACATATAAATATTTTGC
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GATATATGTTCAAGAGGGGTCGATATCTGTCGGTATACTATTAAACGAAGAGACATAACTTTGTTGAGTCAAAAAGTCTATCGTAAATTCGGGGAATAGGATACCGAAGAAAAACCC
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>2025-SEQ-1629 | StxOp842_Stx2d_115 | subject (1241 bp)

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CTGGTCATTGTATTACCACTGAACTCCAATTAAACGCCAGATATGATGAAACAGTGAGTGACAGTGAATTTGCAATTCGGGAACGTTCCAGCGCTGCGACACGTTGCAGAGTGGTATAACTG
CTGTCGTTGTGATGGAACCGTTGTGACACCGGGCAGTATATATGTGTAATACTGAAAAACGCGTAGAAAGTATTTGTTGCCGTATTAACGAACCCGGGCCACATATAAATATTTTGC
TCAATAATCAGACGAAGATGGTCAAACGCGCCTGATAGACATCAAGCCCTCGTATATCCACAGCAAAATAACTGCCCGCGGGGTTGGTTAATAACAGACACCGATGTGGTCCCTGA
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>2025-SEQ-1632 | StxOp429_Stx2a_16 | query (1241 bp)

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>2025-SEQ-1632 | StxOp429_Stx2a_16 | subject (1241 bp)

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>2025-SEQ-1632 | StxOp9_Stx1a_15 | query (1227 bp)

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CGATGCGACTATTAATAATGTTTTTTCATTTTACCCCTCAACTGCTAATAGTTTCTGCGCATCAGAATTGCCCCAGAGTGGATGAATCCACAATATTTTATTGTGCGTAATCCCACG
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TCCAAAAGAAATTTCTTCTACACGAACAGAGTCTTGTCCATGATAGTCAGGCAGGACGCTACTCAACCTTCCCAAGTTCAATGTAAGATCAACATCTTCAGCAGTCAATACATAAGAAGC
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>2025-SEQ-1632 | StxOp9_Stx1a_15 | subject (1227 bp)

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>2025-SEQ-1667 | StxOp2209_Stx1d_4 | query (1227 bp)

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>2025-SEQ-1667 | StxOp2209_Stx1d_4 | subject (1227 bp)

TCAACGAAAAATAACCTCGCTGAATCCCCCTCCATTATGACAAGCAGTAGCTTAAATAGTTACCGTCATCCCCGTAATTTGTGCACCTAGAAGAAGAGACTGAAGATTCATCTGTTAGTAAATAATCTTTTATCTGCCACTTTAACTGTAAGAGTATCGTCATCATTTATATTTTGTATACTCCACCTTTCCGGTTACACAATCTGGCGCTGCCAGAACACTTGCTGAAAGAAATGAAAGTGAACACAGCTTAAATAATACTTTTTTTCATTTTACCCTTCAACTACTGATAGTTCTGCGGAATCAAAATTTGCCCCAGTGTAGATGAGTCCACAATATTTTATTGTGAGTAATCCCACGCCACTTCCATCTGTCTGGGCACATAGAAGGAAACTCATCAGGTGTCATTCTGGCAACTCGTGATGCATGATGATGGCAATTTAATATTAATGCCACGCTTCCCAGAATTGCATTAACTCTCCAAAAGAAATTTCTTCCAACACGAAACGGAGGCTTGTCATGATAGTCGGGCAGAAATACTACTCAGTCTTCCCCAATTCAATGTAAGATCAACATCTTCAGCAGTCAATTACATAAGAACGTCCTACTGAGATCATCAAGTGTGTACGAAATCCCTCTGGATTTGGCGAAACCCGAAAGCTTTCAGTGTGCACAGTAACAAACCGTAACATTGCTCTTGCTACAGACTGCGTCAGTCAGGTGCCGTATACGACATCAAAATCCAGATAAGAAGTAGTCAGCGAAATGGCGATTTATCTGCATCCCCGTACGACTGATCCCCGCAACACGCTGTAACGTGGTATAGTACTGTCAACGAGACAAAGTAAACCGCTTTGTACCAGGAAGGTAAAGTGTGAAAAATCAGCAAAAGCTAAAAACATTGTTTGTCTGTTAAACAAATCCTGTACATATAAATTATTTGTTCAACAATCAGTCGTAGATTGTTAAACCGCTCTTCTCTGTGGTTCTAACCCCATGATATCGACTGCAAAATAAATTATCGCTGTGCCACTATCAATCATCAGTAAAGATGTACCTCCCGATGAAATAGTCTGTAA TGCGCTACCTATTGCGAGGCGAATGACATTACGCGAATCAACATACTTCTTGTCTGTTGAGAAATCTAAGGTAACCTCCTTAGCAATCGCATTAACTGAAAAGATAACGAAGAAAAATGTAGCGCCCTAAAAATCATTATTTTCAT

>2025-SEQ-1668 | StxOp1355_Stx2b_194 | query (1236 bp)

TTAATTAACCTGCATTCAGCAAAATCCTGAACCTGACGCACAGGTATTTGATTGTGATTGTTACCGTCATTCCTGTTAACTGTGCGCTTTGCAGTAGCGGTTGCAGATTCAGCGGTTAGTCCAGTACTCTTTCCCGGCCACTTTTACTGTGAATGTATCATTTCTCATTTAATCTGTTATTTCTCCGGATGTATTTAAAGAGTAGAGCCCTGCGATTACAGAAAAGCAGCTGCGGTATTACTTTCCCATAAAGTATTGTTTATAGCCGCTACAAATATCTTCTTCATATTTAACTCCTGTTATTTCTCCGGATGTATTTAAAGAGTAGAGCCCTGCGATTACAGAAAAGCAGCTGCGGTATTACTTTCCCATAAAGTATTGTTTATCCTTATAACTGGCTGTCTGCGCAGTTATCTGACATTTCTGTTGTATCTCTTCACTCACGGCGCGAACGGAACGTCGCCCTGATGATGGCAATTCAGTATAACCGCCACTGTGCCAGTATCGCTGATATATTATTAAGGATATTCGCCCCACTCTGACACCCACTCTCCCCGAAACTCCGGGAAGCACATTGCTGATTTCTCCCCAGTTCAGTGTGAGGTCCTACTTTCCGGTGTTCATCGTATAAACAGGAGCAGTTTCAGACAGTGCCAGACGAAATTTCTCTGTATTTGCCCTGAACCGTAAGGCTTCTGCTGTGACAGTGACAAAACGCGAGAATGCTCTGGATGCATCTCTGGTCATGGCATTTCCACTAAACTCCAATTAAACGCCAGATGATGAAACGAGTAGTGAATTTGATCTTCAGTTCGGGAACGTTCCAGCGCTGCGACACGTTGCAGAGTGGTATAACTGCTGTGTCGTTGTCATGGAACAGTTGTACACCGGGCAGTGTATATGTGCAAAATCTGAAAAATCTGTAGAAGTATTGTTGCCGTATTAAACGAATCCGGCCACATATAAATTATTTTGTCTCAATAACAGACGAAGATGGTCAAAACGCGCTGATAGACATCAAGCCCTCGTATATCCACAGCAAAATAACTGCCCGTGGGGTGTGGTTAATAACAGATACCGATGTGGTCCCCGTGAGATATGTTCAAGTGGGGTTCGATATTTCTGTCCGTATACAATTTAACGAAGAGACATAAATTTGTTGAGTCGAAAAGTCTATCTGTAATTTCCGGGAATAGGATACCGGAAGAAAGCCCAAGTAA CAGGCACAGTACCCATTTTAAACAATATACACTTCAT

>2025-SEQ-1668 | StxOp1355_Stx2b_194 | subject (1236 bp)

TTAATTAACCTGCATTCAGCAAAATCCTGAACCTGACGCACAGGTATTTGATTGTGATTGTTACCGTCATTCCTGTTAACTGTGCGCTTTGCAGTAGCGGTTGCAGATTCAGCGGTTAGTCCAGTACTCTTTCCCGGCCACTTTTACTGTGAATGTATCATTTCTCATTTAATCTGTTATTTCTCCGGATGTATTTAAAGAGTAGAGCCCTGCGATTACAGAAAAGCAGCTGCGGTATTACTTTCCCATAAAGTATTGTTTATAGCCGCTACAAATATCTTCTTCATATTTAACTCCTGTTATTTCTCCGGATGTATTTAAAGAGTAGAGCCCTGCGATTACAGAAAAGCAGCTGCGGTATTACTTTCCCATAAAGTATTGTTTATCCTTATAACTGGCTGTCTGCGCAGTTATCTGACATTTCTGTTGTATCTCTTCACTCACGGCGCGAACGGAACGTCGCCCTGATGATGGCAATTCAGTATAACCGCCACTGTGCCAGTATCGCTGATATATTATTAAGGATATTCGCCCCACTCTGACACCCACTCTCCCCGAAACTCCGGGAAGCACATTGCTGATTTCTCCCCAGTTCAGTGTGAGGTCCTACTTTCCCGTGTTCATCGTATAAACAGGAGCAGTTTCAGACAGTGCCAGACGAAATTTCTCTGTATTTGCCCTGAACCGTAAGGCTTCTGCTGTGACAGTGACAAAACGCGAGAATGCTCTGGATGCATCTCTGGTCATGGCATTTCCACTAAACTCCAATTAAACGCCAGATATGATGAAACCAAGTGAAGTACGACTGATTTGCAATTCGGGAACGTTCCAGCGCTGCGACACGTTGCAGAGTGGTATAACTGCTGTGTCGTTGTCATGGAACAGTTGTACACCGGGCAGTGTATATGTGCAAAATCTGAAAAATCTGTAGAAGTATTGTTGCCGTATTAAACGAATCCGGCCACATATAAATTATTTTGTCTCAATAACAGACGAAGATGGTCAAAACGCGCTGATAGACATCAAGCCCTCGTATATCCACAGCAAAATAACTGCCCGTGGGGTGTGGTTAATAACAGATACCGATGTGGTCCCCGTGAGATATGTTCAAGTGGGGTTCGATATTTCTGTCCGTATACAATTTAACGAAGAGACATAAATTTGTTGAGTCGAAAAGTCTATCTGTAATTTCCGGGAATAGGATACCGGAAGAAAGCCCAAGTAA CAGGCACAGTACCCATTTTAAACAATATACACTTCAT

>2025-SEQ-1668 | StxOp399_Stx1c_12 | query (1228 bp)

ATGAAAAATAAATTTTATAGAGTGTAACTTTTTTCTTGTATCTTTCTGTGTTAATGTGGTTGCGAAGGAATTTACCTTAGATTTCTCGACAGCAAAAGACGTATGTAGATTCTGCTGAATGTATCTCGCTCTGCAATAGGTACTCCATTACAGACTATTTTCATCAGGAGGTACGCTTTTACTGATGATTGATAGTGGCACAGGGGATAATTTGTTTGCAGTTGATGTACAGAGGGATAGATCCAGAGGAAGGGCGGTTTAAATAACTACGGCTTATTGTTGAACGAATAATTTATATGTGACAGGATTTGTTAAACAGGACAATAATGTTTTTTATCGCTTTTGCTGATTTTTACATGTTACCTTTCTGGTACAACCTGCGGTTACATTGTCTGGTGACAGTAGCTATACCACGTTACAGCGTGTGCGGGGATCAGTCTGACGGGGATGCAGATAAATCGCCATTCTGTTGACTACTTCTTATCTGGATTTAATGTCGATAGCGGAACCTCACTGACGCGAGTCTGTGGCAAGAGCGATGTTACGGTTTGTACTGTGACAGCTGAAGCTTTACGTTTTCGGCAAAATTCAGAGGGGATTCGTACAACACTTTGATGATCTCAGTGGGCGTTCTTATGTAAATGACTGCTGAAGATGTTGATCTACGTTGAACCTGGGGAAGGTTGAGTAGTGTCTGCTGACTATCATGGACAAGACTCTGTCTGCTGTTGGAAGAAATTTCTTTTGAAGTGTAAATGCAATTTCTGGGTAGCGGGAATTAATCACTGAAATGTGCATCATCATGTCATCGCGAGTTGCCAGAATTTGACCTAATGAGTTTCTCTTCTATGTGCCGAGATGGAAGAGTGCGTGGGATACGCACAATAAAATTTGTGGGACTCATCCACTCTGGGGCAATTTTGATACGCAGGGCTATTAGCAGTTGAGGGGGTAAATGAAAAAATATTATTAATAGCTGCATCACTTTTCATTTTTCAGCAAGTGTGCTGGCTGCGCCAGATTGTGTAACCTGGGAAGGTGGAGTATACAAAATATAATGATGACGATACCTTTACAGTTAAAGTGGGAGATAAGAAATTTATTACTAACAGATGGAATCTTCAGTCTCTTCTTCTCAGTGCACAAATACGGGGATGACGGTAACCATTAACCTAATGCCTGTCAATGAGGAGGGATTCACTGATTTTCGTTG

>2025-SEQ-1668 | StxOp399_Stx1c_12 | subject (1228 bp)

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GTCATTGCGCTCTGCAATAGGTACTCCATTACAGACTATTTTCATCAGGAGGTACGTCTTTACTGATGATTGATAGTGGCAGAGGGGATAATTTGTTTGCAGTTGATGTCAGAGGGATAGAT
CCAGAGGAAGGGCGGTTTTAATAATCTACGGCTTATTGTTGAACGAAATAATTTATATGTGACAGGATTTGTTAAACAGGACAAATAATGTTTTTATCGCTTTTGCTGATTTTTACATGTT
ACCTTTCTGGTACAACCTGCGGTTACATTGTCTGGTGACAGTAGCTATACCACGTTACAGCGTGTTCGGGGGATCAGTCGTACGGGATGACAGATAAATGCCATTGTTGACTACTTCT
TATCTGGATTTAATGTCGCATAGCGGAACCTCACTGACGCAGTCTGTGGCAAGAGCGATGTTACGGTTTGTTACTGTGACAGCTGAAGCTTTACGTTTTCGGCAAATTCAGAGGGGATTT
CGTACAACACTTGTGATCTCAGTGGGCGTTCTTATGTAACTGACTGCTGAAGATGTTGATCTTACGTTGAACCTGGGGAAGGTTGAGTAGTGTCTCGCTGACTATCATGGACAAGACTCT
GTTGCGTGTGGGAAGATTTCTTTTGGAAAGTGTTAATGCAATTCCTGGGTAGCGTGGCAATTAATACTGAATTTGTCATCATCATGCATCGCAGGTTGCCAGAATTTGACCTAATGAGTTTCCT
TCTATGTGCCGGTAGATGGAAGAGTGCCTGGGATTACGCACAATAAAATATTTGCGGACTCATCCACTCTGGGGGCAATTTTGATACGACGGGCTATTAGCAGTTGAGGGGGTAAAT
GAAAAAATAATTATTAATAGTCATCATCTTTCACTTTTTTCAGCAAGTGTGCTGGCTGCGCCAGATTGTGTAACCTGGGAAGGTGGAGTATACAAAATAAATGATGACGATACCTTTAC
AGTTAAAGTGGGAGATAAAGAATTATTTACTAACAGATGGAATCTTCAGTCTCTTCTTCTCAGTGCACAAATTACGGGGATGACGGTAACCATTAACCTAATGCCTGTCATAATGGAGG
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>2025-SEQ-1668 | StxOp419_Stx2a_1 | query (1241 bp)

TCAGTCATTATTAACCTGCACCTTCAGCAAAATCCGGAGCCTGATTACAGGTAAGTGGATTTGATTGTGACAGTCATTCTGTCAACTGAGCACTTTGCAGTAACGGTTGCAGATTCAGCG
ACTGGTCCAGTATCTTTCCCGTCAACCTTCACCTGTAAATGTGTCATCCTCATTATACTTGGAAAACTCAATTTTACCTTTAGCACAATCCGCCGCCATTGCATTAAACAGAAGCTAATGC
AAATTAACCCCGCCATAAACATCTTCTTCATGCTTAACTCCTTTATTTACCGGTTGATATAAAAACTGTGACTTTTCTGTTCAGAAAACGCTGCAGCTGATTTACTTTCCCATAAATGATTG
TTTATTTTTATAACGGGCGTGTGCGCAGTTATCTGACATTTCTGGTTGACTCTCTTCATTACAGCGCGCAACAGAACCGGCCCTGATGATGGCAATTCAGTATAACGGCCACAGTCCCC
AGTATCGCTGATATATTATTAAGGATATCTCCCCACTCTGACACCATCTCTCCCGGATACCTCCGGAAGCACATTGCTGATTTCGCCCCAGTTCCAGAGTGAGGTCACGCTCTCCGGC
GTCACTGATATACACAGGAGCAGTTTCAGACAGTGGCTGACGAAATTTCTCTGTATCTGCCTGAAAGCGTAAGGCTTCTGCTGTGACAGTGACAAAAACGCAAGCTGCTCTGGATGCATCT
CTGGTCATTGTATTACCACTGAACCTCAATTAACGCCAGATATGATGAAACCAAGTGAAGTGCAGCTGATTGTCATTCCGGAACGTTCCAGCGCTGCGACACGTTGCAGAGTGGTATAACTG
CTGTCCGTTGTGATGGAAACCGTTGTGCACACGGGCACTGATATATGTTGTAAGATCTGAAACACGGTAGAAAGTATTTGTTGCCGTATTAACGAACCCGGCCACATATAAATTTATTTG
TCAATAATCAGACGAAGATGGTCAAACCGCGCTGATAGACATCAAGCCCTCGTATATCCACAGCAAAATAACTGCCCGTGGGGTGTGGTTAATAACAGACACCGATGTGGTCCCTGA
GATATATGTTCAAGAGGGGTCGATATCTCTGTCCGTATACCTTTAACGAAGAGACATAACTTTGTTGGGTCGAAAAGTCTATCGTAAACTCCCGGAATAGGATACCGAAGAAAAACCC
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>2025-SEQ-1668 | StxOp419_Stx2a_1 | subject (1241 bp)

TCAGTCATTATTAACCTGCACCTTCAGCAAAATCCGGAGCCTGATTACAGGTAAGTGGATTTGATTGTGACAGTCATTCTGTCAACTGAGCACTTTGCAGTAACGGTTGCAGATTCAGCG
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AAATTAACCCCGCCATAAACATCTTCTTCATGCTTAACTCCTTTATTTACCGGTTGATATAAAAACTGTGACTTTCTGTTTCAGAAAACGCTGCAGCTGATTTACTTTCCCATAAATGATTG
TTTATTTTTATAACGGGCGTGTGCGCAGTTATCTGACATTTCTGGTTGACTCTCTTCATTACAGCGCGCAACAGAACCGGCCCTGATGATGGCAATTCAGTATAACGGCCACAGTCCCC
AGTATCGCTGATATATTATTAAGGATATCTCCCCACTCTGACACCATCTCTCCCGGATACCTCCGGAAGCACATTGCTGATTTCGCCCCAGTTCCAGAGTGAGGTCACGCTCTCCGGC
GTCACTGATATACACAGGAGCAGTTTCAGACAGTGGCTGACGAAATTTCTCTGTATCTGCCTGAAAGCGTAAGGCTTCTGCTGTGACAGTGACAAAAACGCAAGCTGCTCTGGATGCATCT
CTGGTCATTGTATTACCACTGAACCTCAATTAACGCCAGATATGATGAAACCAAGTGAAGTGCAGCTGATTGTCATTCCGGAACGTTCCAGCGCTGCGACACGTTGCAGAGTGGTATAACTG
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TCAATAATCAGACGAAGATGGTCAAACCGCGCTGATAGACATCAAGCCCTCGTATATCCACAGCAAAATAACTGCCCGTGGGGTGTGGTTAATAACAGACACCGATGTGGTCCCTGA
GATATATGTTCAAGAGGGGTCGATATCTCTGTCCGTATACCTTTAACGAAGAGACATAACTTTGTTGGGTCGAAAAGTCTATCGTAAACTCCCGGAATAGGATACCGAAGAAAAACCC
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>2025-SEQ-1668 | StxOp670_Stx2b_109 | query (1236 bp)

ATGAAGTGATATTGTTAAAAATGGATACTGTGCTGTTGCTGGGCTTTTCTTCGGTATCCTATTCCCGGGAATTTACGATAGACTTTTGCAGCTCAACAAAGTTATGTCTCTTCGTTAAAT
AGTATACGGACAGAAATATCGACTCCTCTTGAGCATATATCTCAGGGACACATCGGTGTCTGTTATTAAACCATACCCACCAGGGCAGTTATTTTGCTGTGGATATACGAGGGCTTGAT
GTCTATCAGGCGCGTTTTGACCATCTTCGCTGATTGTTGAGCAAAATAATTTATATGTGGCGGGATTGTTAATACGGCAACAAATACTTTCTACAGATTTTCTGATTTTGACATATA
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TATCTGGCGTTAATGGAGTTTAGTGGAAATGCCATGACCAGAGATGCATCCAGAGCAGTTTTCGCTTTTGTCATCTGTACAGCAGAGAAGCCTTACGGTTTCAGGCAAAATACAGAGAGAATTT
CGTCTGGCACTGTCTGAAACTGCTCCTGTTTATACGATGACACCGGAAGAAGTGGACCTCACACTGAACTGGGGGAGAATCAGCAATGTGCTTCCGAGTTTCGGGGAGAGGGTGGTGT
AGAGTGGGGCGAATATCCTTTAATAATATATCAGCGATACTGGGCACAGTGGCGGTTACTGAAATGCCATCATCAGGGGGCGGTTCCGTTTCGCGCCGTGAATGAAGAGATACAACCA
GAATGTGCAGATAACTGGCGACAGGCCAGTTATAAGGATAAACAATACTTTATGGGAAAGTAATACCCGACGTGCTTTTCTGAAATCGCAGGGGCCACTCTTTAAATACATCCGGAGAATAA
CAGGAGTTAAATATGAAGAAGATATTTGTAGCGGCTTTATTTGCTTTTGTGTTCTGTTAATGCAATGGCAGCTGATTGTGCAAAAGGTAAAAATTGAGTTCTCTAAGTATAATGAGAATGAT
ACATTCACAGTAAAAGTGGCCGGGAAAGAGTACTGGACTAACCGCTGGAATCTGCAACCGCTACTGCAAGCGCACAGTTAACAGGAATGACGGTAACAATCAAAATCAAAATACCTGTGCG
TCAGGTTACAGGATTTGCTGAAGTGCAGTTTAATTA

>2025-SEQ-1668 | StxOp670_Stx2b_109 | subject (1236 bp)

ATGAAGTGATATTGTTAAAAATGGATACTGTGCTGTTGCTGGGCTTTTCTTCGGTATCCTATTCCCGGGAATTTACGATAGACTTTTGCAGCTCAACAAAGTTATGTCTCTTCGTTAAAT
AGTATACGGACAGAAATATCGACTCCTCTTGAGCATATATCTCAGGGGACACATCGGTGTCTGTTATTAAACCATACCCACCAGGGCAGTTATTTTGCTGTGGATATACGAGGGCTTGAT
GTCTATCAGGCGCGTTTTGACCATCTTCGCTGATTGTTGAGCAAAATAATTTATATGTGGCGGGATTGTTAATACGGCAACAAATACTTTCTACAGATTTTCTGATTTTGACATATA
TCAGTGCCCGGTGTGACAACCTGTTTCCATGACAACGGACAGCAGTTATACCACTCTGCAACGTGTGCGACGCGTGGAACGTTCCGGAATGCAAAATCAGTCGTCACTCACTGGTTTCATCA
TATCTGGCGTTAATGGAGTTTAGTGGAAATGCCATGACCAGAGATGCATCCAGAGCAGTTTTCGCTTTTGTCATCTGTACAGCAGAGAAGCCTTACGGTTTCAGGCAAAATACAGAGAGAATTT
CGTCTGGCACTGTCTGAAACTGCTCCTGTTTATACGATGACACCGGAAGAAGTGGACCTCACACTGAACTGGGGGAGAATCAGCAATGTGCTTCCGAGTTTCGGGGAGAGGGTGGTGT
AGAGTGGGGCGAATATCCTTTAATAATATATCAGCGATACTGGGCACAGTGGCGGTTACTGAAATGCCATCATCAGGGGGCGGTTCCGTTTCGCGCCGTGAATGAAGAGATACAACCA
GAATGTGCAGATAACTGGCGACAGGCCAGTTATAAGGATAAACAATACTTTATGGGAAAGTAATACCCGACGTGCTTTTCTGAAATCGCAGGGGCCACTCTTTAAATACATCCGGAGAATAA
CAGGAGTTAAATATGAAGAAGATATTTGTAGCGGCTTTATTTGCTTTTGTGTTCTGTTAATGCAATGGCAGCTGATTGTGCAAAAGGTAAAAATTGAGTTCTCTAAGTATAATGAGAATGAT
ACATTCACAGTAAAAGTGGCCGGGAAAGAGTACTGGACTAACCGCTGGAATCTGCAACCGCTACTGCAAGCGCACAGTTAACAGGAATGACGGTAACAATCAAAATCAAAATACCTGTGCG
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>2025-SEQ-1675 | StxOp1039_Stx1d_4 | query (1227 bp)

TCAACGAAAAATAACCTCGCTGAATCCCCCTCCATTATGACAAGCAGTGGTCTTAATAGTTACCGTCATCCCCGTAATTTGTGCACCTAGAGAAGAGAGACTGAAGATTCATCTGTTAGTAAATAATCTTTTATCTGCCACTTTAACTGTAAGAGTATCGTCATCATTATATTTTGTATACTCCACCTTTCCGGTTACACAATCTGGCGCTGCCAGAACACTTGCTGAAAGAAATGAAAGTGAAACAGCTATTAAATAACTTTTTTTCATTTTACCCCTCAACTACTGATAGTTCTGCGAATCAAAATTGCCCCAGTGTAGATGAGTCCACAATATTTTATTGTGAGTAATCCACGCCACTTCCATCTGTGCGGCACATAGAAGGAAACTCATCAGGTGTCATTCTGGCAACTCGTGATGCATGATGATGGCAATTTAATATTAATGCCACGCTTCCCAGAATTGCATTAACTCCCAAAAGAAATCTTCCAACACGAAACGGAGTCTTGTCATGATAGTCGGGCAGAATACTACTCAGTCTTCCCAATTCAATGTAAGATCAACATCTTCAGCAGTCAATTACATAAGAAGCTCCACTGAGATCATCAAGTGTGTACGAAATCCCTCTGGATTTGGCGAAACCCGAAAGCTTCAGCTGTGCACAGTAACAAACCGTAACATTGCTCTTGCTACGAGCTGCCTCAGTGAGGTCCCGCTATACGACATCAAAATCCAGATAAGAAGTAGTCAGCGAATGGCGATTTATCTGCATCCCCGTACGACTGATCCCCGCAACACGCTGTAACGTGGTATAGTCTACTGTCAACGAGACAAATGTAACCGCTTTGTACCAGGAAGGTAAACGTGTGAAAAATCAGCAAAAGCGATAAAAAACATTGTTTGTCTGTTAAACAAATCCTGTACATATAAATATTTTCGTTCAACAAATCAGTCGTAGATTGTTAAACCGCTCTTCTCTGTGGTTCTAACCCCATGATATCGACTGCAAAATAAATTATCGCTGTGCCACTATCAATCATCAGTAAAGATGTACCTCCCGATGAAATAGTCTGTAA TGCGCTACCTATTGACAGAGCGAATGACATTACGCGAATCAACATACTTCTTTGTCTGTTGAGAAATCTAAGGTAAACTCCTTAGCAATCGCATTAACTGAAAAGATAACGAAGAAAAATGTAGCGCCCTAAAAATCATTATTTTCAT

>2025-SEQ-1675 | StxOp1039_Stx1d_4 | subject (1227 bp)

TCAACGAAAAATAACCTCGCTGAATCCCCCTCCATTATGACAAGCAGTGGTCTTAATAGTTACCGTCATCCCCGTAATTTGTGCACCTAGAGAAGAGAGACTGAAGATTCATCTGTTAGTAAATAATCTTTTATCTGCCACTTTAACTGTAAGAGTATCGTCATCATTATATTTTGTATACTCCACCTTTCCGGTTACACAATCTGGCGCTGCCAGAACACTTGCTGAAAGAAATGAAAGTGAAACAGCTATTAAATAACTTTTTTTCATTTTACCCCTCAACTACTGATAGTTCTGCGAATCAAAATTGCCCCAGTGTAGATGAGTCCACAATATTTTATTGTGAGTAATCCACGCCACTTCCATCTGTGCGGCACATAGAAGGAAACTCATCAGGTGTCATTCTGGCAACTCGTGATGCATGATGATGGCAATTTAATATTAATGCCACGCTTCCCAGAATTGCATTAACTCCCAAAAGAAATCTTCCAACACGAAACGGAGTCTTGTCATGATAGTCGGGCAGAATACTACTCAGTCTTCCCAATTCAATGTAAGATCAACATCTTCAGCAGTCAATTACATAAGAAGCTCCACTGAGATCATCAAGTGTGTACGAAATCCCTCTGGATTTGGCGAAACCCGAAAGCTTCAGCTGTGCACAGTAACAAACCGTAACATTGCTCTTGCTACGAGCTGCCTCAGTGAGGTTCGGCTATACGACATCAAAATCCAGATAAGAAGTAGTCAGCGAATGGCGATTTATCTGCATCCCCGTACGACTGATCCCCGCAACACGCTGTAACGTGGTATAGTCTACTGTCAACGAGACAAATGTAACCGCTTTGTACCAGGAAGGTAAACGTGTGAAAAATCAGCAAAAGCGATAAAAAACATTGTTTGTCTGTTAAACAAATCCTGTACATATAAATATTTTCGTTCAACAAATCAGTCGTAGATTGTTAAACCGCTCTTCTCTGTGGTTCTAACCCCATGATATCGACTGCAAAATAAATTATCGCTGTGCCACTATCAATCATCAGTAAAGATGTACCTCCCGATGAAATAGTCTGTAA TGCGCTACCTATTGACAGAGCGAATGACATTACGCGAATCAACATACTTCTTTGTCTGTTGAGAAATCTAAGGTAAACTCCTTAGCAATCGCATTAACTGAAAAGATAACGAAGAAAAATGTAGCGCCCTAAAAATCATTATTTTCAT

>2025-SEQ-1688 | StxOp932_Stx2h_49 | query (1236 bp)

ATGAAGTGATATTTGTTAAAAATGGATGCTGTGCTGTTACTGGGCTTTTTCTTCGGTATCCTATTCCCGGGAGTTTACGATAGACTTTTCGACTCAACAAAGTTATGTCTCTTCGTTAAATAGTATACGGACAGCGATATCGACCCCTCTTGAACATATATCTCAGGGAACCAATCGGTATCTGTTATTAATCATACGTCAACAGGAAGTTATATTACTGTGCATATACGAGGGCTTGATGTCTATCAGGCGCGCTTTGACCATCTTCGCCGTGATTGTTGAGCAAAATAAATTTATATGTGGCAGGGTTCGTTAATACGGCAACAAATACTTTCTACAGATTTTCAGATTTTGCACATATATACAGTGCCCGGTGTGACAACCTGTTTCCATGACAACGGACAGCAGTTATACCACTCTGCAACGTGTGCGACGCTGGAACGTTCCGGAATGCAAAATCAGTGTCACTCACTGGTTTCATCATATCTTGCGCTTAATGGAGTTTCAGTGGTAATGCCATGACCAGAGAGGCTTCCAGGGCTGTCCGTGCGTTTGTGTACTGTTACAGCCGAAGCCTTACGGTTCAGGCAAAATACAGAGAGAAATTTCTGTCAGGCGCTATCTGAAGCAGCTCCTGTTTATACGATGACACCGGAAGCAGTGAGCCTTACTCTGAACTGGGGGAGAAATCAGCAATGTGCTTCCGGAGTTTCCGGGAGAGGATGGTGTGAGAGTGGGAGAAATACCTTTAATAATATATCAGCAATCTGGGTACTGTGGCGGTTATATGAAATGCCATCATCAGGGGGCAGCTTCTGTTTCGCTCCGTTAATGAAGAGAAACAGCCCGAATGTGCAGATAACTGGCGACAGGCCAGTTATAAAAAATAAATAATTTTATGGGAAGTAAATACAGCAGCAGCTTTTCTGAACCGAAGATCTCATTCTTTATATGCAACTGGTGAATGACAGGAGTTAAACATGAAGAAGATGTTTATGGTGGCTTTATTTGCGTTAGTTTCTGTTAATGCAATGGCAGCGGAATGCGCTAAAGGTAAGTTGAGTTCTCTAAATATAATGGGGATAATACATTCACAGTAAAAGTGGCCGGGAAAGAAATCTGGACTAACCGCTGGAATCTGCAACCGTTACTGCAAAAGTGACAGTTGACAGGAATGACTGTAAACATCAAAATCCATTACCTGTGATTCAGGCTCCGGATTTGCTGAAGTGCAGTTTAATTAG

>2025-SEQ-1688 | StxOp932_Stx2h_49 | subject (1236 bp)

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>2025-SEQ-1691 | StxOp399_Stx1c_12 | query (1228 bp)

TCAACGGAAAAATAACCTCGCTGAATCCCCCTCCATTATGACAAGGCATTAGTTTTAATGTTTACCGTCATCCCCGTAATTTGTGCACCTAGAGAAGAGAGACTGAAGATTCATCTGTTAGTAAATAATCTTTTATCTCCCACTTTAACTGTAAGAGTATCGTCATCATTATATTTTGTATACTCCACCTTCCCAAGTTACACAATCTGGCGCAGCCAGCACACTTGCTGAAAAAAATGAAAGTGATGCGACGCTATTAAATAATTTTTTTCATTTTACCCCTCAACTGCTAACTGCTACGTAATAGCCCAATGCCCCAGAGTGATGAGTCCACAATATTTTATTGTGCGTAATCCACGCCACTCTTCCATCTACCGGGCACATAGAAGGAAACTCATTAGGTACAATTCTGGCAACTCGCGATGCATGATGATGACAATTCAGTATTAATGCCACGCTACCCAGAATTGCATTAAACATCTCCAAAAAGAAATCTTCCAACACGAAACAGAGTCTTGTCATGATAGTCAGGCAAGGACACTACTCAACCTTCCCAAGTTCAACGTAAGATCAACATCTTCAGCAGTCAATTACATAAGAAGC6CCCACTGAGATCATCAAGTGTGTACGAAATCCCTCTGAAATTTGCCGAAACGTAAGAGCTTCAGCTGTACACAGTAACAAACCGTAACATCTGCTCTTGCCACAGACTGCCTCAGTGAGGTTCCCGCTATGCGACATTAATCCAGATAAGAAGTAGTCAACGAATGGCGATTTATCTGCATCCCCGTACGACTGATCCCCGCAACACGCTGTAACGTGGTATAGTCTACTGTCAACGAGACAATGTAACCGCAGTTAGTACGAGGAAGTTAACATGTGAAAAAGTAAACGTAACATGTGAAAAAATCAGCAAAAGCGATAAAATATTGTTCTGTTAAACAAATCCTGTACATATAAATATTTTCGTTCAACAAATAAGCGGTAGATTATTAACCGCCCTTCTCTGAGTCTATCCCTCTGACATCAACTGCAAAACAAATTTATCCCTGTGCCACTATCAATCATCAGTAAAGACGTACCTCCTGATGAAATAGTCTGTAATGGAGTACCTATTGACAGAGCGAATGACATTACGCGAATCTACATACGCTTTTGTCTGTGAGAAATCTAAGGTAATTCCTTCGCAACCCACATTACAGAAAAAGATAACAAAGAAAAAAGTTATGACACTCTAAAAATTAATTTTCAT

>2025-SEQ-1691 | StxOp399_Stx1c_12 | subject (1228 bp)

TCAACGGAAAAATAACCTCGCTGAATCCCCCTCCATTATGACAGGCATTAGTTTTAATGGTTACCGTCATCCCCGTAATTTGTGCAC TGAGAAGAAGAGACTGAAGATTCATCTGTTAGTAAATAATCTTTATCTCCCACTTTAAGCTGTAAGAGTATCGTCATCATTATTTTGTATAC TCCACCTTCCCAAGTTACACAATCTGGCGCAGCCAGCACACTTGCTGAAAAAAATGAAAGTGATGACAGCTTAATAATAATTTTTTTCATTTTACCCCCCTCAACTGCTCAATAGGCCCTGCGTATCAAAATTGCCCCAGAGTGGATGAGTCCCAATAATTTTTATTGTGCGTAATCCCA GCACTCTTCCATCTACCGGGCACATAGAAGGAACTCATTAGGTACAATTTCTGGCAACTCGCGATGCATGATGATGACAATTCAGTATTAATGCCACGCTACCCAGAATTCGATTAAACATTCCAAAAGAAATCTTCCAAACACGAACAGAGTCTTGTCCATGATAGTCAGGCAGGACACTACTCAACCTTCCCGAGTTCAACGTAAGATCAACATCTTCAGCAGTCATTACATAAGAACGCCACTTGAGATCATCAAGTGTGTGACGAAATCCCCCTGGAATTTGCCGAAAACGTAAAGCTTCAGCTGTACACAGTAACAAACCGTAACATCGCTCTTGCCACAGACTGCGTCAGTGAGGTTCCGCTATGCGACATTAATCCAGATAGAAGTAGTCAACGAATGGCGATTATCTGTCATCCCCGTACGACTGATCCCGCAACACGCTGTAAACGTGGTATAGCTACTGTACCAGACAAGTAACCGCAGGTAACATGTGAAAAATCAGCAAAGCGATAAAAAACATTATTTGCTGTTAACAAATCCTGTCACATATAAATTTATTCGTTCAACAATAAGCCGTAGATTATTAACCGCCCTTCTCTGGATCTATCCCTCTGACATCAACTGCAAAACAAATATCCCTGTGCCACTATCAATCATCAGTAAAGACGTACCTCCTGATGAAATAGTCTGTATGGAGTACCTATTGCAGAGCGAATGACATTACGCGAATCTACATACGCTTTTGCTGTGAGAAATCTAAGGTAAATTCCTTCGCAACCCACATTACAGAAAAAGATAACAAAGAAAAAAGTTAGCACTCTAAAAATTAATTTTTTCAT

>2025-SEQ-1703 | StxOp1043_Stx2a_1 | query (1241 bp)

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>2025-SEQ-1703 | StxOp1043_Stx2a_1 | subject (1241 bp)

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>2025-SEQ-1703 | StxOp702_Stx2c_17 | query (1241 bp)

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>2025-SEQ-1703 | StxOp702_Stx2c_17 | subject (1241 bp)

ATGAAGTGATATATTATTAATG6GTAAGTGTGCTGTTACTG6GCTTTTCTTCGGTATCCTATTCCCGGGAATTTACGATAGACTTTTGCAGTCAACAAAGTTATGTATCTTCGTTAAAT AGTATACGGACAGAGATATCGACCCCTCTTGAACATATATCTCAGGGACACATCGGTGTCTGTTATTAAACCACACCCACCAGGCGAGTTATTTTGCTGTGGATATACGAGGGCTTGAT GTCTATCAGGCGCGTTTTGACCATCTTCGCTGATTATTGAGCAAAATAAATTTATATGTTGGCGGGTTCGTTAATACGGCAACAAATACTTTCTACCGTTTTTCAGATTTTACACATATA TCAGTGCCCGGTGTGACAACGGTTTCCATGACAACGGACAGCAGTTATACCACTCTGCAACGTGTGCGACGCGTGGAACGTTCCGGAATGCAAAATCAGTCGTCACTCACTGGTTTCATCA TATCTGCGGTAAATGGAGTTCAAGTGGTAATACAATGACCAGAGATGCATCCAGAGCAGTTCTGCGTTTTGTCACTGTGACAGCAGAAAGCCTTACGCTTCAGGCAGATACAGAGAGAAATTT CGTCAGGCACTGTCTGAAACTGCTCCTGTGTATACGATGACGCCGGGAGACGTGGACCTCACTCTGAACTG6GGGCGAATCAGCAATGTGCTTCGGAGTATCGGGAGAGGATGGTGTG AGAGTG6GGGAGAAATCCTTTAATAATATATCGGCGTACTG6GCACTGTG6CGGTTATACCTGTAATGTCATCATCAGGGGCGCGTTCTGTTGCGCGCGTGAATGAAGATAGTCAACCA GAATGTGAGATAACTGCGCAGAGGCCGCTTATAAAAAATAACAAATACATTATGGGAAGTAATACAGCTGCAGCGTTTTCTGAAACAGAAAGTCACAGTTTTTATATACAACGGGTAAATAA AGGAGTTAAGTATGAAGAGATGTTTATG6CGGTTTTATTTGCAATTAGTTTCTGTTAATGCAATGGCGGCGGATTGCGCTAAAGGTAAATTTGAGTTTTTCCAAGTATAATGAGAATGATA CATTCACAGTAAAGTG6CGGAAAGAGTACTG6ACCAAGTCGCTG6AATCTGCAACCGTTACTGCAAAAGTGTCAAGTTGACAGGAATGACTGTCAACAATCAAAATCCAGTACCTGTGAAT CAGGCTCCGGATTGCTGAAGTGCAAGTTAATAATGACTGA

>2025-SEQ-1763 | StxOp3298_Stx2b_23 | query (1236 bp)

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CGCTGATATATTATTAAGGATATTTCGCCCCACTCTGACACCCCCCTCTCCCCGAAACTCCGGAAGCACATTGCTGATTCTCCCCCAGTTCAGTGTGAGGTTCCACTTCTCCGGTGTCTAT
CGGTATACACAGGAGCAGTTTCAGACAGTGGCTGACGAAATCTCTCTGTATCTGCTTGAAGCGTGAAGGCTTTGCTGTGACAGTGCACAAAACGCAGAACTGCTCTGGATGCATCTCTGGT
CATTGTATTACCACTGAACTCCATTAAACGCCAGATATGATGAAACAGTGAAGTACGACTGATTTCGATTCCGGAACGTTCCAGCGCTGCGACACGTTGCAGAGTGGTATAACTGCTGTC
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>2025-SEQ-1763 | StxOp3298_Stx2b_23 | subject (1236 bp)

TTAATTAAACTGCACCTTCAGCAAACTCCTGAACCTGACGCACAGGTATTTGATTGTTGTTACCGTCATTCTCTGTTAACTGTGCGCTTTGCAGTAGCGGTTGCAGATTCCAGCGGTTAGT
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AGCCGCTACAAAATATCTTCTTCATATTTAACTCCTGTTATTCTCGGGATGTAATTTAAAGAGTGGGCCCTGCGATTACAGAAAAGCAGCTGCGGTATTACTTTCCCATAAAGTATTGTTTAT
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CGCTGATATATTATTAAGGATATTTCGCCCCACTCTGACACCCCCCTCTCCCCGAAACTCCGGAAGCACATTGCTGATTCTCCCCCAGTTCAGTGTGAGGTTCCACTTCTTCCGGTGTCTAT
CGTATACACAGGAGCAGTTTCAGACAGTGGCTGACGAAATCTCTCTGTATCTGCTTGAAGCGTGAAGGCTTTGCTGTGACAGTGCACAAAACGCAGAACTGCTCTGGATGCATCTCTGGT
CATTGTATTACCACTGAACTCCATTAAACGCCAGATATGATGAAACAGTGAAGTACGACTGATTTCGATTCCGGAACGTTCCAGCGCTGCGACACGTTGCAGAGTGGTATAACTGCTGTC
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>2025-SEQ-1763 | StxOp4389_Stx2d_112 | query (1241 bp)

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GAATGTGAGATAACTGGCGACAGGCCCTTATAAAAAATAAACAATACATTATGGGAAGTAAATACAGCAGCAGCGTTTCTGAAACAGAAAATCAGAGTCTTTATATACAACGGGTGAATAA
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CATTCACAGTAAAAGTGGCCGGGAAAGAACTGGACCACTGCGTGGGAATCTGCAACCGTTACTGCAAAAGTGTCTAGCTGACAGGAATGACTGTCACAATCAAAATCCAGTACCTGTGAAT
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>2025-SEQ-1763 | StxOp4389_Stx2d_112 | subject (1241 bp)

ATGAAGTGATATATTATTAATGGGTACTGTGCTGTTACTGGGTTTTTCTTCGGTATCCTATTTCCCGGGAGTTTATGATAGACTTTTCGACCCAACAAAGTTATGTCTCTCGTTAAAT
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GAATGTGAGATAACTGGCGACAGGCCCTTATAAAAAATAAACAATACATTATGGGAAGTAAATACAGCAGCAGCGTTTCTGAAACAGAAAATCAGAGTCTTTATATACAACGGGTGAATAA
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>2025-SEQ-1771 | StxOp936_Stx2j_104 | query (1234 bp)

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TGGCATTTACCACGGAACCTCAATTAACTCCAGATATGATGAAACAGTGAAGTACGCGGTGATTTCGATTCCGGAGCGTTCCAGCGCTGCGATACGTTGCAGAGTGGTATAACTGCTGTCCG
TGTGCTGGAACCGGTTGTGCACACCTGGCAGTATATGCGCAAAATGCGAAAGTATTTGTTACCGTATTAACAAACCCGGCCACATATAAATTTATCTGCTCAATGA
TCAGACGAAGATGGTCAAGCGCTCCTGATAAACATCAAGCCCCGTATATCCACGATAATAAATCTCTGGTGGTGTATAATTAATAACAGATACCGATGTGGTTCCTGAGATATAT
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>2025-SEQ-1771 | StxOp936_Stx2j_104 | subject (1234 bp)

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AACCGCCATAAACATCTTTCTTCACGTTTTAACTCCTGTCACTTAATTATATAAGACTGAGACTTTTCGGTTAGAAAAACCTGCTGCCGTATAAATTTCCCATAAATGTATTTATTT
TGATAACCGGCCTGTGCCAGTTATCTGACATTTCTCTTGATATCTTTATTCCAGGAGCGAACAGAAAGTGCTCCCTGATGATGGCAGTTTCAGTATAACCGCCACAGTACCCAGTATTG
CTGATATATTATTAAGGTTTATTGCCCCACTCTGACGCCATCCTCTCCCGGAAATTCGGGAAGCATTGCTGATTCTCCCCAGTTTCAGAGTGAGGTTCCACGTCTTCCGGTGTCAATCG
TATAAACAGGAGCAGCTTCAGATAGCGCCGGAGCAAAATTTCTCTGTATTGCTGAAAGCTTCGGCTGTGACAGTGACAAAAACGACGAGCAGCCCTGGACCGCGCTCTGGTCA
TGGCATTACCACGGAACTCCATTAACTCCAGATATGATGAAACAGTGAGTGACGGCTGATTGTCATTCCGGAGCGTTCCAGCGCTGCGATACGTTGACAGAGTGGTATAACTGCTGTCCG
TTGTCATTGGAACCGTTGTGCACACCTGGCAGTGTATATGCGCAAAATCCGAAATCTGTAAAGATTTTGTACCGTATTAAACAAACCGGCGACATATAAATTTATCTGCTCAATGA
TCAGACGAAGATGGTCAAAGCGCTCTGTATAAACATCAAGCCCCGTATATCCACGATAATATAACTTCTCGTGGTGGTATAATTATAACAGATACCGATGTGGTCCCTGAGATATAT
TTTCAAGAGGGGTCGATATCGCTGTACGTATATTCAATGAAGAGAGATAAATTTGTTGAGTCGAAAAGTCTATAGTAACTCCCGGGAACAGGATACCGAAGAAAAACCCAGTAACA
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>2025-SEQ-1785 | StxOp927_Stx2g_147 | query (1242 bp)

ATGAAGTGATATTGTTTAAATGGGTACTGTGCTGTTACTGGGCTTTTCTTCGGTATCCTATTTCCCGGAATTTACGATAGACTTTTCGACTCAACAAAGTTATGTATCTTCGTTAAAT
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TCAGTGCCCGGTTGTGACAACCTGTTTCCATGACAACGGACAGCAGTTATACCACGCGACGACCTGTGTCGACGCGTGGAACTGTTCCGGAATGCAAAATCAGTCTGCTCACTCACTGGTTTCATCA
TATCTGGCGTTAATGGAGTTTCAGTGGTAATACAAATGACCAGAGACGCATCCAGAGCAGTTCTTCGTTTTGCTCACTGTTACAGCAGAAGCCTTACGGTTTCAGGCAAAATACAGAGAGAAATTT
CGTCTGGCAGCTGTCTGAAACTGCTCCTGTTTATACGATGACGCGGGAAGACGTGGACCTCACTCTGAACTGGGGGAGAATCAGCAATGTGCTTCCGGAGTATCGGGGAGAGGATAGTGTC
AGAGTGGGGAGAATATCCTTAAATAATATATCAGCAATACTGGGTACTGTAGCGGTTATCTGAAATGCCATCATCAGGGGACGCGTTCCGGTTCGCTACGTGAATGAAGAGATGCAACCA
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AAGGAGTTAAGCATGAAGAAGATGTTTATGGCGGTTTTATTGTCATTGGTTTCTGTTAATGCAATGGCGCGGATTGTGCTAAAGGTAATAATGAGTTTTCCAATATAATGGGGTAAC
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>2025-SEQ-1785 | StxOp927_Stx2g_147 | subject (1242 bp)

ATGAAGTGATATTGTTTAAATGGGTACTGTGCTGTTACTGGGCTTTTCTTCGGTATCCTATTTCCCGGAATTTACGATAGACTTTTCGACTCAACAAAGTTATGTATCTTCGTTAAAT
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TCAGTGCCCGGTTGTGACAACCTGTTTCCATGACAACGGACAGCAGTTATACCACGCGACGACCTGTGTCGACGCGTGGAACTGTTCCGGAATGCAAAATCAGTCTGCTCACTCACTGGTTTCATCA
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AGAGTGGGGAGAATATCCTTAAATAATATATCAGCAATACTGGGTACTGTAGCGGTTATCTGAAATGCCATCATCAGGGGACGCGTTCCGGTTCGCTACGTGAATGAAGAGATGCAACCA
GAAATGTCAGATAAAGTGGCGACAGGCCAGTTATAAAAAATAACAATACATTATGGGAAAGTAATACAGCAGCAGCGCTTCTGAACAGAAAGTCTCAGTCATTATATACAACGGGTGAATGA
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ACATTTACTGTAAAGGTTGACGGGAAAGAATCTGGACTAACCGGTGGAATTTGCAGCCGTTGTTACAAGTGACAGTTAAACAGGAATGACCGTAACAATCAAAATCCAATACCTGTGGA
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>2025-SEQ-1786 | StxOp1486_Stx2d_112 | query (1241 bp)

TCAGTCATTATTAACCTGCACCTTCAGCAAAATCCGGAGCCTGATTCACAGGTACTGGATTTGATTGTGACAGTCATTCTGTCAACTGAGCAGCTTTGCAGTAACGGTTGCAGATTCCAGCG
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AAATAAAACCGCCATAAACATCTTCTTCATACTTAACTCCTTTATTACCCGTTGTATATAAAGACTGTGACTTTCTGTTTCAGAAACGCTGCTGCTGTATTACTTTCCCATAAATGATTTG
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AGTATCGCTGATATATTATTAAGGATATCTCCCACTCTGACACCATCCTCTCCCGGATACTCCGGAAGCAGATTGCTGATTTCGCCCCAGTTTCAGAGTGAGGTTCCAGCTCTCCCGGC
GTCATCGTATACAGAGGAGCAGTTTCAGACAGTGCCCTGACGAAATCTCTCTGATCTGCCTGAAGCGTAAGGCTTCTGCTGTGACAGTGACAAAAACGAGAACTGCTCTGGATGCATCT
CTGGTCATTGTATTACCACTGAACTCCATTAAACGCCAGATATGATGAAACCAAGTGAAGTGAACGATGATTGTCATTCCGGAACTTCCAGCGCTGCGACACGTTGCAGAGTGGTATAACTG
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GATATATGTTCAAGAGGGGTCGATATCTCTGTCGGTAACTATTAAACGAAGAGACATAACTTTGTTGGGTCGAAAAGTCTATCATAACTCCCGGGAATAGGATACCGAAGAAAAACCC
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>2025-SEQ-1786 | StxOp1486_Stx2d_112 | subject (1241 bp)

TCAGTCATTATTAACCTGCACCTTCAGCAAAATCCGGAGCCTGATTCACAGGTACTGGATTTGATTGTGACAGTCATTCTGTCAACTGAGCAGCTTTGCAGTAACGGTTGCAGATTCCAGCG
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AGTATCGCTGATATATTATTAAGGATATCTCCCACTCTGACACCATCCTCTCCCGGATACTCCGGAAGCAGATTGCTGATTTCGCCCCAGTTTCAGAGTGAGGTTCCAGCTCTCCCGGC
GTATCGTATACACAGGAGCAGTTTCAGACAGTGCCCTGACGAAATCTCTCTGATCTGCTGTAAGCTGCTGAAAGCGTAAGGCTTCTGCTGTGACAGTGACAAAAACGAGAACTGCTCTGGATGCATCT
CTGGTCATTGTATTACCACTGAACTCCATTAAACGCCAGATATGATGAAACCAAGTGAAGTGAACGATGATTGTCATTCCGGAACTTCCAGCGCTGCGACACGTTGCAGAGTGGTATAACTG
GTGTCGGTTGTGATGGAACCGTTGTACACCGGGGCACTGATATATGTGTAATAATCTGAAAGAACGGTAGAAAGTATTGTTGCCGTATTAACGAACCCGGCCACATATAAATTTATTTTGC
TCAATAATCAGACGAAGATGGTCAAAACGCGCCTGATAGACATCAAGCCCTGATATATGTTGTAATAATCTGAAAGAACGGTAGAAAGTATTGTTGCCGTATTAACGAACCCGGCCACATATAAATTTATTTTGC
GATATATGTTCAAGAGGGGTCGATATCTCTGTCGGTAACTATTAAACGAAGAGACATAACTTTGTTGGGTCGAAAAGTCTATCATAACTCCCGGGAATAGGATACCGAAGAAAAACCC
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>2025-SEQ-1786 | StxOp3298_Stx2b_23 | query (1236 bp)

TTAATTAAACTGCACCTTCAGCAAACTCCTGAACCTGACGCACAGGTATTTGATTGTTGTTACCGTCATTCTGTTAACTGTGCGCTTTGCAGTAGCGGTTGCAGATTCCAGCGGTTAGT
CCAGTACTCTTTCCCGGCCACTTTTACTGTGAATGTATCATTCTCATTATACTTAGAGAACTCAATTTTACCTTTTGACAATCAGCTGCCATTGCATTAAACAGAAAACAAAAGCAATAA
AGCCGCTACAAAATATCTTCTTCATATTTAACTCCTGTTATTCTCCGGATGTAATTTAAAGAGTGGGCCCTGCGATTTCAGAAAAGCAGCTGCGGTATTACTTTCCCATAAAGATTGTTTAT
CCTTATAACTGGCCTGTGCGCAGTTATCTGACATTCTGGTTGTATCTCTTCATTACGGCGCGAACGGAACGCGCCCCGTGATGATGGCAATTTCAGTATAACCGCCACTGTGCCAGTAT
CGCTGATATATTATTAAGGATATTTCGCCCCACTCTGACACCCCCCTCTCCCCGAAACTCCGGAAGCACATTGCTGATTCTCCCCCAGTTCAGTGTGAGGTTCCACTTCTCCGGTGTCTAT
CGTATACACAGGAGCAGTTTCAGACAGTGGCTGACGAAATTCCTCTGTATCTGCCTGGAAGCGTAAAGGCTTTGCTGTGACAGTGCACAAAACGACAGACTGCTCTGGATGCATCTCTGGT
CATTGTATTACCACTGAACTCCATTAAACGCCAGATATGATGAAACAGTGAAGTACGACTGATTTCGATTCCGGGAACGTTCCAGCGCTGCGACACGTTGCAGAGTGGTATAACTGCTGTC
GCTTGTGATGGAACCGGTTGTACACCGGGCAGTATATGTTGTAATCTGAAAACGCGTAGAAAGTATTTGTTGCCGTATTAAACGAACCCAGCCACATATAAATTTATTTGCTCAAT
AATCAGACGAAGATGGTCAAAACGCGCTGATAGACATCAAGCCCTCGTATATCCACAGCAAAATAACTGCCCGTGGGGTGTGGTTAATAACAGACACCGATGTGGTCCCCCTGAGATAT
ATGTTTCAAGAGGGGTGATATCTCTGTCCGTACTATTTAACGAAGAGACATAAATTTGTTGGGTGCAAAAGTCTATCATAAATCCCGGGAATAGGATACCGAAGAAAAACCCAGTAA
CAGGCACAGTACCCATTTAAATAATATACACTTCAT

>2025-SEQ-1786 | StxOp3298_Stx2b_23 | subject (1236 bp)

TTAATTAAACTGCACCTTCAGCAAACTCCTGAACCTGACGCACAGGTATTTGATTGTTGTTACCGTCATTCTGTTAACTGTGCGCTTTGCAGTAGCGGTTGCAGATTCCAGCGGTTAGT
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AGCCGCTACAAAATATCTTCTTCATATTTAACTCCTGTTATTCTCCGGATGTAATTTAAAGAGTGGGCCCTGCGATTTCAGAAAAGCAGCTGCGGTATTACTTTCCCATAAAGATTGTTTAT
CCTTATAACTGGCCTGTGCGCAGTTATCTGACATTCTGGTTGTATCTCTTCATTACGGCGCGAACGGAACGCGCCCCGTGATGATGGCAATTTCAGTATAACCGCCACTGTGCCAGTAT
CGCTGATATATTATTAAGGATATTTCGCCCCACTCTGACACCCCCCTCTCCCCGAAACTCCGGAAGCACATTGCTGATTCTCCCCCAGTTCAGTGTGAGGTTCCACTTCTTCCGGTGTCTAT
CGTATACACAGGAGCAGTTTCAGACAGTGGCTGACGAAATTCCTCTGTATCTGCCTGGAAGCGTAAAGGCTTTGCTGTGACAGTGCACAAAACGACAGACTGCTCTGGATGCATCTCTGGT
CATTGTATTACCACTGAACTCCATTAAACGCCAGATATGATGAAACAGTGAAGTACGACTGATTTCGATTCCGGAACGTTCCAGCGCTGCGACACGTTGCAGAGTGGTATAACTGCTGTC
CGTTGTGATGGAACCGGTTGTACACCGGGCAGTATATGTTGTAATCTGAAAACGCGTAGAAAGTATTTGTTGCCGTATTAAACGAACCCAGCCACATATAAATTTATTTGCTCAAT
AATCAGACGAAGATGGTCAAAACGCGCTGATAGACATCAAGCCCTCGTATATCCACAGCAAAATAACTGCCCGTGGGGTGTGGTTAATAACAGACACCGATGTGGTCCCCCTGAGATAT
ATGTTTCAAGAGGGGTGATATCTCTGTCCGTACTATTTAACGAAGAGACATAAATTTGTTGGGTGCAAAAGTCTATCATAAATCCCGGGAATAGGATACCGAAGAAAAACCCAGTAA
CAGGCACAGTACCCATTTAAATAATATACACTTCAT

>2025-SEQ-1788 | StxOp826_Stx2d_9 | query (1241 bp)

ATGAAGTGATATATTATTAATGGGTACTGTGCTGTTACTGGGTTTTTCTTCGGTATCCTATTCGGGAGTTTACGATAGACTTTTCGACCCAACAAAGTTATGTCTCTCGTTAAAT
ACTATACGGACAGAAAATATCGACTCCTCTTGAGCATATATCTCAGGGGACCACATCGGTGCTGTTATTAACACACCCACCAGGCGAGTTATTTGCTGTGGATATACGAGGGCTTGAT
GTCTATCAGGCGCGTTTTGACCATCTTCGTCTGATTATTGAGCAAAATAATTTATATGTGGCCGGGTTCTGTTAATACGGCAACAAATACTTTCTACCGTTTTTCAGATTTTACACATATA
TCAGTGCCCGGTGTGACAACGGTTTTCCATGACAACGGACAGCAGTTATACCACTCTGCAACGTGTGCGACGCTGGAACGTTCCGGAATGCAAAATCAGTCTGCTACTACTGGTTTCATCA
TATCTGGCGTTAATGGAGTTTCAGTGGTAAATACAATGACCAGAGATGCATCCAGAGCAGTTCTGCGGTTTTGTCACGTGTGACAGCAGAAGCCTTACGCTTCAGGCAGATACAGAGAGAAATTT
CGTCAGGCACTGTCTGAACTGCTCCTGTGTATACGATGACGCGGGAGACGTGGACCTCACTCTGAACTGGGGGCGAATCAGCAATGTGCTTCGGAGTATCGGGAGAGGATGGTGTG
AGAGTGGGAGAAATATCCTTTAATAATATATCAGCGTACTGGGTACTGTGGCCGTTTACTGAAATGCCATCATCAGGGGCGCGTTCTGTTGCGCGCGTGAATGAAGAGAGTCAACCA
GAATGTGATGATGAAAGAGATGTTTATGCGCGTTTTATTTGCAATTAGTTTCTGTTAATGCAATGGCGCGGATTGCGCTAAAGGTAATAATGAGTTTTCCAAGTATAATGAGAAATGATA
CATTCACAGTAAAGTGGCCGGAAGAGAGTACTGGACCACTGCGTGGAATCTGCAACCGTTACTGCAAGAGTGTCAAGTGCAGGAAATGACTGTGCAATCAAAATCCAGTACCTGTGAAT
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>2025-SEQ-1788 | StxOp826_Stx2d_9 | subject (1241 bp)

ATGAAGTGATATATTATTAATGGGTACTGTGCTGTTACTGGGTTTTTCTTCGGTATCCTATTCGGGAGTTTACGATAGACTTTTCGACCCAACAAAGTTATGTCTCTCGTTAAAT
ACTATACGGACAGAAAATATCGACTCCTCTTGAGCATATATCTCAGGGGACCACATCGGTGCTGTTATTAACACACCCACCAGGCGAGTTATTTGCTGTGGATATACGAGGGCTTGAT
GTCTATCAGGCGCGTTTTGACCATCTTCGTCTGATTATTGAGCAAAATAATTTATATGTGGCCGGGTTCTGTTAATACGGCAACAAATACTTTCTACCGTTTTTCAGATTTTACACATATA
TCAGTGCCCGGTGTGACAACGGTTTTCCATGACAACGGACAGCAGTTATACCACTCTGCAACGTGTGCGACGCTGGAACGTTCCGGAATGCAAAATCAGTCTGCTACTACTGGTTTCATCA
TATCTGGCGTTAATGGAGTTTCAGTGGTAAATACAATGACCAGAGATGCATCCAGAGCAGTTCTGCGTTTTGTCACGTGTGACAGCAGAAGCCTTACGCTTCAGGCAGATACAGAGAGAAATTT
CGTCAGGCACTGTCTGAACTGCTCCTGTGTATACGATGACGCGGGAGACGTGGACCTCACTCTGAACTGGGGGCGAATCAGCAATGTGCTTCGGAGTATCGGGAGAGGATGGTGTG
AGAGTGGGAGAAATATCCTTTAATAATATATCAGCGTACTGGGTACTGTGGCCGTTTACTGAAATGCCATCATCAGGGGCGCGTTCTGTTGCGCGCGTGAATGAAGAGAGTCAACCA
GAATGTGATGATGAAAGAGATGTTTATGCGCGTTTTATTTGCAATTAGTTTCTGTTAATGCAATGGCGCGGATTGCGCTAAAGGTAATAATGAGTTTTCCAAGTATAATGAGAAATGATA
CATTCACAGTAAAGTGGCCGGAAGAGTACTGGACCACTGCGTGGAATCTGCAACCGTTACTGCAAGTGTCAAGTGCAGGAAATGACTGTGCAATCAAAATCCAGTACCTGTGAAT
CAGGCTCCGGATTTGCTGAAGTGCAGTTAATAATGACTGA

>2025-SEQ-1789 | StxOp1036_Stx1c_12 | query (1229 bp)

ATGAAAAATAAATTTTATGAGTGTCTAACTTTTTCTTGTATCTTTTCTGTTAATGTGGTTGCGAAGGAATTTACCTTAGATTTCTCGACAGCAAAAGACGTATGTAGATTTCGCTGAAT
GTCAATTGCTCTGCAATAGGTACTCCATTACAGACTATTTTCATCAGGAGGTACGCTTTTACTGATGATTGATAGTGGCAGCAGGGGATAATTTGTTTGCAGTTGATGTGAGAGGATAGAT
CCAGAGGAAGGGCGGTTTTAATAATCTACGGCTTATGTTGACAGAAATAATTTATATGTGACAGGATTTGTTAAACAGGACAAATAATGTTTTTTATCGCTTTTCTGATTTTTACATGTT
ACCTTTCTCGGTACAACCTGCGGTTACATTGTCTGGTGACAGTAGCTATACCACGTTACAGCGTGTGCGGGGATCAGTCTGACGGGGATGCAGATAAATCGCCATTCTGTTGACTACTTCT
TATCTGGATTTAATGTCGCATAGCGGAACCTCACTGACGCACTGTGGCAAGAGCGATGTTACGGTTTTGTTACTGTGACAGCTGAAGCTTTACGTTTTTCGGCAAAATTCAGAGGGGATTT
CGTACAACACTTTGATGATCTCAGTGGCGGTTTTATGTTAATGACTGCTGAAGATGTTGATCTTACGTTGAACTGGGGAAGGTTGAGTAGTGTCCTGCTGACTATCATGGACAAGACTCT
TTCGTTGTTGGAAGAATTTCTTTTGGAAAGTGTAAATGCAATTTCTGGGTAGCGGGAATTAATCACTGAATTTGTCATCATCATGTCATCGCGAGTTGCCAGAATTTGACCTAATGAGTTTCTCT
GCTATGTGCCCGTGAAGTGAAGAGTGGTGGGATTACGCACAATAAATATTGTGGGACTCATCACTCTGGGGGCAATTTTGATACGCAGGGCTATTAGCAATTTGAGGGTGGGTAACAAAA
TGAAAAAATAATTATTAATAGCTGCATCACTTTTCATTTTTCAGCAAGTGTGCTGGCTGCGCCAGATTGTGTAAGTGGGAAGGTGGAGTATACAAAAATAATGATGACGATACCTTTA
CAGTTAAAGTGGGAGATAAAGAAATTTATTTACTAACAGATGGAATCTTCAGTCTCTTCTCTCAAGTGCACAAATACGGGGATGACGGTAACCATTTAAACTAAATGCCCTGTCAATAATGGAG
GGGGATTACGCGAGGTTATTTTCCGTTGA

>2025-SEQ-1789 | StxOp1036_Stx1c_12 | subject (1229 bp)

ATGAAAAATAAATTTTTAGAGTGCCTAACCTTTTTCTTTGTTATCTTTTCGTGTTAATGTGGTTGCGAAGGAATTTACCTTAGATTTCTCGACAGCAAAGACGTATGTAGATTGCTGTAAT
GTCAATTGCGCTCTGCAATAGGTACTCCATTACAGACTATTTTCATCAGGAGGTACGCTCTTTACTGATGATTGATAGTGGCAGAGGGGATAATTTGTTTGCAGTTGATGTCAGAGGGGATAGAT
CCAGAGGAAGGGCGGTTTTAATAATCTACGGCTTATTGTTGAACGAAATAATTTATGTTGACAGGATTTGTTAACAGGACAAATAATGTTTTTATCGCTTTGCTGATTTTTTCACATGTT
ACCTTTCTGGTACAACCTGCGGTTACATTGTCTGGTGACAGTACGTATACCACGTTACAGCGTGTTCGGGGGATCAGTCGTACGGGATGACAGATAAATGCCATTGTTGACTACTTCT
TATCTGGATTTAATGTCGCATAGCGGAACCTCACTGACGCAGTCTGTGGCAAGAGCGATGTTACGGTTTGTTACTGTGACAGCTGAAGCTTTACGTTTTTCGGCAAATTCAGAGGGGATTT
CGTACAACACTTGTGATCTCAGTGGGCGTTCTTATGTAAATGACTGCTGAAGATGTTGATCTTACGTTTGAACTGGGGAAGGTTGAGTAGTGTCTCGCTGACTATCATGGACAAGACTCT
GTTGCGTGTGGAAGAATTTCTTTTGGAAAGTGTAAATGCAATCTCTGGGTAGCGTGGCAATTAATACTGAAATGTCATCATCATGCATCGCAGGTTGCCAGAATTTGACCTAATGAGTTTCCT
TCTATGTGCCCGGTGGAAGAGTGGCGTGGGATTACGCACAATAAAATTTTGGGGACTCATCCACTCTGGGGGCAATTTTGATACGCAGGGCTATTAGCAGTTGAGGGGGGTA
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CAGTTAAAGTGGGAGATAAAGAATTATTACTAACAGATGGAATCTTCAGTCTCTTCTCTCAGTGCACAAATACGGGGATGACGGTAACCATTAACCTAATGCCTGTCTAATAATGGAG
GGGGATTACGCGAGGTTATTTTCCGTTGA

>2025-SEQ-1790 | StxOp933_Stx2i_43 | query (1242 bp)

TCAGTCATTATTAACTGCACTTCAGCAAAATCCAGAGCCTGAGTCACAGGTATTGGATTGATTGTTACGGTCATTCTGTTAACTGTGCACTTTGTAACAACGGCTGCAAAATCCACCG
GTTAGTCCAGTATCTTTCCCGTCAACCTTTATAGTAAATGTGTTATCCCCATTATATTGGAAAACTCAATTTTACCTTTAGCACAATCCGCCGCCATTGCATTAAACAGAAACCAATGC
AAATTAACCCGCCATAAACATCTTCTTCATGCTTAACTCCTTTTCATTACCCGTTGATATATAATGACTGAGACTTTCTGTTTCAGAAAGGCTGCTGCTGATTACTTTCCCATAAATTTATT
GTTTATTTTATAACCGGGCCTGTGCGCCAGTTATCTGACATTGTGGTTGACTCTCTTCATTACCGGCGCGAACAAAAACGCGCGCCCTGATGATGGCAATTACAGTATAACGGCCACAGTACC
AAGTATCGCTGATATATTATTAAGGATATTCTCCCACTCTGACACCATCCTCTCCCGGACTCTCGGGAAGCAGATTGCTGATTGCGCCCAAGTTTCAGAGTGAGGTTCCACGTCTTCCGG
CGTCATCGTATAAACAGGAGCTGTTTCAGACAGTGCCTGACGAAATTTCTCTCTGATCTG6CCTGAACCGTAAG6CCTTCGGCTGTGACGGTTACAAAACGACGAGCAGCCCTAGAAGCCTC
TCTGGTCAATTGTATTACCACTGAACTCCATTAAACGCCAGATATGATGAAACAGTGAGTGACGACTGATTGCAATTCGGGAACGTTCCAGCGCTGCGACACGTTGACAGAGTGGTATAACT
GCTGTCCGTTGATCTGGAAATAGTTGTGACACCGGGCAATGATATATGTGCAAAATCTGAAAACTCTGAGAAAGTATTTGTTGCTGATTAAACAAATCCAGCCACATATAAATTTATTCTG
TTCAATAATCAGACGAAGATGGTCAAAACGCTCCTGATAAACATCAAGCCCTCGTATACCTACGGAATATAAATTCCTGGTGGTGTATGATTAAACGGATACCGATGTAGTCCCTG
AGATATATGTTCAAGAGGGGTCGATATCGCTGTCCGTATACATTATAACGAAGATACATAAATTTGTTGAGTCGAAAAGTCTATCGTAACTCCTGGGAATAGGATACCGAAGAAAAACC
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>2025-SEQ-1790 | StxOp933_Stx2i_43 | subject (1242 bp)

TCAGTCATTATTAACTGCACTTCAGCAAAATCCAGAGCCTGAGTCACAGGTATTGGATTGATTGTTACGGTCATTCTGTTAACTGTGCACTTTGTAACAACGGCTGCAAAATCCACCG
GTTAGTCCAGTATCTTTCCCGTCAACCTTTATAGTAAATGTGTTATCCCCATTATATTGGAAAACTCAATTTTACCTTTAGCACAATCCGCCGCCATTGCATTAAACAGAAACCAATGC
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CGTCACTCGTATAAACAGGAGCTGTTTCAGACAGTGCCTGACGAAATTTCTCTCTGATCTG6CCTGAACCGTAAG6CCTTCGGCTGTGACGGTTACAAAACGACGAGCAGCCCTAGAAGCCTC
TCTGGTCAATTGTATTACCACTGAACTCCATTAAACGCCAGATATGATGAAACAGTGAGTGACGACTGATTGCAATTCGGGAACGTTCCAGCGCTGCGACACGTTGACAGAGTGGTATAACT
GCTGTCCGTTGTCATTGGAATAGTTGTGACACCGGGCAATGATATATGTGCAAAATCTGAAAACTCTGAGAAAGTATTTGTTGCTGATTAAACAAATCCAGCCACATATAAATTTATTCTG
TTCAATAATCAGACGAAGATGGTCAAAACGCTCCTGATAAACATCAAGCCCTCGTATACCTACGGAATATAAATTCCTGGTGGTGTATGATTAAACGGATACCGATGTAGTCCCTG
AGATATATGTTCAAGAGGGGTCGATATCGCTGTCCGTATACATTATAACGAAGATACATAAATTTGTTGAGTCGAAAAGTCTATCGTAACTCCTGGGAATAGGATACCGAAGAAAAACC
CAGTAACAGGCACAGTATCCACTTTAAACAATATACACTTCAT

>2025-SEQ-1791 | StxOp1571_Stx2e_146 | query (1236 bp)

TCAGTTAACTTCACCTGGGCAAGCCTGAGCCTGAACCTGACAGGTATTAGATATGATTGTTACAGTCATCCCTGTGACGTGAGCACTTTGTAACAATGGCTGCAAAATCCACTCTGTTCTG
CCAGTATTCTCTTCCCTGACACCTTCACAGTAAAGGTATTATCCTCATTTACTTTGGAAAACTCAATTTTACCTTTAGCACAATCCGCCGCCATTGCATTAAACAGAAACCAATGCAAAATAA
AACCAGCTATAAACATCTTCTTCATTCTTAACTCCTTTTCATTACCAAGTTGATATAAAGACTGTAACCTTTCTGTTTCAGAAACGCTGCTGCTGATTACTTTCCCATAAATGTATTGTTAT
TTTTATAACGGGCTGTGCGCCAGTTATCTGACATTCTG6TTGACTCTCTTCATTACCGGCGCGAACAGAACGCGCGCCCTGATGATGGCAATTACAGTATAACGGCCACAGTACCAAGTAT
CGCTGATATATTATTAAGGATATTCTCCCACTCTGACACCAAGCCTCTCCCGGATACTCCGGAAGCAGATTGCTGATTCTCCCCCAGTTTCAGAGTGAGGTTCCACGTCTTCCGGCGTCAT
CGTATAAACAGGAGCAGTTTCAGACAGTGCCAGACGAAATTTCTCTGTATTTGCTTGAACCGTAAG6CCTTCGCTGTGACAGTGACAAAACGAGAACTGCTCTTGATGATCTCTGGT
CATTGTATTACCACTGAACTCCATTAAACGCCAGATATGATGAAACAGTGAGTGACGACTGATTGCAATTCGGGAACGTTCCAGCGCTGCGACACGTTGACAGAGTGGTATAACTGCTGTC
CGTTGTCATGGAATAGTTGTGACACCGGGCAATGATATATGTGCAAAATCTGAAAACTCTGTAGAAAGTATTTGTTGCTGATTAAACAAATCCAGCCACATATAAATTTATTCTGTTCAAT
AATCAGACGAAGATGGTCAAAACGCTCCTGGTAAACATCAAGCCCTCGTATACCTACGGAATATAAATTCCTGGTGGTGTATGATTAAACGGATACCGATGTAGTCCCTGAGATAT
ATGTTCAAGAGGGGTCGATATCGCTGTCCGTATACTATTTAACGAAGATACATAAATTTGTTGAGTCGAAAAGTCTATCGTAACTCCTGGGAATAGGATACCGAAGAAAAACCCAGTAA
CAGGCACAGTATCCACTTTAAACAATATACACTTCAT

>2025-SEQ-1791 | StxOp1571_Stx2e_146 | subject (1236 bp)

TCAGTTAACTTCACCTGGGCAAGCCTGAGCCTGAACCTGACAGGTATTAGATATGATTGTTACAGTCATCCCTGTGACGTGAGCACTTTGTAACAATGGCTGCAAAATCCACTCTGTTCTG
CCAGTATTCTCTTCCCTGACACCTTCACAGTAAAGGTATTATCCTCATTTACTTTGGAAAACTCAATTTTACCTTTAGCACAATCCGCCGCCATTGCATTAAACAGAAACCAATGCAAAATAA
AACCAGCTATAAACATCTTCTTCATTCTTAACTCCTTTTCATTACCAAGTTGATATAAAGACTGTAACCTTTCTGTTTCAGAAACGCTGCTGCTGATTACTTTCCCATAAATGTATTGTTAT
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CGCTGATATATTATTAAGGATATTCTCCCACTCTGACACCAAGCCTCTCCCGGATACTCCGGAAGCAGATTGCTGATTCTCCCCCAGTTTCAGAGTGAGGTTCCACGTCTTCCGGCGTCAT
CGTATAAACAGGAGCAGTTTCAGACAGTGCCAGACGAAATTTCTCTGTATTTGCTTGAACCGTAAG6CCTTCGCTGTGACAGTGACAAAACGAGAACTGCTCTTGATGATCTCTGGT
CATTGTATTACCACTGAACTCCATTAAACGCCAGATATGATGAAACAGTGAGTGACGACTGATTGCAATTCGGGAACGTTCCAGCGCTGCGACACGTTGACAGAGTGGTATAACTGCTGTC
CGTTGTCATGGAATAGTTGTGACACCGGGCAATGATATATGTGCAAAATCTGAAAACTCTGTAGAAAGTATTTGTTGCTGATTAAACAAATCCAGCCACATATAAATTTATTCTGTTCAAT
AATCAGACGAAGATGGTCAAAACGCTCCTGGTAAACATCAAGCCCTCGTATACCTACGGAATATAAATTCCTGGTGGTGTATGATTAAACGGATACCGATGTAGTCCCTGAGATAT
ATGTTCAAGAGGGGTCGATATCGCTGTCCGTATACTATTTAACGAAGATACATAAATTTGTTGAGTCGAAAAGTCTATCGTAACTCCTGGGAATAGGATACCGAAGAAAAACCCAGTAA
CAGGCACAGTATCCACTTTAAACAATATACACTTCAT

>2025-SEQ-1792 | StxOp3320_Stx2b_101 | query (1236 bp)

ATGAAGTGATATATTGTTAAAAATGGGTACTGTGCCTGTTACTGGGCTTTTCTTCGGTATCCTATTTCCCGGGAATTTACGATAGACTTTTCGACTCAACAAAGTTATGTCTCTTCGTTAAAT
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TATCTGGCGTTAATGGAGTTTAGTGGAATGCCATGACCAGAGATGCATTCAGAGCAGTTCTGCGTTTTGTCACTGTGCACAGCAGAAGCCTTACGGTTTCAGGCGAGATACAGAGAGAATTT
CGTCTGGCACTGTCTGAAACTGCTCCTGTTTATACGATGACACCGGAAGAAGTGGACCTTCACACTGAACTGGGGGAGAATCAGCAATGTGCTTCGGAGTTTCGGGGAGAGGTTGGTGC
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>2025-SEQ-1792 | StxOp3320_Stx2b_101 | subject (1236 bp)

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CGTCTGGCACTGTCTGAAACTGCTCCTGTTTATACGATGACACCGGAAGAAGTGGACCTTCACACTGAACTGGGGAGAATCAGCAATGTGCTTCGGAGTTTCGGGGAGAGGGTGGTGC
AGAGTGGGGCGAATATCCTTTAATAATATATCAGCGATACTGGGACAGTGGCGGTTATACTGAATTGCCATCATCAGGGGCGCGGTTCCGTTCCGCCCGTGAATGAAGAGATACAACCA
GAATGTGCAGATAACTGGCGACAGGCCAGTTATAGGATAAACAATACTTTATGGGAAGTAAATACCCGACGCTGCTTTCTGAAATCGCAGGGCCCACTTTTAAATACATCCGGAGAATAA
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>2025-SEQ-1792 | StxOp399_Stx1c_12 | query (1228 bp)

ATGAAAAATAAATTTTTAGAGTGCTAACTTTTTTCTTTGTTATCTTTTTCTGTTAATGTGGTTGCGAAGGAATTTACCTTAGATTTCTCGACAGCAAAGACGTATGTAGATTTCGCTGAAT
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TATCTGGATTTAATGTCGCATAGCGGAACCTCACTGACGCGAGTCTGTGGCAAGAGCGATGTTACGGTTTTGTTACTGTGACAGCTGAAGCTTTACGTTTTTCGGCAAAATCAGAGGGGATTT
CGTACAACACTGTGATGATCTCAGTGGGCGTTCTTATGTAAATGACTGCTGAAGATGTTGATCTTACGTTGAACCTGGGGAAGGTTGAGTAGTGCTTCGCTGACTATCATGGACAAGACTCT
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>2025-SEQ-1792 | StxOp399_Stx1c_12 | subject (1228 bp)

ATGAAAAATAAATTTTTAGAGTGCTAACTTTTTTCTTTGTTATCTTTTTCTGTTAATGTGGTTGCGAAGGAATTTACCTTAGATTTCTCGACAGCAAAGACGTATGTAGATTTCGCTGAAT
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CGTACAACACTTGATGATCTCAGTGGGCGTTCTTATGTAAATGACTGCTGAAGATGTTGATCTTACGTTGAACCTGGGGAAGGTTGAGTAGTGCTTCGCTGACTATCATGGACAAGACTCT
GTTGCTGTTGGAAGAAATTTCTTTTGGAAAGTGTTAATGCAATTTCTGGGATGCGTGCCATTAATGCTAATGCTATCATGCAATGCGAGTTGCCAGAAATGACCTAATGAGTTTCCT
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>2025-SEQ-1793 | StxOp933_Stx2i_43 | query (1242 bp)

ATGAAGTGATATATTGTTAAAGTGGATACTGTGCCTGTTACTGGGTTTTTCTTCGGTATCCTATTTCCAGGAGTTTACGATAGACTTTTCGACTCAACAAAGTTATGTATCTTCGTTAAAT
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GTCTATCAGGAGCGTTTTGACCATCTTCGCTGATTATTGAAACGAAATAAATTTATATTGTGGCTGGATTGTTAAATACGACAAACAAATACTTTCTACAGATTTTCAGATTTTGCACATATA
TCATTGCCCGGTGTGACAACCTATTTCCATGACAACGGACAGCAGTTATACCACTCTGCAACGCTGTGCGACGCTGGAACGTTCCGGAATGCAAAATCAGTCGTCACTCACTGTTTCATCA
TATCTGGCGTTAATGGAGTTTCAAGTGAATACAAATGACCAGAGAGGCTTCTAGGGGCTTCCTGCGGTTTTGTAACCGTGCACAGCCGAAGCCTTACGGTTTCAGGCGAGATACAGAGAGAATTT
CGTCAAGCACTGTCTGAAACAGCTCCTGTTTATACGATGACGCGGGAAGAGCGTGGACCTCACTCTGAACTGGGGGCGAATCAGCAATGTGCTTCGGAGTATCGGGGAGAGGATGGTGC
AGAGTGGGGAGAATATCCTTTAATAATATATCAGCGATACTTGGTACTGTGGCGTTATACTGAATTGCCATCATCAGGGGCGCGGTTTTGTTCCGCCCGTGAATGAAGAGAGTCAACCA
CAATGTGCAGATAACTGGCGACAGGCCGTTATAAAAATAACAATAAATTTATGGGAAGTAAATACAGCAGCAGCGCTTCTGAAACGAAAGTCTCAGTCATTATATACAACGGGTGAATGA
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TCAAGGCTCTGGATTTGCTGAAAGTGCAGTTTAATAATGACTGA

>2025-SEQ-1793 | StxOp933_Stx2i_43 | subject (1242 bp)

ATGAAGTGATATATTGTTAAAGTGGATACTGTGCCTGTTACTGGGTTTTCTTCGGTATCCTATTTCCAGGAGTTTACGATAGACTTTTCGACTCAACAAAGTTATGTATCTTCGTTAAAT
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GTTTATCAGGAGCGTTTTGACCATCTTCGCTCGATTATTGACGAAATAAATTTATATGTGGCTGGATTGTTAAATACGACAACAAATACTTTCTACAGATTTTCAGATTTTGACATATA
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TATCTGGCGTTAATGGAGTTCAAGTGGTAATACAAATGACCAGAGAGGCTTCTAGGGCTGTCCCTGCGTTTTGTAACCGTGCACAGCCGAAGCCTTACGGTTCAAGGAGATACAGAGAGAAATTT
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AGAGTGGGAGAAATACCTTTAATAATATATCAGCGATACTTGGTACTGTGGCGTTTATCTGAAATGCCATCATCAGGGCGCGGCTTTTGTTCGCGCCGTGAATGAAGAGAGTCAACCA
CAATGTCAGATAAAGTGGCGACAGGCCGTTATAAAAATAAACAAATAAATTTGGGAAAGTAATACAGCAGCAGCCTTTCTGAACAGAAAGTCTCAGTCATTATATACAACGGGTGAATGA
AAGGAGTTAAGCATGAAGAAGATGTTTATGGCGGTTTTATTGCAATTGGTTCTGTGAATGCAATGGCGCGGATTGTGCTAAAGGTAATAATGGGATAAC
ACATTTACTATAAAGGTTGACGGGAAAGAACTGGAAGTACCGGTGGAATTTGACAGCGTTGTACAAAGTGACAGTTAACAGGAATGACCGTAACAATCAATCAATACCTGTGAC
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>2025-SEQ-1795 | StxOp1089_Stx2a_1 | query (1241 bp)

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AGTATACGGACAGAGATATCGACCCCTCTTGAACATATATCTCAGGGGACCACATCGGTGCTGTGTTAATACACACACCCACCGGGCAGTTATTTTGTCTGGATATACGAGGGCTTGAT
GTCTATACGGGCGGTTTTGACCATCTTCGCTCGATTATTGAGCAAAATAAATTTATATGTGGCCGGGTTTCGTTAATACGGCAACAAATACTTTCTACCGTTTTTCAGATTTTACACATATA
TCAGTGCCCGGTGTGACAACGCGTTTCCATGACAACGGACAGCAGTTATACCACTCTGCAACGTGTGCGACGCTGGAACGTTCCGGAATGCAAAATCAGTCGTCACTCACTGGTTTCATCA
TATCTGGCGTTAATGGAGTTCAAGTGGTAATACAAATGACCAGAGATGCATCCAGAGCAGTTCTGCGTTTTTGTCACTGTGCACAGCAGAAGCCTTACGCTTCAGGCGAGATACAGAGAGAAATTT
CGTCAGGCACTGTCTGAAACTGCTCCTGTGTATACGATGACGCCGGGAGACGTGGACCTCACTCTGAACTGGGGGCGAATCAGCAATGTGCTTCCGGAGTATCGGGGAGAGGATGGTGTC
AGAGTGGGAGAAATACCTTTAATAATATATCAGCGATACTGGGGACTGTGGCGTTTATCTGAATGCCATCATCAGGGGGCGCGTTCTGTTGCGCGCGTGAATGAAGAGAGTCAACCA
GAATGTCAGATAAAGTGGCGACAGGCCCTGTTATAAAAATAAACAAATACATTATGGGAAAGTAATACAGCTGCAGCGTTTTCTGAACAGAAAGTACAGTTTTTATATACAACGGGTAAATAA
AGGAGTTAAGCATGAAGAAGATGTTTATGGCGGTTTTATTGCAATTAGCTCTCTGTTAATGCAATGGCGCGGATTGTGCTAAAGGTAATAATGAGTTTTCCAAAGTATAATGAGGATGACA
CATTTACAGTGAAGGTTGACGGGAAAGAACTGGAACAGTCGCTGGAATCTGCAACCGTTACTGCAAAAGTGCTCAGTTGACAGGAATGACTGTCACAATCAAAATCCAGTACCTGTGAAT
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>2025-SEQ-1795 | StxOp1089_Stx2a_1 | subject (1241 bp)

ATGAAGTGATATATTATTTAAATGGGACTGTGCCTGTTACTGGGTTTTCTTCGGTATCCTATTTCCGGGAGTTTACGATAGACTTTTCGACCCCAACAAAGTTATGTCTCTTCGTTAAAT
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GTCTATACGGGCGGTTTTGACCATCTTCGCTCGATTATTGAGCAAAATAAATTTATATGTGGCCGGGTTTCGTTAATACGGCAACAAATACTTTCTACCGTTTTTCAGATTTTACACATATA
TCAGTGCCCGGTGTGACAACGCGTTTCCATGACAACGGACAGCAGTTATACCACTCTGCAACGTGTGCGACGCTGGAACGTTCCGGAATGCAAAATCAGTCGTCACTCACTGGTTTCATCA
TATCTGGCGTTAATGGAGTTCAAGTGGTAATACAAATGACCAGAGATGCATCCAGAGCAGTTCTGCGTTTTTGTCACTGTGCACAGCAGAAGCCTTACGCTTCAGGCGAGATACAGAGAGAAATTT
CGTCAGGCACTGTCTGAAACTGCTCCTGTGTATACGATGACGCCGGGAGACGTGGACCTCACTCTGAACTGGGGGCGAATCAGCAATGTGCTTCCGGAGTATCGGGGAGAGGATGGTGTC
AGAGTGGGAGAAATACCTTTAATAATATATCAGCGATACTGGGGACTGTGGCGTTTATCTGAATGCCATCATCAGGGGGCGCGTTCTGTTGCGCGCGTGAATGAAGAGAGTCAACCA
GAATGTCAGATAAAGTGGCGACAGGCCCTGTTATAAAAATAAACAAATACATTATGGGAAAGTAATACAGCTGCAGCGTTTTCTGAACAGAAAGTACAGTTTTTATATACAACGGGTAAATAA
AGGAGTTAAGCATGAAGAAGATGTTTATGGCGGTTTTATTGCAATTAGCTCTCTGTTAATGCAATGGCGCGGATTGTGCTAAAGGTAATAATGAGTTTTCCAAAGTATAATGAGGATGACA
CATTTACAGTGAAGGTTGACGGGAAAGAACTGGAACAGTCGCTGGAATCTGCAACCGTTACTGCAAAAGTGCTCAGTTGACAGGAATGACTGTCACAATCAAAATCCAGTACCTGTGAAT
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>2025-SEQ-1796 | StxOp3_Stx1a_9 | query (1227 bp)

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CGATGCAAGCTATTAATAATGTTTTTTTCATTTTACCCTCAACTGCTAATAGTTTCTGCGCATCAGAATTGCCCCCAAGATGGATGAATCCCAACAATTTTTATTGTGCGTAATCCCACG
GACTCTTCCATCTGCCGGACACATAGAAGGAAACTCATCAGATGCCATTTCTGGCAACTCGCGATGCATGATGATGACAATTCAGTATTAATGCCACGCTTCCCAGAATTGCATTAAATGCT
TCCAAAAGAAATTTCTTCTACACGAACAGAGTCTTGTCCATGATAATCAGGCGAGGACACTACTCAACCTTCCCAGTTCAATGTAAGATCAACATCTTCAGCAGTCATTACATAAGAAGC
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TCCACTATGCGACATTAATCCAGATAAGAAGTAGTCAACGAATGCGGATTTATCTGCATCCCCGTACGACTGATCCCTGCAACACGCTGTAACGTGGTATAGCTACTGTCCACAGACAA
TGTAAACCGCTGTTGTACTGGAAGGTAACTGTGAAAAATCAGCAAAAGCGATAAAAAACATTATTTGTCTGTTAAACAAATCCTGTGCATATAAATTTTCGTTCAACAAATAAGCCG
TAGATTATTAACCGCCCTTCCCTCTGGATCTATCCCTCTGACATCAACTGCAAAACAAATTTCCCTGTGCGCACTATCAATCATCAGTAAAGACGTACCTCCTGATGAAATAGTCTGTAA
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TAGCACTCTAAAAATAATTATTTTCAT

>2025-SEQ-1796 | StxOp3_Stx1a_9 | subject (1227 bp)

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CGATGCAAGCTATTAATAATGTTTTTTTCATTTTACCCTCAACTGCTAATAGTTTCTGCGCATCAGAATTGCCCCCAAGATGGATGAATCCCAACAATTTTTATTGTGCGTAATCCCACG
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TCCAAAAGAAATTTCTTCTACACGAACAGAGTCTTGTCCATGATAATCAGGCGAGGACACTACTCAACCTTCCCAGTTCAATGTAAGATCAACATCTTCAGCAGTCATTACATAAGAAGC
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TCCACTATGCGCATTAATCCAGATAAGAAGTAGTCAACGAATGCGGATTTATCTGCATCCCCGTACGACTGATCCCTGCAACACGCTGTAACGTGGTATAGCTACTGTCCACAGACAA
TGTAAACCGCTGTTGTACTGGAAGGTAACTGTGAAAAATCAGCAAAAGCGATAAAAAACATTATTTGTCTGTTAAACAAATCCTGTGCATATAAATTTTCGTTCAACAAATAAGCCG
TAGATTATTAACCGCCCTTCCCTCTGGATCTATCCCTCTGACATCAACTGCAAAACAAATTTCCCTGTGCGCACTATCAATCATCAGTAAAGACGTACCTCCTGATGAAATAGTCTGTAA
TGGAGTACCTATTGACAGAGCGAATGACATTACGCAATCTACATACGCTTTTGCAAGTGCAGAAAGTCTAAGGTAATTCCTTCGCAACCACTAATTAAGTAAAAAGATAACAAAGAAAAAGT
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>2025-SEQ-1798 | StxOp419_Stx2a_1 | query (1241 bp)

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GTCTATCAGGCGCGTTTTGACCATCTTCGCTGATTATTGAGCAAAATAAATTTATATGTGGCCGGGTTTCGTTAATACGGCAACAAATACTTTCTACCGTTTTTCAGATTTTACACATATA
TCAGTGCCCGGTGTGACAACGGTTTCCATGACAACGGACAGCAGTTATACCACTCTGCAACGTGTGCGACGCTGGAACGTTCCGGAATGCAAAATCAGTCGTCACTCACTGGTTTCATCA
TATCTGGCGTTAATGGAGTTTCAGTGGTAATACAATGACCAGAGATGCATCCAGAGCAGTTCTGCGTTTTGTCACTGTGCACAGCAGAAGCCTTACGCTTCAGGCAGATACAGAGAGAATTT
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AGAGTGGGAGAAATACCTTTAATAATATATCAGCGATACTGGGGACTGTGGCGTTTATACCTGAATTGCATCATCAGGGGCGCGGTTCTGTTCCGCCCGTGAATGAAGAGAGTCAACCA
GAATGTGCAGATAACTGGCGACAGGCCCGTTATAAAAAATAACAATACATTATGGGAAAGTAATACAGCTGCAGCGTTTTCTGAAACAGAAAGTCACAGTTTTTATATACAACGGGTAAATAA
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>2025-SEQ-1798 | StxOp419_Stx2a_1 | subject (1241 bp)

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GTCTATCAGGCGCGTTTTGACCATCTTCGCTGATTATTGAGCAAAATAAATTTATATGTGGCCGGGTTTCGTTAATACGGCAACAAATACTTTCTACCGTTTTTCAGATTTTACACATATA
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TATCTGGCGTTAATGGAGTTTCAGTGGTAATACAATGACCAGAGATGCATCCAGAGCAGTTCTGCGTTTTGTCACTGTGCACAGCAGAAGCCTTACGCTTCAGGCAGATACAGAGAGAATTT
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AGAGTGGGAGAAATACCTTTAATAATATATCAGCGATACTGGGGACTGTGGCGTTTATACCTGAATTGCATCATCAGGGGCGCGGTTCTGTTCCGCCCGTGAATGAAGAGAGTCAACCA
GAATGTGCAGATAACTGGCGACAGGCCCGTTATAAAAAATAACAATACATTATGGGAAAGTAATACAGCTGCAGCGTTTTCTGAAACAGAAAGTCACAGTTTTTATATACAACGGGTAAATAA
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>2025-SEQ-1800 | StxOp1448_Stx2c_152 | query (1241 bp)

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>2025-SEQ-1800 | StxOp1448_Stx2c_152 | subject (1241 bp)

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>2025-SEQ-1800 | StxOp3625_Stx2b_215 | query (1236 bp)

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>2025-SEQ-1800 | StxOp3625_Stx2b_215 | subject (1236 bp)

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CGTATACACAGGAGCAGTTTCAGACAGTGGCTGACGAAATTTCTCTGTATCTGCCTGAAGCGTAAGGCTTCTGCTGTGACAGTGCACAAAACGCAGAACTGCTCTGGATGCATCTCTGGT
CATTGTATTACCACTGAACTCCATTAAACGCCAGATATGATGAAACAGTGAAGTACGACTGATTTCGATTCCGGAACGTTCCAGCGCTGCGACACGTTGCAGAGTGGTATAACTGCTGTC
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>2025-SEQ-1964 | StxOp3298_Stx2b_23 | query (1236 bp)

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AGCCGCTACAAATATCTTCTTCATATTTAACTCCTGTATTCTCCGGATGTATTTAAAGAGTGGGCCCTGCGATTTCAGAAAAGCAGCTGCGGTATTACTTTCCCATAAAGTATTGTTTAT
CCTTATAACTGGCCTGTGCGCAGTTATCTGACATTCTGGTTGTATCTCTTCATTACGCGCGCAACGGAACGCGCCCCGTGATGATGGCAATTTCAGTATAACCGCCACTGTGCCAGTAT
CGCTGATATATTATTAAGGATATTTCGCCCCACTCTGACACCCCCCTCTCCCCGAAACTCCGGAAGCACATTGCTGATTCTCCCCCAGTTCAGTGTGAGGTTCCACTTCTTCCGGTGTCTAT
CGTATACACAGGAGCAGTTTCAGACAGTGGCTGACGAAATTTCTCTGTATCTGCCTGAAGCGTAAGGCTTCTGCTGTGACAGTGCACAAAACGCAGAACTGCTCTGGATGCATCTCTGGT
CATTGTATTACCACTGAACTCCATTAAACGCCAGATATGATGAAACAGTGAAGTACGACTGATTTCGATTCCGGAACGTTCCAGCGCTGCGACACGTTGCAGAGTGGTATAACTGCTGTC
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AATCAGACGAAGATGGTCAAAACGCGCTGATAGACATCAAGCCCTCGTATATCCACAGCAAAATAACTGCCCGTGGGGTGTGGTTAATAACAGACACCGATGTGGTCCCCTGAGATAT
ATGTTCAAGAGGGGTGATATCTCTGTCCGTATACTATTTAACGAAGAGACATAAATTTGTTGGGTGAAAAGTCTATCATAAACTCCCGGGAATAGGATACCGAAGAAAAACCCAGTAA
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>2025-SEQ-1964 | StxOp3298_Stx2b_23 | subject (1236 bp)

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CGCTGATATATTATTAAGGATATTTCGCCCCACTCTGACACCCCCCTCTCCCCGAAACTCCGGAAGCACATTGCTGATTCTCCCCCAGTTCAGTGTGAGGTTCCACTTCTTCCGGTGTCTAT
CGTATACACAGGAGCAGTTTCAGACAGTGGCTGACGAAATTTCTCTGTATCTGCCTGAAGCGTAAGGCTTCTGCTGTGACAGTGCACAAAACGCAGAACTGCTCTGGATGCATCTCTGGT
CATTGTATTACCACTGAACTCCATTAAACGCCAGATATGATGAAACAGTGAAGTACGACTGATTTCGATTCCGGAACGTTCCAGCGCTGCGACACGTTGCAGAGTGGTATAACTGCTGTC
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ATGTTCAAGAGGGGTGATATCTCTGTCCGTATACTATTTAACGAAGAGACATAAATTTGTTGGGTGAAAAGTCTATCATAAACTCCCGGGAATAGGATACCGAAGAAAAACCCAGTAA
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>2025-SEQ-1964 | StxOp4178_Stx2c_569 | query (1241 bp)

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CATTCACAGTAAAAGTG6CCGGAAGAGTACTGGACCACTGCGTGGAATCTGCAACCGTTACTGCAAAAGTGTCAAGTGTACAGGAATGACTGTCAACAATCAAAATCCAGTACCTGTGAAT
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>2025-SEQ-1964 | StxOp4178_Stx2c_569 | subject (1241 bp)

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>DRR178135 | StxOp1089_Stx2a_1 | copy 1 of 2 | query (1241 bp)

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>DRR178135 | StxOp1089_Stx2a_1 | copy 1 of 2 | subject (1241 bp)

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>DRR178135 | StxOp1089_Stx2a_1 | copy 2 of 2 | query (1241 bp)

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>DRR178135 | StxOp1089_Stx2a_1 | copy 2 of 2 | subject (1241 bp)

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>DRR178135 | StxOp1577_Stx1a_9 | query (1227 bp)

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>DRR178135 | StxOp1577_Stx1a_9 | subject (1227 bp)

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>DRR401569 | StxOp1089_Stx2a_1 | query (1241 bp)

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>DRR401569 | StxOp1089_Stx2a_1 | subject (1241 bp)

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>DRR401569 | StxOp22_Stx1a_35 | query (1227 bp)

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>DRR401569 | StxOp22_Stx1a_35 | subject (1227 bp)

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>DRR401569 | StxOp717_Stx2c_130 | query (1241 bp)

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>DRR401569 | StxOp717_Stx2c_130 | subject (1241 bp)

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>DRR401577 | StxOp1089_Stx2a_1 | query (1241 bp)

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>DRR401577 | StxOp1089_Stx2a_1 | subject (1241 bp)

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>DRR401577 | StxOp1577_Stx1a_9 | query (1227 bp)

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>DRR401577 | StxOp1577_Stx1a_9 | subject (1227 bp)

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>ERR3330139 | StxOp922_Stx2f_726 | query (1236 bp)

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>ERR3330139 | StxOp922_Stx2f_726 | subject (1236 bp)

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>GCF_002303275 | StxOp417_Stx1e_13 | query (1227 bp)

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>GCF_002303275 | StxOp417_Stx1e_13 | subject (1227 bp)

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>SRR10541011 | StxOp932_Stx2h_49 | query (1236 bp)

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>SRR10541011 | StxOp932_Stx2h_49 | subject (1236 bp)

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>SRR10911972 | StxOp948_Stx2m_48 | query (1236 bp)

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>SRR10911972 | StxOp948_Stx2m_48 | subject (1236 bp)

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>SRR11485261 | StxOp1301_Stx2a_654 | query (1241 bp)

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>SRR11485261 | StxOp1301_Stx2a_654 | subject (1241 bp)

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>SRR11485261 | StxOp846_Stx2d_144 | query (1241 bp)

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>SRR11485261 | StxOp846_Stx2d_144 | subject (1241 bp)

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>SRR11550988 | StxOp399_Stx1c_12 | query (1228 bp)

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>SRR11550988 | StxOp399_Stx1c_12 | subject (1228 bp)

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>SRR12207194 | StxOp429_Stx2a_16 | query (1241 bp)

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>SRR12207194 | StxOp429_Stx2a_16 | subject (1241 bp)

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>SRR12207194 | StxOp9_Stx1a_15 | query (1227 bp)

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>SRR12207194 | StxOp9_Stx1a_15 | subject (1227 bp)

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>SRR12506954 | StxOp1515_Stx2d_115 | query (1241 bp)

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[illegible]

>SRR1272522 | StxOp432_Stx2a_30 | query (1241 bp)

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>SRR1272522 | StxOp432_Stx2a_30 | subject (1241 bp)

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>SRR1272883 | StxOp1365_Stx2c_7 | query (1241 bp)

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>SRR1272883 | StxOp1365_Stx2c_7 | subject (1241 bp)

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>SRR12844969 | StxOp829_Stx2d_27 | query (1241 bp)

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>SRR12844969 | StxOp829_Stx2d_27 | subject (1241 bp)

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>SRR13068663 | StxOp1157_Stx2a_6 | query (1241 bp)

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>SRR13068663 | StxOp1157_Stx2a_6 | subject (1241 bp)

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>SRR13068663 | StxOp1711_Stx1a_9 | query (1227 bp)

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>SRR13068663 | StxOp1711_Stx1a_9 | subject (1227 bp)

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>SRR13132130 | StxOp953_Stx2n_126 | query (1236 bp)

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>SRR13132130 | StxOp953_Stx2n_126 | subject (1236 bp)

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>SRR13209800 | StxOp13_Stx1a_19 | query (1227 bp)

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>SRR13209800 | StxOp13_Stx1a_19 | subject (1227 bp)

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>SRR13209800 | StxOp675_Stx2b_116 | query (1236 bp)

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>SRR13209800 | StxOp675_Stx2b_116 | subject (1236 bp)

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>SRR13209817 | StxOp1510_Stx2d_115 | query (1241 bp)

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>SRR13209817 | StxOp1510_Stx2d_115 | subject (1241 bp)

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>SRR13209817 | StxOp1512_Stx2d_115 | query (1241 bp)

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>SRR13209817 | StxOp1512_Stx2d_115 | subject (1241 bp)

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>SRR13209822 | StxOp1541_Stx2d_134 | query (1241 bp)

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>SRR13209822 | StxOp1541_Stx2d_134 | subject (1241 bp)

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>SRR13240070 | StxOp1173_Stx2a_16 | query (1241 bp)

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>SRR13240070 | StxOp1173_Stx2a_16 | subject (1241 bp)

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>SRR13240070 | StxOp4_Stx1a_9 | query (1227 bp)

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>SRR13240070 | StxOp4_Stx1a_9 | subject (1227 bp)

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>SRR13240070 | StxOp848_Stx2d_164 | query (1241 bp)

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>SRR13240070 | StxOp848_Stx2d_164 | subject (1241 bp)

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>SRR13428524 | StxOp3_Stx1a_9 | query (1227 bp)

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>SRR13428524 | StxOp3_Stx1a_9 | subject (1227 bp)

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>SRR13428524 | StxOp702_Stx2c_17 | query (1241 bp)

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>SRR13428524 | StxOp702_Stx2c_17 | subject (1241 bp)

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>SRR13450255 | StxOp1173_Stx2a_16 | query (1241 bp)

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>SRR13450255 | StxOp1173_Stx2a_16 | subject (1241 bp)

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>SRR13450255 | StxOp1228_Stx2a_18 | query (1241 bp)

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>SRR13450255 | StxOp1228_Stx2a_18 | subject (1241 bp)

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>SRR13557757 | StxOp1089_Stx2a_1 | query (1241 bp)

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>SRR13557757 | StxOp1089_Stx2a_1 | subject (1241 bp)

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>SRR13605977 | StxOp1574_Stx2g_39 | query (1242 bp)

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>SRR13605977 | StxOp1574_Stx2g_39 | subject (1242 bp)

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>SRR13605977 | StxOp1594_Stx1a_9 | query (1227 bp)

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>SRR13605977 | StxOp1594_Stx1a_9 | subject (1227 bp)

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>SRR13674450 | StxOp1361_Stx2c_7 | query (1241 bp)

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>SRR13674450 | StxOp1361_Stx2c_7 | subject (1241 bp)

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>SRR13796384 | StxOp1041_Stx1d_4 | query (1227 bp)

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>SRR13796384 | StxOp1041_Stx1d_4 | subject (1227 bp)

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>SRR13957227 | StxOp913_Stx2e_102 | query (1236 bp)

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>SRR13957227 | StxOp913_Stx2e_102 | subject (1236 bp)

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>SRR14071719 | StxOp955_Stx2o_545 | query (1236 bp)

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>SRR14071719 | StxOp955_Stx2o_545 | subject (1236 bp)

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>SRR14750050 | StxOp432_Stx2a_30 | query (1241 bp)

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>SRR14750050 | StxOp432_Stx2a_30 | subject (1241 bp)

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>SRR14898862 | StxOp415_Stx1d_4 | query (1227 bp)

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>SRR14934092 | StxOp898_Stx2e_25 | query (1236 bp)

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>SRR14934092 | StxOp898_Stx2e_25 | subject (1236 bp)

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>SRR14934097 | StxOp898_Stx2e_25 | query (1236 bp)

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>SRR14934097 | StxOp898_Stx2e_25 | subject (1236 bp)

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>SRR15076491 | StxOp1339_Stx2b_109 | query (1236 bp)

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>SRR15076491 | StxOp1339_Stx2b_109 | subject (1236 bp)

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>SRR15076491 | StxOp403_Stx1c_26 | query (1228 bp)

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>SRR15076491 | StxOp403_Stx1c_26 | subject (1228 bp)

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>SRR15497538 | StxOp914_Stx2e_133 | query (1236 bp)

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>SRR15497538 | StxOp914_Stx2e_133 | subject (1236 bp)

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>SRR15497538 | StxOp916_Stx2e_146 | query (1236 bp)

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>SRR15497538 | StxOp916_Stx2e_146 | subject (1236 bp)

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>SRR15949660 | StxOp399_Stx1c_12 | query (1228 bp)

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>SRR15949660 | StxOp399_Stx1c_12 | subject (1228 bp)

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>SRR16071833 | StxOp915_Stx2e_136 | query (1236 bp)

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>SRR16071833 | StxOp915_Stx2e_136 | subject (1236 bp)

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>SRR1610028 | StxOp419_Stx2a_1 | query (1241 bp)

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>SRR1610028 | StxOp419_Stx2a_1 | subject (1241 bp)

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>SRR1724591 | StxOp1577_Stx1a_9 | query (1227 bp)

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>SRR17923069 | StxOp4737_Stx2l_465 | query (1241 bp)

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>SRR18191506 | StxOp1512_Stx2d_115 | query (1241 bp)

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>SRR18191506 | StxOp1512_Stx2d_115 | subject (1241 bp)

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>SRR18191506 | StxOp4_Stx1a_9 | query (1227 bp)

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>SRR18191506 | StxOp4_Stx1a_9 | subject (1227 bp)

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>SRR18191510 | StxOp9_Stx1a_15 | query (1227 bp)

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>SRR18191510 | StxOp9_Stx1a_15 | subject (1227 bp)

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>SRR18191519 | StxOp1577_Stx1a_9 | query (1227 bp)

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>SRR18191519 | StxOp1577_Stx1a_9 | subject (1227 bp)

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>SRR18191528 | StxOp1283_Stx2a_170 | query (1241 bp)

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>SRR18191528 | StxOp1283_Stx2a_170 | subject (1241 bp)

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>SRR18191528 | StxOp1708_Stx1a_9 | query (1227 bp)

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>SRR18191528 | StxOp1708_Stx1a_9 | subject (1227 bp)

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>SRR18191538 | StxOp1577_Stx1a_9 | query (1227 bp)

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>SRR18191538 | StxOp1577_Stx1a_9 | subject (1227 bp)

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>SRR18191541 | StxOp1036_Stx1c_12 | query (1229 bp)

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>SRR18191541 | StxOp1036_Stx1c_12 | subject (1229 bp)

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>SRR18191549 | StxOp840_Stx2d_99 | subject (1241 bp)

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>SRR18191551 | StxOp1036_Stx1c_12 | query (1229 bp)

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>SRR18191551 | StxOp1036_Stx1c_12 | subject (1229 bp)

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>SRR18191553 | StxOp71_Stx1a_89 | query (1227 bp)

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>SRR18191553 | StxOp71_Stx1a_89 | subject (1227 bp)

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>SRR18191553 | StxOp883_Stx2d_471 | copy 1 of 2 | query (1241 bp)

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>SRR18191553 | StxOp883_Stx2d_471 | copy 1 of 2 | subject (1241 bp)

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>SRR18191553 | StxOp883_Stx2d_471 | copy 2 of 2 | query (1241 bp)

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>SRR18191553 | StxOp883_Stx2d_471 | copy 2 of 2 | subject (1241 bp)

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>SRR18191566 | StxOp457_Stx2a_181 | query (1241 bp)

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>SRR18191566 | StxOp457_Stx2a_181 | subject (1241 bp)

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>SRR18191571 | StxOp1594_Stx1a_9 | query (1227 bp)

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>SRR18191571 | StxOp1594_Stx1a_9 | subject (1227 bp)

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>SRR18191580 | StxOp1040_Stx1d_4 | query (1227 bp)

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>SRR18191580 | StxOp1040_Stx1d_4 | subject (1227 bp)

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>SRR18191598 | StxOp702_Stx2c_17 | query (1241 bp)

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>SRR18191598 | StxOp702_Stx2c_17 | subject (1241 bp)

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>SRR18191601 | StxOp826_Stx2d_9 | query (1241 bp)

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>SRR18191601 | StxOp826_Stx2d_9 | subject (1241 bp)

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GAATGTGAGATAAAGTGGCGACAGGCCCGTTATAAAAAATAACAATACATTATGGGAAAGTAATACAGCAGCAGCGTTTCTGAAACAGAAAGTCACAGTCTTTATATACAACGGGTGAATAA
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>SRR18191603 | StxOp933_Stx2i_43 | query (1242 bp)

TCAGTCATTATTAAACTGCATTTACGCAAAATCCAGAGCCTGAGTCACAGGATTGGATTGGATTGTTACGGTCATTCTGTTAACTGTGCATTTGTAACACAGGCTGCAAAATTCACCG
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>SRR18191603 | StxOp933_Stx2i_43 | subject (1242 bp)

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>SRR18191605 | StxOp1168_Stx2a_16 | query (1241 bp)

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TCAGTGCCCGATGTGACAACG6TTTTCCATGACAACGGACAGCAGTTATACCACTCTGCAACG6TGTGCGACGCTGGAACGTTCCGGGAATGCAAAATCAGTCGTCACTCACTGGTTTCATCA
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AGAGTGGGGAGAATATCCTTTAATAATATATCAGCGATACTGGGTACTGTGGCGTTATACTGAATTGCCATCATCAGGGGGCGGTTCTGTTGCGCGCGTGAATGAAGAGAGTCAACCA
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>SRR18191605 | StxOp1168_Stx2a_16 | subject (1241 bp)

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CGTCAGGCACTGTCTGAAACTGCTCCTGTGTATACCATGACGCCGGGAGACG6TGGACCTCACTCTGAACTGGGGGCGAATCAGCAATGTGCTTCCGGAGTATCGGGGAGAGGATGGTGTG
AGAGTGGGAGAAATATCCTTTAATAATATATCAGCGATACTGGGTACTGTGGCGTTATACTGAATTGCCATCATCAGGGGGCGGTTCTGTTGCGCGCGTGAATGAAGAGAGTCAACCA
GAATGTGAGATAACTGGCGACAGGGCCGTTATAAAAATAAACAAATACATTATGGGAAAGTAATACAGCTGCAGCGTTTTCTGAAACAGAAAGTACAGTTTTTATATACAACGGGTAAATA
AGGAGTTAAGTATGAAGAAGATGTTTATG6CGGTTTTATTTGCAATTAGTTTCTGTTAATGCAATGGCGGGGATTTG6CTAAAGGTAATAATGAGTTTTCCAAAGTATAATGAGGATGACA
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>SRR18191606 | StxOp1035_Stx1c_12 | query (1228 bp)

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CGTACAACACTTGATGATCTCAGTGGGCGTTCTTATGTAATGACTGCTGAAGATGTTGATCTTACGTTGAACCTGGGGAAGGTTGAGTAGTGTCTGCTGACTATCATGGACAAGACTCT
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GAAAAAAATATTATTAATAGCTGCATCACTTTCAATTTTTTTCAGCAAGTG6CTG6CTGCGCCAGATTTGTGTAACCTGGGAAGGTGGAGTATACAAAAATATAATGATGACGATACCTTTAC
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>SRR18191606 | StxOp1035_Stx1c_12 | subject (1228 bp)

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>SRR18191610 | StxOp1321_Stx2a_6 | copy 1 of 2 | query (1241 bp)

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CTGTCCGTTGTGATCGGAAACCGTTGTCACACCGGGCAGTATATGTTGTAATACTGAAAAACGGTAGAAAGTATTGTTGCCGTATTAAACGAACCGGCCACATATAAATTTATTTTGC
TCAATAATCAGACGAAGATGGTCAAAACGCGCTGATAGACATCAAGCCCTCGTATATCCACAGCAAAATAACTGCCCGTGGGGTGTGGTTAATAACAGACACCGATGTGGTCCCTGA
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>SRR18191610 | StxOp1321_Stx2a_6 | copy 1 of 2 | subject (1241 bp)

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AGTATCGCTGATATATTATTAAAGGATATCTCCCACTCTGACACCATCTCTCCCGGATACTCCGGAAGCACATTGCTGATTGCGCCCAAGTTTCAGAGTGAGGTCACAGTCTCCCGG
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CTGTCCGTTGTGATCGGAAACCGTTGTCACACCGGGCAGTATATGTTGTAATACTGAAAAACGGTAGAAAGTATTGTTGCCGTATTAAACGAACCGGCCACATATAAATTTATTTTGC
TCAATAATCAGACGAAGATGGTCAAAACGCGCTGATAGACATCAAGCCCTCGTATATCCACAGCAAAATAACTGCCCGTGGGGTGTGGTTAATAACAGACACCGATGTGGTCCCTGA
GATATATGTTCAAGAGGGGTGATATCTCTGTCCGTACTATTAAACGAAGAGACATAACTTTGTTGGGTGCAAAAGTCTATCGTAAACTCCCGGGAATAGGATACCGAAGAAAAACCC
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>SRR18191610 | StxOp1321_Stx2a_6 | copy 2 of 2 | query (1241 bp)

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AGAGTGGGAGAAATATCCTTTAATAATATATCAGCGCACTGGGTCATCTGTGGCGTTACTGAACTATCAGGGGCGCGCTCTGTTCCGCGCGTGAATGAAGAGAGTCAACCA
GAATGTGAGATAACTGGCGACAGGCCGTTATAAAAATAAACAATACATTATGGGAAGTAAATACAGCAGCAGCGTTTCTGAAACAGAAAGTCACAGTTTTTATATACAACGGGTAAATAA
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>SRR18191610 | StxOp1321_Stx2a_6 | copy 2 of 2 | subject (1241 bp)

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>SRR18191616 | StxOp1339_Stx2b_109 | query (1236 bp)

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AGCCGCTACAGAAATATCTTTTCATATTTAACTCCTGTTATTCTCCGGGTGTTATTAAAGAGTGAAGCCTGCGATTTCAGAAAGCAGCTGCGGATTACTTTCCCATAAAGATTGTTTAT
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AATCAGACGAAGATGGTCAAAACGCGCTGATAGACATCAAGCCCTCGTATATCCACAGCAAAATAACTGCCCGTGGGGTATGGTTAATAACAGACACCGATGTGGTCCCTGAGATAT
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>SRR18191616 | StxOp1339_Stx2b_109 | subject (1236 bp)

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>SRR18191616 | StxOp399_Stx1c_12 | query (1228 bp)

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GTTTCGTGTTGGAAGAATTTCTTTTGAAGTGTAAATGCAATTTCTGGGTAGCGTGGCATTAATACTGAATGTGCATCATATGCATCGCGAGTTGCCAGAATTTGACCTAATGAGTTTCCT
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>SRR18191616 | StxOp399_Stx1c_12 | subject (1228 bp)

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CGTACAACACTTGTATGATCTCAGTGGCGGTTCTTATGTAATGACTGCTGAAGATGTTGATCTTACGTTGAACTGGGGAAGGTTGAGTAGTGCTTCCTGCTGACTATCATGGACAAGACTCT
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>SRR18191618 | StxOp1530_Stx2d_115 | query (1241 bp)

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>SRR18191618 | StxOp1530_Stx2d_115 | subject (1241 bp)

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>SRR18191628 | StxOp1168_Stx2a_16 | query (1241 bp)

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>SRR18191628 | StxOp1168_Stx2a_16 | subject (1241 bp)

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>SRR18191628 | StxOp4_Stx1a_9 | query (1227 bp)

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CGTACAACACTGGATGATCTCAGTGGGCGTTCTTATGTAAAGTACTGCTGAAGATGTTGATCTTACATTGAACCTGGGGAAGGTTGAGTAGTGCTCTGCCCTGACTATCATGGACAAGACTCT
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>SRR18191628 | StxOp4_Stx1a_9 | subject (1227 bp)

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GTTAAAGTGGGTGATAAAGAATTTATTAACACAGATGGAATCTTCAGTCTCTTCTTCTCAGTGCACAAATTACGGGGATGACTGTAACCATTAAGTAACTAATGCTGTCTAATGGAGGG
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>SRR18191635 | StxOp1494_Stx2d_115 | query (1241 bp)

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TCAATAATCAGACGAAGATGGTCAAACGCGCTGATAGACATCAAGCCCTCGTATATCCACAGCAAAATAACTGCCCGCGGGGTGTGGTTAATAACAGACACCGATGTGGTCCCTGA
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>SRR18191635 | StxOp1494_Stx2d_115 | subject (1241 bp)

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>SRR18191635 | StxOp3298_Stx2b_23 | query (1241 bp)

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>SRR18191635 | StxOp3298_Stx2b_23 | subject (1241 bp)

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>SRR18191644 | StxOp1339_Stx2b_109 | query (1236 bp)

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>SRR18191644 | StxOp1339_Stx2b_109 | subject (1236 bp)

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>SRR18191644 | StxOp13_Stx1a_19 | query (1227 bp)

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>SRR18191644 | StxOp13_Stx1a_19 | subject (1227 bp)

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>SRR18191644 | StxOp675_Stx2b_116 | query (1236 bp)

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>SRR18191644 | StxOp675_Stx2b_116 | subject (1236 bp)

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>SRR1871147 | StxOp1157_Stx2a_6 | copy 1 of 2 | query (1241 bp)

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>SRR1871147 | StxOp1157_Stx2a_6 | copy 1 of 2 | subject (1241 bp)

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>SRR1871147 | StxOp1157_Stx2a_6 | copy 2 of 2 | query (1241 bp)

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>SRR1871147 | StxOp1157_Stx2a_6 | copy 2 of 2 | subject (1241 bp)

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>SRR19368884 | StxOp1540_Stx2d_132 | query (1241 bp)

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>SRR19368884 | StxOp1540_Stx2d_132 | subject (1241 bp)

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>SRR1944449 | StxOp716_Stx2c_106 | query (1241 bp)

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>SRR1944449 | StxOp716_Stx2c_106 | subject (1241 bp)

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>SRR1950298 | StxOp675_Stx2b_116 | query (1236 bp)

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>SRR1950298 | StxOp675_Stx2b_116 | subject (1236 bp)

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>SRR1981027 | StxOp1344_Stx2b_109 | query (1236 bp)

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>SRR1999111 | StxOp1133_Stx2a_6 | query (1241 bp)

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>SRR1999111 | StxOp1133_Stx2a_6 | subject (1241 bp)

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>SRR1999997 | StxOp923_Stx2g_26 | subject (1242 bp)

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>SRR2015112 | StxOp1577_Stx1a_9 | query (1227 bp)

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>SRR2015112 | StxOp1577_Stx1a_9 | subject (1227 bp)

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>SRR2015125 | StxOp4_Stx1a_9 | query (1227 bp)

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>SRR2015125 | StxOp4_Stx1a_9 | subject (1227 bp)

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>SRR2015133 | StxOp1448_Stx2c_152 | query (1241 bp)

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>SRR2015133 | StxOp1448_Stx2c_152 | subject (1241 bp)

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>SRR2038672 | StxOp1181_Stx2a_16 | query (1241 bp)

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>SRR2038672 | StxOp1181_Stx2a_16 | subject (1241 bp)

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>SRR2056392 | StxOp1250_Stx2a_127 | query (1241 bp)

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>SRR2056392 | StxOp1250_Stx2a_127 | subject (1241 bp)

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>SRR2095948 | StxOp1250_Stx2a_127 | query (1241 bp)

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>SRR2095948 | StxOp1250_Stx2a_127 | subject (1241 bp)

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>SRR2102939 | StxOp935_Stx2j_103 | query (1234 bp)

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>SRR2102939 | StxOp935_Stx2j_103 | subject (1234 bp)

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>SRR21085779 | StxOp432_Stx2a_30 | query (1241 bp)

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>SRR21085779 | StxOp432_Stx2a_30 | subject (1241 bp)

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>SRR21085779 | StxOp4728_Stx2k_40 | query (1241 bp)

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>SRR21085779 | StxOp4728_Stx2k_40 | subject (1241 bp)

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>SRR2120919 | StxOp1173_Stx2a_16 | query (1241 bp)

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>SRR2120919 | StxOp1173_Stx2a_16 | subject (1241 bp)

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>SRR21504745 | StxOp1575_Stx2l_21 | query (1241 bp)

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>SRR21504745 | StxOp1575_Stx2l_21 | subject (1241 bp)

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>SRR21507356 | StxOp1477_Stx2d_38 | query (1241 bp)

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>SRR21507356 | StxOp1477_Stx2d_38 | subject (1241 bp)

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>SRR21507356 | StxOp1580_Stx1a_9 | query (1227 bp)

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>SRR21507356 | StxOp1580_Stx1a_9 | subject (1227 bp)

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>SRR21620895 | StxOp1040_Stx1d_4 | query (1227 bp)

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>SRR21620895 | StxOp1040_Stx1d_4 | subject (1227 bp)

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>SRR21634505 | StxOp679_Stx2b_135 | query (1236 bp)

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>SRR21634505 | StxOp679_Stx2b_135 | subject (1236 bp)

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>SRR2177988 | StxOp1173_Stx2a_16 | query (1241 bp)

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>SRR2177988 | StxOp1173_Stx2a_16 | subject (1241 bp)

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>SRR2178103 | StxOp1173_Stx2a_16 | query (1241 bp)

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>SRR2178103 | StxOp1173_Stx2a_16 | subject (1241 bp)

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>SRR23421918 | StxOp1250_Stx2a_127 | query (1241 bp)

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>SRR23421918 | StxOp1250_Stx2a_127 | subject (1241 bp)

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>SRR23421918 | StxOp13_Stx1a_19 | query (1227 bp)

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>SRR23421918 | StxOp13_Stx1a_19 | subject (1227 bp)

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>SRR23828388 | StxOp1575_Stx2l_21 | query (1241 bp)

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>SRR23828388 | StxOp1575_Stx2l_21 | subject (1241 bp)

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>SRR24225801 | StxOp932_Stx2h_49 | query (1236 bp)

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>SRR24225801 | StxOp932_Stx2h_49 | subject (1236 bp)

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>SRR2480649 | StxOp1035_Stx1c_12 | query (1228 bp)

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>SRR2480649 | StxOp1035_Stx1c_12 | subject (1228 bp)

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>SRR2481365 | StxOp935_Stx2j_103 | query (1234 bp)

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>SRR2481365 | StxOp935_Stx2j_103 | subject (1234 bp)

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>SRR2544681 | StxOp1160_Stx2a_11 | query (1241 bp)

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>SRR2544681 | StxOp1160_Stx2a_11 | subject (1241 bp)

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>SRR2549981 | StxOp1548_Stx2d_164 | query (1241 bp)

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>SRR2549981 | StxOp1548_Stx2d_164 | subject (1241 bp)

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>SRR2549981 | StxOp1708_Stx1a_9 | query (1227 bp)

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>SRR2549981 | StxOp1708_Stx1a_9 | subject (1227 bp)

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>SRR25642116 | StxOp1448_Stx2c_152 | query (1241 bp)

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>SRR25642116 | StxOp1448_Stx2c_152 | subject (1241 bp)

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>SRR25642116 | StxOp3298_Stx2b_23 | query (1236 bp)

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>SRR25642116 | StxOp3298_Stx2b_23 | subject (1236 bp)

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>SRR26362796 | StxOp1217_Stx2a_18 | query (1241 bp)

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>SRR26362796 | StxOp1217_Stx2a_18 | subject (1241 bp)

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>SRR26362796 | StxOp4_Stx1a_9 | query (1227 bp)

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>SRR26362796 | StxOp4_Stx1a_9 | subject (1227 bp)

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>SRR26411587 | StxOp1089_Stx2a_1 | query (1241 bp)

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>SRR26411587 | StxOp1089_Stx2a_1 | subject (1241 bp)

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>SRR26411587 | StxOp1168_Stx2a_16 | query (1241 bp)

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>SRR26411587 | StxOp1168_Stx2a_16 | subject (1241 bp)

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>SRR27958430 | StxOp1168_Stx2a_16 | query (1241 bp)

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>SRR27958430 | StxOp1168_Stx2a_16 | subject (1241 bp)

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>SRR27958430 | StxOp4_Stx1a_9 | query (1227 bp)

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>SRR27958430 | StxOp4_Stx1a_9 | subject (1227 bp)

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>SRR27967202 | StxOp1168_Stx2a_16 | query (1241 bp)

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>SRR27967202 | StxOp1168_Stx2a_16 | subject (1241 bp)

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>SRR28356459 | StxOp1574_Stx2g_39 | query (1242 bp)

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>SRR28356459 | StxOp1574_Stx2g_39 | subject (1242 bp)

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>SRR28356459 | StxOp1586_Stx1a_9 | query (1227 bp)

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>SRR28356459 | StxOp1586_Stx1a_9 | subject (1227 bp)

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>SRR28420203 | StxOp1096_Stx2a_6 | query (1241 bp)

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>SRR28420203 | StxOp1096_Stx2a_6 | subject (1241 bp)

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>SRR28420203 | StxOp39_Stx1a_54 | query (1227 bp)

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>SRR28420203 | StxOp39_Stx1a_54 | subject (1227 bp)

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>SRR28618069 | StxOp1124_Stx2a_6 | query (1241 bp)

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>SRR28618069 | StxOp1124_Stx2a_6 | subject (1241 bp)

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>SRR28618069 | StxOp4_Stx1a_9 | query (1227 bp)

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>SRR28723267 | StxOp1173_Stx2a_16 | query (1241 bp)

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>SRR28837514 | StxOp943_Stx2k_40 | query (1241 bp)

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>SRR28837514 | StxOp943_Stx2k_40 | subject (1241 bp)

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>SRR28994774 | StxOp2848_Stx2a_183 | query (1241 bp)

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>SRR28994774 | StxOp2848_Stx2a_183 | subject (1241 bp)

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>SRR28994774 | StxOp4_Stx1a_9 | query (1227 bp)

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>SRR28994774 | StxOp4_Stx1a_9 | subject (1227 bp)

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>SRR29067298 | StxOp943_Stx2k_40 | query (1241 bp)

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>SRR29067298 | StxOp943_Stx2k_40 | subject (1241 bp)

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>SRR3068663 | StxOp939_Stx2j_225 | query (1234 bp)

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>SRR3136876 | StxOp1168_Stx2a_16 | query (1241 bp)

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>SRR3136876 | StxOp1168_Stx2a_16 | subject (1241 bp)

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>SRR3136876 | StxOp966_Stx1a_18 | query (1227 bp)

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>SRR3136876 | StxOp966_Stx1a_18 | subject (1227 bp)

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>SRR3180475 | StxOp1089_Stx2a_1 | query (1241 bp)

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>SRR3180475 | StxOp1089_Stx2a_1 | subject (1241 bp)

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>SRR3180475 | StxOp1577_Stx1a_9 | query (1227 bp)

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>SRR3180475 | StxOp1577_Stx1a_9 | subject (1227 bp)

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>SRR3180697 | StxOp4647_Stx2f_725 | query (1236 bp)

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>SRR3180697 | StxOp4647_Stx2f_725 | subject (1236 bp)

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>SRR3180701 | StxOp918_Stx2f_89 | query (1236 bp)

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>SRR3180701 | StxOp918_Stx2f_89 | subject (1236 bp)

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>SRR3189185 | StxOp1089_Stx2a_1 | query (1241 bp)

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>SRR3189185 | StxOp1089_Stx2a_1 | subject (1241 bp)

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>SRR3189185 | StxOp3_Stx1a_9 | query (1227 bp)

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>SRR3189185 | StxOp3_Stx1a_9 | subject (1227 bp)

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>SRR3191403 | StxOp1577_Stx1a_9 | query (1227 bp)

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>SRR3191403 | StxOp1577_Stx1a_9 | subject (1227 bp)

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>SRR3191403 | StxOp702_Stx2c_17 | query (1241 bp)

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>SRR3191403 | StxOp702_Stx2c_17 | subject (1241 bp)

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>SRR3195851 | StxOp702_Stx2c_17 | query (1241 bp)

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>SRR3195851 | StxOp702_Stx2c_17 | subject (1241 bp)

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>SRR3213947 | StxOp1040_Stx1d_4 | query (1227 bp)

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>SRR3213947 | StxOp1040_Stx1d_4 | subject (1227 bp)

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>SRR3231803 | StxOp1089_Stx2a_1 | query (1241 bp)

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>SRR3231803 | StxOp1089_Stx2a_1 | subject (1241 bp)

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>SRR3231803 | StxOp3_Stx1a_9 | query (1227 bp)

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>SRR3231803 | StxOp3_Stx1a_9 | subject (1227 bp)

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>SRR3278959 | StxOp432_Stx2a_30 | query (1241 bp)

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>SRR3278959 | StxOp432_Stx2a_30 | subject (1241 bp)

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>SRR3394639 | StxOp1173_Stx2a_16 | subject (1241 bp)

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>SRR3394639 | StxOp1259_Stx2a_128 | query (1241 bp)

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>SRR3394830 | StxOp1036_Stx1c_12 | query (1229 bp)

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>SRR3394830 | StxOp1036_Stx1c_12 | subject (1229 bp)

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>SRR3794154 | StxOp806_Stx2c_642 | subject (1241 bp)

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>SRR3794154 | StxOp866_Stx2d_350 | query (1241 bp)

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>SRR3794154 | StxOp866_Stx2d_350 | subject (1241 bp)

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>SRR4176409 | StxOp399_Stx1c_12 | query (1228 bp)

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>SRR4176409 | StxOp399_Stx1c_12 | subject (1228 bp)

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>SRR4176409 | StxOp653_Stx2b_23 | query (1236 bp)

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>SRR4176409 | StxOp653_Stx2b_23 | subject (1236 bp)

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>SRR4430014 | StxOp1041_Stx1d_4 | query (1227 bp)

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>SRR4430014 | StxOp1041_Stx1d_4 | subject (1227 bp)

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>SRR5264142 | StxOp1448_Stx2c_152 | query (1241 bp)

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>SRR5264142 | StxOp1448_Stx2c_152 | subject (1241 bp)

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>SRR5264142 | StxOp3298_Stx2b_23 | query (1236 bp)

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>SRR5264142 | StxOp3298_Stx2b_23 | subject (1236 bp)

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>SRR5434351 | StxOp1040_Stx1d_4 | query (1227 bp)

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>SRR5434351 | StxOp1040_Stx1d_4 | subject (1227 bp)

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>SRR5616352 | StxOp1038_Stx1d_4 | query (1227 bp)

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>SRR5616352 | StxOp1038_Stx1d_4 | subject (1227 bp)

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>SRR5642675 | StxOp941_Stx2j_705 | query (1234 bp)

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>SRR5642675 | StxOp941_Stx2j_705 | subject (1234 bp)

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>SRR5832137 | StxOp1168_Stx2a_16 | query (1241 bp)

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>SRR6321371 | StxOp918_Stx2f_89 | query (1236 bp)

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>SRR6321371 | StxOp918_Stx2f_89 | subject (1236 bp)

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>SRR6767327 | StxOp938_Stx2j_223 | query (1234 bp)

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>SRR6767327 | StxOp938_Stx2j_223 | subject (1234 bp)

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>SRR6903006 | StxOp955_Stx2o_545 | query (1236 bp)

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>SRR6903006 | StxOp955_Stx2o_545 | subject (1236 bp)

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>SRR7163944 | StxOp949_Stx2m_547 | subject (1236 bp)

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>SRR7235296 | StxOp4746_Stx2o_41 | query (1236 bp)

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>SRR7235296 | StxOp4746_Stx2o_41 | subject (1236 bp)

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>SRR7612274 | StxOp943_Stx2k_40 | query (1241 bp)

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>SRR7612274 | StxOp943_Stx2k_40 | subject (1241 bp)

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>SRR7738059 | StxOp953_Stx2n_126 | query (1236 bp)

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CAAC
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>SRR7738059 | StxOp953_Stx2n_126 | subject (1236 bp)

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>SRR7767477 | StxOp953_Stx2n_126 | query (1236 bp)

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CAAC
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>SRR7767477 | StxOp953_Stx2n_126 | subject (1236 bp)

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CAAC
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>SRR7767490 | StxOp4596_Stx2e_36 | query (1236 bp)

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>SRR7767490 | StxOp4596_Stx2e_36 | subject (1236 bp)

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>SRR7826316 | StxOp1089_Stx2a_1 | query (1241 bp)

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>SRR7826316 | StxOp3_Stx1a_9 | query (1227 bp)

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>SRR7826361 | StxOp21_Stx1a_33 | query (1227 bp)

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>SRR7896248 | StxOp842_Stx2d_115 | subject (1241 bp)

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>SRR7896259 | StxOp1208_Stx2a_16 | query (1241 bp)

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>SRR7896259 | StxOp1208_Stx2a_16 | subject (1241 bp)

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>SRR7896259 | StxOp1510_Stx2d_115 | copy 1 of 2 | query (1241 bp)

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>SRR7896259 | StxOp1510_Stx2d_115 | copy 1 of 2 | subject (1241 bp)

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>SRR7896259 | StxOp1510_Stx2d_115 | copy 2 of 2 | query (1241 bp)

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>SRR7896259 | StxOp1510_Stx2d_115 | copy 2 of 2 | subject (1241 bp)

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>SRR7958537 | StxOp942_Stx2j_706 | query (1234 bp)

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>SRR7958537 | StxOp942_Stx2j_706 | subject (1234 bp)

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>SRR8931387 | StxOp1036_Stx1c_12 | query (1229 bp)

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>SRR8931387 | StxOp1036_Stx1c_12 | subject (1229 bp)

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>SRR8931387 | StxOp1348_Stx2b_110 | query (1236 bp)

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>SRR8931387 | StxOp1348_Stx2b_110 | subject (1236 bp)

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>SRR8931387 | StxOp167_Stx1a_191 | query (1227 bp)

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>SRR8931387 | StxOp167_Stx1a_191 | subject (1227 bp)

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>SRR9007492 | StxOp948_Stx2m_48 | query (1236 bp)

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>SRR9007492 | StxOp948_Stx2m_48 | subject (1236 bp)

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>SRR9833314 | StxOp937_Stx2j_108 | query (1234 bp)

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>SRR9833314 | StxOp937_Stx2j_108 | subject (1234 bp)

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>SRR9972670 | StxOp1168_Stx2a_16 | query (1241 bp)

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>SRR9972670 | StxOp1168_Stx2a_16 | subject (1241 bp)

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Supplementary Table S3.1. Sequence Read Archive accessions and isolate metadata for the 497 CFIA STEC genomes.

All 497 genomes were deposited under BioProject PRJNA454819. Genome ID is the identifier used throughout Chapter 3; genomes retrieved as public assemblies are identified by their GenBank assembly accession (GCA_*, n = 35).

Isolation source gives the seven commodity categories used in Chapter 3. Isolation source detail separates lamb from other meat, and is the basis of the lamb subset analysed in Chapter 4.

Stx subtype(s) lists the distinct subtypes detected in each genome; STxOP_nt operon(s) gives the reference operon for each copy, so a genome carrying a duplicated operon lists that identifier twice.

497 rows, 10 columns.

Genome ID	CFIA isolate	BioProject	BioSample	SRA run accession	Serotype	Isolation source	Isolation source detail	Stx subtype(s)	STxOP_nt operon(s)
2013-SEQ-0045	CFIACB20130349	PRJNA454819	SAMN13334359	SRR22858767	OUT:H18	Nuts/Seeds	Nuts/Seeds	Stx1a; Stx2a	StxOp1168_Stx2a_16; StxOp966_Stx1a_18
2013-SEQ-0046	CFIACB20130338	PRJNA454819	SAMN13334348	SRR22858863	OUT:H18	Nuts/Seeds	Nuts/Seeds	Stx1a; Stx2a	StxOp1168_Stx2a_16; StxOp966_Stx1a_18
2013-SEQ-0047	CFIACB20130339	PRJNA454819	SAMN13334349	SRR22858874	OUT:H18	Nuts/Seeds	Nuts/Seeds	Stx1a; Stx2a	StxOp1168_Stx2a_16; StxOp966_Stx1a_18
2013-SEQ-0048	CFIACB20130340	PRJNA454819	SAMN13334350	SRR22858890	OUT:H18	Nuts/Seeds	Nuts/Seeds	Stx1a; Stx2a	StxOp966_Stx1a_18; StxOp1168_Stx2a_16
2013-SEQ-0109	CFIACB20120240	PRJNA454819	SAMN09668397	SRR7629310	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2013-SEQ-0110	CFIACB20120241	PRJNA454819	SAMN09668398	SRR7629295	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2013-SEQ-0111	CFIACB20130130	PRJNA454819	SAMN09668408	SRR7629318	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2014-SEQ-0093	CFIACB20100227	PRJNA454819	SAMN13333857	SRR22858879	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2014-SEQ-0094	CFIACB20100228	PRJNA454819	SAMN13333858	SRR22858885	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2014-SEQ-0095	CFIACB20100229	PRJNA454819	SAMN13333859	SRR22858886	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2014-SEQ-0096	CFIACB20100230	PRJNA454819	SAMN13333860	SRR22858887	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2014-SEQ-0148	CFIACB20120182	PRJNA454819	SAMN09668376	SRR7536943	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2014-SEQ-0149	CFIACB20120201	PRJNA454819	SAMN13334070	SRR22858897	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2014-SEQ-0179	CFIACB20120186	PRJNA454819	SAMN09668385	SRR7629283	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2014-SEQ-0181	CFIACB20120216	PRJNA454819	SAMN09668394	SRR7629285	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2014-SEQ-0182	CFIACB20120244	PRJNA454819	SAMN09668404	SRR7629287	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2014-SEQ-0183	CFIACB20120252	PRJNA454819	SAMN09668406	SRR7629319	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2014-SEQ-0184	CFIACB20130141	PRJNA454819	SAMN09668415	SRR7629327	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2014-SEQ-0186	CFIACB20130344	PRJNA454819	SAMN13334354	SRR22858762	O157:H7	Meat	Meat	Stx1a; Stx2c	StxOp3_Stx1a_9; StxOp702_Stx2c_17
2014-SEQ-0245	CFIACB20130343	PRJNA454819	SAMN13334353	SRR22858813	O185:H7	Nuts/Seeds	Nuts/Seeds	Stx2c	StxOp716_Stx2c_106
2014-SEQ-0246	CFIACB20130342	PRJNA454819	SAMN13334352	SRR22858902	O185:H7	Nuts/Seeds	Nuts/Seeds	Stx2c	StxOp716_Stx2c_106
2014-SEQ-0252	CFIACB20120192	PRJNA454819	SAMN09668392	SRR7629317	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2014-SEQ-0349	CFIACB20140388	PRJNA454819	SAMN19011647	SRR14750047	OUT:H23	Vegetable	Vegetable	Stx2j	StxOp936_Stx2j_104
2014-SEQ-0350	CFIACB20130347	PRJNA454819	SAMN13334357	SRR22858765	O113:H21	Meat	Meat	Stx2d	StxOp827_Stx2d_10
2014-SEQ-0351	CFIACB20130348	PRJNA454819	SAMN13334358	SRR22858766	O113:H21	Meat	Meat	Stx2d	StxOp827_Stx2d_10
2014-SEQ-0445	CFIACB20140238	PRJNA454819	SAMN13334591	SRR22858770	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2014-SEQ-0446	CFIACB20140243	PRJNA454819	SAMN13334596	SRR22858776	O157:H7	Meat	Meat	Stx2c	StxOp702_Stx2c_17
2014-SEQ-0447	CFIACB20140245	PRJNA454819	SAMN13334598	SRR22858778	O157:H7	Meat	Meat	Stx2c	StxOp702_Stx2c_17
2014-SEQ-0448	CFIACB20140247	PRJNA454819	SAMN13334600	SRR22858780	O157:H7	Meat	Meat	Stx2c	StxOp702_Stx2c_17
2014-SEQ-0449	CFIACB20140250	PRJNA454819	SAMN13334602	SRR22858783	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2014-SEQ-0450	CFIACB20140251	PRJNA454819	SAMN13334603	SRR22858784	O157:H7	Meat	Meat	Stx2c	StxOp702_Stx2c_17
2014-SEQ-0453	CFIACB20140257	PRJNA454819	SAMN13334609	SRR22858790	O157:H7	Environment	Environment	Stx2c	StxOp702_Stx2c_17
2014-SEQ-0455	CFIACB20140260	PRJNA454819	SAMN13334612	SRR22858794	O157:H7	Environment	Environment	Stx1a; Stx2c	StxOp4081_Stx2c_106; StxOp3_Stx1a_9
2014-SEQ-0456	CFIACB20130252	PRJNA454819	SAMN09668422	SRR7629321	O157:H7	Dairy	Dairy	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2014-SEQ-0457	CFIACB20130254	PRJNA454819	SAMN09668424	SRR7629286	O157:H7	Dairy	Dairy	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2014-SEQ-0757	CFIACB20140303	PRJNA454819	SAMN13334648	SRR22858796	O146:H32	Vegetable	Vegetable	Stx2a	StxOp594_Stx2a_567

Genome ID	CFIA isolate	BioProject	BioSample	SRA run accession	Serotype	Isolation source	Isolation source detail	Stx subtype(s)	STxOP_nt operon(s)
2014-SEQ-0758	CFIACB20140304	PRJNA454819	SAMN13334649	SRR22858797	O146:H32	Vegetable	Vegetable	Stx2a	StxOp594_Stx2a_567
2014-SEQ-1296	CFIACB20150310	PRJNA454819	SAMN13334685	SRR22858803	O159:H19	Nuts/Seeds	Nuts/Seeds	Stx2a	StxOp432_Stx2a_30
2014-SEQ-1297	CFIACB20150311	PRJNA454819	SAMN13334686	SRR22858805	O159:H19	Nuts/Seeds	Nuts/Seeds	Stx2a	StxOp432_Stx2a_30
2014-SEQ-1298	CFIACB20150312	PRJNA454819	SAMN13334687	SRR22858806	O159:H19	Nuts/Seeds	Nuts/Seeds	Stx2a	StxOp432_Stx2a_30
2014-SEQ-1300	CFIACB20150314	PRJNA454819	SAMN13334688	SRR22858807	O159:H19	Nuts/Seeds	Nuts/Seeds	Stx2a	StxOp432_Stx2a_30
2014-SEQ-1301	CFIACB20150315	PRJNA454819	SAMN13334689	SRR22858808	O159:H19	Nuts/Seeds	Nuts/Seeds	Stx2a	StxOp432_Stx2a_30
2014-SEQ-1302	CFIACB20090190	PRJNA454819	SAMN13333594	SRR22858857	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2014-SEQ-1303	CFIACB20090191	PRJNA454819	SAMN13333595	SRR22858858	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2014-SEQ-1304	CFIACB20090193	PRJNA454819	SAMN13333596	SRR22858859	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2014-SEQ-1305	CFIACB20090196	PRJNA454819	SAMN13333599	SRR22858860	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2014-SEQ-1306	CFIACB20090198	PRJNA454819	SAMN13333600	SRR22858861	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2014-SEQ-1307	CFIACB20090199	PRJNA454819	SAMN13333601	SRR22858862	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2014-SEQ-1308	CFIACB20090201	PRJNA454819	SAMN13333602	SRR22858864	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2014-SEQ-1309	CFIACB20090227	PRJNA454819	SAMN13333615	SRR22858865	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2014-SEQ-1312	CFIACB20100182	PRJNA454819	SAMN13333830	SRR22858867	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2014-SEQ-1314	CFIACB20100190	PRJNA454819	SAMN13333834	SRR22858869	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2014-SEQ-1315	CFIACB20100193	PRJNA454819	SAMN13333836	SRR22858870	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2015-SEQ-0036	CFIACB20100212	PRJNA454819	SAMN13333849	SRR22858871	O157:H7	Meat	Meat	Stx2c	StxOp702_Stx2c_17
2015-SEQ-0037	CFIACB20100213	PRJNA454819	SAMN13333850	SRR22858872	O157:H7	Meat	Meat	Stx2c	StxOp702_Stx2c_17
2015-SEQ-0039	CFIACB20100215	PRJNA454819	SAMN13333852	SRR22858875	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2015-SEQ-0040	CFIACB20110113	PRJNA454819	SAMN13333970	SRR22858888	O157:H7	Meat	Meat	Stx2c	StxOp702_Stx2c_17
2015-SEQ-0044	CFIACB20120176	PRJNA454819	SAMN13334061	SRR22858894	O157:H7	Meat	Meat	Stx1a; Stx2c	StxOp1577_Stx1a_9; StxOp702_Stx2c_17
2015-SEQ-0046	CFIACB20120199	PRJNA454819	SAMN13334068	SRR22858896	O157:H7	Meat	Meat	Stx2c	StxOp702_Stx2c_17
2015-SEQ-0048	CFIACB20120210	PRJNA454819	SAMN13334075	SRR22858898	O157:H7	Meat	Meat	Stx2c	StxOp702_Stx2c_17
2015-SEQ-0049	CFIACB20120211	PRJNA454819	SAMN13334076	SRR22858899	O157:H7	Meat	Meat	Stx2c	StxOp702_Stx2c_17
2015-SEQ-0051	CFIACB20120219	PRJNA454819	SAMN13334082	SRR22858880	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2015-SEQ-0056	CFIACB20130259	PRJNA454819	SAMN09668427	SRR7629323	O157:H7	Dairy	Dairy	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2015-SEQ-0113	CFIACB20150316	PRJNA454819	SAMN10746426	SRR8448492	OUT:H26	Meat	Meat	Stx2e	StxOp1568_Stx2e_36
2015-SEQ-0655	CFIACB20150322	PRJNA454819	SAMN10746420	SRR8448488	O121:H10	Meat	Meat	Stx2e	StxOp4587_Stx2e_25
2015-SEQ-0656	CFIACB20150324	PRJNA454819	SAMN13334691	SRR22858809	O121:H10	Meat	Meat	Stx2e	StxOp4587_Stx2e_25
2015-SEQ-0657	CFIACB20150325	PRJNA454819	SAMN13334692	SRR22858814	O121:H10	Meat	Meat	Stx2e	StxOp4587_Stx2e_25
2015-SEQ-0658	CFIACB20150326	PRJNA454819	SAMN10746424	SRR8448484	O36:HUT	Meat	Meat	Stx2e	StxOp4587_Stx2e_25
2015-SEQ-0659	CFIACB20150327	PRJNA454819	SAMN10746427	SRR8448493	OUT:H38	Meat	Meat	Stx2e	StxOp898_Stx2e_25
2015-SEQ-0725	CFIACB20150328	PRJNA454819	SAMN10746425	SRR8448485	O91:H14	Meat	Meat	Stx1a	StxOp4_Stx1a_9
2015-SEQ-0877	CFIACB20130272	PRJNA454819	SAMN13334344	SRR22858881	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2015-SEQ-0878	CFIACB20130273	PRJNA454819	SAMN13334345	SRR22858883	O157:H7	Meat	Meat	Stx2c	StxOp702_Stx2c_17
2015-SEQ-0879	CFIACB20140198	PRJNA454819	SAMN13334553	SRR22858768	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2015-SEQ-0886	CFIACB20150336	PRJNA454819	SAMN13334695	SRR22858815	O91:H14	Meat	Meat	Stx1a	StxOp4_Stx1a_9
2015-SEQ-0887	CFIACB20150337	PRJNA454819	SAMN13334696	SRR22858816	O91:H14	Meat	Meat	Stx1a	StxOp4_Stx1a_9
2015-SEQ-0888	CFIACB20150335	PRJNA454819	SAMN10746428	SRR8448482	O91:H14	Meat	Meat	Stx1a	StxOp4_Stx1a_9
2015-SEQ-0889	CFIACB20150339	PRJNA454819	SAMN13334698	SRR22858817	O91:H14	Meat	Meat	Stx1a	StxOp4_Stx1a_9
2015-SEQ-0890	CFIACB20150340	PRJNA454819	SAMN13334699	SRR22858818	O91:H14	Meat	Meat	Stx1a	StxOp4_Stx1a_9
2015-SEQ-0891	CFIACB20140202	PRJNA454819	SAMN13334557	SRR22858769	O157:H7	Meat	Meat	Stx2c	StxOp702_Stx2c_17
2015-SEQ-0964	CFIACB20150226	PRJNA454819	SAMN13334678	SRR22858799	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2015-SEQ-1027	CFIACB20150236	PRJNA454819	SAMN13334680	SRR22858800	O157:H7	Meat	Meat	Stx2c	StxOp4081_Stx2c_106
2015-SEQ-1048	CFIACB20150247	PRJNA454819	SAMN13334682	SRR22858801	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2015-SEQ-1131	CFIACB20150253	PRJNA454819	SAMN13334684	SRR22858802	O157:H7	Vegetable	Vegetable	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9

Genome ID	CFIA isolate	BioProject	BioSample	SRA run accession	Serotype	Isolation source	Isolation source detail	Stx subtype(s)	STxOP_nt operon(s)
2015-SEQ-1141	CFIACB20150349	PRJNA454819	SAMN10746429	SRR8448483	O163:H19	Meat	Meat	Stx1a	StxOp4_Stx1a_9
2015-SEQ-1142	CFIACB20150351	PRJNA454819	SAMN10746430	SRR8448481	O86:H32	Meat	Meat	Stx2e	StxOp4587_Stx2e_25
2015-SEQ-1454	CFIACB20150357	PRJNA454819	SAMN13334701	SRR22858820	O132:H18	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2015-SEQ-1455	CFIACB20150356	PRJNA454819	SAMN10746421	SRR8448489	O132:H18	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2015-SEQ-1456	CFIACB20150359	PRJNA454819	SAMN13334703	SRR22858821	O132:H18	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2015-SEQ-1555	CFIACB20160248	PRJNA454819	SAMN10746422	SRR8448486	O91:H14	Meat	Meat	Stx1a	StxOp4_Stx1a_9
2015-SEQ-1635	CFIACB20160170	PRJNA454819	SAMN13334711	SRR22858823	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2015-SEQ-1636	CFIACB20160172	PRJNA454819	SAMN13334713	SRR22858825	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3231_Stx2a_1126; StxOp3_Stx1a_9
2015-SEQ-1638	CFIACB20160174	PRJNA454819	SAMN13334715	SRR22858826	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2015-SEQ-1639	CFIACB20160175	PRJNA454819	SAMN13334716	SRR22858827	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2015-SEQ-1640	CFIACB20160176	PRJNA454819	SAMN13334717	SRR22858828	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2015-SEQ-1643	CFIACB20160179	PRJNA454819	SAMN13334720	SRR22858811	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2015-SEQ-1849	CFIACB20160185	PRJNA454819	SAMN13334721	SRR22858812	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2015-SEQ-1850	CFIACB20160186	PRJNA454819	SAMN13334722	SRR22858771	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2015-SEQ-1868	CFIACB20160187	PRJNA454819	SAMN13334723	SRR22858782	O157:H7	Meat	Meat	Stx2a	StxOp1043_Stx2a_1
2016-BUR-0134	CFIACB20000006	PRJNA454819	SAMN13333295	SRR22858882	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2016-BUR-0136	CFIACB20020028	PRJNA454819	SAMN13333330	SRR22858836	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2016-BUR-0137	CFIACB20000007	PRJNA454819	SAMN13333296	SRR22858884	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2016-BUR-0138	CFIACB20000008	PRJNA454819	SAMN13333297	SRR22858832	O157:H7	Other	Other	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2016-BUR-0140	CFIACB20080071	PRJNA454819	SAMN13333401	SRR22858853	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2016-BUR-0141	CFIACB20080072	PRJNA454819	SAMN13333402	SRR22858854	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2016-SEQ-0016	CFIACB20160262	PRJNA454819	SAMN13334730	SRR22858831	O115:H10	Meat	Meat	Stx1a	StxOp4_Stx1a_9
2016-SEQ-0053	CFIACB20160269	PRJNA454819	SAMN11373400	SRR22715108	O87:H16	Meat	Meat	Stx2b	StxOp3298_Stx2b_23
2016-SEQ-0054	CFIACB20160268	PRJNA454819	SAMN13334731	SRR18215052	O176:H4	Meat	Meat	Stx1c	StxOp1037_Stx1c_26
2016-SEQ-0055	CFIACB20160267	PRJNA454819	SAMN11030301	SRR22715083	O113:H21	Meat	Meat	Stx2d	StxOp827_Stx2d_10
2016-SEQ-0057	CFIACB20160270	PRJNA454819	SAMN11030302	SRR22715082	O113:H21	Meat	Meat	Stx2d	StxOp827_Stx2d_10
2016-SEQ-0118	CFIACB20160277	PRJNA454819	SAMN10989390	SRR22715063	O113:H4	Meat	Meat	Stx2d	StxOp840_Stx2d_99
2016-SEQ-0129	CFIACB20160190	PRJNA454819	SAMN13334725	SRR22858793	O157:H7	Meat	Meat	Stx2c	StxOp702_Stx2c_17
2016-SEQ-0130	CFIACB20160191	PRJNA454819	SAMN13334726	SRR22858804	O157:H7	Meat	Meat	Stx2c	StxOp702_Stx2c_17
2016-SEQ-0243	CFIACB20160280	PRJNA454819	SAMN13334732	SRR22858727	O132:H18	Nuts/Seeds	Nuts/Seeds	Stx2a	StxOp1089_Stx2a_1
2016-SEQ-0250	CFIACB20080069	PRJNA454819	SAMN13333399	SRR18215057	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2016-SEQ-0251	CFIACB20080070	PRJNA454819	SAMN13333400	SRR22858851	O157:H7	Meat	Meat	Stx2c	StxOp702_Stx2c_17
2016-SEQ-0252	CFIACB20120266	PRJNA454819	SAMN19011646	SRR14750046	O177:NM	Vegetable	Vegetable	Stx2h	StxOp932_Stx2h_49
2016-SEQ-0254	CFIACB20130337	PRJNA454819	SAMN13334347	SRR22858852	O174:H21	Nuts/Seeds	Nuts/Seeds	Stx2c	StxOp1364_Stx2c_7
2016-SEQ-0255	CFIACB20130336	PRJNA454819	SAMN13334346	SRR22858841	O168:H8	Nuts/Seeds	Nuts/Seeds	Stx2a	StxOp419_Stx2a_1
2016-SEQ-0354	CFIACB20160289	PRJNA454819	SAMN11030305	SRR18215056	O178:H19	Meat	Meat	Stx1a; Stx2a	StxOp1250_Stx2a_127; StxOp4_Stx1a_9
2016-SEQ-0355	CFIACB20160290	PRJNA454819	SAMN11373509	SRR22715047	O178:H19	Meat	Meat	Stx1a; Stx2a	StxOp1250_Stx2a_127; StxOp4_Stx1a_9
2016-SEQ-0396	CFIACB20120177	PRJNA454819	SAMN09668380	SRR7629315	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2016-SEQ-0420	CFIACB20160291	PRJNA454819	SAMN10989389	SRR22715064	O142:H38	Nuts/Seeds	Nuts/Seeds	Stx1d	StxOp1038_Stx1d_4
2016-SEQ-0836	CFIACB20160152	PRJNA454819	SAMN11030467	SRR22715118	O157:H7	Meat	Meat	Stx2c	StxOp702_Stx2c_17
2016-SEQ-1059	CFIACB20130341	PRJNA454819	SAMN13334351	SRR22858901	O168:H8	Nuts/Seeds	Nuts/Seeds	Stx2a	StxOp419_Stx2a_1
2016-SEQ-1089	CFIACB20170241	PRJNA454819	SAMN11030308	SRR22715076	O109:H5	Meat	Meat	Stx1a	StxOp15_Stx1a_21
2016-SEQ-1338	CFIACB20170256	PRJNA454819	SAMN11373398	SRR22715110	O8:H2	Meat	Meat	Stx2b	StxOp669_Stx2b_105
2016-SEQ-1339	CFIACB20170257	PRJNA454819	SAMN11373396	SRR22715112	O8:H2	Meat	Meat	Stx2b	StxOp669_Stx2b_105
2016-SEQ-1340	CFIACB20170258	PRJNA454819	SAMN11373397	SRR22715111	O8:H2	Meat	Meat	Stx2b	StxOp669_Stx2b_105
2016-SEQ-1341	CFIACB20170259	PRJNA454819	SAMN11373399	SRR22715109	O8:H2	Meat	Meat	Stx2b	StxOp669_Stx2b_105
2016-SEQ-1342	CFIACB20170260	PRJNA454819	SAMN13334742	SRR22858697	O8:H2	Meat	Meat	Stx2b	StxOp669_Stx2b_105

Genome ID	CFIA isolate	BioProject	BioSample	SRA run accession	Serotype	Isolation source	Isolation source detail	Stx subtype(s)	STxOP_nt operon(s)
2016-SEQ-1356	CFIACB20170134	PRJNA454819	SAMN10103178	SRR7890019	O8:H28	Flour	Flour	Stx2a	StxOp432_Stx2a_30
2016-SEQ-1373	CFIACB20170128	PRJNA454819	SAMN09468191	SRR7435871	O121:H19	Flour	Flour	Stx2a	StxOp419_Stx2a_1
2017-BUR-0001	CFIACB20170177	PRJNA454819	SAMN09468183	SRR7435865	O8:H28	Flour	Flour	Stx2a	StxOp432_Stx2a_30
2017-BUR-0049	CFIACB20170191	PRJNA454819	SAMN09468184	SRR7435873	O159:H19	Flour	Flour	Stx2a	StxOp432_Stx2a_30
2017-BUR-0060	CFIACB20170339	PRJNA454819	SAMN13334750	SRR22858703	O171:H2	Meat	Meat	Stx2a	StxOp1287_Stx2a_183
2017-BUR-0073	CFIACB20170184	PRJNA454819	SAMN09468185	SRR7435869	O121:H19	Flour	Flour	Stx2a	StxOp419_Stx2a_1
2017-BUR-0074	CFIACB20170185	PRJNA454819	SAMN13334737	SRR22858693	O121:H19	Flour	Flour	Stx2a	StxOp419_Stx2a_1
2017-BUR-0075	CFIACB20170186	PRJNA454819	SAMN13334738	SRR22858694	O121:H19	Flour	Flour	Stx2a	StxOp419_Stx2a_1
2017-BUR-0077	CFIACB20170188	PRJNA454819	SAMN13334739	SRR22858695	O121:H19	Flour	Flour	Stx2a	StxOp419_Stx2a_1
2017-CAL-0117	CFIACB20040036	PRJNA454819	SAMN13333350	SRR22858842	O157:H7	Meat	Meat	Stx1a	StxOp1577_Stx1a_9
2017-CAL-0118	CFIACB20040035	PRJNA454819	SAMN13333349	SRR22858840	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2017-CAL-0119	CFIACB20040033	PRJNA454819	SAMN13333347	SRR22858838	O157:H7	Meat	Meat	Stx2a	StxOp1043_Stx2a_1
2017-CAL-0120	CFIACB20040032	PRJNA454819	SAMN13333346	SRR22858837	O157:H7	Meat	Meat	Stx2c	StxOp702_Stx2c_17
2017-CAL-0121	CFIACB20040034	PRJNA454819	SAMN13333348	SRR22858839	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2017-CAL-0123	CFIACB20050045	PRJNA454819	SAMN13333360	SRR22858843	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2017-CAL-0125	CFIACB20070057	PRJNA454819	SAMN13333371	SRR22858844	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2017-CAL-0126	CFIACB20070058	PRJNA454819	SAMN13333372	SRR22858845	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2017-CAL-0127	CFIACB20070059	PRJNA454819	SAMN13333373	SRR22858846	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2017-CAL-0128	CFIACB20070060	PRJNA454819	SAMN13333374	SRR22858847	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1081_Stx2a_1; StxOp3_Stx1a_9
2017-CAL-0226	CFIACB20070062	PRJNA454819	SAMN13333376	SRR22858848	O157:H7	Meat	Meat	Stx1a; Stx2c	StxOp3_Stx1a_9; StxOp702_Stx2c_17
2017-CAL-0227	CFIACB20070063	PRJNA454819	SAMN13333377	SRR22858849	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2017-CAL-0229	CFIACB20070064	PRJNA454819	SAMN13333378	SRR22858850	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2017-CAL-0230	CFIACB20080073	PRJNA454819	SAMN13333403	SRR22858855	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2017-CAL-0231	CFIACB20080074	PRJNA454819	SAMN13333404	SRR22858856	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2017-CAL-0262	CFIACB20170169	PRJNA454819	SAMN09468187	SRR7435861	O121:H19	Flour	Flour	Stx2a	StxOp419_Stx2a_1
2017-CAL-0287	CFIACB20170348	PRJNA454819	SAMN09468188	SRR7435875	O8:H28	Flour	Flour	Stx2a	StxOp432_Stx2a_30
2017-CAL-0335	CFIACB20170179	PRJNA454819	SAMN09468189	SRR7435870	O88:H25	Flour	Flour	Stx1a; Stx2a	StxOp1157_Stx2a_6; StxOp4_Stx1a_9
2017-CAL-0427	CFIACB20180572	PRJNA454819	SAMN13334882	SRR22858717	O76:H19	Meat	Meat	Stx1c	StxOp4754_Stx1c_708
2017-DAR-0087	CFIACB20170345	PRJNA454819	SAMN13334752	SRR22858704	O91:H14	Meat	Meat	Stx1a; Stx2b	StxOp675_Stx2b_116; StxOp13_Stx1a_19
2017-DAR-0088	CFIACB20170347	PRJNA454819	SAMN13334753	SRR22858705	O76:H19	Meat	Meat	Stx1c	StxOp4754_Stx1c_708
2017-GTA-0025	CFIACB20170178	PRJNA454819	SAMN09468195	SRR7435876	O8:H28	Flour	Flour	Stx2a	StxOp432_Stx2a_30
2017-GTA-0042	CFIACB20170343	PRJNA454819	SAMN09468196	SRR7435864	O187:H52	Flour	Flour	Stx1c	StxOp1036_Stx1c_12
2017-GTA-0058	CFIACB20170170	PRJNA454819	SAMN10989255	SRR22715070	O157:H7	Meat	Meat	Stx1a; Stx2c	StxOp702_Stx2c_17; StxOp1577_Stx1a_9
2017-GTA-0059	CFIACB20170171	PRJNA454819	SAMN10989256	SRR22715069	O157:H7	Meat	Meat	Stx1a; Stx2c	StxOp1577_Stx1a_9; StxOp702_Stx2c_17
2017-GTA-0109	CFIACB20170180	PRJNA454819	SAMN13334736	SRR22858692	O157:H7	Meat	Meat	Stx1a; Stx2c	StxOp702_Stx2c_17; StxOp3_Stx1a_9
2017-GTA-0110	CFIACB20170181	PRJNA454819	SAMN11030293	SRR22715055	O157:H7	Meat	Meat	Stx1a; Stx2c	StxOp702_Stx2c_17; StxOp3_Stx1a_9
2017-GTA-0164	CFIACB20100224	PRJNA454819	SAMN13333854	SRR22858876	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2017-GTA-0165	CFIACB20100225	PRJNA454819	SAMN13333855	SRR22858877	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2017-GTA-0166	CFIACB20100226	PRJNA454819	SAMN13333856	SRR22858878	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2017-GTA-0167	CFIACB20110144	PRJNA454819	SAMN13333987	SRR22858893	O157:H7	Nuts/Seeds	Nuts/Seeds	Stx2a	StxOp1043_Stx2a_1
2017-GTA-0244	CFIACB20180574	PRJNA454819	SAMN13334883	SRR22858718	O78:H16	Meat	Meat	Stx1a	StxOp1580_Stx1a_9
2017-GTA-0247	CFIACB20180575	PRJNA454819	SAMN13334884	SRR22858719	O174:H21	Meat	Meat	Stx2c	StxOp1365_Stx2c_7
2017-GTA-0248	CFIACB20180576	PRJNA454819	SAMN13334885	SRR22858720	O174:H21	Meat	Meat	Stx2c	StxOp1365_Stx2c_7
2017-GTA-0249	CFIACB20180577	PRJNA454819	SAMN13334886	SRR22858721	O174:H21	Meat	Meat	Stx2c	StxOp1365_Stx2c_7
2017-GTA-0253	CFIACB20180223	PRJNA454819	SAMN10989146	SRR22715125	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2017-HCLON-0011	CFIACB20160171	PRJNA454819	SAMN13334712	SRR22858824	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2017-HCLON-0013	CFIACB20160139	PRJNA454819	SAMN11030468	SRR22715117	O185:H7	Meat	Meat	Stx2c	StxOp702_Stx2c_17

Genome ID	CFIA isolate	BioProject	BioSample	SRA run accession	Serotype	Isolation source	Isolation source detail	Stx subtype(s)	STxOP_nt operon(s)
2017-MER-0036	CFIACB20160173	PRJNA454819	SAMN19011643	SRR14750051	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2017-MER-0037	CFIACB20160177	PRJNA454819	SAMN13334718	SRR22858829	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2017-MER-0049	CFIACB20160142	PRJNA454819	SAMN11373447	SRR22715051	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2017-MER-0052	CFIACB20170176	PRJNA454819	SAMN10989258	SRR22715067	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2017-MER-0170	CFIACB20170287	PRJNA454819	SAMN10989393	SRR22715060	O113:H21	Meat	Meat	Stx2d	StxOp827_Stx2d_10
2017-MER-0171	CFIACB20170269	PRJNA454819	SAMN09468190	SRR7435872	O8:H28	Flour	Flour	Stx2a	StxOp432_Stx2a_30
2017-MER-0172	CFIACB20160250	PRJNA454819	SAMN10746423	SRR8448487	O163:H19	Meat	Meat	Stx1a	StxOp4_Stx1a_9
2017-MER-0173	CFIACB20150352	PRJNA454819	SAMN10746419	SRR8448491	O91:H14	Meat	Meat	Stx1a	StxOp4_Stx1a_9
2017-MER-0187	CFIACB20130346	PRJNA454819	SAMN13334356	SRR22858764	O157:H7	Meat	Meat	Stx2c	StxOp702_Stx2c_17
2017-MER-0214	CFIACB20170162	PRJNA454819	SAMN10989252	SRR22715088	O157:H7	Meat	Meat	Stx1a; Stx2c	StxOp702_Stx2c_17; StxOp1577_Stx1a_9
2017-MER-0215	CFIACB20170163	PRJNA454819	SAMN10989253	SRR22715087	O157:H7	Meat	Meat	Stx1a; Stx2c	StxOp702_Stx2c_17; StxOp1577_Stx1a_9
2017-MER-0216	CFIACB20170164	PRJNA454819	SAMN13334734	SRR22858728	O157:H7	Meat	Meat	Stx1a; Stx2c	StxOp702_Stx2c_17; StxOp1577_Stx1a_9
2017-MER-0217	CFIACB20170165	PRJNA454819	SAMN13334735	SRR22858729	O157:H7	Meat	Meat	Stx1a; Stx2c	StxOp1577_Stx1a_9; StxOp702_Stx2c_17
2017-MER-0250	CFIACB20170154	PRJNA454819	SAMN10989242	SRR22715089	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2017-MER-0306	CFIACB20100189	PRJNA454819	SAMN13333833	SRR22858868	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2017-MER-0335	CFIACB20160192	PRJNA454819	SAMN13334727	SRR22858819	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2017-MER-0336	CFIACB20160193	PRJNA454819	SAMN13334728	SRR22858830	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2017-MER-0377	CFIACB20110141	PRJNA454819	SAMN13333984	SRR22858891	O157:H7	Meat	Meat	Stx2a	StxOp1043_Stx2a_1
2017-MER-0388	CFIACB20170307	PRJNA454819	SAMN11373512	SRR22715043	O128:H2	Meat	Meat	Stx1c; Stx2b	StxOp1348_Stx2b_110; StxOp399_Stx1c_12
2017-MER-0403	CFIACB20160155	PRJNA454819	SAMN13334710	SRR22858822	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2017-MER-0414	CFIACB20150153	PRJNA454819	SAMN10746418	SRR8448490	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2017-MER-0420	CFIACB20170330	PRJNA454819	SAMN11030306	SRR22715079	O91:H14	Meat	Meat	Stx1a	StxOp4_Stx1a_9
2017-MER-0516	CFIACB20130253	PRJNA454819	SAMN09668423	SRR7629316	O157:H7	Dairy	Dairy	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2017-MER-0517	CFIACB20130257	PRJNA454819	SAMN09668425	SRR7629290	O157:H7	Dairy	Dairy	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2017-MER-0518	CFIACB20130258	PRJNA454819	SAMN09668426	SRR7629328	O157:H7	Dairy	Dairy	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2017-MER-0519	CFIACB20130260	PRJNA454819	SAMN09668428	SRR7629288	O157:H7	Dairy	Dairy	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2017-MER-0520	CFIACB20130261	PRJNA454819	SAMN09668429	SRR7629292	O157:H7	Dairy	Dairy	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2017-MER-0539	CFIACB20100214	PRJNA454819	SAMN13333851	SRR22858873	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2017-MER-0542	CFIACB20110122	PRJNA454819	SAMN13333978	SRR22858889	O157:H7	Meat	Meat	Stx2c	StxOp702_Stx2c_17
2017-MER-0543	CFIACB20120213	PRJNA454819	SAMN13334077	SRR22858900	O157:H7	Vegetable	Vegetable	Stx1a; Stx2a	StxOp1577_Stx1a_9; StxOp419_Stx2a_1
2017-NML-0030	CFIACB20140239	PRJNA454819	SAMN13334592	SRR22858772	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2017-NML-0031	CFIACB20140240	PRJNA454819	SAMN13334593	SRR22858773	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2017-NML-0032	CFIACB20140241	PRJNA454819	SAMN13334594	SRR22858774	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2017-NML-0033	CFIACB20140242	PRJNA454819	SAMN13334595	SRR22858775	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2017-NML-0035	CFIACB20140244	PRJNA454819	SAMN13334597	SRR22858777	O157:H7	Meat	Meat	Stx2c	StxOp702_Stx2c_17
2017-NML-0037	CFIACB20140246	PRJNA454819	SAMN13334599	SRR22858779	O157:H7	Meat	Meat	Stx2c	StxOp702_Stx2c_17
2017-NML-0039	CFIACB20140248	PRJNA454819	SAMN13334601	SRR22858781	O157:H7	Meat	Meat	Stx2c	StxOp702_Stx2c_17
2017-NML-0043	CFIACB20140252	PRJNA454819	SAMN13334604	SRR22858785	O157:H7	Meat	Meat	Stx2c	StxOp702_Stx2c_17
2017-NML-0045	CFIACB20140254	PRJNA454819	SAMN13334606	SRR22858787	O157:H7	Environment	Environment	Stx2c	StxOp702_Stx2c_17
2017-NML-0046	CFIACB20140255	PRJNA454819	SAMN13334607	SRR22858788	O157:H7	Environment	Environment	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2017-NML-0047	CFIACB20140256	PRJNA454819	SAMN13334608	SRR22858789	O157:H7	Environment	Environment	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2017-NML-0050	CFIACB20140259	PRJNA454819	SAMN13334611	SRR22858792	O157:H7	Environment	Environment	Stx2c	StxOp4303_Stx2c_1247
2017-NML-0052	CFIACB20140261	PRJNA454819	SAMN13334613	SRR22858795	O157:H7	Environment	Environment	Stx2c	StxOp4081_Stx2c_106
2017-PAC-0024	CFIACB20140258	PRJNA454819	SAMN13334610	SRR22858791	O157:H7	Environment	Environment	Stx2c	StxOp4081_Stx2c_106
2017-SEQ-0032	CFIACB20170153	PRJNA454819	SAMN09468194	SRR7435863	O121:H19	Flour	Flour	Stx2a	StxOp419_Stx2a_1
2017-SEQ-0041	CFIACB20170155	PRJNA454819	SAMN09468186	SRR7435862	O121:H19	Flour	Flour	Stx2a	StxOp419_Stx2a_1
2017-SEQ-0050	CFIACB20170158	PRJNA454819	SAMN09468182	SRR7435866	O8:H28	Flour	Flour	Stx2a	StxOp432_Stx2a_30

Genome ID	CFIA isolate	BioProject	BioSample	SRA run accession	Serotype	Isolation source	Isolation source detail	Stx subtype(s)	STxOP_nt operon(s)
2017-SEQ-0059	CFIACB20170346	PRJNA454819	SAMN09468193	SRR7435867	O8:H28	Flour	Flour	Stx2a	StxOp432_Stx2a_30
2017-SEQ-0069	CFIACB20170175	PRJNA454819	SAMN09468197	SRR7435874	O8:H28	Flour	Flour	Stx2a	StxOp432_Stx2a_30
2017-SEQ-0083	CFIACB20170266	PRJNA454819	SAMN10989391	SRR22715062	O55:H12	Meat	Meat	Stx1a	StxOp1580_Stx1a_9
2017-SEQ-0084	CFIACB20170267	PRJNA454819	SAMN13334743	SRR22858698	O116:H16	Meat	Meat	Stx2g	StxOp927_Stx2g_147
2017-SEQ-0168	CFIACB20170161	PRJNA454819	SAMN09468192	SRR7435868	O121:H19	Flour	Flour	Stx2a	StxOp419_Stx2a_1
2017-SEQ-0209	CFIACB20170336	PRJNA454819	SAMN10989394	SRR22715059	OUT:H29	Meat	Meat	Stx2d	StxOp1486_Stx2d_112
2017-SEQ-0210	CFIACB20170337	PRJNA454819	SAMN11373508	SRR22715048	OUT:H29	Meat	Meat	Stx2d	StxOp1486_Stx2d_112
2017-SEQ-0212	CFIACB20170338	PRJNA454819	SAMN13334749	SRR18215055	O118:H2	Meat	Meat	Stx1a	StxOp1577_Stx1a_9
2017-SEQ-0277	CFIACB20170270	PRJNA454819	SAMN11030313	SRR18215054	O55:H12	Meat	Meat	Stx1a	StxOp1580_Stx1a_9
2017-SEQ-0383	CFIACB20170276	PRJNA454819	SAMN13334744	SRR22858700	O55:H12	Meat	Meat	Stx1a	StxOp1580_Stx1a_9
2017-SEQ-0384	CFIACB20170274	PRJNA454819	SAMN11373406	SRR22715053	O55:H12	Meat	Meat	Stx1a	StxOp1580_Stx1a_9
2017-SEQ-0385	CFIACB20170275	PRJNA454819	SAMN11373404	SRR22715103	O55:H12	Meat	Meat	Stx1a	StxOp1580_Stx1a_9
2017-SEQ-0386	CFIACB20170277	PRJNA454819	SAMN11373402	SRR22715106	O8:H19	Nuts/Seeds	Nuts/Seeds	Stx2a	StxOp432_Stx2a_30
2017-SEQ-0388	CFIACB20170279	PRJNA454819	SAMN11373511	SRR22715044	OUT:H20	Meat	Meat	Stx1a; Stx2a	StxOp1173_Stx2a_16; StxOp969_Stx1a_19
2017-SEQ-0389	CFIACB20170280	PRJNA454819	SAMN11373403	SRR22715105	O78:H16	Meat	Meat	Stx1a	StxOp1580_Stx1a_9
2017-SEQ-0456	CFIACB20170192	PRJNA454819	SAMN13334740	SRR22858696	O121:H19	Flour	Flour	Stx2a	StxOp419_Stx2a_1
2017-SEQ-0458	CFIACB20170286	PRJNA454819	SAMN11373395	SRR22715113	O91:H21	Meat	Meat	Stx1a; Stx2a	StxOp16_Stx1a_24; StxOp1096_Stx2a_6
2017-SEQ-0460	CFIACB20170288	PRJNA454819	SAMN11373401	SRR22715107	O91:H21	Meat	Meat	Stx1a; Stx2a	StxOp1096_Stx2a_6; StxOp16_Stx1a_24
2017-SEQ-0521	CFIACB20170195	PRJNA454819	SAMN11030294	SRR22715086	O157:H7	Meat	Meat	None detected	None detected
2017-SEQ-0522	CFIACB20170196	PRJNA454819	SAMN11030295	SRR22715085	O157:H7	Meat	Meat	None detected	None detected
2017-SEQ-0617	CFIACB20170293	PRJNA454819	SAMN11030307	SRR22715077	O171:H2	Meat	Meat	Stx2d	StxOp841_Stx2d_100
2017-SEQ-0618	CFIACB20170294	PRJNA454819	SAMN10989392	SRR22715061	O113:H4	Meat	Meat	Stx1a; Stx2d	StxOp840_Stx2d_99; StxOp4_Stx1a_9
2017-SEQ-0657	CFIACB20170296	PRJNA454819	SAMN13334745	SRR22858701	O113:H21	Meat	Meat	Stx2a	StxOp439_Stx2a_107
2017-SEQ-0694	CFIACB20170299	PRJNA454819	SAMN10989388	SRR22715065	OUT:H11	Meat	Meat	Stx2a	StxOp1173_Stx2a_16
2017-SEQ-0695	CFIACB20170300	PRJNA454819	SAMN10989387	SRR22715066	OUT:H11	Meat	Meat	Stx2a	StxOp1173_Stx2a_16
2017-SEQ-0730	CFIACB20170306	PRJNA454819	SAMN13334747	SRR22858702	O5:H19	Meat	Meat	Stx1c; Stx2b	StxOp653_Stx2b_23; StxOp399_Stx1c_12
2017-SEQ-0908	CFIACB20170314	PRJNA454819	SAMN11030300	SRR22715084	O113:H21	Meat	Meat	Stx2d	StxOp827_Stx2d_10
2017-SEQ-0911	CFIACB20170317	PRJNA454819	SAMN11030304	SRR22715080	O55:H12	Meat	Meat	Stx1a	StxOp1580_Stx1a_9
2017-SEQ-1066	CFIACB20170310	PRJNA454819	SAMN11030312	SRR22715072	O109:H5	Meat	Meat	Stx1a	StxOp1594_Stx1a_9
2017-SEQ-1600	CFIACB20180432	PRJNA454819	SAMN11030311	SRR22715073	O185:H7	Meat	Meat	Stx2c	StxOp716_Stx2c_106
2017-SEQ-1601	CFIACB20180433	PRJNA454819	SAMN11030309	SRR22715075	O185:H7	Meat	Meat	Stx2c	StxOp716_Stx2c_106
2017-SEQ-1617	CFIACB20170303	PRJNA454819	SAMN11030314	SRR22715071	O76:H19	Meat	Meat	Stx1c	StxOp4754_Stx1c_708
2017-SEQ-1652	CFIACB20180434	PRJNA454819	SAMN11373407	SRR22715052	O91:H14	Meat	Meat	Stx1a; Stx2b	StxOp675_Stx2b_116; StxOp13_Stx1a_19
2017-SEQ-1688	CFIACB20180578	PRJNA454819	SAMN13334887	SRR22858722	O17/O77:H18	Dairy	Dairy	Stx2d	StxOp829_Stx2d_27
2017-SEQ-1689	CFIACB20180579	PRJNA454819	SAMN13334888	SRR22858723	O17/O77:H18	Dairy	Dairy	Stx2d	StxOp829_Stx2d_27
2017-SEQ-1690	CFIACB20180580	PRJNA454819	SAMN13334889	SRR22858746	O17/O77:H18	Dairy	Dairy	Stx2d	StxOp829_Stx2d_27
2017-SEQ-1696	CFIACB20180221	PRJNA454819	SAMN11373322	SRR22715116	O91:H14	Meat	Meat	Stx1a; Stx2b	StxOp675_Stx2b_116; StxOp13_Stx1a_19
2017-SEQ-1697	CFIACB20180222	PRJNA454819	SAMN11373323	SRR22715114	O91:H14	Meat	Meat	Stx1a; Stx2b	StxOp675_Stx2b_116; StxOp13_Stx1a_19
2017-SEQ-1842	CFIACB20180236	PRJNA454819	SAMN10989158	SRR22715124	O116:H16	Meat	Meat	Stx2a	StxOp419_Stx2a_1
2017-SEQ-1843	CFIACB20180237	PRJNA454819	SAMN10989159	SRR22715123	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2017-SEQ-1844	CFIACB20180238	PRJNA454819	SAMN10989160	SRR22715121	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2018-BUR-0050	CFIACB20190220	PRJNA454819	SAMN40646265	SRR28490841	O76:H19	Meat	Meat	Stx1c	StxOp4754_Stx1c_708
2018-BUR-0121	CFIACB20190221	PRJNA454819	SAMN40646227	SRR28490883	O128:H2	Meat	Meat	Stx1c; Stx2b	StxOp1348_Stx2b_110; StxOp4754_Stx1c_708
2018-BUR-0122	CFIACB20190222	PRJNA454819	SAMN40646290	SRR28490861	O113:H4	Meat	Meat	Stx1a; Stx2d	StxOp840_Stx2d_99; StxOp4_Stx1a_9
2018-GTA-0012	CFIACB20180272	PRJNA454819	SAMN10989055	SRR22715068	O103:H25	Meat	Meat	Stx1a	StxOp1577_Stx1a_9
2018-GTA-0049	CFIACB20180584	PRJNA454819	SAMN13334893	SRR22858751	O146:H8	Meat	Meat	Stx1c	StxOp1036_Stx1c_12
2018-GTA-0051	CFIACB20180585	PRJNA454819	SAMN13334894	SRR22858752	O146:H8	Meat	Meat	Stx1c	StxOp1036_Stx1c_12

Genome ID	CFIA isolate	BioProject	BioSample	SRA run accession	Serotype	Isolation source	Isolation source detail	Stx subtype(s)	STxOP_nt operon(s)
2018-GTA-0101	CFIACB20180588	PRJNA454819	SAMN13334896	SRR22858754	O128:H2	Meat	Meat	Stx1c; Stx2b	StxOp1348_Stx2b_110; StxOp399_Stx1c_12
2018-GTA-0107	CFIACB20180619	PRJNA454819	SAMN15300860	SRR12041535	O8:H19	Flour	Flour	Stx2a	StxOp432_Stx2a_30
2018-GTA-0110	CFIACB20180583	PRJNA454819	SAMN13334892	SRR22858750	O146:H8	Meat	Meat	Stx1c	StxOp1036_Stx1c_12
2018-GTA-0113	CFIACB20180587	PRJNA454819	SAMN13334895	SRR22858753	O123/O186:H10	Meat	Meat	Stx1c; Stx2b	StxOp1339_Stx2b_109; StxOp399_Stx1c_12
2018-GTA-0114	CFIACB20180589	PRJNA454819	SAMN13334897	SRR22858755	O75:H8	Meat	Meat	Stx1c; Stx2b	StxOp1336_Stx2b_109; StxOp399_Stx1c_12
2018-GTA-0121	CFIACB20180620	PRJNA454819	SAMN15300861	SRR12041534	O159:H19	Flour	Flour	Stx2a	StxOp432_Stx2a_30
2018-GTA-0123	CFIACB20180593	PRJNA454819	SAMN13334901	SRR22858756	O178:H7	Meat	Meat	Stx1c	StxOp4754_Stx1c_708
2018-GTA-0124	CFIACB20180594	PRJNA454819	SAMN13334902	SRR22858757	O178:H7	Meat	Meat	Stx1c	StxOp399_Stx1c_12
2018-GTA-0125	CFIACB20180595	PRJNA454819	SAMN13334903	SRR22858758	O178:H7	Meat	Meat	Stx1c	StxOp399_Stx1c_12
2018-GTA-0126	CFIACB20180596	PRJNA454819	SAMN13334904	SRR22858759	O178:H7	Meat	Meat	Stx1c	StxOp399_Stx1c_12
2018-GTA-0150	CFIACB20180622	PRJNA454819	SAMN15300862	SRR12041533	OUT:H2	Flour	Flour	Stx2c	StxOp1450_Stx2c_152
2018-GTA-0200	CFIACB20180623	PRJNA454819	SAMN15300863	SRR12041532	OUT:H19	Flour	Flour	Stx2a	StxOp432_Stx2a_30
2018-MER-0007	CFIACB20140253	PRJNA454819	SAMN13334605	SRR22858786	O157:H7	Environment	Environment	Stx2c	StxOp702_Stx2c_17
2018-MER-0070	CFIACB20180251	PRJNA454819	SAMN10989176	SRR22715099	O103:H2	Meat	Meat	Stx1a	StxOp1577_Stx1a_9
2018-MER-0083	CFIACB20180384	PRJNA454819	SAMN10988985	SRR22715134	O103:H25	Flour	Flour	Stx1a	StxOp4_Stx1a_9
2018-MER-0084	CFIACB20180385	PRJNA454819	SAMN13334836	SRR22858735	O103:H25	Flour	Flour	Stx1a	StxOp4_Stx1a_9
2018-MER-0085	CFIACB20180386	PRJNA454819	SAMN10988986	SRR22715133	O103:H25	Flour	Flour	Stx1a	StxOp4_Stx1a_9
2018-MER-0108	CFIACB20180560	PRJNA454819	SAMN13334877	SRR22858713	OUT:H29	Meat	Meat	Stx2c	StxOp1448_Stx2c_152
2018-SEQ-0096	CFIACB20180275	PRJNA454819	SAMN10989064	SRR22715057	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2018-SEQ-0097	CFIACB20180276	PRJNA454819	SAMN10989065	SRR22715078	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2018-SEQ-0225	CFIACB20180277	PRJNA454819	SAMN10989066	SRR22715115	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2018-SEQ-0226	CFIACB20180278	PRJNA454819	SAMN10989067	SRR22715104	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2018-SEQ-0227	CFIACB20180279	PRJNA454819	SAMN10989068	SRR22715045	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2018-SEQ-0228	CFIACB20180280	PRJNA454819	SAMN10989069	SRR22715132	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2018-SEQ-0229	CFIACB20180281	PRJNA454819	SAMN10989070	SRR22715131	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2018-SEQ-0230	CFIACB20180282	PRJNA454819	SAMN10989071	SRR22715130	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2018-SEQ-0231	CFIACB20180283	PRJNA454819	SAMN10989072	SRR22715129	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2018-SEQ-0232	CFIACB20180284	PRJNA454819	SAMN10989073	SRR22715128	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2018-SEQ-0233	CFIACB20180285	PRJNA454819	SAMN11030263	SRR22715058	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2018-SEQ-0240	CFIACB20180582	PRJNA454819	SAMN13334891	SRR22858748	O108:H21	Meat	Meat	Stx1c	StxOp1036_Stx1c_12
2018-SEQ-0241	CFIACB20180606	PRJNA454819	SAMN13334914	SRR22858761	O108:H21	Meat	Meat	Stx1c	StxOp1036_Stx1c_12
2018-SEQ-0293	CFIACB20180485	PRJNA454819	SAMN13334854	SRR22858738	O128:H2	Meat	Lamb	Stx1c; Stx2b	StxOp1348_Stx2b_110; StxOp4754_Stx1c_708
2018-SEQ-0340	CFIACB20180484	PRJNA454819	SAMN11373510	SRR22715046	OUT:H10	Meat	Meat	Stx2d	StxOp826_Stx2d_9
2018-SEQ-0371	CFIACB20180300	PRJNA454819	SAMN10989083	SRR22715127	O26:H32	Meat	Meat	None detected	None detected
2018-SEQ-0375	CFIACB20180304	PRJNA454819	SAMN13334807	SRR22858707	O157:H7	Meat	Meat	Stx1a; Stx2c	StxOp1687_Stx1a_9; StxOp702_Stx2c_17
2018-SEQ-0399	CFIACB20180305	PRJNA454819	SAMN13334808	SRR22858730	O91:H14	Meat	Meat	Stx1a; Stx2b	StxOp675_Stx2b_116; StxOp13_Stx1a_19
2018-SEQ-0400	CFIACB20180306	PRJNA454819	SAMN13334809	SRR22858731	O128:H2	Meat	Meat	Stx1c; Stx2b	StxOp1348_Stx2b_110; StxOp4754_Stx1c_708
2018-SEQ-0589	CFIACB20180504	PRJNA454819	SAMN11030303	SRR22715081	O104:H7	Meat	Lamb	Stx1c	StxOp1036_Stx1c_12
2018-SEQ-0617	CFIACB20180501	PRJNA454819	SAMN13334858	SRR22858739	O91:H14	Meat	Lamb	Stx1a; Stx2b	StxOp1329_Stx2b_101; StxOp13_Stx1a_19
2018-SEQ-0618	CFIACB20180502	PRJNA454819	SAMN13334859	SRR22858740	O91:H14	Meat	Lamb	Stx1a; Stx2b	StxOp1329_Stx2b_101; StxOp13_Stx1a_19
2018-SEQ-0619	CFIACB20180503	PRJNA454819	SAMN13334860	SRR22858741	O104:H7	Meat	Lamb	Stx1c	StxOp1036_Stx1c_12
2018-SEQ-0642	CFIACB20180323	PRJNA454819	SAMN10989116	SRR22715126	O91:H14	Meat	Lamb	Stx1a; Stx2b	StxOp1346_Stx2b_110; StxOp13_Stx1a_19
2018-SEQ-0643	CFIACB20180324	PRJNA454819	SAMN13334817	SRR22858733	O91:H14	Meat	Lamb	Stx1a; Stx2b	StxOp675_Stx2b_116; StxOp13_Stx1a_19
2018-SEQ-0644	CFIACB20180325	PRJNA454819	SAMN13334818	SRR22858734	O91:H14	Meat	Lamb	Stx1a; Stx2b	StxOp675_Stx2b_116; StxOp13_Stx1a_19
2018-SEQ-0771	CFIACB20180322	PRJNA454819	SAMN11373500	SRR22715049	O75:H8	Meat	Lamb	Stx1c; Stx2b	StxOp1339_Stx2b_109; StxOp4754_Stx1c_708
2018-SEQ-0777	CFIACB20180346	PRJNA454819	SAMN10989006	SRR22715122	O104:H7	Meat	Lamb	Stx1c	StxOp1036_Stx1c_12
2018-SEQ-0778	CFIACB20180347	PRJNA454819	SAMN10989007	SRR22715095	O104:H7	Meat	Lamb	Stx1c	StxOp1036_Stx1c_12

Genome ID	CFIA isolate	BioProject	BioSample	SRA run accession	Serotype	Isolation source	Isolation source detail	Stx subtype(s)	STxOP_nt operon(s)
2018-SEQ-0897	CFIACB20180514	PRJNA454819	SAMN13334862	SRR22858742	O174:H21	Meat	Meat	Stx2c	StxOp1365_Stx2c_7
2018-SEQ-0898	CFIACB20180515	PRJNA454819	SAMN13334863	SRR22858744	O174:H21	Meat	Meat	Stx2c	StxOp1365_Stx2c_7
2018-SEQ-1017	CFIACB20180532	PRJNA454819	SAMN13334865	SRR22858745	OUT:H16	Meat	Lamb	Stx1c	StxOp1036_Stx1c_12
2018-SEQ-1060	CFIACB20180513	PRJNA454819	SAMN11030310	SRR22715074	O174:H21	Meat	Meat	Stx2c	StxOp1365_Stx2c_7
2018-SEQ-1061	CFIACB20180538	PRJNA454819	SAMN13334867	SRR22858708	O91:H14	Meat	Lamb	Stx1c; Stx2b	StxOp1346_Stx2b_110; StxOp399_Stx1c_12
2018-SEQ-1062	CFIACB20180539	PRJNA454819	SAMN13334868	SRR22858709	O128:H2	Meat	Lamb	Stx1c; Stx2b	StxOp1348_Stx2b_110; StxOp399_Stx1c_12
2018-SEQ-1090	CFIACB20180529	PRJNA454819	SAMN40646305	SRR28490886	O174:H21	Meat	Meat	Stx2c	StxOp1365_Stx2c_7
2018-SEQ-1270	CFIACB20180540	PRJNA454819	SAMN13334869	SRR22858710	O8:H9	Meat	Lamb	Stx2e	StxOp1571_Stx2e_146
2018-SEQ-1323	CFIACB20180561	PRJNA454819	SAMN13334878	SRR22858714	O91:H14	Meat	Lamb	Stx1a; Stx2b	StxOp678_Stx2b_122; StxOp13_Stx1a_19
2018-SEQ-1324	CFIACB20180562	PRJNA454819	SAMN13334879	SRR22858715	O91:H14	Meat	Lamb	Stx1a; Stx2b	StxOp678_Stx2b_122; StxOp13_Stx1a_19
2018-SEQ-1328	CFIACB20180411	PRJNA454819	SAMN13334845	SRR22858737	O157:H7	Dairy	Dairy	Stx2a	StxOp1089_Stx2a_1
2018-SEQ-1382	CFIACB20180550	PRJNA454819	SAMN13334871	SRR22858711	O128:H2	Meat	Lamb	Stx1c; Stx2b	StxOp1348_Stx2b_110; StxOp400_Stx1c_22
2018-SEQ-1383	CFIACB20180551	PRJNA454819	SAMN13334872	SRR18215050	O176:H4	Meat	Lamb	Stx1c; Stx2b	StxOp1339_Stx2b_109; StxOp1037_Stx1c_26
2018-SEQ-1392	CFIACB20180410	PRJNA454819	SAMN13334844	SRR22858736	O157:H7	Dairy	Dairy	Stx2a	StxOp1089_Stx2a_1
2018-SEQ-1598	CFIACB20190167	PRJNA454819	SAMN40646276	SRR28490829	O128:H2	Meat	Lamb	Stx1c	StxOp2108_Stx1c_27
2018-SEQ-1637	CFIACB20190171	PRJNA454819	SAMN40646260	SRR28490846	O3:H21	Meat	Lamb	Stx1c; Stx2b	StxOp1348_Stx2b_110; StxOp2167_Stx1c_500
2018-STH-0001	CFIACB20180597	PRJNA454819	SAMN13334905	SRR22858724	O91:H14	Meat	Meat	Stx1a; Stx2b	StxOp675_Stx2b_116; StxOp13_Stx1a_19
2018-STH-0003	CFIACB20180598	PRJNA454819	SAMN13334906	SRR22858725	O91:H14	Meat	Meat	Stx1a; Stx2b	StxOp675_Stx2b_116; StxOp13_Stx1a_19
2018-STH-0004	CFIACB20180599	PRJNA454819	SAMN13334907	SRR22858726	O91:H14	Meat	Meat	Stx1a; Stx2b	StxOp675_Stx2b_116; StxOp13_Stx1a_19
2018-STH-0049	CFIACB20180244	PRJNA454819	SAMN10989170	SRR22715119	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2018-STH-0050	CFIACB20180246	PRJNA454819	SAMN10989172	SRR22715101	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2018-STH-0051	CFIACB20180245	PRJNA454819	SAMN10989171	SRR22715102	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2018-STH-0052	CFIACB20180247	PRJNA454819	SAMN10989173	SRR22715100	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2018-STH-0053	CFIACB20180248	PRJNA454819	SAMN10989169	SRR22715120	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2018-STH-0054	CFIACB20180249	PRJNA454819	SAMN11030272	SRR22715056	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2018-STH-0056	CFIACB20180261	PRJNA454819	SAMN10989186	SRR22715092	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2018-STH-0057	CFIACB20180262	PRJNA454819	SAMN10989187	SRR22715091	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2018-STH-0058	CFIACB20180263	PRJNA454819	SAMN10989188	SRR22715090	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2018-STH-0060	CFIACB20180252	PRJNA454819	SAMN10989177	SRR22715098	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2018-STH-0061	CFIACB20180253	PRJNA454819	SAMN10989178	SRR22715097	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2018-STH-0062	CFIACB20180254	PRJNA454819	SAMN10989179	SRR22715096	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2018-STH-0063	CFIACB20180255	PRJNA454819	SAMN10989180	SRR22715094	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2018-STH-0064	CFIACB20180257	PRJNA454819	SAMN10989181	SRR22715093	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2018-STH-0065	CFIACB20180600	PRJNA454819	SAMN13334908	SRR22858699	O128:H2	Meat	Meat	Stx1c; Stx2b	StxOp1348_Stx2b_110; StxOp1037_Stx1c_26
2018-STH-0066	CFIACB20180601	PRJNA454819	SAMN13334909	SRR22858732	O128:H2	Meat	Meat	Stx1c; Stx2b	StxOp1348_Stx2b_110; StxOp1037_Stx1c_26
2018-STH-0067	CFIACB20180602	PRJNA454819	SAMN13334910	SRR22858743	O128:H2	Meat	Meat	Stx1c; Stx2b	StxOp675_Stx2b_116; StxOp399_Stx1c_12
2018-STH-0258	CFIACB20190230	PRJNA454819	SAMN40646251	SRR28490897	O91:H14	Meat	Meat	Stx1a; Stx2b	StxOp675_Stx2b_116; StxOp1593_Stx1a_9
2018-STH-0259	CFIACB20190231	PRJNA454819	SAMN40646304	SRR28490887	O183:H18	Meat	Meat	Stx1a; Stx2a	StxOp1168_Stx2a_16; StxOp1645_Stx1a_9
2018-STH-0260	CFIACB20190232	PRJNA454819	SAMN40646237	SRR28490872	O128:H2	Meat	Meat	Stx1c; Stx2a; Stx2b	StxOp1173_Stx2a_16; StxOp1348_Stx2b_110; StxOp400_Stx1c_22
2019-BUR-0117	CFIACB20190265	PRJNA454819	SAMN40646228	SRR28490881	O128:H2	Meat	Meat	Stx1c; Stx2b	StxOp671_Stx2b_110; StxOp4754_Stx1c_708
2019-BUR-0118	CFIACB20190266	PRJNA454819	SAMN40646271	SRR28490834	O159:H19	Meat	Meat	Stx2a; Stx2k	StxOp943_Stx2k_40; StxOp432_Stx2a_30
2019-CAL-0097	CFIACB20010012	PRJNA454819	SAMN13333304	SRR22858833	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2019-CAL-0098	CFIACB20010013	PRJNA454819	SAMN13333305	SRR22858834	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2019-CAL-0099	CFIACB20010014	PRJNA454819	SAMN13333306	SRR22858835	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2019-DAR-0015	CFIACB20180603	PRJNA454819	SAMN13334911	SRR22858716	OUT:H21	Meat	Lamb	Stx2i	StxOp933_Stx2i_43
2019-DAR-0017	CFIACB20180604	PRJNA454819	SAMN13334912	SRR22858749	OUT:H21	Meat	Lamb	Stx2i	StxOp933_Stx2i_43
2019-DAR-0018	CFIACB20180605	PRJNA454819	SAMN13334913	SRR22858760	OUT:H21	Meat	Lamb	Stx2i	StxOp933_Stx2i_43

Genome ID	CFIA isolate	BioProject	BioSample	SRA run accession	Serotype	Isolation source	Isolation source detail	Stx subtype(s)	STxOP_nt operon(s)
2019-DAR-0020	CFIACB20190223	PRJNA454819	SAMN40646292	SRR28490859	O91:H14	Meat	Meat	Stx1a; Stx2b	StxOp675_Stx2b_116; StxOp13_Stx1a_19
2019-DAR-0250	CFIACB20180611	PRJNA454819	SAMN40646226	SRR28490853	O75:H8	Meat	Meat	Stx1a; Stx2b	StxOp1336_Stx2b_109; StxOp13_Stx1a_19
2019-DAR-0251	CFIACB20180612	PRJNA454819	SAMN40646280	SRR28490824	O146:H8	Meat	Meat	Stx1c	StxOp1036_Stx1c_12
2019-DAR-0257	CFIACB20180618	PRJNA454819	SAMN40646242	SRR28490866	O91:H14	Meat	Meat	Stx1a; Stx2b	StxOp675_Stx2b_116; StxOp13_Stx1a_19
2019-DAR-0258	CFIACB20190302	PRJNA454819	SAMN40646298	SRR28490852	O15:H16	Meat	Meat	Stx2g	StxOp923_Stx2g_26
2019-DAR-0260	CFIACB20190304	PRJNA454819	SAMN40646231	SRR28490878	O128:H2	Meat	Meat	Stx1c; Stx2b	StxOp1348_Stx2b_110; StxOp4754_Stx1c_708
2019-DAR-0262	CFIACB20190306	PRJNA454819	SAMN40646253	SRR28490895	O6:H10	Meat	Meat	Stx1c	StxOp1036_Stx1c_12
2019-DAR-0263	CFIACB20190307	PRJNA454819	SAMN40646297	SRR28490854	O98:H25	Meat	Meat	Stx1a	StxOp1577_Stx1a_9
2019-DAR-0290	CFIACB20200103	PRJNA454819	SAMN40646223	SRR28490838	O38:H26	Meat	Meat	Stx1c; Stx2b	StxOp1348_Stx2b_110; StxOp4754_Stx1c_708
2019-DAR-0291	CFIACB20200104	PRJNA454819	SAMN40646243	SRR28490906	O91:H14	Meat	Meat	Stx2b	StxOp675_Stx2b_116
2019-DAR-0295	CFIACB20200108	PRJNA454819	SAMN40646286	SRR28490818	O146:H8	Meat	Meat	Stx1c	StxOp1036_Stx1c_12
2019-MER-0039	CFIACB20130345	PRJNA454819	SAMN13334355	SRR22858763	O157:H7	Meat	Meat	Stx1a; Stx2c	StxOp3_Stx1a_9; StxOp702_Stx2c_17
2019-MER-0040	CFIACB20110143	PRJNA454819	SAMN13333986	SRR22858892	O157:H7	Meat	Meat	Stx2a	StxOp1043_Stx2a_1
2019-MER-0054	CFIACB20190157	PRJNA454819	SAMN26419928	SRR18215049	O5:H19	Meat	Lamb	Stx1c; Stx2b	StxOp3320_Stx2b_101; StxOp4754_Stx1c_708
2019-MER-0057	CFIACB20190189	PRJNA454819	SAMN40646229	SRR28490880	O128:H2	Meat	Lamb	Stx1c; Stx2b	StxOp4754_Stx1c_708; StxOp1348_Stx2b_110
2019-MER-0102	CFIACB20180264	PRJNA454819	SAMN13334804	SRR22858706	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2019-MER-0123	CFIACB20190159	PRJNA454819	SAMN40646235	SRR28490874	O128:H2	Meat	Lamb	Stx1c; Stx2b	StxOp1348_Stx2b_110; StxOp399_Stx1c_12
2019-MER-0137	CFIACB20190194	PRJNA454819	SAMN40734924	SRR28538989	OUT:H21	Meat	Lamb	Stx2i	StxOp933_Stx2i_43
2019-MER-0138	CFIACB20190195	PRJNA454819	SAMN40646258	SRR28490848	OUT:H21	Meat	Lamb	Stx2i	StxOp933_Stx2i_43
2019-MER-0194	CFIACB20190179	PRJNA454819	SAMN40646282	SRR28490822	O146:H8	Meat	Lamb	Stx1c	StxOp1036_Stx1c_12
2019-MER-0212	CFIACB20090264	PRJNA454819	SAMN13333645	SRR22858866	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2019-MER-0213	CFIACB20120189	PRJNA454819	SAMN13334062	SRR22858895	O157:H7	Meat	Meat	Stx2a	StxOp1043_Stx2a_1
2019-MER-0216	CFIACB20160178	PRJNA454819	SAMN13334719	SRR22858810	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2019-MER-0224	CFIACB20180383	PRJNA454819	SAMN13334835	SRR12039817	O103:H25	Flour	Flour	Stx1a	StxOp4_Stx1a_9
2019-MER-0225	CFIACB20180558	PRJNA454819	SAMN13334875	SRR18215051	O176:H4	Meat	Lamb	Stx1c; Stx2b	StxOp1339_Stx2b_109; StxOp1037_Stx1c_26
2019-MER-0227	CFIACB20180559	PRJNA454819	SAMN13334876	SRR22858712	O174:H21	Meat	Meat	Stx2c	StxOp1365_Stx2c_7
2019-MER-0269	CFIACB20200013	PRJNA454819	SAMN40646246	SRR28490903	O91:H14	Meat	Meat	Stx1a; Stx2b	StxOp675_Stx2b_116; StxOp13_Stx1a_19
2019-SEQ-0093	CFIACB20190183	PRJNA454819	SAMN40646278	SRR28490826	OUT:H8	Meat	Lamb	Stx1c	StxOp1036_Stx1c_12
2019-SEQ-0137	CFIACB20190190	PRJNA454819	SAMN40646284	SRR28490820	O146:H8	Meat	Lamb	Stx1c	StxOp1036_Stx1c_12
2019-SEQ-0145	CFIACB20190191	PRJNA454819	SAMN40646238	SRR28490870	OUT:H2	Meat	Lamb	Stx1c; Stx2b	StxOp1348_Stx2b_110; StxOp400_Stx1c_22
2019-SEQ-0212	CFIACB20190192	PRJNA454819	SAMN40646236	SRR28490873	O128:H2	Meat	Lamb	Stx1c; Stx2b	StxOp1348_Stx2b_110; StxOp4754_Stx1c_708
2019-SEQ-0254	CFIACB20170273	PRJNA454819	SAMN11373405	SRR22715054	O55:H12	Meat	Meat	Stx1a	StxOp1580_Stx1a_9
2019-SEQ-0300	CFIACB20150154	PRJNA454819	SAMN13334673	SRR22858798	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
2019-SEQ-0328	CFIACB20190226	PRJNA454819	SAMN40646294	SRR28490857	O88:H25	Meat	Meat	Stx1a; Stx2a	StxOp1157_Stx2a_6; StxOp4_Stx1a_9
2019-SEQ-0356	CFIACB20190197	PRJNA454819	SAMN40646262	SRR28490844	O5:H19	Meat	Lamb	Stx1c	StxOp4754_Stx1c_708
2019-SEQ-0385	CFIACB20190199	PRJNA454819	SAMN40646239	SRR28490869	O128:H2	Meat	Lamb	Stx1c; Stx2b	StxOp1348_Stx2b_110; StxOp4754_Stx1c_708
2019-SEQ-0398	CFIACB20190168	PRJNA454819	SAMN40646267	SRR28490839	O76:H19	Meat	Lamb	Stx1c	StxOp4754_Stx1c_708
2019-SEQ-0516	CFIACB20190201	PRJNA454819	SAMN40646303	SRR28490888	O109:H5	Meat	Meat	Stx1a	StxOp1594_Stx1a_9
2019-SEQ-0782	CFIACB20190214	PRJNA454819	SAMN40646224	SRR28490827	OUT:H26	Meat	Lamb	Stx1c; Stx2b	StxOp1339_Stx2b_109; StxOp4754_Stx1c_708
2019-SEQ-1011	CFIACB20190258	PRJNA454819	SAMN40646240	SRR28490868	OUT:H2	Meat	Meat	Stx1c; Stx2b	StxOp1348_Stx2b_110; StxOp399_Stx1c_12
2019-SEQ-1012	CFIACB20190259	PRJNA454819	SAMN40646255	SRR28490893	O176:H4	Meat	Meat	Stx1c	StxOp1037_Stx1c_26
2019-SEQ-1083	CFIACB20190289	PRJNA454819	SAMN40646263	SRR28490843	OUT:H19	Meat	Meat	Stx1c; Stx2b	StxOp1339_Stx2b_109; StxOp4754_Stx1c_708
2019-SEQ-1198	CFIACB20190299	PRJNA454819	SAMN40646225	SRR28490864	O38:H26	Meat	Meat	Stx1c; Stx2b	StxOp1348_Stx2b_110; StxOp399_Stx1c_12
2019-SEQ-1223	CFIACB20200006	PRJNA454819	SAMN40646299	SRR28490851	OUT:H2	Meat	Meat	Stx2c	StxOp1364_Stx2c_7
2019-SEQ-1242	CFIACB20200007	PRJNA454819	SAMN40646291	SRR28490860	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2019-SEQ-1265	CFIACB20200036	PRJNA454819	SAMN40646296	SRR28490855	O98:H25	Meat	Meat	Stx1a	StxOp1577_Stx1a_9
2019-SEQ-1266	CFIACB20200037	PRJNA454819	SAMN40646295	SRR28490856	O8:H23	Meat	Meat	Stx1a	StxOp1580_Stx1a_9

Genome ID	CFIA isolate	BioProject	BioSample	SRA run accession	Serotype	Isolation source	Isolation source detail	Stx subtype(s)	STxOP_nt operon(s)
2019-SEQ-1270	CFIACB20200041	PRJNA454819	SAMN40646234	SRR28490875	O128:H2	Meat	Meat	Stx1c; Stx2b	StxOp1348_Stx2b_110; StxOp4754_Stx1c_708
2019-SEQ-1525	CFIACB20200052	PRJNA454819	SAMN40646264	SRR28490842	O5:H19	Meat	Meat	Stx1c	StxOp4754_Stx1c_708
2019-STH-0001	CFIACB20190215	PRJNA454819	SAMN40646289	SRR28490862	O104:H7	Meat	Meat	Stx1c	StxOp1036_Stx1c_12
2019-STH-0002	CFIACB20190216	PRJNA454819	SAMN40646307	SRR28490884	O132:H18	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
2020-CAL-0073	CFIACB20210043	PRJNA454819	SAMN40646230	SRR28490879	O128:H2	Meat	Meat	Stx1c; Stx2b	StxOp1348_Stx2b_110; StxOp402_Stx1c_25
2020-DAR-0002	CFIACB20200159	PRJNA454819	SAMN40646277	SRR28490828	O181:H16	Meat	Meat	Stx1c; Stx2b	StxOp3673_Stx2b_508; StxOp399_Stx1c_12
2020-DAR-0004	CFIACB20200173	PRJNA454819	SAMN40646245	SRR28490904	O91:H14	Meat	Meat	Stx1a; Stx2b	StxOp675_Stx2b_116; StxOp13_Stx1a_19
2020-DAR-0005	CFIACB20200174	PRJNA454819	SAMN40646244	SRR28490905	O91:H14	Meat	Meat	Stx2b	StxOp675_Stx2b_116
2020-DAR-0008	CFIACB20200177	PRJNA454819	SAMN40646254	SRR28490894	O176:H4	Meat	Meat	Stx1c; Stx2b	StxOp1339_Stx2b_109; StxOp1037_Stx1c_26
2020-DAR-0009	CFIACB20200178	PRJNA454819	SAMN40646288	SRR28490863	O5:H19	Meat	Meat	Stx1c; Stx2b	StxOp3320_Stx2b_101; StxOp4754_Stx1c_708
2020-DAR-0029	CFIACB20210012	PRJNA454819	SAMN40646273	SRR28490832	O87:H16	Meat	Meat	Stx2b	StxOp3564_Stx2b_118; StxOp1339_Stx2b_109
2020-DAR-0032	CFIACB20210015	PRJNA454819	SAMN40646266	SRR28490840	O76:H19	Meat	Meat	Stx2b	StxOp1344_Stx2b_109
2020-DAR-0036	CFIACB20210019	PRJNA454819	SAMN40646232	SRR28490877	O128:H2	Meat	Meat	Stx1c; Stx2b	StxOp1348_Stx2b_110; StxOp399_Stx1c_12
2020-SEQ-0005	CFIACB20200082	PRJNA454819	SAMN40646308	SRR28490882	OUT:H29	Meat	Meat	Stx2b	StxOp3298_Stx2b_23
2020-SEQ-0075	CFIACB20200113	PRJNA454819	SAMN40646219	SRR28490907	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2020-SEQ-0076	CFIACB20200114	PRJNA454819	SAMN40646220	SRR28490871	O157:H7	Meat	Meat	Stx2a	StxOp1043_Stx2a_1
2020-SEQ-0219	CFIACB20200140	PRJNA454819	SAMN40646247	SRR28490902	O91:H14	Meat	Meat	Stx1a; Stx2b	StxOp675_Stx2b_116; StxOp13_Stx1a_19
2020-SEQ-0289	CFIACB20200033	PRJNA454819	SAMN40646222	SRR28490849	O38:H26	Meat	Meat	Stx1c; Stx2b	StxOp1348_Stx2b_110; StxOp4754_Stx1c_708
2020-SEQ-0292	CFIACB20200125	PRJNA454819	SAMN40646287	SRR28490865	OUT:H8	Meat	Meat	Stx1c; Stx2b	StxOp1336_Stx2b_109; StxOp399_Stx1c_12
2020-SEQ-0302	CFIACB20200148	PRJNA454819	SAMN40646221	SRR28490901	O157:H7	Meat	Meat	Stx2a	StxOp1043_Stx2a_1
2020-SEQ-0314	CFIACB20200149	PRJNA454819	SAMN40646269	SRR28490836	O128:H2	Meat	Meat	Stx1c; Stx2b	StxOp677_Stx2b_120; StxOp1037_Stx1c_26
2020-SEQ-0325	CFIACB20200162	PRJNA454819	SAMN40646250	SRR28490898	O91:H14	Meat	Meat	Stx1a; Stx2b	StxOp675_Stx2b_116; StxOp13_Stx1a_19
2020-SEQ-0326	CFIACB20200163	PRJNA454819	SAMN40646241	SRR28490867	O91:H14	Meat	Meat	Stx1a; Stx2b	StxOp675_Stx2b_116; StxOp13_Stx1a_19
2020-SEQ-0387	CFIACB20200190	PRJNA454819	SAMN40646275	SRR28490830	OUT:H10	Meat	Meat	Stx1c; Stx2b	StxOp1339_Stx2b_109; StxOp401_Stx1c_23
2021-BUR-0024	CFIACB20210042	PRJNA454819	SAMN40646261	SRR28490845	O171:H2	Meat	Meat	Stx2d	StxOp841_Stx2d_100
2021-DAR-0007	CFIACB20210118	PRJNA454819	SAMN40646281	SRR28490823	O146:H8	Meat	Lamb	Stx1c	StxOp1036_Stx1c_12
2021-DAR-0014	CFIACB20210070	PRJNA454819	SAMN40646301	SRR28490890	O5:H9	Meat	Meat	Stx1a	StxOp9_Stx1a_15
2021-DAR-0015	CFIACB20210071	PRJNA454819	SAMN40646233	SRR28490876	O128:H2	Meat	Meat	Stx1c; Stx2b	StxOp1348_Stx2b_110; StxOp4754_Stx1c_708
2021-DAR-0023	CFIACB20210077	PRJNA454819	SAMN40646268	SRR28490837	O128:H2	Meat	Meat	Stx1c; Stx2b	StxOp1348_Stx2b_110; StxOp4754_Stx1c_708
2021-DAR-0024	CFIACB20210078	PRJNA454819	SAMN40646270	SRR28490835	O9:H30	Meat	Lamb	Stx2i	StxOp933_Stx2i_43
2021-GTA-0018	CFIACB20210083	PRJNA454819	SAMN40646306	SRR28490885	O174:H21	Meat	Meat	Stx2c	StxOp1365_Stx2c_7
2021-GTA-0022	CFIACB20210086	PRJNA454819	SAMN40646249	SRR28490899	O91:H14	Meat	Lamb	Stx1a; Stx2b	StxOp1348_Stx2b_110; StxOp13_Stx1a_19
2021-GTA-0023	CFIACB20210082	PRJNA454819	SAMN40646283	SRR28490821	OUT:H8	Meat	Lamb	Stx1c	StxOp1036_Stx1c_12
2021-GTA-0024	CFIACB20210085	PRJNA454819	SAMN40646300	SRR28490850	OUT:H2	Meat	Meat	Stx2d	StxOp841_Stx2d_100
2021-MER-0003	CFIACB20200188	PRJNA454819	SAMN40646256	SRR28490892	O153:H25	Meat	Meat	Stx2b	StxOp1339_Stx2b_109
2021-MER-0004	CFIACB20200189	PRJNA454819	SAMN40646279	SRR28490825	OUT:H8	Meat	Meat	Stx1c	StxOp1036_Stx1c_12
2021-MER-0007	CFIACB20190260	PRJNA454819	SAMN40646252	SRR28490896	O91:H14	Meat	Meat	Stx1a; Stx2b	StxOp675_Stx2b_116; StxOp13_Stx1a_19
2021-MER-0013	CFIACB20170190	PRJNA454819	SAMN40646218	SRR28490908	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
2021-MIN-0009	CFIACB20190239	PRJNA454819	SAMN40646259	SRR28490847	O8:H19	Meat	Lamb	Stx2i	StxOp933_Stx2i_43
2021-MIN-0249	CFIACB20190172	PRJNA454819	SAMN26419929	SRR18215048	O8:H25	Meat	Lamb	Stx2i	StxOp933_Stx2i_43
2021-MIN-0250	CFIACB20180511	PRJNA454819	SAMN11373501	SRR18215053	O9:H30	Meat	Lamb	Stx2i	StxOp933_Stx2i_43
2021-MIN-0251	CFIACB20180512	PRJNA454819	SAMN11373499	SRR22715050	OUT:H21	Meat	Lamb	Stx2i	StxOp933_Stx2i_43
2021-MIN-0252	CFIACB20200034	PRJNA454819	SAMN40646257	SRR28490891	OUT:H21	Meat	Lamb	Stx2i	StxOp933_Stx2i_43
2021-SEQ-0001	CFIACB20200200	PRJNA454819	SAMN40646274	SRR28490831	O123:H10	Meat	Meat	Stx1c; Stx2b	StxOp1339_Stx2b_109; StxOp4754_Stx1c_708
2021-SEQ-0006	CFIACB20200208	PRJNA454819	SAMN40646293	SRR28490858	O55:H12	Meat	Meat	Stx1a	StxOp1580_Stx1a_9
2021-SEQ-0074	CFIACB20200207	PRJNA454819	SAMN40646272	SRR28490833	O104:H7	Meat	Meat	Stx1c	StxOp1036_Stx1c_12
2021-SEQ-0097	CFIACB20200206	PRJNA454819	SAMN40646248	SRR28490900	O91:H14	Meat	Meat	Stx1a; Stx2b	StxOp1352_Stx2b_116; StxOp13_Stx1a_19

Genome ID	CFIA isolate	BioProject	BioSample	SRA run accession	Serotype	Isolation source	Isolation source detail	Stx subtype(s)	STxOP_nt operon(s)
2021-SEQ-0388	CFIACB20210105	PRJNA454819	SAMN40646302	SRR28490889	OUT:H29	Meat	Meat	Stx2b; Stx2c	StxOp1448_Stx2c_152; StxOp3298_Stx2b_23
2021-SEQ-0504	CFIACB20190305	PRJNA454819	SAMN40646285	SRR28490819	O91:H14	Meat	Meat	Stx1a; Stx2b	StxOp675_Stx2b_116; StxOp13_Stx1a_19
CFIACB20180581	CFIACB20180581	PRJNA454819	SAMN13334890	SRR22858747	OUT:H21	Meat	Lamb	Stx2i	StxOp933_Stx2i_43
GCA_012139605.1	CFIACB20120179	PRJNA454819	SAMN09668382	SRR7629331	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
GCA_012139625.1	CFIACB20120247	PRJNA454819	SAMN09668403	SRR7629332	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
GCA_012139645.1	CFIACB20120242	PRJNA454819	SAMN09668399	SRR7629329	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
GCA_012139665.1	CFIACB20120254	PRJNA454819	SAMN09668420	SRR7629330	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
GCA_012139705.1	CFIACB20120251	PRJNA454819	SAMN09668417	SRR7629324	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
GCA_012139725.1	CFIACB20120246	PRJNA454819	SAMN09668402	SRR7629320	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
GCA_012139885.1	CFIACB20130140	PRJNA454819	SAMN09668414	SRR7629326	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
GCA_012139945.1	CFIACB20120180	PRJNA454819	SAMN09668383	SRR7629322	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
GCA_012140065.1	CFIACB20120269	PRJNA454819	SAMN09668389	SRR7629284	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
GCA_012140085.1	CFIACB20120250	PRJNA454819	SAMN09668405	SRR7629325	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
GCA_012140205.1	CFIACB20130131	PRJNA454819	SAMN09668409	SRR7629314	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
GCA_012140245.1	CFIACB20120243	PRJNA454819	SAMN09668400	SRR7629308	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
GCA_012140285.1	CFIACB20120245	PRJNA454819	SAMN09668401	SRR7629304	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
GCA_012140305.1	CFIACB20120271	PRJNA454819	SAMN09668391	SRR7629307	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
GCA_012140325.1	CFIACB20120249	PRJNA454819	SAMN09668419	SRR7629313	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
GCA_012140365.1	CFIACB20120253	PRJNA454819	SAMN09668407	SRR7629306	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
GCA_012140385.1	CFIACB20120268	PRJNA454819	SAMN09668388	SRR7629303	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
GCA_012140405.1	CFIACB20120191	PRJNA454819	SAMN09668393	SRR7629305	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
GCA_012140425.1	CFIACB20130142	PRJNA454819	SAMN09668416	SRR7629300	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
GCA_012140445.1	CFIACB20120221	PRJNA454819	SAMN09668396	SRR7629302	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
GCA_012140465.1	CFIACB20130133	PRJNA454819	SAMN09668411	SRR7629298	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
GCA_012140485.1	CFIACB20130135	PRJNA454819	SAMN09668412	SRR7629301	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
GCA_012140505.1	CFIACB20130132	PRJNA454819	SAMN09668410	SRR7629299	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
GCA_012140605.1	CFIACB20120248	PRJNA454819	SAMN09668418	SRR7629294	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
GCA_012140625.1	CFIACB20120178	PRJNA454819	SAMN09668381	SRR7629297	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
GCA_012140665.1	CFIACB20130134	PRJNA454819	SAMN09668413	SRR7629291	O157:H7	Meat	Meat	Stx2a	StxOp1089_Stx2a_1
GCA_012140725.1	CFIACB20120181	PRJNA454819	SAMN09668384	SRR7629289	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
GCA_012141085.1	CFIACB20120187	PRJNA454819	SAMN09668386	SRR7629311	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
GCA_012141125.1	CFIACB20120188	PRJNA454819	SAMN09668387	SRR7629312	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
GCA_012141225.1	CFIACB20120270	PRJNA454819	SAMN09668390	SRR7629309	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
GCA_012148805.1	CFIACB20120183	PRJNA454819	SAMN09668377	SRR7536942	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
GCA_012149105.1	CFIACB20120184	PRJNA454819	SAMN09668378	SRR7536941	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
GCA_012149165.1	CFIACB20120185	PRJNA454819	SAMN09668379	SRR7536940	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp1089_Stx2a_1; StxOp3_Stx1a_9
GCA_012250705.1	CFIACB20120220	PRJNA454819	SAMN09668395	SRR7629293	O157:H7	Meat	Meat	Stx1a; Stx2a	StxOp3_Stx1a_9; StxOp1089_Stx2a_1
GCA_013061965.1	CFIACB20120255	PRJNA454819	SAMN09668421	SRR7629296	O157:H7	Meat	Meat	None detected	None detected

Supplementary Table S4.1. Metadata and assembly characteristics of the 66 stx2i-positive *Escherichia coli* genomes.

The seven CFIA lamb isolates and the 59 publicly available genomes retained after curation, with their accessions, serotype, sequence type, provenance, assembly statistics, NCBI Pathogen Detection SNP cluster and the stx2i operon assigned by KMA.

66 rows, 18 columns.

Assembly	Dataset	CFIA isolate	BioSample	Illumina reads	Serotype	Sequence type	Country of origin	Imported from	Isolation source	Source category	Collection year	Contigs	N50 (bp)	Genome size (bp)	Illumina coverage (x)	SNP cluster	stx2i operon (KMA)
2021-MIN-0251	CFIA	OLC3375	SAMN11373499	SRR22715050	O-untypeable:H21	ST155	Canada	New Zealand	Raw Ground Lamb	Ovis aries	2018	3	5096570	5203767	63.3		StxOp933_Stx2i_43
2021-MIN-0252	CFIA	CFIAFB20200034	SAMN40646257	SRR28490891	O-untypeable:H21	ST155	Canada	New Zealand	Raw Ground Lamb	Ovis aries	2020	7	5059451	5271659	166.4	PDS000129627.5	StxOp933_Stx2i_43
2022-MIN-0036	CFIA	CFIAFB20180581	SAMN13334890	SRR22858747	O-untypeable:H21	ST155	Canada	New Zealand	Raw Ground Lamb	Ovis aries	2018	21	701209	5181639	159	PDS000129627.5	StxOp933_Stx2i_43
2021-MIN-0009	CFIA	CFIAFB20190239	SAMN40646259	SRR28490847	O8:H19	ST162	Canada	New Zealand	Raw Ground Lamb	Ovis aries	2019	12	2732064	5305006	154.9		StxOp933_Stx2i_43
2021-MIN-0249	CFIA	OLC3504	SAMN26419929	SRR18215048	O8:H25	ST8660	Canada	New Zealand	Raw Ground Lamb	Ovis aries	2019	4	5075290	5177897	128		StxOp933_Stx2i_43
2021-MIN-0250	CFIA	OLC3374	SAMN11373501	SRR18215053	O9:H30	ST1294	Canada	New Zealand	Raw Ground Lamb	Ovis aries	2018	3	5065598	5188405	97.1		StxOp933_Stx2i_43
2022-MIN-0037	CFIA	CFIAFB20210075	SAMN54529050	SRR36737647	O9:H30	ST1294	Canada	New Zealand	Raw Ground Lamb	Ovis aries	2021	5	3089149	5134902	145.2		StxOp933_Stx2i_43
GCA_036292925.1	Public		SAMN37069730	SRR25724210	O-untypeable:H21	ST1248	New Zealand		feces	Environmental	2020	153	110685	5114367		PDS000168800.3	StxOp933_Stx2i_43
GCA_036299585.1	Public		SAMN37069514	SRR25724374	O-untypeable:H21	ST1248	New Zealand		feces	Environmental	2020	128	126935	5121233		PDS000168800.3	StxOp933_Stx2i_43
GCA_038420345.1	Public		SAMN41027598	SRR28752976	O-untypeable:H21	ST155	United Kingdom		food	Unknown Food	2024	204	133288	5325869		PDS000129627.5	StxOp933_Stx2i_43
GCA_039685295.1	Public		SAMN41444113	SRR29074270	O-untypeable:H21	ST155	United Kingdom		food	Unknown Food	2023	221	88115	5211820		PDS000129627.5	StxOp933_Stx2i_43
GCA_012153565.1	Public		SAMN09634209	SRR7501292	O100:H30	ST993	United Kingdom		environmental	Environmental	2017	155	78798	5064891		PDS000024140.20	StxOp933_Stx2i_43
GCA_012153625.1	Public		SAMN09633700	SRR7500930	O100:H30	ST993	United Kingdom		food	Unknown Food	2017	160	78798	5001263		PDS000024140.20	StxOp933_Stx2i_43
GCA_017125235.1	Public		SAMN17799185	SRR13633165	O100:H30	ST993	United Kingdom		food	Unknown Food	2021	176	75816	5072456		PDS000024140.20	StxOp933_Stx2i_43
GCA_017125255.1	Public		SAMN17799155	SRR13633150	O100:H30	ST993	United Kingdom		food	Unknown Food	2021	176	82799	5006878		PDS000024140.20	StxOp933_Stx2i_43
GCA_018927075.1	Public		SAMN19680875	SRR14794503	O100:H30	ST993	United Kingdom		human	Human Clinical	2021	160	74110	4999806		PDS000183839.1	StxOp4689_Stx2i_235
GCA_019436045.1	Public		SAMN20516166	SRR15319335	O100:H30	ST993	United Kingdom		food	Unknown Food	2021	168	73254	5054699		PDS000024140.20	StxOp933_Stx2i_43
GCA_020370835.1	Public		SAMN06006283	SRR5006141	O100:H30	ST993	United Kingdom		human	Human Clinical	2016	173	77199	5019061		PDS000024140.20	StxOp933_Stx2i_43
GCA_021078905.1	Public		SAMN20192700	SRR15115831	O100:H30	ST993	Ireland		sheep recto-anal junction	Ovis aries	2019	141	92728	5072969		PDS000024140.20	StxOp933_Stx2i_43
GCA_021079095.1	Public		SAMN20192693	SRR15115839	O100:H30	ST993	Ireland		sheep recto-anal junction	Ovis aries	2018	354	24807	5012290		PDS000100862.1	StxOp933_Stx2i_43
GCA_021079545.1	Public		SAMN20192674	SRR15115860	O100:H30	ST993	Ireland		sheep recto-anal junction	Ovis aries	2019	134	87466	4979496		PDS000097197.2	StxOp933_Stx2i_43
GCA_021079625.1	Public		SAMN20192669	SRR15115925	O100:H30	ST993	Ireland		sheep recto-anal junction	Ovis aries	2019	323	76997	5143382		PDS000097185.2	StxOp933_Stx2i_43
GCA_021080295.1	Public		SAMN20192636	SRR15115961	O100:H30	ST993	Ireland		sheep recto-anal junction	Ovis aries	2019	136	86914	5015677		PDS000097185.2	StxOp933_Stx2i_43
GCA_021081115.1	Public		SAMN20192598	SRR15115887	O100:H30	ST993	Ireland		sheep recto-anal junction	Ovis aries	2018	174	118168	5136186		PDS000100878.1	StxOp933_Stx2i_43
GCA_021082345.1	Public		SAMN20192564	SRR15115893	O100:H30	ST993	Ireland		sheep recto-anal junction	Ovis aries	2018	138	111581	5002130		PDS000100872.1	StxOp933_Stx2i_43
GCA_021082805.1	Public		SAMN20192541	SRR15115919	O100:H30	ST993	Ireland		sheep recto-anal junction	Ovis aries	2018	138	91088	5069700		PDS000100858.1	StxOp933_Stx2i_43
GCA_025327895.1	Public		SAMN30899955	SRR21607564	O100:H30	ST993	United Kingdom		human	Human Clinical	2022	169	69425	5035910		PDS000024140.20	StxOp933_Stx2i_43

Assembly	Dataset	CFIA isolate	BioSample	Illumina reads	Serotype	Sequence type	Country of origin	Imported from	Isolation source	Source category	Collection year	Contigs	N50 (bp)	Genome size (bp)	Illumina coverage (x)	SNP cluster	stx2i operon (KMA)
GCA_038017815.1	Public		SAMN40859897	SRR28576417	O100:H30	ST993	United Kingdom		food	Unknown Food	2024	197	82908	5124074		PDS000024140.20	StxOp933_Stx2i_43
GCA_038086625.1	Public		SAMN40923829	SRR28621926	O100:H30	ST993	United Kingdom		food	Unknown Food	2024	210	76086	5121688		PDS000024140.20	StxOp933_Stx2i_43
GCA_039686495.1	Public		SAMN41449217	SRR29077784	O100:H30	ST993	United Kingdom		food	Unknown Food	2023	208	67571	5008602		PDS000024140.20	StxOp933_Stx2i_43
GCA_039694335.1	Public		SAMN41444163	SRR29070937	O100:H30	ST993	United Kingdom		food	Unknown Food	2023	214	63087	5032784		PDS000199344.1	StxOp933_Stx2i_43
GCA_039735685.1	Public		SAMN41465198	SRR29091375	O100:H30	ST993	United Kingdom		food	Unknown Food	2023	188	75816	5152944		PDS000024140.20	StxOp933_Stx2i_43
GCA_039775235.1	Public		SAMN41481858	SRR29118840	O100:H30	ST993	United Kingdom		food	Unknown Food	2023	232	59834	5105242		PDS000024140.20	StxOp933_Stx2i_43
GCA_039778795.1	Public		SAMN41482030	SRR29119989	O100:H30	ST993	United Kingdom		food	Unknown Food	2023	262	45403	5099583		PDS000024140.20	StxOp933_Stx2i_43
GCA_039787335.1	Public		SAMN41500593	SRR29138391	O100:H30	ST993	United Kingdom		food	Unknown Food	2023	217	59089	5008283			StxOp933_Stx2i_43
GCA_039865785.1	Public		SAMN41516343	SRR29155112	O100:H30	ST993	United Kingdom		human	Human Clinical	2023	197	69642	5066661		PDS000183839.1	StxOp933_Stx2i_43
GCA_039879905.1	Public		SAMN41550020	SRR29188554	O100:H30	ST993	United Kingdom		human	Human Clinical	2023	213	64629	5133960		PDS000024140.20	StxOp933_Stx2i_43
GCA_039947755.1	Public		SAMN41574626	SRR29208759	O100:H30	ST993	United Kingdom		human	Human Clinical	2023	240	60536	5002339		PDS000183839.1	StxOp933_Stx2i_43
GCA_040279925.1	Public		SAMN41992278	SRR29499143	O100:H30	ST993	United Kingdom		human	Human Clinical	2024	208	66055	5032480			StxOp933_Stx2i_43
GCA_041047495.1	Public		SAMN43015297	SRR30118389	O100:H30	ST993	United Kingdom		human	Human Clinical	2024	196	75816	5022352		PDS000024140.20	StxOp933_Stx2i_43
GCA_041545175.1	Public		SAMN43373094	SRR30407660	O100:H30	ST993	United Kingdom		human	Human Clinical	2024	194	67199	5005039		PDS000024140.20	StxOp933_Stx2i_43
GCA_042177385.1	Public		SAMN43879893	SRR30764002	O100:H30	ST993	United Kingdom		human	Human Clinical	2024	228	74030	5021882		PDS000199344.1	StxOp933_Stx2i_43
GCA_042441235.1	Public		SAMN43948175	SRR30813966	O100:H30	ST993	United Kingdom		human	Human Clinical	2024	206	70529	5003763			StxOp933_Stx2i_43
GCA_052635535.1	Public		SAMN50709792	SRR35059917	O100:H30	ST993	United Kingdom		human	Human Clinical	2025	215	67201	5050955			StxOp933_Stx2i_43
GCA_012560085.1	Public		SAMN06006011	SRR5005391	O8:H19	ST162	United Kingdom		human	Human Clinical	2016	202	97517	5148618			StxOp933_Stx2i_43
GCA_025988885.1	Public		SAMN26036523	SRR18191603	O8:H19	ST162	France		Dairy product	Dairy	2009	3	5022134	5132730			StxOp933_Stx2i_43
GCA_017076875.1	Public		SAMN12996567	SRR10253926	O8:H25	ST8660	New Zealand		periphyton	Environmental	2018	128	136293	5062309			StxOp933_Stx2i_43
GCA_021492705.1	Public		SAMN25002964	SRR17622291	O8:H25	ST8660	United Kingdom		human	Human Clinical	2022	182	130960	5323170			StxOp933_Stx2i_43
GCA_023211155.1	Public		SAMN26797296	SRR18965643	O8:H25	ST8660	United Kingdom		feces	Human Clinical	2014	141	158163	5145427		PDS000177475.1	StxOp933_Stx2i_43
GCA_023768585.1	Public		SAMN28861568	SRR19539327	O8:H25	ST8660	United Kingdom		human	Human Clinical	2022	172	84849	4939098			StxOp933_Stx2i_43
GCA_037035515.1	Public		SAMN40255785	SRR28220657	O8:H25	ST8660	United Kingdom		human	Human Clinical	2023	201	92865	5240843			StxOp933_Stx2i_43
GCA_039997695.1	Public		SAMN41594209	SRR29226700	O8:H25	ST8660	United Kingdom		food	Unknown Food	2023	157	109714	5044115			StxOp933_Stx2i_43
GCA_042710215.1	Public		SAMN39582411	SRR27704196	O8:H25	ST8660	New Zealand		clinical	Human Clinical	2022	128	126866	5130797			StxOp933_Stx2i_43
GCA_042935545.1	Public		SAMEA115376134	ERR12723348	O8:H25	ST8660	Belgium		human	Human Clinical	2023	91	259017	5156296		PDS000177475.1	StxOp933_Stx2i_43
GCA_043574615.1	Public		SAMN43267956	SRR30307958	O8:H25	ST8660	Denmark		not collected	Human Clinical	2022	247	47609	5066315			StxOp933_Stx2i_43
GCA_045501555.1	Public		SAMN39582387	SRR27704273	O8:H25	ST8660	New Zealand		clinical	Human Clinical	2017	105	193198	5084238			StxOp933_Stx2i_43
GCA_012193385.1	Public		SAMN09380381	SRR7277734	O8:H9	ST23	United Kingdom		human	Human Clinical	2017	151	96406	5022525			StxOp933_Stx2i_43
GCA_051797185.1	Public		SAMN50428753	SRR34847042	O8:H9	ST23	United Kingdom		human	Human Clinical	2025	144	113860	5095794			StxOp933_Stx2i_43
GCA_029481885.1	Public		SAMEA4964526	ERR2834321	O9:H10	ST88	Norway		Oyster	Oyster	2016	106	185657	5142435		PDS000140473.1	StxOp933_Stx2i_43

Assembly	Dataset	CFIA isolate	BioSample	Illumina reads	Serotype	Sequence type	Country of origin	Imported from	Isolation source	Source category	Collection year	Contigs	N50 (bp)	Genome size (bp)	Illumina coverage (x)	SNP cluster	stx2i operon (KMA)
GCA_040935505.2	Public		SAMN42828538	SRR30000023	O9:H19	ST162	United Kingdom		human	Human Clinical	2024	210	107518	5065515		PDS000189319.2	StxOp933_Stx2i_43
GCA_016773735.1	Public		SAMN17526174	SRR13518973	O9:H30	ST1294	Switzerland		raw meat-based diets for companion animals	Raw Animal Feed	2020	117	128921	4991306		PDS000177433.1	StxOp933_Stx2i_43
GCA_028221245.1	Public		SAMEA112275639	ERR10712336	O9:H30	ST1294	Netherlands		feces	Human Clinical	2020	141	91362	5091779			StxOp933_Stx2i_43
GCA_028240895.1	Public		SAMEA112275358	ERR10712055	O9:H30	ST1294	Netherlands		feces	Human Clinical	2018	163	86673	5127973			StxOp933_Stx2i_43
GCA_039951795.1	Public		SAMN41574618	SRR29208720	O9:H30	ST1294	United Kingdom		human	Human Clinical	2023	241	84674	4960429		PDS000205861.1	StxOp933_Stx2i_43
GCA_044682155.1	Public		SAMN44596646	SRR31237447	O9:H30	ST1294	United Kingdom		human	Human Clinical	2024	193	79185	4918943		PDS000205861.1	StxOp933_Stx2i_43
GCA_044986905.1	Public		SAMN39582592	SRR27704367	O9:H30	ST1294	New Zealand		clinical	Human Clinical	2022	148	94586	4985226			StxOp933_Stx2i_43

Supplementary Table S4.2. Publicly available assemblies retrieved for the stx2i genome dataset and their curation outcome.

The 99 assemblies retrieved from the NCBI Pathogen Detection MicroBIGG-E and Isolates browsers, with the stx element reported by MicroBIGG-E, the operon identified in the corresponding raw reads by KMA, and whether the assembly was retained or excluded.

99 rows, 16 columns.

Assembly	BioSample	Illumina reads	Strain	CFIA isolate	Serotype (ECTyper)	Sequence type	Contigs	N50 (bp)	Genome size (bp)	Location	Isolation source	SNP cluster	MicroBIGG-E stx element call	stx2i operon (KMA)	Curation outcome
GCA_012153565.1	SAMN09634209	SRR7501292	420503		O100:H30	ST993	155	78798	5064891	United Kingdom: United Kingdom	environmental	PDS000024140.20	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_012153625.1	SAMN09633700	SRR7500930	419105		O100:H30	ST993	160	78798	5001263	United Kingdom: United Kingdom	food	PDS000024140.20	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_012193385.1	SAMN09380381	SRR7277734	440785		O8:H9	ST23	151	96406	5022525	United Kingdom: United Kingdom	human	0	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_012560085.1	SAMN06006011	SRR5005391	287552		O8:H19	ST162	202	97517	5148618	United Kingdom: United Kingdom	human	0	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_016773735.1	SAMN17526174		ATC-41-3		O9:H30	ST1294	117	128921	4991306	Switzerland: Zurich	raw meat-based diets for companion animals	PDS000177433.1	stx2i operon (complete)		Retained
GCA_017076875.1	SAMN12996567	SRR10253926			O8:H25	ST8660	128	136293	5062309	New Zealand: Makakahi River, Hamua Bridge	periphyton	0	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_017125235.1	SAMN17799185	SRR13633165	1084158		O100:H30	ST993	176	75816	5072456	United Kingdom: United Kingdom	food	PDS000024140.20	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_017125255.1	SAMN17799155	SRR13633150	1084157		O100:H30	ST993	176	82799	5006878	United Kingdom: United Kingdom	food	PDS000024140.20	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_018927075.1	SAMN19680875	SRR14794503	1203565		O100:H30	ST993	160	74110	4999806	United Kingdom: United Kingdom	human	PDS000183839.1	stx2i operon (complete)	StxOp4689_Stx2i_235	Retained
GCA_019436045.1	SAMN20516166	SRR15319335	1289784		O100:H30	ST993	168	73254	5054699	United Kingdom: United Kingdom	food	PDS000024140.20	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_020370835.1	SAMN06006283	SRR5006141	293071		O100:H30	ST993	173	77199	5019061	United Kingdom: United Kingdom	human	PDS000024140.20	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_021078905.1	SAMN20192700		24_B_17_2_4		O100:H30	ST993	141	92728	5072969	Ireland: Mayo	recto-anal junction	PDS000024140.20	stx2i operon (complete)		Retained
GCA_021079095.1	SAMN20192693		6_A_4_1_1		O100:H30	ST993	354	24807	5012290	Ireland: Donegal	recto-anal junction	PDS000100862.1	stx2i operon (complete)		Retained
GCA_021079545.1	SAMN20192674		22_B_26_1_1		O100:H30	ST993	134	87466	4979496	Ireland: Galway	recto-anal junction	PDS000097197.2	stx2i operon (complete)		Retained
GCA_021079625.1	SAMN20192669		20_B_28_4_1		O100:H30	ST993	323	76997	5143382	Ireland: Galway	recto-anal junction	PDS000097185.2	stx2i operon (complete)		Retained
GCA_021080295.1	SAMN20192636		20_B_33_1_1		O100:H30	ST993	136	86914	5015677	Ireland: Galway	recto-anal junction	PDS000097185.2	stx2i operon (complete)		Retained
GCA_021081115.1	SAMN20192598		11_A_12_2_1		O100:H30	ST993	174	118168	5136186	Ireland: Donegal	recto-anal junction	PDS000100878.1	stx2i operon (complete)		Retained
GCA_021082345.1	SAMN20192564		05_A_5_5_1		O100:H30	ST993	138	111581	5002130	Ireland: Wicklow	recto-anal junction	PDS000100872.1	stx2i operon (complete)		Retained
GCA_021082805.1	SAMN20192541		08_A_26_1_2		O100:H30	ST993	138	91088	5069700	Ireland: Carlow	recto-anal junction	PDS000100858.1	stx2i operon (complete)		Retained
GCA_021492705.1	SAMN25002964	SRR17622291	1560496		O8:H25	ST8660	182	130960	5323170	United Kingdom: United Kingdom	human	0	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_023211155.1	SAMN26797296	SRR18965643	548		O8:H25	ST8660	141	158163	5145427	United Kingdom: Scotland	feces	PDS000177475.1	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_023768585.1	SAMN28861568	SRR19539327	1640989		O8:H25	ST8660	172	84849	4939098	United Kingdom: United Kingdom	human	0	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_025327895.1	SAMN30899955	SRR21607564	1704278		O100:H30	ST993	169	69425	5035910	United Kingdom: United Kingdom	human	PDS000024140.20	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_025988885.1	SAMN26036523		09QMA303-1A		O8:H19	ST162	3	5022134	5132730	France	Dairy product	0	stx2i operon (complete)		Retained
GCA_028221245.1	SAMEA112275639	ERR10712336			O9:H30	ST1294	141	91362	5091779	Netherlands	feces	0	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_028240895.1	SAMEA112275358	ERR10712055			O9:H30	ST1294	163	86673	5127973	Netherlands	feces	0	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_029481885.1	SAMEA4964526	ERR2834321			O9:H10	ST88	106	185657	5142435	Norway	Oyster	PDS000140473.1	stx2i operon (complete)	StxOp933_Stx2i_43	Retained

Assembly	BioSample	Illumina reads	Strain	CFIA isolate	Serotype (ECTyper)	Sequence type	Contigs	N50 (bp)	Genome size (bp)	Location	Isolation source	SNP cluster	MicroBIGG-E stx element call	stx2i operon (KMA)	Curation outcome
GCA_036292925.1	SAMN37069730		AGR5468		-:H21	ST1248	153	110685	5114367	New Zealand: Kai Iwi	feces	PDS000168800.3	stx2i operon (complete)		Retained
GCA_036299585.1	SAMN37069514		AGR4630		-:H21	ST1248	128	126935	5121233	New Zealand: Kai Iwi	feces	PDS000168800.3	stx2i operon (complete)		Retained
GCA_037035515.1	SAMN40255785	SRR28220657	1829105		O8:H25	ST8660	201	92865	5240843	United Kingdom: United Kingdom	human	0	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_038017815.1	SAMN40859897	SRR28576417	01465088		O100:H30	ST993	197	82908	5124074	United Kingdom: United Kingdom	food	PDS000024140.20	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_038086625.1	SAMN40923829	SRR28621926	01466934		O100:H30	ST993	210	76086	5121688	United Kingdom: United Kingdom	food	PDS000024140.20	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_038420345.1	SAMN41027598	SRR28752976	01474328		-:H21	ST155	204	133288	5325869	United Kingdom: United Kingdom	food	PDS000129627.5	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_039685295.1	SAMN41444113	SRR29074270	01337037		-:H21	ST155	221	88115	5211820	United Kingdom: United Kingdom	food	PDS000129627.5	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_039686495.1	SAMN41449217	SRR29077784	01331529		O100:H30	ST993	208	67571	5008602	United Kingdom: United Kingdom	food	PDS000024140.20	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_039694335.1	SAMN41444163	SRR29070937	01336033		O100:H30	ST993	214	63087	5032784	United Kingdom: United Kingdom	food	PDS000199344.1	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_039735685.1	SAMN41465198	SRR29091375	01291951		O100:H30	ST993	188	75816	5152944	United Kingdom: United Kingdom	food	PDS000024140.20	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_039775235.1	SAMN41481858	SRR29118840	01274437		O100:H30	ST993	232	59834	5105242	United Kingdom: United Kingdom	food	PDS000024140.20	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_039778795.1	SAMN41482030	SRR29119989	01277576		O100:H30	ST993	262	45403	5099583	United Kingdom: United Kingdom	food	PDS000024140.20	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_039787335.1	SAMN41500593	SRR29138391	01260925		O100:H30	ST993	217	59089	5008283	United Kingdom: United Kingdom	food	0	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_039865785.1	SAMN41516343	SRR29155112	01239766		O100:H30	ST993	197	69642	5066661	United Kingdom: United Kingdom	human	PDS000183839.1	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_039879905.1	SAMN41550020	SRR29188554	01211874		O100:H30	ST993	213	64629	5133960	United Kingdom: United Kingdom	human	PDS000024140.20	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_039947755.1	SAMN41574626	SRR29208759	01198109		O100:H30	ST993	240	60536	5002339	United Kingdom: United Kingdom	human	PDS000183839.1	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_039951795.1	SAMN41574618	SRR29208720	01198108		O9:H30	ST1294	241	84674	4960429	United Kingdom: United Kingdom	human	PDS000205861.1	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_03997695.1	SAMN41594209	SRR29226700	01189777		O8:H25	ST8660	157	109714	5044115	United Kingdom: United Kingdom	food	0	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_040279925.1	SAMN41992278	SRR29499143	01540777				208	66055	5032480	United Kingdom: United Kingdom	human	0	stx2_operon (PARTIAL); stx2_operon (PARTIAL_CONTIG_END); stxA2 (PARTIAL_CONTIG_ENDP); stxB2 (EXACTP)	StxOp933_Stx2i_43	Retained
GCA_040935505.2	SAMN42828538	SRR30000023	01571067		O9:H19	ST162	210	107518	5065515	United Kingdom: United Kingdom	human	PDS000189319.2	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_041047495.1	SAMN43015297	SRR30118389	01575887		O100:H30	ST993	196	75816	5022352	United Kingdom: United Kingdom	human	PDS000024140.20	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_041545175.1	SAMN43373094	SRR30407660	01591502		O100:H30	ST993	194	67199	5005039	United Kingdom: United Kingdom	human	PDS000024140.20	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_042177385.1	SAMN43879893	SRR30764002	01605180		O100:H30	ST993	228	74030	5021882	United Kingdom: United Kingdom	human	PDS000199344.1	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_042441235.1	SAMN43948175	SRR30813966	01608955		O100:H30	ST993	206	70529	5003763	United Kingdom: United Kingdom	human	0	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_042710215.1	SAMN39582411	SRR27704196	O8/O30:H25		O8:H25	ST8660	128	126866	5130797	New Zealand	clinical	0	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_042935545.1	SAMEA115376134	ERR12723348	EH4141		O8:H25	ST8660	91	259017	5156296	Belgium	human	PDS000177475.1	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_043574615.1	SAMN43267956	SRR30307958			O8:H25	ST8660	247	47609	5066315	Denmark	not collected	0	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_044682155.1	SAMN44596646	SRR31237447	01633873		O9:H30	ST1294	193	79185	4918943	United Kingdom: United Kingdom	human	PDS000205861.1	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_044986905.1	SAMN39582592	SRR27704367	O9:H30		O9:H30	ST1294	148	94586	4985226	New Zealand	clinical	0	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_045501555.1	SAMN39582387	SRR27704273	O8		O8:H25	ST8660	105	193198	5084238	New Zealand	clinical	0	stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_051797185.1	SAMN50428753	SRR34847042	01792145							United Kingdom: United Kingdom	human		stx2i operon (complete)	StxOp933_Stx2i_43	Retained

Assembly	BioSample	Illumina reads	Strain	CFIA isolate	Serotype (ECTyper)	Sequence type	Contigs	N50 (bp)	Genome size (bp)	Location	Isolation source	SNP cluster	MicroBIGG-E stx element call	stx2i operon (KMA)	Curation outcome
GCA_052635535.1	SAMN50709792	SRR35059917	01803611							United Kingdom: United Kingdom	human		stx2i operon (complete)	StxOp933_Stx2i_43	Retained
GCA_021079575.1	SAMN20192672		22_B_18_1_1				302	280040	5752183	Ireland: Roscommon	recto-anal junction	PDS000100880.1	stx2_operon (PARTIAL_CONTIG_END); stxB2 (EXACTX)		Excluded after KMA: stx1c at 100% identity, 68x depth; stx2i at 99.84%, 4x depth
GCA_027231485.1	SAMN13334911	SRR22858716	CFIACB20180603		-:H21	ST155	177	127128	5180457	Canada	Raw Ground Lamb	PDS000129627.5	stx2i operon (complete)		Excluded: CFIA deposition not resolved against the laboratory database
GCA_042949965.1	SAMN40646270	SRR28490835	CFIACB20210078	CFIACB20210078	O9:H30	ST1294	139	118172	5030392	Canada	Raw Ground Lamb	0	stx2i operon (complete)		Excluded: CFIA isolate represented by CFIACB20210075
GCA_021079515.1	SAMN20192675		24_B_1_1_2		O100:H17	ST3101	328	95959	6142888	Ireland: Galway	recto-anal junction	PDS000097200.2	stx2i operon (complete)		Excluded: contaminated (ConFindR)
GCA_911104275.1	SAMEA8949066				O100:H30	ST993	138	91088	5069700	Ireland	sheep recto-anal junction	PDS000100858.1	stx2i operon (complete)		Excluded: likely duplicate deposition
GCA_911104315.1	SAMEA8949089				O100:H30	ST993	138	111581	5002130	Ireland	sheep recto-anal junction	PDS000100872.1	stx2i operon (complete)		Excluded: likely duplicate deposition
GCA_911104325.1	SAMEA8949123				O100:H30	ST993	174	118168	5136243	Ireland	sheep recto-anal junction	PDS000100878.1	stx2i operon (complete)		Excluded: likely duplicate deposition
GCA_911104485.1	SAMEA8949194				O100:H30	ST993	325	76997	5144744	Ireland	sheep recto-anal junction	PDS000097185.2	stx2i operon (complete)		Excluded: likely duplicate deposition
GCA_911104685.1	SAMEA8949161				O100:H30	ST993	136	86914	5015677	Ireland	sheep recto-anal junction	PDS000097185.2	stx2i operon (complete)		Excluded: likely duplicate deposition
GCA_911104795.1	SAMEA8949198				O100:H30	ST993	495	76729	5270113	Ireland	sheep recto-anal junction	PDS000097197.2	stx2i operon (complete)		Excluded: likely duplicate deposition
GCA_911104835.1	SAMEA8949199				O100:H30	ST993	136	87466	4980855	Ireland	sheep recto-anal junction	PDS000097197.2	stx2i operon (complete)		Excluded: likely duplicate deposition
GCA_911104865.1	SAMEA8949200				O100:H17	ST3101	331	95959	6144909	Ireland	sheep recto-anal junction	PDS000097200.2	stx2i operon (complete)		Excluded: likely duplicate deposition
GCA_911104875.1	SAMEA8949218				O100:H30	ST993	362	24807	5021650	Ireland	sheep recto-anal junction	PDS000100862.1	stx2i operon (complete)		Excluded: likely duplicate deposition
GCA_911105015.1	SAMEA8949197						304	280040	5753752	Ireland	sheep recto-anal junction	PDS000100880.1	stx2_operon (PARTIAL_CONTIG_END); stxB2 (EXACTX)		Excluded: likely duplicate deposition
GCA_911198995.1	SAMEA8949211						300	155451	5528340	Ireland	sheep recto-anal junction	PDS000100811.1	stx2_operon (PARTIAL_CONTIG_END); stxA2 (PARTIAL_CONTIG_ENDX)		Excluded: likely duplicate deposition
GCA_963892125.1	SAMEA115018428				O9:H30	ST1294	117	128921	4991306	not collected		PDS000177433.1	stx2i operon (complete)		Excluded: likely duplicate deposition
GCA_027231285.1	SAMN13334913	SRR22858760	CFIACB20180605	CFIACB20180605	-:H21	ST155	164	87465	5141200	Canada	Raw Ground Lamb	PDS000129627.5	stx2i operon (complete)		Excluded: likely duplicate deposition of a CFIA isolate
GCA_027231345.1	SAMN13334912	SRR22858749	CFIACB20180604	CFIACB20180604	-:H21	ST155	166	99387	5141066	Canada	Raw Ground Lamb	PDS000129627.5	stx2i operon (complete)		Excluded: likely duplicate deposition of a CFIA isolate

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GCA_042951185.1	SAMN40646258	SRR28490848	CFIABF20190195	OLC3603	-:H21	ST155	132	177527	5145830	Canada	Raw Ground Lamb	PDS000129627.5	stx2i operon (complete)		Excluded: likely duplicate deposition of a CFIA isolate
GCA_042986525.1	SAMN40734924	SRR28538989	CFIABF20190194	OLC3602	-:H21	ST155	196	139198	5148595	Canada	Raw Ground Lamb	PDS000129627.5	stx2i operon (complete)		Excluded: likely duplicate deposition of a CFIA isolate
GCA_012392875.1	SAMN07629747	SRR6057747	PNUSAE009590				125	119434	4960454	USA		0	stxA2 (PARTIAL_CONTIG_ENDP)		Excluded: no complete stx2i operon call in MicroBIGG-E
GCA_012772735.1	SAMN05374548	SRR3829562	2015C-4851				105	119387	5241946	USA:ME	Manure/Bedding	0	stxA2 (PARTIAL_CONTIG_ENDP)		Excluded: no complete stx2i operon call in MicroBIGG-E
GCA_021079345.1	SAMN20192686		30_B_30_1_1				300	155451	5528340	Ireland: Donegal	recto-anal junction	PDS000100811.1	stx2_operon (PARTIAL_CONTIG_END); stxA2 (PARTIAL_CONTIG_ENDX)		Excluded: no complete stx2i operon call in MicroBIGG-E
GCA_025391635.1	SAMEA4644856	ERR7420565					200	89927	5236274	United Kingdom	Caecal Content	PDS000119125.1	stxA2 (PARTIAL_CONTIG_ENDP)		Excluded: no complete stx2i operon call in MicroBIGG-E
GCA_028217835.2	SAMEA112275716	ERR10712413					207	70349	5204517	Netherlands	manure	0	stxA2 (BLASTP)		Excluded: no complete stx2i operon call in MicroBIGG-E
GCA_037034625.2	SAMN40255891	SRR28220722	1831264				111	110727	4893081	United Kingdom: United Kingdom	human	0	stx2_operon (PARTIAL); stx2_operon (PARTIAL_CONTIG_END); stxB2 (EXACTP)		Excluded: no complete stx2i operon call in MicroBIGG-E
GCA_043762375.2	SAMD00808593	DRR589110	STEC_AC466				125	137660	5076902	Japan	stool	PDS000194155.3	stx2_operon (PARTIAL); stx2_operon (PARTIAL_CONTIG_END); stxB2 (EXACTP)		Excluded: no complete stx2i operon call in MicroBIGG-E
GCA_043775135.2	SAMD00808272	DRR588789	STEC_AC145				125	122608	5063682	Japan	stool	PDS000194155.3	stx2_operon (PARTIAL); stx2_operon (PARTIAL_CONTIG_END); stxB2 (EXACTP)		Excluded: no complete stx2i operon call in MicroBIGG-E
GCA_051180815.1	SAMN49760864	SRR34335669	PNUSAE211900				77	175806	4832019	USA		0	stx2_operon (PARTIAL_CONTIG_END); stxB2 (EXACTP)		Excluded: no complete stx2i operon call in MicroBIGG-E
GCA_051231105.1	SAMN49824977	SRR34401938	PNUSAE212229				89	132940	4824911	USA		0	stx2_operon (PARTIAL); stx2_operon (PARTIAL_CONTIG_END); stxB2 (EXACTP)		Excluded: no complete stx2i operon call in MicroBIGG-E
GCA_048062135.1	SAMN44956905		PSS2-5				161	150554	5076198	South Korea: Gyeonggi-do		PDS000220345.1	stx2_operon (PARTIAL_CONTIG_END); stxB2 (EXACTX)		Excluded: partial stx2i operon, no raw reads available
GCA_048062175.1	SAMN44956906		PSS2-6				172	156117	5078380	South Korea: Gyeonggi-do		PDS000220345.1	stx2_operon (PARTIAL_CONTIG_END); stxB2 (EXACTX)		Excluded: partial stx2i operon, no raw reads available
GCA_021079565.1	SAMN20192673		22_B_24_2_1		O100:H30	ST993	495	76729	5269917	Ireland: Galway	recto-anal junction	PDS000097197.2	stx2i operon (complete)		Excluded: poor assembly quality
GCA_039993315.1	SAMN41594206	SRR29225638	01191057		O100:H30	ST993	420	26416	5046639	United Kingdom: United Kingdom	food	PDS000024140.20	stx2i operon (complete)		Excluded: poor assembly quality
GCA_042949905.1	SAMN40646259	SRR28490847	CFIABF20190239	CFIABF20190239	O8:H19	ST162	12	2732064	5305006	Canada	Raw Ground Lamb	0	stx2i operon (complete)	StxOp933_Stx2i_43	Excluded: superseded by hybrid assembly 2021-MIN-0009

Assembly	BioSample	Illumina reads	Strain	CFIA isolate	Serotype (ECTyper)	Sequence type	Contigs	N50 (bp)	Genome size (bp)	Location	Isolation source	SNP cluster	MicroBIGG-E stx element call	stx2i operon (KMA)	Curation outcome
GCA_032267725.1	SAMN26419929	SRR18215048	CFIAFB20190172	OLC3504	O8:H25	ST8660	4	5075290	5177897	Canada	Raw Ground Lamb	0	stx2i operon (complete)	StxOp933_Stx2i_43	Excluded: superseded by hybrid assembly 2021-MIN-0249
GCA_027656605.1	SAMN11373501	SRR18215053	CFIAFB20180511	OLC3374	O9:H30	ST1294	3	5065598	5188405	Canada	Raw Ground Lamb	0	stx2i operon (complete)	StxOp933_Stx2i_43	Excluded: superseded by hybrid assembly 2021-MIN-0250
GCA_026888975.1	SAMN11373499	SRR22715050	CFIAFB20180512	OLC3375	--H21	ST155	3	5096570	5203767	Canada	Raw Ground Lamb	0	stx2i operon (complete)	StxOp933_Stx2i_43	Excluded: superseded by hybrid assembly 2021-MIN-0251
GCA_042950755.1	SAMN40646257	SRR28490891	CFIAFB20200034	CFIAFB20200034	--H21	ST155	7	5059451	5271659	Canada	Raw Ground Lamb	PDS000129627.5	stx2i operon (complete)	StxOp933_Stx2i_43	Excluded: superseded by hybrid assembly 2021-MIN-0252
GCA_027231365.1	SAMN13334890	SRR22858747	CFIAFB20180581	CFIAFB20180581	--H21	ST155	21	701209	5181639	Canada	Raw Ground Lamb	PDS000129627.5	stx2i operon (complete)	StxOp933_Stx2i_43	Excluded: superseded by hybrid assembly 2022-MIN-0036